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**Specific Gamma-Ray Dose  
Constants for Nuclides  
Important to Dosimetry and  
Radiological Assessment**

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**MASTER**

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**SPECIFIC GAMMA-RAY DOSE CONSTANTS FOR NUCLIDES  
IMPORTANT TO DOSIMETRY AND RADIOLOGICAL ASSESSMENT**

Laurie M. Unger\* and D. K. Trubey

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## PREFACE TO THE REVISED EDITION

The specific dose equivalent constants were recalculated when an error in the computer program was discovered. The error, omission of the third term in the polynomial representation of the flux-to-dose equivalent factor between 0.5 and 5 MeV, resulted in the dose factor being too high by 2% at 0.5 MeV, 0% at 1 MeV, and increasing to nearly 9% at 5 MeV. This is to be compared to the correct polynomial which gives results within 3% of the recommended standard values. The effect on the specific dose constant was much smaller, generally much less than 2%. The greatest error noted was for  $^{49}\text{Ca}$  which was too high by 4.5%.

Although errors of this magnitude are of little practical significance, the results have been corrected for this edition since a reprinting was necessary anyway.

Some other changes were made also. These included correcting a typographical error in the  $^{137}\text{Cs}$  value and inserting a space where data were too crowded together.

It may be noted that in calculating the lead shielding thickness, buildup was not included. The only justification for this is that these thicknesses were meant to be rough indication of the shielding problem and the buildup in lead at these energies is quite small.

Revised March 1982

## SPECIFIC GAMMA-RAY DOSE CONSTANTS FOR NUCLIDES IMPORTANT TO DOSIMETRY AND RADIOLOGICAL ASSESSMENT

The unshielded gamma-ray dose-equivalent rate at 1 meter from a point source, i.e., the specific gamma-ray dose constant, is a useful quantity in radiation protection applications. Recently, an extensive compilation of the nuclear data required to compute this constant has become available<sup>1</sup> for approximately 500 nuclides important to dosimetry and radiological assessment applications, and it has been used to compute a table of the specific gamma-ray dose constant. In addition, the half-life, mean attenuation coefficient, and thickness for 95% attenuation with a lead shield have been computed.

The data were computed on a different basis from earlier tabulations, e.g., Ref. 2. The dose equivalent rate is given in SI units as mSv/h for a unit source of 1 MBq. To convert to the previous common normalization, one may note that 1 mCi is equal to 37 MBq and 1 Sv is equal to 100 rem. That is, to convert data in units of (mSv/h)/MBq to (mrem/h)/ $\mu$ Ci, multiply by 3.7.

A second difference is that the conversion of gamma-ray flux density was taken to be:

$$\ln D(E) = A + B(\ln E) + C(\ln E)^2 + F(\ln E)^3 \quad (\text{rem/h})(\text{cm}^2\text{-s}) \quad (1)$$

where the energy is in units of MeV, and the constants are given in Table 1 taken from Ref. 3.

The specific gamma-ray dose constant,  $\Gamma$ , was summed over  $n$  according to

$$\Gamma = (1/4\pi R^2) \sum S_i D(E_i) \quad (\text{rem/h per Bq}) \quad (2)$$

where  $R = 100$  cm,

$n$  = number of gamma-rays emitted by the nuclide,

$S_i$  = emission probability of each gamma ray,

$E_i$  = energy of the gamma-ray (MeV),

$D(E_i)$  = dose rate per unit flux density from Eq. 1.

Only gamma-rays of energy greater than 0.01 MeV were included.

The mean attenuation coefficient was determined by first computing the thickness of lead required for 95% attenuation, i.e.,

$$0.05 \Gamma = \sum \Gamma_i \exp(-\mu(E_i) t) \quad (3)$$

where  $\Gamma_i = S_i D(E_i)/(4\pi R^2)$

$\mu(E_i)$  = linear attenuation coefficient from Ref. 4 for lead,

$t$  = thickness required (cm),

The value of  $t$  was determined by an iterative procedure, i.e., Newton's method. The value of the mean attenuation coefficient was determined by solving the following equation for  $\mu$ .

$$0.05 = \exp(-\mu t). \quad (4)$$

The results, presented in Table 2, were computed using computer-readable data from Ref. 1 available as DLC-80/DRALIST from the Radiation Shielding Information Center. The computations were performed using a computer program, SPEC-GAM, programmed by the first author for the ORNL Engineering Physics Information Centers (EPIC) Data General Eclipse S/130 computer in FORTRAN.

### ACKNOWLEDGEMENT

The authors are grateful to J. B. Zipperer, of the EPIC staff, for assistance in implementing the program on the Eclipse, and for his suggestions in regard to programming Newton's method.

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2. D. Nachtigall, "Table of Specific Gamma-ray Constants," Verlag Karl Thiernig KG, Munich (1969).
3. "Neutron and Gamma-ray Flux-to-Dose-Rate Factors," ANSI/ANS-6.1.1-1977, American Nuclear Society, La Grange Park, IL 60525, \$12
4. J. H. Hubbell, "Photon Cross Sections, Attenuation Coefficients, and Energy Absorption Coefficients From 10 keV to 100 GeV," NSRDS-NBS 29, National Bureau of Standards (1969).

TABLE I

## Gamma-Ray-Flux-to-Dose-Rate Conversion Factors

## Polynomial Coefficients in Analytic Form

$$\ln D(E) = A + Bx + Cx^2 + Fx^3$$

$$D(E) = (\text{rem/h})(\text{cm}^2\text{-s}), E = \text{Photon energy in MeV and } x = \ln E$$

| Photon Energy<br>(MeV) | A       | B        | C         | F        |
|------------------------|---------|----------|-----------|----------|
| 0.01 to 0.03           | -20.477 | -1.7454  |           |          |
| 0.03 to 0.5            | -13.626 | -0.57117 | -1.0954   | -0.24897 |
| 0.5 to 5.0             | -13.133 | 0.72008  | -0.033603 |          |
| 5.0 to 15.0            | -12.791 | 0.28309  | 0.10873   |          |

TABLE 2

## Specific Gamma-ray Dose Constants

Units: E - keV,  $\Gamma$  - (mSv/h)/MBq, T - cm,  $\mu$  - cm<sup>-1</sup>

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|--------|
| 7Be     | 53.4d     | 477.6  | 0.1042 | 9.292-6  | 1.594 | 1.879 | 42K     | 12.4h     | 312.7  | 0.0032 | 3.846-5  | 4.962 | 0.604  |
| 11C     | 20.5m     | 511.0  | 1.9950 | 1.908-4  | 1.768 | 1.695 |         |           | 1525.0 | 0.1790 |          |       |        |
| 13N     | 10.0m     | 511.0  | 1.9960 | 1.909-4  | 1.768 | 1.695 |         |           | 1446.0 | 0.0014 |          |       |        |
| 16N     | 7.1s      | 1755.0 | 0.0013 | 3.984-4  | 6.051 | 0.495 | 43K     | 22.6h     | 184.0  | 0.0027 | 1.803-4  | 1.937 | 1.546  |
|         |           | 2741.0 | 0.0076 |          |       |       |         |           | 220.6  | 0.0411 |          |       |        |
|         |           | 6129.0 | 0.6900 |          |       |       |         |           | 372.8  | 0.8727 |          |       |        |
|         |           | 7115.0 | 0.0500 |          |       |       |         |           | 396.9  | 0.1143 |          |       |        |
|         |           | 6334.0 | 0.0016 |          |       |       |         |           | 404.3  | 0.0011 |          |       |        |
| 15O     | 2.0m      | 511.0  | 1.9980 | 1.911-4  | 1.768 | 1.695 |         |           | 593.4  | 0.1103 |          |       |        |
| 18F     | 1.8h      | 511.0  | 1.9350 | 1.851-4  | 1.768 | 1.695 |         |           | 617.5  | 0.8051 |          |       |        |
| 22Na    | 2.6y      | 1274.0 | 0.9994 | 3.590-4  | 3.517 | 0.852 |         |           | 800.8  | 0.0015 |          |       |        |
|         |           | 511.0  | 1.7980 |          |       |       |         |           | 990.2  | 0.0033 |          |       |        |
| 24Na    | 15.0h     | 1368.0 | 1.0000 | 5.120-4  | 5.692 | 0.526 |         |           | 1015.0 | 0.0016 |          |       |        |
|         |           | 2754.0 | 0.9986 |          |       |       |         |           | 1022.0 | 0.0188 |          |       |        |
|         |           | 3824.0 | 0.0006 |          |       |       |         |           | 1394.0 | 0.0010 |          |       |        |
| 27Mg    | 9.5m      | 170.7  | 0.0084 | 1.448-4  | 3.447 | 0.869 | 45Ca    | 162.7d    | 12.4   | 0.0000 | 8.07-12  | 0.004 | 823.3  |
|         |           | 843.8  | 0.7180 |          |       |       |         |           |        |        |          |       |        |
|         |           | 1014.0 | 0.2800 |          |       |       | 47Ca    | 4.5d      | 489.2  | 0.0674 | 1.577-4  | 4.368 | 0.686  |
| 28Mg    | 20.9h     | 30.6   | 0.6600 | 2.371-4  | 3.926 | 0.763 |         |           | 530.4  | 0.0010 |          |       |        |
|         |           | 400.7  | 0.3660 |          |       |       |         |           | 767.0  | 0.0019 |          |       |        |
|         |           | 941.4  | 0.3830 |          |       |       |         |           | 807.9  | 0.0689 |          |       |        |
|         |           | 1342.0 | 0.5260 |          |       |       |         |           | 1297.0 | 0.7490 |          |       |        |
|         |           | 1373.0 | 0.0470 |          |       |       |         |           | 1542.0 | 0.0003 |          |       |        |
|         |           | 1589.0 | 0.0420 |          |       |       |         |           |        |        |          |       |        |
|         |           | 1620.0 | 0.0030 |          |       |       |         |           |        |        |          |       |        |
|         |           | 717.6  | 0.0013 |          |       |       | 49Ca    | 8.7m      | 856.1  | 0.0013 | 3.459-4  | 6.344 | 0.472  |
| 26Al    | 7.2+5y    | 1130.0 | 0.0250 | 3.995-4  | 4.597 | 0.652 |         |           | 1144.0 | 0.0011 |          |       |        |
|         |           | 1809.0 | 0.9976 |          |       |       |         |           | 1409.0 | 0.0063 |          |       |        |
|         |           | 2938.0 | 0.0024 |          |       |       |         |           | 2229.0 | 0.0019 |          |       |        |
|         |           | 511.0  | 1.6360 |          |       |       |         |           | 2372.0 | 0.0049 |          |       |        |
| 28Al    | 2.2m      | 1779.0 | 1.0000 | 2.358-4  | 5.467 | 0.548 |         |           | 3084.0 | 0.9210 |          |       |        |
| 31Si    | 2.6h      | 1266.0 | 0.0007 | 1.306-7  | 4.431 | 0.676 |         |           | 4072.0 | 0.0700 |          |       |        |
| 38Cl    | 37.2m     | 1642.0 | 0.3250 | 1.912-4  | 5.687 | 0.527 |         |           | 4738.0 | 0.0021 |          |       |        |
|         |           | 2167.0 | 0.4400 |          |       |       |         |           | 947.0  | 0.0018 |          |       |        |
|         |           | 3809.0 | 0.0002 |          |       |       | 44Sc    | 3.9h      | 1157.0 | 0.9988 | 3.573-4  | 3.272 | 0.915  |
| 41Ar    | 1.8h      | 1294.0 | 0.9916 | 1.877-4  | 4.492 | 0.667 |         |           | 1500.0 | 0.0091 |          |       |        |
|         |           | 1677.0 | 0.0005 |          |       |       |         |           | 2656.0 | 0.0011 |          |       |        |
| 40K     | 1.28+9y   | 1461.0 | 0.1067 | 2.197-5  | 4.843 | 0.619 |         |           | 2375.0 | 0.0000 |          |       |        |
|         |           |        |        |          |       |       |         |           | 511.0  | 1.8870 |          |       |        |
|         |           |        |        |          |       |       | 46Sc    | 83.8d     | 889.2  | 0.9998 | 3.154-4  | 3.808 | 0.787  |
|         |           |        |        |          |       |       |         |           | 1120.0 | 0.9999 |          |       |        |
|         |           |        |        |          |       |       |         |           | 2010.0 | 0.0000 |          |       |        |
|         |           |        |        |          |       |       | 46mSc   | 18.7s     | 142.5  | 0.6270 | 1.809-5  | 0.123 | 24.378 |
|         |           |        |        |          |       |       |         |           |        |        |          |       |        |
|         |           |        |        |          |       |       | 47Sc    | 3.4d      | 159.4  | 0.6800 | 2.170-5  | 0.162 | 18.513 |
|         |           |        |        |          |       |       |         |           |        |        |          |       |        |
|         |           |        |        |          |       |       | 48Sc    | 1.8d      | 175.4  | 0.0747 | 5.112-4  | 4.113 | 0.728  |
|         |           |        |        |          |       |       |         |           | 983.5  | 1.0000 |          |       |        |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 1038.0 | 0.9750 |          |       |        |         |           | 884.5  | 0.0037 |          |       |       |
|         |           | 1213.0 | 0.0238 |          |       |        |         |           | 511.0  | 0.5880 |          |       |       |
|         |           | 1312.0 | 1.0000 |          |       |        |         |           |        |        |          |       |       |
| 49Sc    | 57.4m     | 1739.0 | 0.0006 | 1.393-7  | 5.391 | 0.556  | 52Mn    | 21.4m     | 377.7  | 0.0168 | 3.866-4  | 3.767 | 0.795 |
|         |           |        |        |          |       |        |         |           | 1434.0 | 0.9822 |          |       |       |
|         |           |        |        |          |       |        |         |           | 1727.0 | 0.0022 |          |       |       |
| 44Ti    | 47.3y     | 67.8   | 0.9192 | 3.909-5  | 0.104 | 28.726 |         |           | 1644.0 | 0.0017 |          |       |       |
|         |           | 78.4   | 0.9762 |          |       |        |         |           | 511.0  | 1.9320 |          |       |       |
|         |           | 147.0  | 0.0010 |          |       |        |         |           |        |        |          |       |       |
| 45Ti    | 3.1h      | 720.3  | 0.0015 | 1.629-4  | 1.775 | 1.687  | 54Mn    | 312.7d    | 834.8  | 0.9998 | 1.381-4  | 3.216 | 0.932 |
|         |           | 1201.0 | 0.0020 |          |       |        | 56Mn    | 2.6h      | 846.8  | 0.9887 | 2.478-4  | 4.523 | 0.662 |
|         |           | 511.0  | 1.6970 |          |       |        |         |           | 1811.0 | 0.2719 |          |       |       |
|         |           |        |        |          |       |        |         |           | 2113.0 | 0.1434 |          |       |       |
| 51Ti    | 5.8m      | 320.1  | 0.9290 | 7.128-5  | 1.420 | 2.110  |         |           | 2523.0 | 0.0099 |          |       |       |
|         |           | 608.6  | 0.0118 |          |       |        |         |           | 2658.0 | 0.0065 |          |       |       |
|         |           | 928.6  | 0.0687 |          |       |        |         |           | 2960.0 | 0.0031 |          |       |       |
| 48V     | 16.0d     | 803.2  | 0.0015 | 4.577-4  | 3.893 | 0.770  |         |           | 3370.0 | 0.0017 |          |       |       |
|         |           | 928.3  | 0.0077 |          |       |        |         |           | 1351.0 | 0.0016 |          |       |       |
|         |           | 944.1  | 0.0776 |          |       |        | 57Mn    | 1.5m      | 14.4   | 0.1063 | 3.027-5  | 1.766 | 1.697 |
|         |           | 983.5  | 1.0000 |          |       |        |         |           | 122.1  | 0.1035 |          |       |       |
|         |           | 1312.0 | 0.9750 |          |       |        |         |           | 136.5  | 0.0143 |          |       |       |
|         |           | 1437.0 | 0.0012 |          |       |        |         |           | 230.3  | 0.0016 |          |       |       |
|         |           | 2240.0 | 0.0241 |          |       |        |         |           | 339.6  | 0.0013 |          |       |       |
|         |           | 2361.0 | 0.0003 |          |       |        |         |           | 352.3  | 0.0155 |          |       |       |
|         |           | 511.0  | 1.0020 |          |       |        |         |           | 366.7  | 0.0029 |          |       |       |
| 52V     | 3.7m      | 1334.0 | 0.0059 | 2.048-4  | 4.786 | 0.626  |         |           | 569.9  | 0.0038 |          |       |       |
|         |           | 1434.0 | 1.0000 |          |       |        |         |           | 692.0  | 0.0409 |          |       |       |
|         |           | 1531.0 | 0.0012 |          |       |        |         |           | 706.4  | 0.0018 |          |       |       |
|         |           | 1006.0 | 0.0012 |          |       |        |         |           | 870.7  | 0.0019 |          |       |       |
| 49Cr    | 42.1m     | 62.3   | 0.1639 | 2.002-4  | 1.696 | 1.766  |         |           | 992.7  | 0.0011 |          |       |       |
|         |           | 90.6   | 0.5320 |          |       |        |         |           | 1260.0 | 0.0024 |          |       |       |
|         |           | 152.9  | 0.3032 |          |       |        |         |           | 1613.0 | 0.0054 |          |       |       |
|         |           | 1451.0 | 0.0012 |          |       |        |         |           | 1725.0 | 0.0012 |          |       |       |
|         |           | 511.0  | 1.8380 |          |       |        |         |           | 952.4  | 0.0010 |          |       |       |
| 51Cr    | 27.7d     | 320.1  | 0.0983 | 6.320-6  | 0.782 | 3.833  | 52Fe    | 8.3h      | 168.7  | 0.9660 | 1.397-4  | 1.611 | 1.859 |
| 52Mn    | 5.6d      | 346.0  | 0.0098 | 5.409-4  | 3.878 | 0.773  |         |           | 511.0  | 1.1200 |          |       |       |
|         |           | 399.6  | 0.0018 |          |       |        | 59Fe    | 44.6d     | 142.7  | 0.0103 | 1.787-4  | 4.249 | 0.705 |
|         |           | 502.0  | 0.0021 |          |       |        |         |           | 192.3  | 0.0311 |          |       |       |
|         |           | 600.2  | 0.0039 |          |       |        |         |           | 334.8  | 0.0026 |          |       |       |
|         |           | 647.4  | 0.0040 |          |       |        |         |           | 1099.0 | 0.5650 |          |       |       |
|         |           | 744.2  | 0.9000 |          |       |        |         |           | 1292.0 | 0.4320 |          |       |       |
|         |           | 848.1  | 0.0332 |          |       |        |         |           | 1228.0 | 0.0009 |          |       |       |
|         |           | 935.5  | 0.9450 |          |       |        | 56Co    | 78.8d     | 733.6  | 0.0019 | 5.145-4  | 4.645 | 0.645 |
|         |           | 1246.0 | 0.0421 |          |       |        |         |           | 787.8  | 0.0031 |          |       |       |
|         |           | 1248.0 | 0.0038 |          |       |        |         |           | 846.8  | 0.9996 |          |       |       |
|         |           | 1334.0 | 0.0507 |          |       |        |         |           | 977.4  | 0.0143 |          |       |       |
|         |           | 1434.0 | 1.0000 |          |       |        |         |           | 996.9  | 0.0014 |          |       |       |
|         |           |        |        |          |       |        |         |           | 1038.0 | 0.1403 |          |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 1140.0 | 0.0013 |          |       |        |         |           | 269.5  | 0.3650 |          |       |       |
|         |           | 1175.0 | 0.0228 |          |       |        |         |           | 480.4  | 0.3650 |          |       |       |
|         |           | 1238.0 | 0.6697 |          |       |        |         |           | 749.9  | 0.4950 |          |       |       |
|         |           | 1335.0 | 0.0012 |          |       |        |         |           | 811.9  | 0.8600 |          |       |       |
|         |           | 1360.0 | 0.0429 |          |       |        |         |           | 1562.0 | 0.1400 |          |       |       |
|         |           | 1443.0 | 0.0017 |          |       |        |         |           |        |        |          |       |       |
|         |           | 1771.0 | 0.1551 |          |       |        | 57Ni    | 1.5d      | 127.2  | 0.1293 | 2.887-4  | 4.399 | 0.681 |
|         |           | 1811.0 | 0.0065 |          |       |        |         |           | 1046.0 | 0.0012 |          |       |       |
|         |           | 1964.0 | 0.0071 |          |       |        |         |           | 1378.0 | 0.7790 |          |       |       |
|         |           | 2015.0 | 0.0303 |          |       |        |         |           | 1757.0 | 0.0709 |          |       |       |
|         |           | 2035.0 | 0.0778 |          |       |        |         |           | 1919.0 | 0.1472 |          |       |       |
|         |           | 2113.0 | 0.0038 |          |       |        |         |           | 2804.0 | 0.0013 |          |       |       |
|         |           | 2213.0 | 0.0039 |          |       |        |         |           | 1193.0 | 0.0050 |          |       |       |
|         |           | 2276.0 | 0.0012 |          |       |        |         |           | 511.0  | 0.8016 |          |       |       |
|         |           | 2599.0 | 0.1689 |          |       |        |         |           |        |        |          |       |       |
|         |           | 3010.0 | 0.0106 |          |       |        | 65Ni    | 2.5h      | 366.3  | 0.0461 | 8.009-5  | 4.571 | 0.655 |
|         |           | 3202.0 | 0.0318 |          |       |        |         |           | 507.8  | 0.0029 |          |       |       |
|         |           | 3253.0 | 0.0779 |          |       |        |         |           | 609.3  | 0.0014 |          |       |       |
|         |           | 3273.0 | 0.0185 |          |       |        |         |           | 1116.0 | 0.1483 |          |       |       |
|         |           | 3451.0 | 0.0093 |          |       |        |         |           | 1482.0 | 0.2350 |          |       |       |
|         |           | 3548.0 | 0.0019 |          |       |        |         |           | 1623.0 | 0.0047 |          |       |       |
|         |           | 1453.0 | 0.0071 |          |       |        |         |           | 1725.0 | 0.0039 |          |       |       |
|         |           | 511.0  | 0.3950 |          |       |        |         |           | 814.8  | 0.0017 |          |       |       |
| 57Co    | 270.9d    | 14.4   | 0.0954 | 4.087-5  | 0.075 | 39.809 | 61Cu    | 3.4h      | 67.4   | 0.0387 | 1.517-4  | 2.001 | 1.497 |
|         |           | 122.1  | 0.8551 |          |       |        |         |           | 283.0  | 0.1230 |          |       |       |
|         |           | 136.5  | 0.1060 |          |       |        |         |           | 373.0  | 0.0212 |          |       |       |
|         |           | 692.0  | 0.0016 |          |       |        |         |           | 529.2  | 0.0041 |          |       |       |
|         |           | 536.0  | 0.0003 |          |       |        |         |           | 588.6  | 0.0118 |          |       |       |
|         |           |        |        |          |       |        |         |           | 656.0  | 0.1049 |          |       |       |
| 58Co    | 70.8d     | 810.8  | 0.9943 | 1.652-4  | 2.973 | 1.008  |         |           | 816.7  | 0.0036 |          |       |       |
|         |           | 153.9  | 0.0074 |          |       |        |         |           | 841.2  | 0.0024 |          |       |       |
|         |           | 1675.0 | 0.0054 |          |       |        |         |           | 908.6  | 0.0119 |          |       |       |
|         |           | 511.0  | 0.2986 |          |       |        |         |           | 1100.0 | 0.0028 |          |       |       |
| 58mCo   | 9.1h      | 24.9   | 0.0004 | 2.637-8  | 0.006 | 528.9  |         |           | 1185.0 | 0.0363 |          |       |       |
|         |           |        |        |          |       |        |         |           | 1198.0 | 0.0082 |          |       |       |
|         |           |        |        |          |       |        |         |           | 511.0  | 1.2290 |          |       |       |
| 60Co    | 5.3y      | 1173.0 | 1.0000 | 3.697-4  | 4.412 | 0.679  | 62Cu    | 9.7m      | 875.7  | 0.0015 | 1.881-4  | 1.784 | 1.679 |
|         |           | 1333.0 | 1.0000 |          |       |        |         |           | 1173.0 | 0.0034 |          |       |       |
|         |           | 693.8  | 0.0002 |          |       |        |         |           | 1966.0 | 0.0009 |          |       |       |
| 60mCo   | 10.5m     | 58.6   | 0.0202 | 9.044-7  | 3.578 | 0.837  |         |           | 511.0  | 1.9560 |          |       |       |
|         |           | 1333.0 | 0.0024 |          |       |        | 64Cu    | 12.7h     | 1346.0 | 0.0049 | 3.514-5  | 1.856 | 1.614 |
|         |           | 947.4  | 0.0001 |          |       |        |         |           | 511.0  | 0.3574 |          |       |       |
| 61Co    | 1.6h      | 67.4   | 0.8500 | 2.286-5  | 1.756 | 1.706  | 67Cu    | 61.9d     | 91.3   | 0.0700 | 2.363-5  | 0.230 |       |
|         |           | 283.0  | 0.0012 |          |       |        |         |           | 93.3   | 0.1610 |          |       |       |
|         |           | 625.6  | 0.0012 |          |       |        |         |           | 184.6  | 0.4870 |          |       |       |
|         |           | 841.2  | 0.0059 |          |       |        |         |           | 208.9  | 0.0012 |          |       |       |
|         |           | 908.6  | 0.0298 |          |       |        |         |           | 300.2  | 0.0080 |          |       |       |
|         |           | 215.6  | 0.0000 |          |       |        |         |           | 393.5  | 0.0022 |          |       |       |
| 56Ni    | 6.1d      | 158.4  | 0.9879 | 2.936-4  | 2.988 | 1.003  |         |           |        |        |          |       |       |



| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
| 62Zn    | 9.3h      | 40.8   | 0.2695 | 8.990-5  | 1.882 | 1.592 |         |           | 3792.0 | 0.0104 |          |       |       |
|         |           | 243.4  | 0.0267 |          |       |       |         |           | 4086.0 | 0.0116 |          |       |       |
|         |           | 247.0  | 0.0201 |          |       |       |         |           | 4296.0 | 0.0356 |          |       |       |
|         |           | 260.5  | 0.0143 |          |       |       |         |           | 4462.0 | 0.0073 |          |       |       |
|         |           | 304.9  | 0.0031 |          |       |       |         |           | 4807.0 | 0.0151 |          |       |       |
|         |           | 349.6  | 0.0048 |          |       |       |         |           | 1349.0 | 0.0105 |          |       |       |
|         |           | 394.1  | 0.0236 |          |       |       |         |           | 511.0  | 1.1050 |          |       |       |
|         |           | 507.6  | 0.1568 |          |       |       |         |           |        |        |          |       |       |
|         |           | 548.4  | 0.1622 |          |       |       |         |           |        |        |          |       |       |
|         |           | 596.7  | 0.2750 |          |       |       | 67Ga    | 3.3d      | 91.3   | 0.0286 | 3.004-5  | 0.628 | 4.769 |
|         |           | 637.4  | 0.0027 |          |       |       |         |           | 93.3   | 0.3570 |          |       |       |
|         |           | 921.2  | 0.0020 |          |       |       |         |           | 184.6  | 0.1971 |          |       |       |
|         |           | 511.0  | 0.1520 |          |       |       |         |           | 208.9  | 0.0224 |          |       |       |
| 65Zn    | 244.4d    | 1116.0 | 0.5075 | 8.924-5  | 4.057 | 0.738 |         |           | 300.2  | 0.1599 |          |       |       |
|         |           | 557.7  | 0.0001 |          |       |       |         |           | 393.5  | 0.0448 |          |       |       |
|         |           | 511.0  | 0.0283 |          |       |       |         |           | 887.7  | 0.0014 |          |       |       |
|         |           |        |        |          |       |       |         |           | 629.4  | 0.0012 |          |       |       |
| 69Zn    | 55.6m     | 413.9  | 0.0030 | 1.168-9  | 1.268 | 2.362 | 68Ga    | 1.1h      | 1077.0 | 0.0329 | 1.763-4  | 1.863 | 1.608 |
| 69mZn   | 13.8h     | 438.6  | 0.9489 | 7.983-5  | 1.391 | 2.153 |         |           | 1883.0 | 0.0014 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1014.0 | 0.0024 |          |       |       |
|         |           |        |        |          |       |       |         |           | 511.0  | 1.7780 |          |       |       |
| 66Ga    | 9.4h      | 448.9  | 0.0011 | 3.422-4  | 4.842 | 0.619 | 72Ga    | 14.1h     | 11.0   | 0.0003 | 3.895-4  | 4.642 | 0.645 |
|         |           | 686.3  | 0.0026 |          |       |       |         |           | 112.5  | 0.0014 |          |       |       |
|         |           | 833.6  | 0.0619 |          |       |       |         |           | 289.5  | 0.0020 |          |       |       |
|         |           | 856.7  | 0.0012 |          |       |       |         |           | 336.6  | 0.0011 |          |       |       |
|         |           | 907.0  | 0.0012 |          |       |       |         |           | 381.2  | 0.0028 |          |       |       |
|         |           | 1039.0 | 0.3880 |          |       |       |         |           | 428.4  | 0.0018 |          |       |       |
|         |           | 1190.0 | 0.0014 |          |       |       |         |           | 587.4  | 0.0012 |          |       |       |
|         |           | 1233.0 | 0.0054 |          |       |       |         |           | 600.8  | 0.0559 |          |       |       |
|         |           | 1333.0 | 0.0127 |          |       |       |         |           | 629.9  | 0.2439 |          |       |       |
|         |           | 1356.0 | 0.0038 |          |       |       |         |           | 735.6  | 0.0036 |          |       |       |
|         |           | 1357.0 | 0.0013 |          |       |       |         |           | 786.4  | 0.0317 |          |       |       |
|         |           | 1357.0 | 0.0019 |          |       |       |         |           | 810.2  | 0.0201 |          |       |       |
|         |           | 1419.0 | 0.0065 |          |       |       |         |           | 834.0  | 0.9565 |          |       |       |
|         |           | 1459.0 | 0.0010 |          |       |       |         |           | 861.1  | 0.0091 |          |       |       |
|         |           | 1508.0 | 0.0059 |          |       |       |         |           | 894.2  | 0.0985 |          |       |       |
|         |           | 1899.0 | 0.0044 |          |       |       |         |           | 924.1  | 0.0014 |          |       |       |
|         |           | 1919.0 | 0.0219 |          |       |       |         |           | 939.4  | 0.0026 |          |       |       |
|         |           | 2174.0 | 0.0012 |          |       |       |         |           | 970.5  | 0.0110 |          |       |       |
|         |           | 2190.0 | 0.0582 |          |       |       |         |           | 999.9  | 0.0080 |          |       |       |
|         |           | 2214.0 | 0.0014 |          |       |       |         |           | 1051.0 | 0.0692 |          |       |       |
|         |           | 2393.0 | 0.0026 |          |       |       |         |           | 1215.0 | 0.0080 |          |       |       |
|         |           | 2423.0 | 0.0199 |          |       |       |         |           | 1231.0 | 0.0144 |          |       |       |
|         |           | 2752.0 | 0.2371 |          |       |       |         |           | 1260.0 | 0.0115 |          |       |       |
|         |           | 2781.0 | 0.0013 |          |       |       |         |           | 1277.0 | 0.0156 |          |       |       |
|         |           | 2934.0 | 0.0022 |          |       |       |         |           | 1464.0 | 0.0356 |          |       |       |
|         |           | 3229.0 | 0.0152 |          |       |       |         |           | 1568.0 | 0.0020 |          |       |       |
|         |           | 3257.0 | 0.0010 |          |       |       |         |           | 1572.0 | 0.0083 |          |       |       |
|         |           | 3381.0 | 0.0145 |          |       |       |         |           | 1597.0 | 0.0424 |          |       |       |
|         |           | 3423.0 | 0.0084 |          |       |       |         |           | 1681.0 | 0.0087 |          |       |       |
|         |           | 3433.0 | 0.0029 |          |       |       |         |           | 1711.0 | 0.0038 |          |       |       |
|         |           | 3767.0 | 0.0014 |          |       |       |         |           | 1838.0 | 0.0020 |          |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---|-------|
|         |           | 1861.0 | 0.0523 |          |       |       |         |           | 714.3  | 0.0707 |          |   |       |
|         |           | 1878.0 | 0.0023 |          |       |       |         |           | 743.7  | 0.0018 |          |   |       |
|         |           | 1920.0 | 0.0016 |          |       |       |         |           | 745.8  | 0.0096 |          |   |       |
|         |           | 1991.0 | 0.0011 |          |       |       |         |           | 749.9  | 0.0087 |          |   |       |
|         |           | 2029.0 | 0.0012 |          |       |       |         |           | 766.7  | 0.0078 |          |   |       |
|         |           | 2109.0 | 0.0103 |          |       |       |         |           | 781.3  | 0.0100 |          |   |       |
|         |           | 2202.0 | 0.2611 |          |       |       |         |           | 784.8  | 0.0130 |          |   |       |
|         |           | 2214.0 | 0.0019 |          |       |       |         |           | 794.3  | 0.0027 |          |   |       |
|         |           | 2491.0 | 0.0748 |          |       |       |         |           | 810.4  | 0.0224 |          |   |       |
|         |           | 2508.0 | 0.1282 |          |       |       |         |           | 813.4  | 0.0013 |          |   |       |
|         |           | 2515.0 | 0.0025 |          |       |       |         |           | 823.1  | 0.0059 |          |   |       |
|         |           | 2622.0 | 0.0013 |          |       |       |         |           | 843.2  | 0.0021 |          |   |       |
|         |           | 2844.0 | 0.0041 |          |       |       |         |           | 875.2  | 0.0077 |          |   |       |
|         |           | 1275.0 | 0.0155 |          |       |       |         |           | 896.5  | 0.0012 |          |   |       |
| 68Ge    | 288d      | 10.3   | 0.0546 | 1.634-5  | 0.002 | 1345. |         |           | 901.0  | 0.0012 |          |   |       |
| 71Ge    | 11.8d     | 10.3   | 0.0552 | 1.653-5  | 0.002 | 1345  |         |           | 907.0  | 0.0094 |          |   |       |
| 77Ge    | 11.3h     | 10.5   | 0.0057 | 1.929-4  | 2.748 | 1.090 |         |           | 913.8  | 0.0036 |          |   |       |
|         |           | 10.5   | 0.0111 |          |       |       |         |           | 923.1  | 0.0068 |          |   |       |
|         |           | 11.7   | 0.0026 |          |       |       |         |           | 925.5  | 0.0071 |          |   |       |
|         |           | 156.4  | 0.0079 |          |       |       |         |           | 928.8  | 0.0103 |          |   |       |
|         |           | 159.1  | 0.0023 |          |       |       |         |           | 939.4  | 0.0028 |          |   |       |
|         |           | 177.3  | 0.0018 |          |       |       |         |           | 996.6  | 0.0010 |          |   |       |
|         |           | 194.8  | 0.0175 |          |       |       |         |           | 1062.0 | 0.0015 |          |   |       |
|         |           | 209.0  | 0.0093 |          |       |       |         |           | 1081.0 | 0.0024 |          |   |       |
|         |           | 211.0  | 0.3049 |          |       |       |         |           | 1085.0 | 0.0598 |          |   |       |
|         |           | 215.5  | 0.2830 |          |       |       |         |           | 1115.0 | 0.0010 |          |   |       |
|         |           | 219.1  | 0.0029 |          |       |       |         |           | 1125.0 | 0.0012 |          |   |       |
|         |           | 254.7  | 0.0021 |          |       |       |         |           | 1152.0 | 0.0019 |          |   |       |
|         |           | 264.4  | 0.5330 |          |       |       |         |           | 1193.0 | 0.0254 |          |   |       |
|         |           | 268.1  | 0.0059 |          |       |       |         |           | 1215.0 | 0.0013 |          |   |       |
|         |           | 337.6  | 0.0023 |          |       |       |         |           | 1242.0 | 0.0039 |          |   |       |
|         |           | 338.7  | 0.0066 |          |       |       |         |           | 1264.0 | 0.0084 |          |   |       |
|         |           | 367.4  | 0.1386 |          |       |       |         |           | 1280.0 | 0.0017 |          |   |       |
|         |           | 416.3  | 0.2159 |          |       |       |         |           | 1309.0 | 0.0048 |          |   |       |
|         |           | 419.7  | 0.0122 |          |       |       |         |           | 1313.0 | 0.0035 |          |   |       |
|         |           | 439.4  | 0.0020 |          |       |       |         |           | 1320.0 | 0.0030 |          |   |       |
|         |           | 461.4  | 0.0125 |          |       |       |         |           | 1368.0 | 0.0331 |          |   |       |
|         |           | 475.4  | 0.0098 |          |       |       |         |           | 1453.0 | 0.0012 |          |   |       |
|         |           | 520.0  | 0.0029 |          |       |       |         |           | 1476.0 | 0.0024 |          |   |       |
|         |           | 558.0  | 0.1588 |          |       |       |         |           | 1479.0 | 0.0013 |          |   |       |
|         |           | 582.5  | 0.0077 |          |       |       |         |           | 1496.0 | 0.0049 |          |   |       |
|         |           | 614.4  | 0.0050 |          |       |       |         |           | 1539.0 | 0.0014 |          |   |       |
|         |           | 624.8  | 0.0018 |          |       |       |         |           | 1574.0 | 0.0065 |          |   |       |
|         |           | 631.8  | 0.0689 |          |       |       |         |           | 1710.0 | 0.0030 |          |   |       |
|         |           | 634.4  | 0.0206 |          |       |       |         |           | 1720.0 | 0.0039 |          |   |       |
|         |           | 673.1  | 0.0053 |          |       |       |         |           | 1727.0 | 0.0015 |          |   |       |
|         |           | 673.1  | 0.0013 |          |       |       |         |           | 1846.0 | 0.0017 |          |   |       |
|         |           | 698.5  | 0.0023 |          |       |       |         |           | 2000.0 | 0.0055 |          |   |       |
|         |           | 705.2  | 0.0011 |          |       |       |         |           | 2077.0 | 0.0023 |          |   |       |
|         |           | 712.4  | 0.0082 |          |       |       |         |           | 2090.0 | 0.0024 |          |   |       |
|         |           |        |        |          |       |       |         |           | 2126.0 | 0.0020 |          |   |       |
|         |           |        |        |          |       |       |         |           | 2342.0 | 0.0047 |          |   |       |
|         |           |        |        |          |       |       |         |           | 1101.0 | 0.0291 |          |   |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|
| 72As    | 1.1d      | 11.0   | 0.0080 | 3.118-4  | 2.708 | 1.106 |
|         |           | 600.8  | 0.0031 |          |       |       |
|         |           | 629.9  | 0.0786 |          |       |       |
|         |           | 786.4  | 0.0047 |          |       |       |
|         |           | 834.0  | 0.7970 |          |       |       |
|         |           | 894.2  | 0.0077 |          |       |       |
|         |           | 1051.0 | 0.0099 |          |       |       |
|         |           | 1215.0 | 0.0021 |          |       |       |
|         |           | 1390.0 | 0.0024 |          |       |       |
|         |           | 1464.0 | 0.0110 |          |       |       |
|         |           | 1476.0 | 0.0051 |          |       |       |
|         |           | 1568.0 | 0.0013 |          |       |       |
|         |           | 1681.0 | 0.0012 |          |       |       |
|         |           | 1711.0 | 0.0024 |          |       |       |
|         |           | 1991.0 | 0.0034 |          |       |       |
|         |           | 2106.0 | 0.0063 |          |       |       |
|         |           | 2109.0 | 0.0027 |          |       |       |
|         |           | 2202.0 | 0.0047 |          |       |       |
|         |           | 2248.0 | 0.0031 |          |       |       |
|         |           | 2508.0 | 0.0032 |          |       |       |
|         |           | 2622.0 | 0.0039 |          |       |       |
|         |           | 2940.0 | 0.0029 |          |       |       |
|         |           | 3804.0 | 0.0010 |          |       |       |
|         |           | 1929.0 | 0.0185 |          |       |       |
|         |           | 511.0  | 1.7610 |          |       |       |

|      |       |      |        |         |       |       |
|------|-------|------|--------|---------|-------|-------|
| 73As | 80.3d | 11.0 | 0.1326 | 3.784-5 | 0.005 | 662.1 |
|      |       | 53.4 | 0.1030 |         |       |       |
|      |       | 13.3 | 0.0009 |         |       |       |

|      |       |        |        |         |       |       |
|------|-------|--------|--------|---------|-------|-------|
| 74As | 17.8d | 11.0   | 0.0222 | 1.456-4 | 2.031 | 1.475 |
|      |       | 595.9  | 0.5985 |         |       |       |
|      |       | 608.4  | 0.0055 |         |       |       |
|      |       | 1204.0 | 0.0029 |         |       |       |
|      |       | 1194.0 | 0.0009 |         |       |       |
|      |       | 511.0  | 0.5923 |         |       |       |
|      |       | 634.8  | 0.1543 |         |       |       |
|      |       | 685.4  | 0.0003 |         |       |       |

|      |      |        |        |         |       |       |
|------|------|--------|--------|---------|-------|-------|
| 76As | 1.1d | 559.1  | 0.4470 | 7.342-5 | 2.894 | 1.035 |
|      |      | 563.2  | 0.0117 |         |       |       |
|      |      | 571.3  | 0.0014 |         |       |       |
|      |      | 657.0  | 0.0608 |         |       |       |
|      |      | 665.3  | 0.0039 |         |       |       |
|      |      | 740.1  | 0.0012 |         |       |       |
|      |      | 771.8  | 0.0012 |         |       |       |
|      |      | 867.6  | 0.0013 |         |       |       |
|      |      | 1130.0 | 0.0014 |         |       |       |
|      |      | 1213.0 | 0.0163 |         |       |       |
|      |      | 1216.0 | 0.0384 |         |       |       |
|      |      | 1229.0 | 0.0139 |         |       |       |
|      |      | 1439.0 | 0.0033 |         |       |       |
|      |      | 1454.0 | 0.0013 |         |       |       |
|      |      | 1788.0 | 0.0033 |         |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 2096.0 | 0.0066 |          |       |       |
|         |           | 2111.0 | 0.0039 |          |       |       |
|         |           | 1152.0 | 0.0065 |          |       |       |
| 77As    | 1.6d      | 87.9   | 0.0020 | 1.691-6  | 1.182 | 2.534 |
|         |           | 161.9  | 0.0013 |          |       |       |
|         |           | 239.0  | 0.0157 |          |       |       |
|         |           | 249.8  | 0.0042 |          |       |       |
|         |           | 520.7  | 0.0061 |          |       |       |
|         |           | 245.5  | 0.0010 |          |       |       |
| 73Se    | 7.1h      | 10.5   | 0.0841 | 2.950-4  | 1.350 | 2.218 |
|         |           | 10.5   | 0.1639 |          |       |       |
|         |           | 11.7   | 0.0379 |          |       |       |
|         |           | 67.0   | 0.7730 |          |       |       |
|         |           | 361.1  | 0.9650 |          |       |       |
|         |           | 509.5  | 0.0106 |          |       |       |
|         |           | 764.4  | 0.0014 |          |       |       |
|         |           | 865.4  | 0.0047 |          |       |       |
|         |           | 901.2  | 0.0014 |          |       |       |
|         |           | 1111.0 | 0.0018 |          |       |       |
|         |           | 1423.0 | 0.0014 |          |       |       |
|         |           | 894.4  | 0.0042 |          |       |       |
|         |           | 511.0  | 1.3170 |          |       |       |

|      |        |       |        |         |       |       |
|------|--------|-------|--------|---------|-------|-------|
| 75Se | 119.8d | 10.5  | 0.1611 | 2.323-4 | 0.330 | 9.079 |
|      |        | 10.5  | 0.3140 |         |       |       |
|      |        | 11.7  | 0.0727 |         |       |       |
|      |        | 66.0  | 0.0102 |         |       |       |
|      |        | 96.7  | 0.0341 |         |       |       |
|      |        | 121.1 | 0.1674 |         |       |       |
|      |        | 136.0 | 0.5920 |         |       |       |
|      |        | 198.6 | 0.0145 |         |       |       |
|      |        | 264.6 | 0.5980 |         |       |       |
|      |        | 279.5 | 0.2524 |         |       |       |
|      |        | 303.9 | 0.0132 |         |       |       |
|      |        | 400.7 | 0.1142 |         |       |       |
|      |        | 332.7 | 0.0010 |         |       |       |

|      |      |       |        |         |       |       |
|------|------|-------|--------|---------|-------|-------|
| 77Br | 2.4d | 11.2  | 0.1543 | 1.919-4 | 0.972 | 3.083 |
|      |      | 11.2  | 0.3003 |         |       |       |
|      |      | 12.5  | 0.0718 |         |       |       |
|      |      | 87.9  | 0.0140 |         |       |       |
|      |      | 138.9 | 0.0013 |         |       |       |
|      |      | 161.9 | 0.0110 |         |       |       |
|      |      | 180.7 | 0.0028 |         |       |       |
|      |      | 200.4 | 0.0121 |         |       |       |
|      |      | 239.0 | 0.2310 |         |       |       |
|      |      | 249.8 | 0.0298 |         |       |       |
|      |      | 270.8 | 0.0032 |         |       |       |
|      |      | 281.7 | 0.0229 |         |       |       |
|      |      | 297.2 | 0.0416 |         |       |       |
|      |      | 303.8 | 0.0118 |         |       |       |



| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 1833.0 | 0.0015 |          |       |       |         |           | 15.0   | 0.0002 |          |       |       |
|         |           | 789.3  | 0.0108 |          |       |       |         |           | 402.6  | 0.4950 |          |       |       |
|         |           |        |        |          |       |       |         |           | 673.9  | 0.0191 |          |       |       |
| 79Kr    | 1.5d      | 11.9   | 0.1491 | 1.628-4  | 0.950 | 3.154 |         |           | 814.2  | 0.0017 |          |       |       |
|         |           | 11.9   | 0.2894 |          |       |       |         |           | 836.4  | 0.0075 |          |       |       |
|         |           | 13.3   | 0.0715 |          |       |       |         |           | 845.4  | 0.0728 |          |       |       |
|         |           | 44.2   | 0.0021 |          |       |       |         |           | 946.6  | 0.0014 |          |       |       |
|         |           | 136.0  | 0.0100 |          |       |       |         |           | 1175.0 | 0.0112 |          |       |       |
|         |           | 180.2  | 0.0010 |          |       |       |         |           | 1338.0 | 0.0065 |          |       |       |
|         |           | 208.5  | 0.0078 |          |       |       |         |           | 1382.0 | 0.0029 |          |       |       |
|         |           | 217.0  | 0.0240 |          |       |       |         |           | 1390.0 | 0.0012 |          |       |       |
|         |           | 261.3  | 0.1270 |          |       |       |         |           | 1531.0 | 0.0036 |          |       |       |
|         |           | 299.5  | 0.0157 |          |       |       |         |           | 1578.0 | 0.0013 |          |       |       |
|         |           | 306.3  | 0.0260 |          |       |       |         |           | 1611.0 | 0.0010 |          |       |       |
|         |           | 344.7  | 0.0024 |          |       |       |         |           | 1741.0 | 0.0205 |          |       |       |
|         |           | 389.0  | 0.0152 |          |       |       |         |           | 1843.0 | 0.0014 |          |       |       |
|         |           | 397.6  | 0.0950 |          |       |       |         |           | 2012.0 | 0.0290 |          |       |       |
|         |           | 523.0  | 0.0025 |          |       |       |         |           | 2408.0 | 0.0021 |          |       |       |
|         |           | 525.3  | 0.0043 |          |       |       |         |           | 2555.0 | 0.0931 |          |       |       |
|         |           | 606.1  | 0.0810 |          |       |       |         |           | 2558.0 | 0.0391 |          |       |       |
|         |           | 832.0  | 0.0126 |          |       |       |         |           | 2811.0 | 0.0032 |          |       |       |
|         |           | 934.8  | 0.0013 |          |       |       |         |           | 3309.0 | 0.0045 |          |       |       |
|         |           | 1026.0 | 0.0016 |          |       |       |         |           | 1620.0 | 0.0065 |          |       |       |
|         |           | 1115.0 | 0.0037 |          |       |       |         |           |        |        |          |       |       |
|         |           | 1332.0 | 0.0044 |          |       |       |         |           |        |        |          |       |       |
|         |           | 788.6  | 0.0055 |          |       |       |         |           |        |        |          |       |       |
|         |           | 511.0  | 0.1421 |          |       |       |         |           |        |        |          |       |       |
| 81Kr    | 2.1+5y    | 11.9   | 0.1486 | 1.172-4  | 0.004 | 795.2 |         |           | 13.3   | 0.0237 | 2.726-4  | 5.380 | 0.557 |
|         |           | 11.9   | 0.2885 |          |       |       |         |           | 13.4   | 0.0459 |          |       |       |
|         |           | 13.3   | 0.0713 |          |       |       |         |           | 15.0   | 0.0120 |          |       |       |
|         |           | 276.0  | 0.0360 |          |       |       |         |           | 27.5   | 0.0206 |          |       |       |
|         |           |        |        |          |       |       |         |           | 122.3  | 0.0020 |          |       |       |
|         |           |        |        |          |       |       |         |           | 166.0  | 0.0310 |          |       |       |
|         |           |        |        |          |       |       |         |           | 196.3  | 0.2599 |          |       |       |
|         |           |        |        |          |       |       |         |           | 240.7  | 0.0025 |          |       |       |
|         |           |        |        |          |       |       |         |           | 311.7  | 0.0011 |          |       |       |
| 83mKr   | 1.8h      | 12.6   | 0.0457 | 3.209-5  | 0.004 | 801.4 |         |           | 334.7  | 0.0015 |          |       |       |
|         |           | 12.7   | 0.0886 |          |       |       |         |           | 362.2  | 0.0225 |          |       |       |
|         |           | 14.1   | 0.0226 |          |       |       |         |           | 390.5  | 0.0064 |          |       |       |
|         |           | 32.2   | 0.0005 |          |       |       |         |           | 421.7  | 0.0013 |          |       |       |
|         |           |        |        |          |       |       |         |           | 471.8  | 0.0073 |          |       |       |
| 85Kr    | 10.7y     | 514.0  | 0.0043 | 4.170-7  | 1.782 | 1.681 |         |           | 677.3  | 0.0024 |          |       |       |
|         |           |        |        |          |       |       |         |           | 788.3  | 0.0053 |          |       |       |
| 85mKr   | 4.5h      | 12.6   | 0.0112 | 4.328-5  | 0.331 | 9.045 |         |           | 790.3  | 0.0012 |          |       |       |
|         |           | 12.7   | 0.0216 |          |       |       |         |           | 834.8  | 0.1298 |          |       |       |
|         |           | 14.1   | 0.0055 |          |       |       |         |           | 850.3  | 0.0017 |          |       |       |
|         |           | 304.9  | 0.1399 |          |       |       |         |           | 862.3  | 0.0067 |          |       |       |
|         |           | 13.3   | 0.0061 |          |       |       |         |           | 944.9  | 0.0029 |          |       |       |
|         |           | 13.4   | 0.0118 |          |       |       |         |           | 985.8  | 0.0131 |          |       |       |
|         |           | 15.0   | 0.0031 |          |       |       |         |           | 990.1  | 0.0014 |          |       |       |
|         |           | 129.8  | 0.0030 |          |       |       |         |           | 1040.0 | 0.0048 |          |       |       |
|         |           | 151.2  | 0.7528 |          |       |       |         |           | 1050.0 | 0.0014 |          |       |       |
|         |           | 581.3  | 0.0002 |          |       |       |         |           | 1141.0 | 0.0128 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1179.0 | 0.0100 |          |       |       |
| 87Kr    | 1.3h      | 13.3   | 0.0003 | 1.153-4  | 4.756 | 0.630 |         |           | 1185.0 | 0.0069 |          |       |       |
|         |           | 13.4   | 0.0007 |          |       |       |         |           | 1210.0 | 0.0014 |          |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---|-------|
|         |           | 1213.0 | 0.0014 |          |       |       |         |           | 674.1  | 0.0023 |          |   |       |
|         |           | 1245.0 | 0.0036 |          |       |       |         |           | 696.2  | 0.0178 |          |   |       |
|         |           | 1251.0 | 0.0112 |          |       |       |         |           | 707.0  | 0.0050 |          |   |       |
|         |           | 1325.0 | 0.0016 |          |       |       |         |           | 710.0  | 0.0078 |          |   |       |
|         |           | 1352.0 | 0.0016 |          |       |       |         |           | 729.6  | 0.0030 |          |   |       |
|         |           | 1370.0 | 0.0148 |          |       |       |         |           | 738.4  | 0.0420 |          |   |       |
|         |           | 1407.0 | 0.0022 |          |       |       |         |           | 747.4  | 0.0011 |          |   |       |
|         |           | 1465.0 | 0.0011 |          |       |       |         |           | 762.9  | 0.0040 |          |   |       |
|         |           | 1518.0 | 0.0215 |          |       |       |         |           | 762.9  | 0.0092 |          |   |       |
|         |           | 1530.0 | 0.1093 |          |       |       |         |           | 776.5  | 0.0112 |          |   |       |
|         |           | 1604.0 | 0.0046 |          |       |       |         |           | 826.7  | 0.0076 |          |   |       |
|         |           | 1686.0 | 0.0066 |          |       |       |         |           | 835.5  | 0.0110 |          |   |       |
|         |           | 1893.0 | 0.0014 |          |       |       |         |           | 857.4  | 0.0029 |          |   |       |
|         |           | 1909.0 | 0.0010 |          |       |       |         |           | 867.1  | 0.0592 |          |   |       |
|         |           | 2030.0 | 0.0453 |          |       |       |         |           | 870.4  | 0.0016 |          |   |       |
|         |           | 2035.0 | 0.0374 |          |       |       |         |           | 904.3  | 0.0718 |          |   |       |
|         |           | 2186.0 | 0.0029 |          |       |       |         |           | 930.9  | 0.0062 |          |   |       |
|         |           | 2196.0 | 0.1318 |          |       |       |         |           | 944.2  | 0.0016 |          |   |       |
|         |           | 2232.0 | 0.0339 |          |       |       |         |           | 953.2  | 0.0011 |          |   |       |
|         |           | 2352.0 | 0.0073 |          |       |       |         |           | 960.4  | 0.0032 |          |   |       |
|         |           | 2392.0 | 0.3460 |          |       |       |         |           | 974.4  | 0.0098 |          |   |       |
|         |           | 2409.0 | 0.0010 |          |       |       |         |           | 997.4  | 0.0066 |          |   |       |
|         |           | 2548.0 | 0.0062 |          |       |       |         |           | 1011.0 | 0.0011 |          |   |       |
|         |           | 2771.0 | 0.0015 |          |       |       |         |           | 1044.0 | 0.0041 |          |   |       |
|         |           | 1000.0 | 0.0189 |          |       |       |         |           | 1076.0 | 0.0024 |          |   |       |
|         |           |        |        |          |       |       |         |           | 1088.0 | 0.0036 |          |   |       |
| 89Kr    | 3.2m      | 13.3   | 0.0012 | 2.589-4  | 4.796 | 0.625 |         |           | 1103.0 | 0.0090 |          |   |       |
|         |           | 13.4   | 0.0023 |          |       |       |         |           | 1108.0 | 0.0292 |          |   |       |
|         |           | 15.0   | 0.0006 |          |       |       |         |           | 1117.0 | 0.0166 |          |   |       |
|         |           | 196.2  | 0.0022 |          |       |       |         |           | 1132.0 | 0.0016 |          |   |       |
|         |           | 197.5  | 0.0182 |          |       |       |         |           | 1163.0 | 0.0021 |          |   |       |
|         |           | 205.0  | 0.0012 |          |       |       |         |           | 1172.0 | 0.0098 |          |   |       |
|         |           | 220.9  | 0.2000 |          |       |       |         |           | 1182.0 | 0.0017 |          |   |       |
|         |           | 264.1  | 0.0066 |          |       |       |         |           | 1186.0 | 0.0018 |          |   |       |
|         |           | 338.2  | 0.0034 |          |       |       |         |           | 1229.0 | 0.0014 |          |   |       |
|         |           | 345.0  | 0.0118 |          |       |       |         |           | 1236.0 | 0.0059 |          |   |       |
|         |           | 356.1  | 0.0414 |          |       |       |         |           | 1274.0 | 0.0136 |          |   |       |
|         |           | 364.9  | 0.0090 |          |       |       |         |           | 1303.0 | 0.0010 |          |   |       |
|         |           | 369.3  | 0.0138 |          |       |       |         |           | 1324.0 | 0.0306 |          |   |       |
|         |           | 402.2  | 0.0032 |          |       |       |         |           | 1335.0 | 0.0013 |          |   |       |
|         |           | 411.4  | 0.0256 |          |       |       |         |           | 1341.0 | 0.0019 |          |   |       |
|         |           | 438.1  | 0.0096 |          |       |       |         |           | 1368.0 | 0.0015 |          |   |       |
|         |           | 466.1  | 0.0080 |          |       |       |         |           | 1372.0 | 0.0013 |          |   |       |
|         |           | 490.8  | 0.0032 |          |       |       |         |           | 1413.0 | 0.0026 |          |   |       |
|         |           | 497.5  | 0.0664 |          |       |       |         |           | 1422.0 | 0.0022 |          |   |       |
|         |           | 498.6  | 0.0114 |          |       |       |         |           | 1461.0 | 0.0012 |          |   |       |
|         |           | 557.3  | 0.0016 |          |       |       |         |           | 1464.0 | 0.0018 |          |   |       |
|         |           | 577.0  | 0.0564 |          |       |       |         |           | 1469.0 | 0.0019 |          |   |       |
|         |           | 585.8  | 0.1660 |          |       |       |         |           | 1473.0 | 0.0688 |          |   |       |
|         |           | 626.2  | 0.0060 |          |       |       |         |           | 1501.0 | 0.0132 |          |   |       |
|         |           | 629.8  | 0.0034 |          |       |       |         |           | 1506.0 | 0.0011 |          |   |       |
|         |           | 665.7  | 0.0011 |          |       |       |         |           | 1530.0 | 0.0332 |          |   |       |
|         |           | 671.4  | 0.0011 |          |       |       |         |           | 1534.0 | 0.0512 |          |   |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|---|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 1555.0 | 0.0015 |          |   |       |         |           | 3827.0 | 0.0014 |          |       |       |
|         |           | 1574.0 | 0.0019 |          |   |       |         |           | 3843.0 | 0.0011 |          |       |       |
|         |           | 1634.0 | 0.0082 |          |   |       |         |           | 3902.0 | 0.0013 |          |       |       |
|         |           | 1644.0 | 0.0034 |          |   |       |         |           | 3923.0 | 0.0041 |          |       |       |
|         |           | 1667.0 | 0.0013 |          |   |       |         |           | 3965.0 | 0.0021 |          |       |       |
|         |           | 1677.0 | 0.0014 |          |   |       |         |           | 3977.0 | 0.0027 |          |       |       |
|         |           | 1684.0 | 0.0013 |          |   |       |         |           | 3996.0 | 0.0014 |          |       |       |
|         |           | 1692.0 | 0.0026 |          |   |       |         |           | 4048.0 | 0.0012 |          |       |       |
|         |           | 1694.0 | 0.0438 |          |   |       |         |           | 4341.0 | 0.0010 |          |       |       |
|         |           | 1721.0 | 0.0022 |          |   |       |         |           | 4439.0 | 0.0013 |          |       |       |
|         |           | 1778.0 | 0.0076 |          |   |       |         |           | 2181.0 | 0.0712 |          |       |       |
|         |           | 1788.0 | 0.0011 |          |   |       |         |           |        |        |          |       |       |
|         |           | 1811.0 | 0.0014 |          |   |       | 90Kr    | 32.3s     | 13.3   | 0.0166 | 2.061-4  | 4.084 | 0.733 |
|         |           | 1837.0 | 0.0012 |          |   |       |         |           | 13.4   | 0.0321 |          |       |       |
|         |           | 1840.0 | 0.0035 |          |   |       |         |           | 15.0   | 0.0084 |          |       |       |
|         |           | 1868.0 | 0.0020 |          |   |       |         |           | 106.1  | 0.0038 |          |       |       |
|         |           | 1880.0 | 0.0016 |          |   |       |         |           | 120.9  | 0.0272 |          |       |       |
|         |           | 1903.0 | 0.0104 |          |   |       |         |           | 121.8  | 0.3211 |          |       |       |
|         |           | 1939.0 | 0.0064 |          |   |       |         |           | 227.8  | 0.0012 |          |       |       |
|         |           | 1967.0 | 0.0013 |          |   |       |         |           | 234.4  | 0.0250 |          |       |       |
|         |           | 1999.0 | 0.0012 |          |   |       |         |           | 242.2  | 0.0959 |          |       |       |
|         |           | 2012.0 | 0.0156 |          |   |       |         |           | 249.3  | 0.0128 |          |       |       |
|         |           | 2021.0 | 0.0024 |          |   |       |         |           | 399.1  | 0.0013 |          |       |       |
|         |           | 2047.0 | 0.0026 |          |   |       |         |           | 356.0  | 0.0010 |          |       |       |
|         |           | 2101.0 | 0.0094 |          |   |       |         |           | 386.5  | 0.0012 |          |       |       |
|         |           | 2160.0 | 0.0053 |          |   |       |         |           | 419.1  | 0.0031 |          |       |       |
|         |           | 2196.0 | 0.0013 |          |   |       |         |           | 429.9  | 0.0014 |          |       |       |
|         |           | 2280.0 | 0.0020 |          |   |       |         |           | 433.5  | 0.0125 |          |       |       |
|         |           | 2377.0 | 0.0080 |          |   |       |         |           | 470.3  | 0.0023 |          |       |       |
|         |           | 2401.0 | 0.0072 |          |   |       |         |           | 476.1  | 0.0013 |          |       |       |
|         |           | 2598.0 | 0.0011 |          |   |       |         |           | 492.6  | 0.0116 |          |       |       |
|         |           | 2645.0 | 0.0042 |          |   |       |         |           | 498.6  | 0.0015 |          |       |       |
|         |           | 2751.0 | 0.0012 |          |   |       |         |           | 539.5  | 0.2947 |          |       |       |
|         |           | 2782.0 | 0.0076 |          |   |       |         |           | 554.4  | 0.0485 |          |       |       |
|         |           | 2794.0 | 0.0068 |          |   |       |         |           | 565.2  | 0.0020 |          |       |       |
|         |           | 2820.0 | 0.0013 |          |   |       |         |           | 569.2  | 0.0058 |          |       |       |
|         |           | 2853.0 | 0.0024 |          |   |       |         |           | 614.4  | 0.0020 |          |       |       |
|         |           | 2866.0 | 0.0174 |          |   |       |         |           | 619.1  | 0.0104 |          |       |       |
|         |           | 2879.0 | 0.0032 |          |   |       |         |           | 626.5  | 0.0027 |          |       |       |
|         |           | 3018.0 | 0.0025 |          |   |       |         |           | 661.2  | 0.0032 |          |       |       |
|         |           | 3029.0 | 0.0027 |          |   |       |         |           | 677.7  | 0.0037 |          |       |       |
|         |           | 3107.0 | 0.0019 |          |   |       |         |           | 690.7  | 0.0038 |          |       |       |
|         |           | 3140.0 | 0.0104 |          |   |       |         |           | 705.5  | 0.0012 |          |       |       |
|         |           | 3172.0 | 0.0010 |          |   |       |         |           | 731.3  | 0.0142 |          |       |       |
|         |           | 3220.0 | 0.0043 |          |   |       |         |           | 925.5  | 0.0021 |          |       |       |
|         |           | 3362.0 | 0.0104 |          |   |       |         |           | 941.9  | 0.0128 |          |       |       |
|         |           | 3371.0 | 0.0062 |          |   |       |         |           | 967.3  | 0.0021 |          |       |       |
|         |           | 3400.0 | 0.0014 |          |   |       |         |           | 980.3  | 0.0018 |          |       |       |
|         |           | 3533.0 | 0.0134 |          |   |       |         |           | 1039.0 | 0.0040 |          |       |       |
|         |           | 3584.0 | 0.0026 |          |   |       |         |           | 1104.0 | 0.0033 |          |       |       |
|         |           | 3718.0 | 0.0084 |          |   |       |         |           | 1119.0 | 0.3730 |          |       |       |
|         |           | 3733.0 | 0.0014 |          |   |       |         |           | 1166.0 | 0.0079 |          |       |       |
|         |           | 3781.0 | 0.0013 |          |   |       |         |           | 1240.0 | 0.0034 |          |       |       |

| Nuclide | Half-life | E      | Prob.  | T       | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | T       | T     | $\mu$ |
|---------|-----------|--------|--------|---------|-------|-------|---------|-----------|--------|--------|---------|-------|-------|
|         |           | 1310.0 | 0.0026 |         |       |       |         |           | 698.3  | 0.0015 |         |       |       |
|         |           | 1341.0 | 0.0015 |         |       |       |         |           | 776.5  | 0.1350 |         |       |       |
|         |           | 1387.0 | 0.0019 |         |       |       |         |           | 1395.0 | 0.0051 |         |       |       |
|         |           | 1424.0 | 0.0281 |         |       |       |         |           | 1634.0 | 0.0033 |         |       |       |
|         |           | 1466.0 | 0.0024 |         |       |       |         |           | 511.0  | 1.9100 |         |       |       |
|         |           | 1538.0 | 0.0925 |         |       |       |         |           |        |        |         |       |       |
|         |           | 1552.0 | 0.0210 |         |       |       | 83Rb    | 86.2d     | 12.6   | 0.1639 | 2.072-4 | 1.366 | 2.194 |
|         |           | 1620.0 | 0.0015 |         |       |       |         |           | 12.7   | 0.3177 |         |       |       |
|         |           | 1658.0 | 0.0127 |         |       |       |         |           | 14.1   | 0.0809 |         |       |       |
|         |           | 1780.0 | 0.0642 |         |       |       |         |           | 520.4  | 0.4590 |         |       |       |
|         |           | 1885.0 | 0.0022 |         |       |       |         |           | 529.6  | 0.3029 |         |       |       |
|         |           | 1900.0 | 0.0018 |         |       |       |         |           | 552.6  | 0.1639 |         |       |       |
|         |           | 1981.0 | 0.0016 |         |       |       |         |           | 790.1  | 0.0067 |         |       |       |
|         |           | 2006.0 | 0.0011 |         |       |       |         |           | 799.4  | 0.0024 |         |       |       |
|         |           | 2127.0 | 0.0132 |         |       |       |         |           | 591.9  | 0.0014 |         |       |       |
|         |           | 2149.0 | 0.0026 |         |       |       |         |           |        |        |         |       |       |
|         |           | 2191.0 | 0.0011 |         |       |       | 84Rb    | 32.9d     | 12.6   | 0.1136 | 2.317-4 | 2.549 | 1.175 |
|         |           | 2417.0 | 0.0018 |         |       |       |         |           | 12.7   | 0.2201 |         |       |       |
|         |           | 2433.0 | 0.0015 |         |       |       |         |           | 14.1   | 0.0561 |         |       |       |
|         |           | 2469.0 | 0.0045 |         |       |       |         |           | 881.5  | 0.6768 |         |       |       |
|         |           | 2727.0 | 0.0084 |         |       |       |         |           | 1016.0 | 0.0032 |         |       |       |
|         |           | 2855.0 | 0.0031 |         |       |       |         |           | 1897.0 | 0.0093 |         |       |       |
|         |           | 2866.0 | 0.0018 |         |       |       |         |           | 511.0  | 0.5434 |         |       |       |
|         |           | 3344.0 | 0.0011 |         |       |       |         |           |        |        |         |       |       |
|         |           | 3855.0 | 0.0012 |         |       |       | 86Rb    | 18.7d     | 1077.0 | 0.0878 | 1.458-5 | 4.010 | 0.747 |
|         |           | 1519.0 | 0.0181 |         |       |       |         |           |        |        |         |       |       |
| 81Rb    | 4.6h      | 12.6   | 0.1579 | 2.253-4 | 1.217 | 2.461 | 88Rb    | 17.8m     | 898.0  | 0.1404 | 8.594-5 | 5.240 | 0.572 |
|         |           | 12.7   | 0.3061 |         |       |       |         |           | 1366.0 | 0.0010 |         |       |       |
|         |           | 14.1   | 0.0780 |         |       |       |         |           | 1382.0 | 0.0074 |         |       |       |
|         |           | 180.2  | 0.0012 |         |       |       |         |           | 1780.0 | 0.0022 |         |       |       |
|         |           | 190.3  | 0.6570 |         |       |       |         |           | 1836.0 | 0.2140 |         |       |       |
|         |           | 243.8  | 0.0020 |         |       |       |         |           | 2111.0 | 0.0012 |         |       |       |
|         |           | 357.4  | 0.0056 |         |       |       |         |           | 2119.0 | 0.0042 |         |       |       |
|         |           | 388.8  | 0.0028 |         |       |       |         |           | 2578.0 | 0.0018 |         |       |       |
|         |           | 446.1  | 0.1895 |         |       |       |         |           | 2678.0 | 0.0196 |         |       |       |
|         |           | 456.7  | 0.0231 |         |       |       |         |           | 2734.0 | 0.0011 |         |       |       |
|         |           | 476.7  | 0.0039 |         |       |       |         |           | 3009.0 | 0.0024 |         |       |       |
|         |           | 510.5  | 0.0046 |         |       |       |         |           | 3219.0 | 0.0021 |         |       |       |
|         |           | 537.6  | 0.0155 |         |       |       |         |           | 3486.0 | 0.0013 |         |       |       |
|         |           | 549.0  | 0.0033 |         |       |       |         |           | 4743.0 | 0.0014 |         |       |       |
|         |           | 568.9  | 0.0039 |         |       |       |         |           | 1475.0 | 0.0033 |         |       |       |
|         |           | 729.1  | 0.0022 |         |       |       |         |           |        |        |         |       |       |
|         |           | 803.7  | 0.0066 |         |       |       | 89Rb    | 15.4m     | 272.4  | 0.0142 | 2.934-4 | 4.742 | 0.632 |
|         |           | 834.7  | 0.0063 |         |       |       |         |           | 289.8  | 0.0054 |         |       |       |
|         |           | 977.2  | 0.0038 |         |       |       |         |           | 657.7  | 0.0998 |         |       |       |
|         |           | 1041.0 | 0.0039 |         |       |       |         |           | 766.8  | 0.0016 |         |       |       |
|         |           | 626.6  | 0.0103 |         |       |       |         |           | 947.7  | 0.0922 |         |       |       |
|         |           | 511.0  | 0.6620 |         |       |       |         |           | 1025.0 | 0.0023 |         |       |       |
|         |           |        |        |         |       |       |         |           | 1032.0 | 0.5800 |         |       |       |
|         |           |        |        |         |       |       |         |           | 1220.0 | 0.0022 |         |       |       |
| 82Rb    | 1.2m      | 12.6   | 0.0075 | 2.076-4 | 1.907 | 1.571 |         |           | 1228.0 | 0.0012 |         |       |       |
|         |           | 12.7   | 0.0146 |         |       |       |         |           | 1248.0 | 0.4234 |         |       |       |
|         |           | 14.1   | 0.0037 |         |       |       |         |           | 1473.0 | 0.0035 |         |       |       |



| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 1501.0 | 0.0020 |          |       |       |         |           | 4135.0 | 0.0746 |          |       |       |
|         |           | 1538.0 | 0.0255 |          |       |       |         |           | 4332.0 | 0.0043 |          |       |       |
|         |           | 1940.0 | 0.0033 |          |       |       |         |           | 4356.0 | 0.0049 |          |       |       |
|         |           | 2007.0 | 0.0238 |          |       |       |         |           | 4366.0 | 0.0883 |          |       |       |
|         |           | 2058.0 | 0.0023 |          |       |       |         |           | 4599.0 | 0.0017 |          |       |       |
|         |           | 2196.0 | 0.1334 |          |       |       |         |           | 4646.0 | 0.0248 |          |       |       |
|         |           | 2280.0 | 0.0018 |          |       |       |         |           | 4974.0 | 0.0023 |          |       |       |
|         |           | 2570.0 | 0.0986 |          |       |       |         |           | 5070.0 | 0.0016 |          |       |       |
|         |           | 2707.0 | 0.0203 |          |       |       |         |           | 5187.0 | 0.0129 |          |       |       |
|         |           | 3509.0 | 0.0115 |          |       |       |         |           | 5254.0 | 0.0026 |          |       |       |
|         |           | 1950.0 | 0.0118 |          |       |       |         |           | 5333.0 | 0.0048 |          |       |       |
|         |           |        |        |          |       |       |         |           | 2428.0 | 0.0259 |          |       |       |
| 90Rb    | 2.6m      | 824.2  | 0.0075 | 2.445-4  | 5.755 | 0.521 | 90mRb   | 4.3m      | 13.3   | 0.0033 | 4.339-4  | 5.096 | 0.588 |
|         |           | 831.7  | 0.3260 |          |       |       |         |           | 13.4   | 0.0064 |          |       |       |
|         |           | 997.9  | 0.0051 |          |       |       |         |           | 15.0   | 0.0017 |          |       |       |
|         |           | 1039.0 | 0.0035 |          |       |       |         |           | 106.9  | 0.0020 |          |       |       |
|         |           | 1061.0 | 0.0779 |          |       |       |         |           | 196.8  | 0.0030 |          |       |       |
|         |           | 1141.0 | 0.0013 |          |       |       |         |           | 314.5  | 0.0084 |          |       |       |
|         |           | 1302.0 | 0.0012 |          |       |       |         |           | 442.3  | 0.0012 |          |       |       |
|         |           | 1326.0 | 0.0015 |          |       |       |         |           | 522.1  | 0.0041 |          |       |       |
|         |           | 1375.0 | 0.0035 |          |       |       |         |           | 551.2  | 0.0087 |          |       |       |
|         |           | 1590.0 | 0.0016 |          |       |       |         |           | 720.7  | 0.0057 |          |       |       |
|         |           | 1632.0 | 0.0019 |          |       |       |         |           | 779.9  | 0.0028 |          |       |       |
|         |           | 1666.0 | 0.0037 |          |       |       |         |           | 824.2  | 0.0746 |          |       |       |
|         |           | 1669.0 | 0.0017 |          |       |       |         |           | 831.7  | 0.9321 |          |       |       |
|         |           | 1804.0 | 0.0067 |          |       |       |         |           | 872.0  | 0.0054 |          |       |       |
|         |           | 1830.0 | 0.0017 |          |       |       |         |           | 921.2  | 0.0023 |          |       |       |
|         |           | 1892.0 | 0.0044 |          |       |       |         |           | 952.4  | 0.0173 |          |       |       |
|         |           | 2139.0 | 0.0036 |          |       |       |         |           | 1014.0 | 0.0026 |          |       |       |
|         |           | 2148.0 | 0.0024 |          |       |       |         |           | 1027.0 | 0.0014 |          |       |       |
|         |           | 2208.0 | 0.0051 |          |       |       |         |           | 1061.0 | 0.0948 |          |       |       |
|         |           | 2216.0 | 0.0059 |          |       |       |         |           | 1141.0 | 0.0082 |          |       |       |
|         |           | 2240.0 | 0.0018 |          |       |       |         |           | 1243.0 | 0.0312 |          |       |       |
|         |           | 2474.0 | 0.0068 |          |       |       |         |           | 1272.0 | 0.0163 |          |       |       |
|         |           | 2477.0 | 0.0012 |          |       |       |         |           | 1299.0 | 0.0021 |          |       |       |
|         |           | 2689.0 | 0.0014 |          |       |       |         |           | 1375.0 | 0.1696 |          |       |       |
|         |           | 2724.0 | 0.0016 |          |       |       |         |           | 1377.0 | 0.0233 |          |       |       |
|         |           | 2789.0 | 0.0010 |          |       |       |         |           | 1392.0 | 0.0045 |          |       |       |
|         |           | 2981.0 | 0.0010 |          |       |       |         |           | 1425.0 | 0.0028 |          |       |       |
|         |           | 3039.0 | 0.0082 |          |       |       |         |           | 1457.0 | 0.0026 |          |       |       |
|         |           | 3081.0 | 0.0017 |          |       |       |         |           | 1460.0 | 0.0020 |          |       |       |
|         |           | 3205.0 | 0.0055 |          |       |       |         |           | 1486.0 | 0.0021 |          |       |       |
|         |           | 3295.0 | 0.0095 |          |       |       |         |           | 1489.0 | 0.0035 |          |       |       |
|         |           | 3304.0 | 0.0098 |          |       |       |         |           | 1577.0 | 0.0012 |          |       |       |
|         |           | 3317.0 | 0.0031 |          |       |       |         |           | 1604.0 | 0.0048 |          |       |       |
|         |           | 3362.0 | 0.0108 |          |       |       |         |           | 1659.0 | 0.0045 |          |       |       |
|         |           | 3383.0 | 0.0746 |          |       |       |         |           | 1666.0 | 0.0459 |          |       |       |
|         |           | 3534.0 | 0.0447 |          |       |       |         |           | 1692.0 | 0.0028 |          |       |       |
|         |           | 3539.0 | 0.0017 |          |       |       |         |           | 1696.0 | 0.0170 |          |       |       |
|         |           | 3627.0 | 0.0014 |          |       |       |         |           | 1739.0 | 0.0197 |          |       |       |
|         |           | 3814.0 | 0.0065 |          |       |       |         |           | 1747.0 | 0.0026 |          |       |       |
|         |           | 4062.0 | 0.0026 |          |       |       |         |           | 1794.0 | 0.0087 |          |       |       |
|         |           | 4087.0 | 0.0028 |          |       |       |         |           |        |        |          |       |       |

| ()      |           |        |        |          |       |       |         |           |        |        |          |       |       |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|         |           | 1838.0 | 0.0085 |          |       |       |         |           | 15.0   | 0.0866 |          |       |       |
|         |           | 1877.0 | 0.0046 |          |       |       |         |           | 514.0  | 0.9927 |          |       |       |
|         |           | 1892.0 | 0.0065 |          |       |       |         |           | 868.5  | 0.0001 |          |       |       |
|         |           | 1903.0 | 0.0014 |          |       |       |         |           |        |        |          |       |       |
|         |           | 1942.0 | 0.0063 |          |       |       | 85mSr   | 1.1h      | 14.1   | 0.0044 | 6.004-5  | 0.334 | 8.972 |
|         |           | 2128.0 | 0.0537 |          |       |       |         |           | 14.2   | 0.0085 |          |       |       |
|         |           | 2139.0 | 0.0012 |          |       |       |         |           | 15.8   | 0.0023 |          |       |       |
|         |           | 2201.0 | 0.0049 |          |       |       |         |           | 231.7  | 0.8472 |          |       |       |
|         |           | 2208.0 | 0.0017 |          |       |       |         |           | 238.7  | 0.0032 |          |       |       |
|         |           | 2245.0 | 0.0030 |          |       |       |         |           | 13.3   | 0.0226 |          |       |       |
|         |           | 2257.0 | 0.0068 |          |       |       |         |           | 13.4   | 0.0437 |          |       |       |
|         |           | 2298.0 | 0.0038 |          |       |       |         |           | 15.0   | 0.0115 |          |       |       |
|         |           | 2311.0 | 0.0030 |          |       |       |         |           | 151.2  | 0.1212 |          |       |       |
|         |           | 2335.0 | 0.0021 |          |       |       |         |           |        |        |          |       |       |
|         |           | 2443.0 | 0.0027 |          |       |       | 87mSr   | 2.8h      | 14.1   | 0.0292 | 8.010-5  | 1.043 | 2.872 |
|         |           | 2497.0 | 0.0074 |          |       |       |         |           | 14.2   | 0.0563 |          |       |       |
|         |           | 2538.0 | 0.0018 |          |       |       |         |           | 15.8   | 0.0151 |          |       |       |
|         |           | 2544.0 | 0.0057 |          |       |       |         |           | 388.4  | 0.8225 |          |       |       |
|         |           | 2592.0 | 0.0066 |          |       |       |         |           | 13.3   | 0.0005 |          |       |       |
|         |           | 2618.0 | 0.0063 |          |       |       |         |           | 13.4   | 0.0010 |          |       |       |
|         |           | 2724.0 | 0.0051 |          |       |       |         |           | 15.0   | 0.0003 |          |       |       |
|         |           | 2741.0 | 0.0015 |          |       |       |         |           |        |        |          |       |       |
|         |           | 2753.0 | 0.1184 |          |       |       | 89Sr    | 50.6d     | 909.1  | 0.0002 | 2.205-8  | 3.493 | 0.858 |
|         |           | 2834.0 | 0.0190 |          |       |       |         |           |        |        |          |       |       |
|         |           | 2900.0 | 0.0011 |          |       |       | 91Sr    | 9.5h      | 261.2  | 0.0044 | 1.116-4  | 3.465 | 0.865 |
|         |           | 2912.0 | 0.0013 |          |       |       |         |           | 272.7  | 0.0025 |          |       |       |
|         |           | 3032.0 | 0.0045 |          |       |       |         |           | 274.7  | 0.0100 |          |       |       |
|         |           | 3039.0 | 0.0027 |          |       |       |         |           | 379.9  | 0.0014 |          |       |       |
|         |           | 3149.0 | 0.0255 |          |       |       |         |           | 620.1  | 0.0172 |          |       |       |
|         |           | 3198.0 | 0.0015 |          |       |       |         |           | 631.3  | 0.0054 |          |       |       |
|         |           | 3215.0 | 0.0014 |          |       |       |         |           | 652.3  | 0.0289 |          |       |       |
|         |           | 3317.0 | 0.1473 |          |       |       |         |           | 652.9  | 0.0780 |          |       |       |
|         |           | 3371.0 | 0.0041 |          |       |       |         |           | 653.0  | 0.0045 |          |       |       |
|         |           | 3383.0 | 0.0043 |          |       |       |         |           | 749.8  | 0.2298 |          |       |       |
|         |           | 3503.0 | 0.0243 |          |       |       |         |           | 761.4  | 0.0056 |          |       |       |
|         |           | 3573.0 | 0.0158 |          |       |       |         |           | 820.8  | 0.0016 |          |       |       |
|         |           | 3621.0 | 0.0060 |          |       |       |         |           | 879.7  | 0.0018 |          |       |       |
|         |           | 3627.0 | 0.0093 |          |       |       |         |           | 925.8  | 0.0374 |          |       |       |
|         |           | 3972.0 | 0.0037 |          |       |       |         |           | 1024.0 | 0.3250 |          |       |       |
|         |           | 4116.0 | 0.0036 |          |       |       |         |           | 1055.0 | 0.0022 |          |       |       |
|         |           | 4193.0 | 0.0088 |          |       |       |         |           | 1141.0 | 0.0012 |          |       |       |
|         |           | 4210.0 | 0.0093 |          |       |       |         |           | 1281.0 | 0.0091 |          |       |       |
|         |           | 4257.0 | 0.0076 |          |       |       |         |           | 1413.0 | 0.0095 |          |       |       |
|         |           | 4454.0 | 0.0121 |          |       |       |         |           | 1474.0 | 0.0016 |          |       |       |
|         |           | 4726.0 | 0.0011 |          |       |       |         |           | 1651.0 | 0.0028 |          |       |       |
|         |           | 2072.0 | 0.0083 |          |       |       |         |           | 1724.0 | 0.0016 |          |       |       |
|         |           |        |        |          |       |       |         |           | 776.3  | 0.0119 |          |       |       |
| 82Sr    | 25.0d     | 13.3   | 0.1676 | 1.065-4  | 0.004 | 842.8 |         |           |        |        |          |       |       |
|         |           | 13.4   | 0.3243 |          |       |       |         |           |        |        |          |       |       |
|         |           | 15.0   | 0.0851 |          |       |       |         |           |        |        |          |       |       |
| 85Sr    | 64.8d     | 13.3   | 0.1707 | 2.038-4  | 1.330 | 2.252 |         |           |        |        |          |       |       |
|         |           | 13.4   | 0.3301 |          |       |       |         |           |        |        |          |       |       |
|         |           |        |        |          |       |       | 92Sr    | 2.7h      | 241.5  | 0.0297 | 1.940-4  | 4.604 | 0.651 |
|         |           |        |        |          |       |       |         |           | 430.6  | 0.0333 |          |       |       |
|         |           |        |        |          |       |       |         |           | 491.3  | 0.0026 |          |       |       |
|         |           |        |        |          |       |       |         |           | 650.7  | 0.0037 |          |       |       |
|         |           |        |        |          |       |       |         |           | 953.3  | 0.0360 |          |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---|-------|
|         |           | 1142.0 | 0.0288 |          |       |       |         |           | 875.7  | 0.2419 |          |   |       |
|         |           | 1384.0 | 0.9000 |          |       |       |         |           | 888.1  | 0.2184 |          |   |       |
|         |           | 664.6  | 0.0017 |          |       |       |         |           | 901.0  | 0.0069 |          |   |       |
| 93Sr    | 7.3m      | 14.9   | 0.0277 | 3.642-4  | 3.833 | 0.782 |         |           | 910.2  | 0.0081 |          |   |       |
|         |           | 15.0   | 0.0534 |          |       |       |         |           | 922.7  | 0.0033 |          |   |       |
|         |           | 16.7   | 0.0147 |          |       |       |         |           | 927.7  | 0.0063 |          |   |       |
|         |           | 166.6  | 0.0062 |          |       |       |         |           | 930.9  | 0.0040 |          |   |       |
|         |           | 168.7  | 0.1821 |          |       |       |         |           | 952.6  | 0.0011 |          |   |       |
|         |           | 260.1  | 0.0732 |          |       |       |         |           | 991.6  | 0.0012 |          |   |       |
|         |           | 285.7  | 0.0027 |          |       |       |         |           | 1032.0 | 0.0010 |          |   |       |
|         |           | 332.0  | 0.0035 |          |       |       |         |           | 1035.0 | 0.0020 |          |   |       |
|         |           | 346.5  | 0.0324 |          |       |       |         |           | 1041.0 | 0.0316 |          |   |       |
|         |           | 377.4  | 0.0147 |          |       |       |         |           | 1055.0 | 0.0034 |          |   |       |
|         |           | 406.7  | 0.0042 |          |       |       |         |           | 1064.0 | 0.0037 |          |   |       |
|         |           | 424.7  | 0.0026 |          |       |       |         |           | 1078.0 | 0.0024 |          |   |       |
|         |           | 428.0  | 0.0015 |          |       |       |         |           | 1094.0 | 0.0174 |          |   |       |
|         |           | 432.7  | 0.0147 |          |       |       |         |           | 1105.0 | 0.0015 |          |   |       |
|         |           | 440.8  | 0.0019 |          |       |       |         |           | 1123.0 | 0.0397 |          |   |       |
|         |           | 446.2  | 0.0233 |          |       |       |         |           | 1137.0 | 0.0019 |          |   |       |
|         |           | 482.0  | 0.0112 |          |       |       |         |           | 1181.0 | 0.0024 |          |   |       |
|         |           | 483.7  | 0.0165 |          |       |       |         |           | 1196.0 | 0.0097 |          |   |       |
|         |           | 486.7  | 0.0012 |          |       |       |         |           | 1215.0 | 0.0247 |          |   |       |
|         |           | 518.5  | 0.0013 |          |       |       |         |           | 1239.0 | 0.0012 |          |   |       |
|         |           | 541.9  | 0.0072 |          |       |       |         |           | 1243.0 | 0.0079 |          |   |       |
|         |           | 545.8  | 0.0039 |          |       |       |         |           | 1266.0 | 0.0110 |          |   |       |
|         |           | 559.9  | 0.0020 |          |       |       |         |           | 1269.0 | 0.0706 |          |   |       |
|         |           | 572.0  | 0.0021 |          |       |       |         |           | 1278.0 | 0.0086 |          |   |       |
|         |           | 586.5  | 0.0044 |          |       |       |         |           | 1309.0 | 0.0040 |          |   |       |
|         |           | 590.3  | 0.6720 |          |       |       |         |           | 1321.0 | 0.0258 |          |   |       |
|         |           | 593.8  | 0.0110 |          |       |       |         |           | 1333.0 | 0.0047 |          |   |       |
|         |           | 596.1  | 0.0132 |          |       |       |         |           | 1335.0 | 0.0067 |          |   |       |
|         |           | 610.9  | 0.0107 |          |       |       |         |           | 1379.0 | 0.0035 |          |   |       |
|         |           | 631.0  | 0.0019 |          |       |       |         |           | 1387.0 | 0.0343 |          |   |       |
|         |           | 633.5  | 0.0011 |          |       |       |         |           | 1434.0 | 0.0089 |          |   |       |
|         |           | 650.6  | 0.0019 |          |       |       |         |           | 1439.0 | 0.0050 |          |   |       |
|         |           | 658.6  | 0.0042 |          |       |       |         |           | 1466.0 | 0.0010 |          |   |       |
|         |           | 663.6  | 0.0163 |          |       |       |         |           | 1469.0 | 0.0052 |          |   |       |
|         |           | 687.8  | 0.0066 |          |       |       |         |           | 1483.0 | 0.0010 |          |   |       |
|         |           | 690.1  | 0.0100 |          |       |       |         |           | 1492.0 | 0.0054 |          |   |       |
|         |           | 692.0  | 0.0022 |          |       |       |         |           | 1520.0 | 0.0032 |          |   |       |
|         |           | 710.4  | 0.2150 |          |       |       |         |           | 1539.0 | 0.0010 |          |   |       |
|         |           | 716.8  | 0.0029 |          |       |       |         |           | 1552.0 | 0.0101 |          |   |       |
|         |           | 718.3  | 0.0148 |          |       |       |         |           | 1610.0 | 0.0019 |          |   |       |
|         |           | 771.2  | 0.0115 |          |       |       |         |           | 1634.0 | 0.0143 |          |   |       |
|         |           | 776.1  | 0.0026 |          |       |       |         |           | 1648.0 | 0.0088 |          |   |       |
|         |           | 782.8  | 0.0022 |          |       |       |         |           | 1669.0 | 0.0016 |          |   |       |
|         |           | 788.7  | 0.0076 |          |       |       |         |           | 1685.0 | 0.0071 |          |   |       |
|         |           | 791.1  | 0.0026 |          |       |       |         |           | 1694.0 | 0.0255 |          |   |       |
|         |           | 795.3  | 0.0023 |          |       |       |         |           | 1699.0 | 0.0329 |          |   |       |
|         |           | 834.9  | 0.0165 |          |       |       |         |           | 1707.0 | 0.0110 |          |   |       |
|         |           | 837.8  | 0.0012 |          |       |       |         |           | 1765.0 | 0.0106 |          |   |       |
|         |           | 858.5  | 0.0072 |          |       |       |         |           | 1775.0 | 0.0016 |          |   |       |
|         |           |        |        |          |       |       |         |           | 1811.0 | 0.0139 |          |   |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---|-------|
|         |           | 1816.0 | 0.0023 |          |       |       |         |           | 702.2  | 0.0025 |          |   |       |
|         |           | 1894.0 | 0.0012 |          |       |       |         |           | 703.3  | 0.1543 |          |   |       |
|         |           | 1908.0 | 0.0017 |          |       |       |         |           | 709.9  | 0.0262 |          |   |       |
|         |           | 1929.0 | 0.0116 |          |       |       |         |           | 719.2  | 0.0022 |          |   |       |
|         |           | 1945.0 | 0.0035 |          |       |       |         |           | 740.8  | 0.0136 |          |   |       |
|         |           | 2011.0 | 0.0012 |          |       |       |         |           | 767.6  | 0.0239 |          |   |       |
|         |           | 2055.0 | 0.0013 |          |       |       |         |           | 772.2  | 0.0032 |          |   |       |
|         |           | 2064.0 | 0.0062 |          |       |       |         |           | 774.4  | 0.2244 |          |   |       |
|         |           | 2105.0 | 0.0031 |          |       |       |         |           | 833.6  | 0.0026 |          |   |       |
|         |           | 2129.0 | 0.0010 |          |       |       |         |           | 826.0  | 0.0330 |          |   |       |
|         |           | 2179.0 | 0.0029 |          |       |       |         |           | 833.7  | 0.0148 |          |   |       |
|         |           | 2230.0 | 0.0153 |          |       |       |         |           | 835.7  | 0.0437 |          |   |       |
|         |           | 2296.0 | 0.0073 |          |       |       |         |           | 883.0  | 0.0025 |          |   |       |
|         |           | 2365.0 | 0.0156 |          |       |       |         |           | 887.4  | 0.0044 |          |   |       |
|         |           | 2416.0 | 0.0011 |          |       |       |         |           | 955.3  | 0.0104 |          |   |       |
|         |           | 2544.0 | 0.0299 |          |       |       |         |           | 971.4  | 0.0027 |          |   |       |
|         |           | 2574.0 | 0.0013 |          |       |       |         |           | 1018.0 | 0.0018 |          |   |       |
|         |           | 2689.0 | 0.0210 |          |       |       |         |           | 1024.0 | 0.0380 |          |   |       |
|         |           | 2828.0 | 0.0017 |          |       |       |         |           | 1077.0 | 0.8250 |          |   |       |
|         |           | 2986.0 | 0.0019 |          |       |       |         |           | 1093.0 | 0.0069 |          |   |       |
|         |           | 3007.0 | 0.0012 |          |       |       |         |           | 1102.0 | 0.0020 |          |   |       |
|         |           | 1802.0 | 0.0230 |          |       |       |         |           | 1133.0 | 0.0030 |          |   |       |
|         |           |        |        |          |       |       |         |           | 1153.0 | 0.3052 |          |   |       |
| 86Y     | 14.7h     | 14.1   | 0.1171 | 6.256-4  | 3.825 | 0.783 |         |           | 1163.0 | 0.0118 |          |   |       |
|         |           | 14.2   | 0.2261 |          |       |       |         |           | 1253.0 | 0.0153 |          |   |       |
|         |           | 15.8   | 0.0607 |          |       |       |         |           | 1270.0 | 0.0065 |          |   |       |
|         |           | 132.3  | 0.0017 |          |       |       |         |           | 1284.0 | 0.0029 |          |   |       |
|         |           | 182.3  | 0.0011 |          |       |       |         |           | 1295.0 | 0.0029 |          |   |       |
|         |           | 187.9  | 0.0126 |          |       |       |         |           | 1296.0 | 0.0054 |          |   |       |
|         |           | 190.8  | 0.0102 |          |       |       |         |           | 1349.0 | 0.0294 |          |   |       |
|         |           | 209.8  | 0.0040 |          |       |       |         |           | 1405.0 | 0.0018 |          |   |       |
|         |           | 235.4  | 0.0040 |          |       |       |         |           | 1415.0 | 0.0033 |          |   |       |
|         |           | 237.9  | 0.0013 |          |       |       |         |           | 1508.0 | 0.0035 |          |   |       |
|         |           | 252.0  | 0.0037 |          |       |       |         |           | 1533.0 | 0.0022 |          |   |       |
|         |           | 264.5  | 0.0054 |          |       |       |         |           | 1536.0 | 0.0012 |          |   |       |
|         |           | 307.0  | 0.0347 |          |       |       |         |           | 1564.0 | 0.0018 |          |   |       |
|         |           | 331.1  | 0.0083 |          |       |       |         |           | 1696.0 | 0.0064 |          |   |       |
|         |           | 370.3  | 0.0082 |          |       |       |         |           | 1712.0 | 0.0017 |          |   |       |
|         |           | 380.4  | 0.0045 |          |       |       |         |           | 1724.0 | 0.0055 |          |   |       |
|         |           | 382.9  | 0.0363 |          |       |       |         |           | 1791.0 | 0.0100 |          |   |       |
|         |           | 426.0  | 0.0031 |          |       |       |         |           | 1802.0 | 0.0165 |          |   |       |
|         |           | 439.5  | 0.0020 |          |       |       |         |           | 1854.0 | 0.1716 |          |   |       |
|         |           | 443.1  | 0.1691 |          |       |       |         |           | 1921.0 | 0.2079 |          |   |       |
|         |           | 444.2  | 0.0064 |          |       |       |         |           | 2017.0 | 0.0013 |          |   |       |
|         |           | 469.2  | 0.0030 |          |       |       |         |           | 2088.0 | 0.0025 |          |   |       |
|         |           | 515.2  | 0.0489 |          |       |       |         |           | 2292.0 | 0.0012 |          |   |       |
|         |           | 580.6  | 0.0479 |          |       |       |         |           | 2482.0 | 0.0012 |          |   |       |
|         |           | 608.3  | 0.0201 |          |       |       |         |           | 2568.0 | 0.0225 |          |   |       |
|         |           | 618.2  | 0.0021 |          |       |       |         |           | 2610.0 | 0.0124 |          |   |       |
|         |           | 627.7  | 0.3259 |          |       |       |         |           | 2642.0 | 0.0017 |          |   |       |
|         |           | 644.8  | 0.0223 |          |       |       |         |           | 2795.0 | 0.0021 |          |   |       |
|         |           | 645.9  | 0.0916 |          |       |       |         |           | 2866.0 | 0.0038 |          |   |       |
|         |           | 689.3  | 0.0017 |          |       |       |         |           | 3070.0 | 0.0012 |          |   |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|--------|
|         |           | 3334.0 | 0.0012 |          |       |       |         |           | 1918.0 | 0.0140 |          |       |        |
|         |           | 1366.0 | 0.0103 |          |       |       |         |           | 2185.0 | 0.0016 |          |       |        |
|         |           | 511.0  | 0.6637 |          |       |       |         |           | 2191.0 | 0.0017 |          |       |        |
|         |           |        |        |          |       |       |         |           | 1413.0 | 0.0033 |          |       |        |
| 87Y     | 3.3d      | 14.1   | 0.1751 | 1.861-4  | 1.205 | 2.486 | 86Zr    | 16.5h     | 14.9   | 0.3310 | 2.383-4  | 0.266 | 11.245 |
|         |           | 14.2   | 0.3380 |          |       |       |         |           | 15.0   | 0.6377 |          |       |        |
|         |           | 15.8   | 0.0908 |          |       |       |         |           | 16.7   | 0.1753 |          |       |        |
|         |           | 484.7  | 0.9394 |          |       |       |         |           | 29.1   | 0.2160 |          |       |        |
|         |           | 511.0  | 0.0032 |          |       |       |         |           | 135.6  | 0.0047 |          |       |        |
| 88Y     | 106.6d    | 14.1   | 0.1744 | 4.788-4  | 4.525 | 0.662 |         |           | 242.8  | 0.9580 |          |       |        |
|         |           | 14.2   | 0.3367 |          |       |       |         |           | 612.0  | 0.0570 |          |       |        |
|         |           | 15.8   | 0.0905 |          |       |       |         |           | 620.6  | 0.0027 |          |       |        |
|         |           | 898.0  | 0.9342 |          |       |       |         |           | 169.4  | 0.0045 |          |       |        |
|         |           | 1836.0 | 0.9938 |          |       |       |         |           |        |        |          |       |        |
|         |           | 2734.0 | 0.0060 |          |       |       | 88Zr    | 83.4d     | 14.9   | 0.1820 | 1.710-4  | 0.842 | 3.559  |
|         |           | 1188.0 | 0.0010 |          |       |       |         |           | 15.0   | 0.3508 |          |       |        |
|         |           | 511.0  | 0.0043 |          |       |       |         |           | 16.7   | 0.0964 |          |       |        |
|         |           |        |        |          |       |       |         |           | 392.9  | 0.9732 |          |       |        |
| 90mY    | 3.2h      | 14.9   | 0.0205 | 1.316-4  | 1.354 | 2.213 | 89Zr    | 3.3d      | 14.9   | 0.1398 | 2.655-4  | 2.867 | 1.045  |
|         |           | 15.0   | 0.0395 |          |       |       |         |           | 15.0   | 0.2694 |          |       |        |
|         |           | 16.7   | 0.0109 |          |       |       |         |           | 16.7   | 0.0741 |          |       |        |
|         |           | 202.5  | 0.9663 |          |       |       |         |           | 909.1  | 0.9904 |          |       |        |
|         |           | 479.5  | 0.9099 |          |       |       |         |           | 1713.0 | 0.0076 |          |       |        |
|         |           | 682.0  | 0.0032 |          |       |       |         |           | 1745.0 | 0.0013 |          |       |        |
| 91Y     | 58.5d     | 1205.0 | 0.0030 | 5.397-7  | 4.297 | 0.697 |         |           | 1642.0 | 0.0017 |          |       |        |
|         |           |        |        |          |       |       |         |           | 511.0  | 0.4588 |          |       |        |
| 91mY    | 49.7m     | 14.9   | 0.0086 | 1.017-4  | 1.961 | 1.527 | 95Zr    | 64.0d     | 724.2  | 0.4366 | 1.254-4  | 2.828 | 1.059  |
|         |           | 15.0   | 0.0165 |          |       |       |         |           | 756.7  | 0.5534 |          |       |        |
|         |           | 16.7   | 0.0045 |          |       |       |         |           |        |        |          |       |        |
|         |           | 557.6  | 0.9508 |          |       |       | 97Zr    | 16.9h     | 202.2  | 0.0010 | 2.905-5  | 3.732 | 0.803  |
| 92Y     | 3.5h      | 448.5  | 0.0234 | 3.963-5  | 3.821 | 0.784 |         |           | 218.7  | 0.0023 |          |       |        |
|         |           | 492.6  | 0.0049 |          |       |       |         |           | 254.1  | 0.0125 |          |       |        |
|         |           | 561.1  | 0.0241 |          |       |       |         |           | 272.3  | 0.0025 |          |       |        |
|         |           | 844.3  | 0.0125 |          |       |       |         |           | 330.4  | 0.0011 |          |       |        |
|         |           | 912.6  | 0.0063 |          |       |       |         |           | 355.4  | 0.0227 |          |       |        |
|         |           | 934.5  | 0.1390 |          |       |       |         |           | 400.4  | 0.0032 |          |       |        |
|         |           | 1132.0 | 0.0024 |          |       |       |         |           | 507.6  | 0.0529 |          |       |        |
|         |           | 1405.0 | 0.0478 |          |       |       |         |           | 513.5  | 0.0051 |          |       |        |
|         |           | 1847.0 | 0.0036 |          |       |       |         |           | 602.5  | 0.0139 |          |       |        |
|         |           | 1647.0 | 0.0015 |          |       |       |         |           | 690.6  | 0.0025 |          |       |        |
|         |           |        |        |          |       |       |         |           | 699.2  | 0.0012 |          |       |        |
| 93Y     | 10.1h     | 15.7   | 0.0003 | 1.388-5  | 4.295 | 0.698 |         |           | 703.8  | 0.0093 |          |       |        |
|         |           | 15.8   | 0.0006 |          |       |       |         |           | 795.7  | 0.0012 |          |       |        |
|         |           | 17.7   | 0.0002 |          |       |       |         |           | 804.5  | 0.0065 |          |       |        |
|         |           | 266.9  | 0.0685 |          |       |       |         |           | 829.8  | 0.0022 |          |       |        |
|         |           | 680.2  | 0.0061 |          |       |       |         |           | 854.9  | 0.0033 |          |       |        |
|         |           | 947.1  | 0.0195 |          |       |       |         |           | 971.4  | 0.0029 |          |       |        |
|         |           | 1203.0 | 0.0010 |          |       |       |         |           | 1021.0 | 0.0121 |          |       |        |
|         |           | 1425.0 | 0.0024 |          |       |       |         |           | 1119.0 | 0.0011 |          |       |        |
|         |           | 1450.0 | 0.0034 |          |       |       |         |           | 1148.0 | 0.0265 |          |       |        |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|--------|
|         |           | 1276.0 | 0.0097 |          |       |       |         |           |        |        |          |       |        |
|         |           | 1363.0 | 0.0135 |          |       |       | 92mNb   | 10.1d     | 15.7   | 0.1834 | 2.413-4  | 3.067 | 0.977  |
|         |           | 1750.0 | 0.0135 |          |       |       |         |           | 15.8   | 0.3527 |          |       |        |
|         |           | 1852.0 | 0.0035 |          |       |       |         |           | 17.7   | 0.0992 |          |       |        |
|         |           | 344.1  | 0.0021 |          |       |       |         |           | 912.6  | 0.0168 |          |       |        |
|         |           |        |        |          |       |       |         |           | 934.5  | 0.9915 |          |       |        |
|         |           |        |        |          |       |       |         |           | 1847.0 | 0.0085 |          |       |        |
| 90Nb    | 14.6h     | 15.7   | 0.1386 | 6.513-4  | 4.763 | 0.629 |         |           |        |        |          |       |        |
|         |           | 15.8   | 0.2665 |          |       |       | 93mNb   | 14.6y     | 16.5   | 0.0323 | 1.421-5  | 0.002 | 1.292  |
|         |           | 17.7   | 0.0749 |          |       |       |         |           | 16.6   | 0.0620 |          |       |        |
|         |           | 132.6  | 0.0414 |          |       |       |         |           | 18.6   | 0.0178 |          |       |        |
|         |           | 141.1  | 0.6900 |          |       |       |         |           |        |        |          |       |        |
|         |           | 329.1  | 0.0011 |          |       |       | 94Nb    | 2.0+4y    | 17.4   | 0.0004 | 2.642-4  | 3.048 | 0.983  |
|         |           | 371.3  | 0.0190 |          |       |       |         |           | 17.5   | 0.0007 |          |       |        |
|         |           | 518.2  | 0.0049 |          |       |       |         |           | 19.6   | 0.0002 |          |       |        |
|         |           | 561.5  | 0.0013 |          |       |       |         |           | 702.6  | 1.0000 |          |       |        |
|         |           | 827.7  | 0.0090 |          |       |       |         |           | 871.1  | 1.0000 |          |       |        |
|         |           | 890.6  | 0.0173 |          |       |       |         |           |        |        |          |       |        |
|         |           | 1052.0 | 0.0023 |          |       |       | 94mNb   | 6.3m      | 16.5   | 0.1231 | 5.481-5  | 0.003 | 1.180  |
|         |           | 1129.0 | 0.9200 |          |       |       |         |           | 16.6   | 0.2363 |          |       |        |
|         |           | 1270.0 | 0.0121 |          |       |       |         |           | 18.6   | 0.0679 |          |       |        |
|         |           | 1470.0 | 0.0042 |          |       |       |         |           | 40.9   | 0.0008 |          |       |        |
|         |           | 1575.0 | 0.0047 |          |       |       |         |           | 871.1  | 0.0047 |          |       |        |
|         |           | 1612.0 | 0.0221 |          |       |       |         |           | 758.5  | 0.0000 |          |       |        |
|         |           | 1658.0 | 0.0031 |          |       |       |         |           |        |        |          |       |        |
|         |           | 1843.0 | 0.0065 |          |       |       | 95Nb    | 35.1d     | 17.4   | 0.0003 | 1.295-4  | 2.930 | 1.023  |
|         |           | 1913.0 | 0.0123 |          |       |       |         |           | 17.5   | 0.0005 |          |       |        |
|         |           | 1985.0 | 0.0063 |          |       |       |         |           | 19.6   | 0.0002 |          |       |        |
|         |           | 2056.0 | 0.0011 |          |       |       |         |           | 765.8  | 0.9981 |          |       |        |
|         |           | 2186.0 | 0.1803 |          |       |       |         |           | 389.2  | 0.0003 |          |       |        |
|         |           | 2222.0 | 0.0063 |          |       |       | 95mNb   | 3.6d      | 16.5   | 0.1183 | 6.390-5  | 0.177 | 16.959 |
|         |           | 2319.0 | 0.8197 |          |       |       |         |           | 16.6   | 0.2271 |          |       |        |
|         |           | 1273.0 | 0.0016 |          |       |       |         |           | 18.6   | 0.0653 |          |       |        |
|         |           | 511.0  | 1.0620 |          |       |       |         |           | 235.7  | 0.2504 |          |       |        |
| 91Nb    | 1.0+4y    | 15.7   | 0.1833 | 8.831-5  | 0.002 | 1330. |         |           | 204.1  | 0.0013 |          |       |        |
|         |           | 15.8   | 0.3524 |          |       |       |         |           | 629.1  | 0.0000 |          |       |        |
|         |           | 17.7   | 0.0991 |          |       |       |         |           |        |        |          |       |        |
|         |           | 511.0  | 0.0033 |          |       |       | 96Nb    | 23.3h     | 17.4   | 0.0003 | 4.109-4  | 3.237 | 0.925  |
| 91mNb   | 61.0d     | 16.5   | 0.1414 | 7.159-5  | 0.810 | 3.699 |         |           | 17.5   | 0.0005 |          |       |        |
|         |           | 16.6   | 0.2714 |          |       |       |         |           | 19.6   | 0.0001 |          |       |        |
|         |           | 18.6   | 0.0780 |          |       |       |         |           | 219.1  | 0.0378 |          |       |        |
|         |           | 104.5  | 0.0058 |          |       |       |         |           | 241.4  | 0.0387 |          |       |        |
|         |           | 15.7   | 0.0063 |          |       |       |         |           | 349.9  | 0.0073 |          |       |        |
|         |           | 15.8   | 0.0120 |          |       |       |         |           | 350.3  | 0.0111 |          |       |        |
|         |           | 17.7   | 0.0034 |          |       |       |         |           | 352.5  | 0.0082 |          |       |        |
|         |           | 1205.0 | 0.0350 |          |       |       |         |           | 369.7  | 0.0012 |          |       |        |
| 92Nb    | 3.6+7y    | 15.7   | 0.1832 | 3.402-4  | 2.734 | 1.096 |         |           | 371.8  | 0.0281 |          |       |        |
|         |           | 15.8   | 0.3524 |          |       |       |         |           | 434.7  | 0.0053 |          |       |        |
|         |           | 17.7   | 0.0991 |          |       |       |         |           | 460.0  | 0.2817 |          |       |        |
|         |           | 561.1  | 0.9970 |          |       |       |         |           | 477.7  | 0.0012 |          |       |        |
|         |           | 934.5  | 0.9992 |          |       |       |         |           | 480.7  | 0.0629 |          |       |        |
|         |           |        |        |          |       |       |         |           | 568.9  | 0.5566 |          |       |        |
|         |           |        |        |          |       |       |         |           | 591.2  | 0.0097 |          |       |        |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|-------|--------|----------|-------|-------|
|         |           | 593.3  | 0.0031 |          |       |       |         |           | 680.1 | 0.0026 |          |       |       |
|         |           | 719.6  | 0.0726 |          |       |       |         |           |       |        |          |       |       |
|         |           | 721.5  | 0.0077 |          |       |       | 101Mo   | 14.6m     | 18.2  | 0.0166 | 2.376-4  | 4.066 | 0.737 |
|         |           | 778.2  | 0.9680 |          |       |       |         |           | 18.4  | 0.0318 |          |       |       |
|         |           | 810.2  | 0.0987 |          |       |       |         |           | 20.6  | 0.0095 |          |       |       |
|         |           | 812.5  | 0.0339 |          |       |       |         |           | 80.9  | 0.0540 |          |       |       |
|         |           | 847.7  | 0.0165 |          |       |       |         |           | 104.7 | 0.0016 |          |       |       |
|         |           | 849.9  | 0.2071 |          |       |       |         |           | 105.9 | 0.0024 |          |       |       |
|         |           | 1091.0 | 0.4937 |          |       |       |         |           | 115.8 | 0.0017 |          |       |       |
|         |           | 1127.0 | 0.0053 |          |       |       |         |           | 187.4 | 0.0048 |          |       |       |
|         |           | 1200.0 | 0.2004 |          |       |       |         |           | 191.9 | 0.1920 |          |       |       |
|         |           | 1441.0 | 0.0040 |          |       |       |         |           | 195.9 | 0.0292 |          |       |       |
|         |           | 1498.0 | 0.0300 |          |       |       |         |           | 212.0 | 0.0052 |          |       |       |
|         |           | 589.2  | 0.0053 |          |       |       |         |           | 221.8 | 0.0010 |          |       |       |
|         |           |        |        |          |       |       |         |           | 317.8 | 0.0024 |          |       |       |
| 97Nb    | 1.2h      | 17.4   | 0.0004 | 1.168-4  | 2.491 | 1.203 |         |           | 327.7 | 0.0022 |          |       |       |
|         |           | 17.5   | 0.0007 |          |       |       |         |           | 333.5 | 0.0079 |          |       |       |
|         |           | 19.6   | 0.0002 |          |       |       |         |           | 352.9 | 0.0014 |          |       |       |
|         |           | 480.9  | 0.0015 |          |       |       |         |           | 367.9 | 0.0011 |          |       |       |
|         |           | 657.9  | 0.9809 |          |       |       |         |           | 370.0 | 0.0016 |          |       |       |
|         |           | 1024.0 | 0.0108 |          |       |       |         |           | 371.6 | 0.0016 |          |       |       |
|         |           | 1269.0 | 0.0016 |          |       |       |         |           | 377.9 | 0.0017 |          |       |       |
|         |           | 1516.0 | 0.0012 |          |       |       |         |           | 379.3 | 0.0023 |          |       |       |
|         |           | 798.3  | 0.0049 |          |       |       |         |           | 381.2 | 0.0031 |          |       |       |
|         |           |        |        |          |       |       |         |           | 398.7 | 0.0092 |          |       |       |
| 97mNb   | 60.0s     | 16.5   | 0.0037 | 1.259-4  | 2.818 | 1.063 |         |           | 408.5 | 0.0163 |          |       |       |
|         |           | 16.6   | 0.0072 |          |       |       |         |           | 421.4 | 0.0042 |          |       |       |
|         |           | 18.6   | 0.0021 |          |       |       |         |           | 432.9 | 0.0011 |          |       |       |
|         |           | 743.4  | 0.5796 |          |       |       |         |           | 448.5 | 0.0070 |          |       |       |
|         |           |        |        |          |       |       |         |           | 469.0 | 0.0012 |          |       |       |
| 91Mo    | 15.5m     | 16.5   | 0.0117 | 1.870-4  | 1.804 | 1.661 |         |           | 497.0 | 0.0014 |          |       |       |
|         |           | 16.6   | 0.0225 |          |       |       |         |           | 499.6 | 0.0136 |          |       |       |
|         |           | 18.6   | 0.0065 |          |       |       |         |           | 505.1 | 0.0134 |          |       |       |
|         |           | 1582.0 | 0.0023 |          |       |       |         |           | 505.9 | 0.1210 |          |       |       |
|         |           | 1637.0 | 0.0033 |          |       |       |         |           | 510.1 | 0.0100 |          |       |       |
|         |           | 2632.0 | 0.0012 |          |       |       |         |           | 512.2 | 0.0179 |          |       |       |
|         |           | 2233.0 | 0.0030 |          |       |       |         |           | 514.1 | 0.0083 |          |       |       |
|         |           | 511.0  | 1.8760 |          |       |       |         |           | 515.8 | 0.0052 |          |       |       |
|         |           |        |        |          |       |       |         |           | 523.8 | 0.0018 |          |       |       |
| 93Mo    | 3500.0y   | 16.5   | 0.1811 | 7.963-5  | 0.002 | 1.292 |         |           | 533.5 | 0.0041 |          |       |       |
|         |           | 16.6   | 0.3477 |          |       |       |         |           | 566.5 | 0.0075 |          |       |       |
|         |           | 18.6   | 0.0999 |          |       |       |         |           | 571.7 | 0.0019 |          |       |       |
|         |           |        |        |          |       |       |         |           | 590.1 | 0.0576 |          |       |       |
| 99Mo    | 2.8d      | 18.2   | 0.0090 | 3.046-5  | 2.571 | 1.165 |         |           | 590.8 | 0.1670 |          |       |       |
|         |           | 18.4   | 0.0173 |          |       |       |         |           | 603.0 | 0.0010 |          |       |       |
|         |           | 20.6   | 0.0052 |          |       |       |         |           | 606.8 | 0.0022 |          |       |       |
|         |           | 40.6   | 0.0088 |          |       |       |         |           | 608.3 | 0.0109 |          |       |       |
|         |           | 140.5  | 0.0379 |          |       |       |         |           | 611.6 | 0.0015 |          |       |       |
|         |           | 181.1  | 0.0623 |          |       |       |         |           | 625.6 | 0.0011 |          |       |       |
|         |           | 366.4  | 0.0137 |          |       |       |         |           | 642.6 | 0.0127 |          |       |       |
|         |           | 739.6  | 0.1280 |          |       |       |         |           | 660.6 | 0.0023 |          |       |       |
|         |           | 778.0  | 0.0448 |          |       |       |         |           | 695.5 | 0.0595 |          |       |       |
|         |           | 822.9  | 0.0013 |          |       |       |         |           | 701.8 | 0.0034 |          |       |       |

| Nuclide | Half-life | E      | Prob. | $\Gamma$ | T | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|-------|----------|---|-------|---------|-----------|--------|--------|----------|-------|-------|
|         | 712.9     | 0.0334 |       |          |   |       |         | 1383.0    | 0.0117 |        |          |       |       |
|         | 732.9     | 0.0027 |       |          |   |       |         | 1395.0    | 0.0062 |        |          |       |       |
|         | 739.5     | 0.0031 |       |          |   |       |         | 1414.0    | 0.0051 |        |          |       |       |
|         | 773.8     | 0.0034 |       |          |   |       |         | 1418.0    | 0.0089 |        |          |       |       |
|         | 775.8     | 0.0011 |       |          |   |       |         | 1430.0    | 0.0014 |        |          |       |       |
|         | 778.2     | 0.0098 |       |          |   |       |         | 1432.0    | 0.0037 |        |          |       |       |
|         | 790.0     | 0.0013 |       |          |   |       |         | 1441.0    | 0.0016 |        |          |       |       |
|         | 804.2     | 0.0102 |       |          |   |       |         | 1486.0    | 0.0011 |        |          |       |       |
|         | 815.2     | 0.0018 |       |          |   |       |         | 1514.0    | 0.0019 |        |          |       |       |
|         | 853.0     | 0.0024 |       |          |   |       |         | 1518.0    | 0.0022 |        |          |       |       |
|         | 859.1     | 0.0011 |       |          |   |       |         | 1520.0    | 0.0024 |        |          |       |       |
|         | 869.7     | 0.0035 |       |          |   |       |         | 1523.0    | 0.0030 |        |          |       |       |
|         | 871.1     | 0.0157 |       |          |   |       |         | 1527.0    | 0.0012 |        |          |       |       |
|         | 877.4     | 0.0315 |       |          |   |       |         | 1530.0    | 0.0028 |        |          |       |       |
|         | 883.3     | 0.0064 |       |          |   |       |         | 1533.0    | 0.0595 |        |          |       |       |
|         | 887.0     | 0.0024 |       |          |   |       |         | 1549.0    | 0.0015 |        |          |       |       |
|         | 888.7     | 0.0023 |       |          |   |       |         | 1590.0    | 0.0029 |        |          |       |       |
|         | 896.3     | 0.0022 |       |          |   |       |         | 1599.0    | 0.0179 |        |          |       |       |
|         | 903.4     | 0.0021 |       |          |   |       |         | 1662.0    | 0.0013 |        |          |       |       |
|         | 933.3     | 0.0077 |       |          |   |       |         | 1662.0    | 0.0056 |        |          |       |       |
|         | 934.2     | 0.0035 |       |          |   |       |         | 1674.0    | 0.0173 |        |          |       |       |
|         | 980.4     | 0.0027 |       |          |   |       |         | 1713.0    | 0.0021 |        |          |       |       |
|         | 987.9     | 0.0016 |       |          |   |       |         | 1755.0    | 0.0036 |        |          |       |       |
|         | 1007.0    | 0.0018 |       |          |   |       |         | 1760.0    | 0.0036 |        |          |       |       |
|         | 1011.0    | 0.0179 |       |          |   |       |         | 1760.0    | 0.0063 |        |          |       |       |
|         | 1011.0    | 0.0052 |       |          |   |       |         | 1768.0    | 0.0015 |        |          |       |       |
|         | 1012.0    | 0.1306 |       |          |   |       |         | 1840.0    | 0.0017 |        |          |       |       |
|         | 1019.0    | 0.0065 |       |          |   |       |         | 1840.0    | 0.0123 |        |          |       |       |
|         | 1020.0    | 0.0048 |       |          |   |       |         | 2028.0    | 0.0011 |        |          |       |       |
|         | 1050.0    | 0.0035 |       |          |   |       |         | 2032.0    | 0.0708 |        |          |       |       |
|         | 1064.0    | 0.0022 |       |          |   |       |         | 2038.0    | 0.0022 |        |          |       |       |
|         | 1066.0    | 0.0017 |       |          |   |       |         | 2041.0    | 0.0215 |        |          |       |       |
|         | 1161.0    | 0.0405 |       |          |   |       |         | 2089.0    | 0.0081 |        |          |       |       |
|         | 1169.0    | 0.0024 |       |          |   |       |         | 2113.0    | 0.0015 |        |          |       |       |
|         | 1184.0    | 0.0020 |       |          |   |       |         | 2115.0    | 0.0046 |        |          |       |       |
|         | 1187.0    | 0.0106 |       |          |   |       |         | 2223.0    | 0.0017 |        |          |       |       |
|         | 1200.0    | 0.0179 |       |          |   |       |         | 1110.0    | 0.0405 |        |          |       |       |
|         | 1210.0    | 0.0013 |       |          |   |       |         |           |        |        |          |       |       |
|         | 1249.0    | 0.0027 |       |          |   |       | 95Tc    | 20.0h     | 17.4   | 0.1908 | 2.089-4  | 2.540 | 1.180 |
|         | 1251.0    | 0.0470 |       |          |   |       |         |           | 17.5   | 0.3655 |          |       |       |
|         | 1260.0    | 0.0016 |       |          |   |       |         |           | 19.6   | 0.1074 |          |       |       |
|         | 1286.0    | 0.0015 |       |          |   |       |         |           | 204.1  | 0.0031 |          |       |       |
|         | 1291.0    | 0.0012 |       |          |   |       |         |           | 604.0  | 0.0030 |          |       |       |
|         | 1293.0    | 0.0021 |       |          |   |       |         |           | 765.8  | 0.9382 |          |       |       |
|         | 1304.0    | 0.0284 |       |          |   |       |         |           | 785.9  | 0.0015 |          |       |       |
|         | 1314.0    | 0.0024 |       |          |   |       |         |           | 869.6  | 0.0032 |          |       |       |
|         | 1326.0    | 0.0017 |       |          |   |       |         |           | 947.7  | 0.0195 |          |       |       |
|         | 1337.0    | 0.0014 |       |          |   |       |         |           | 1074.0 | 0.0374 |          |       |       |
|         | 1339.0    | 0.0018 |       |          |   |       |         |           | 693.7  | 0.0015 |          |       |       |
|         | 1346.0    | 0.0088 |       |          |   |       |         |           |        |        |          |       |       |
|         | 1356.0    | 0.0171 |       |          |   |       | 95mTc   | 61d       | 18.2   | 0.0020 | 1.939-4  | 2.129 | 1.407 |
|         | 1378.0    | 0.0025 |       |          |   |       |         |           | 18.4   | 0.0039 |          |       |       |
|         | 1380.0    | 0.0011 |       |          |   |       |         |           | 20.6   | 0.0012 |          |       |       |



| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | $T$   | $\mu$ | Nuclide | Half-life | E     | Prob.  | $\Gamma$ | $T$   | $\mu$  |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|-------|--------|----------|-------|--------|
|         |           | 38.9   | 0.0000 |          |       |       |         |           | 17.5  | 0.3612 |          |       |        |
|         |           | 17.4   | 0.1881 |          |       |       |         |           | 19.6  | 0.1061 |          |       |        |
|         |           | 17.5   | 0.3603 |          |       |       |         |           |       |        |          |       |        |
|         |           | 19.6   | 0.1058 |          |       |       | 97mTc   | 89d       | 18.2  | 0.1412 | 5.233-5  | 0.003 | 1060   |
|         |           | 204.1  | 0.6192 |          |       |       |         |           | 18.4  | 0.2701 |          |       |        |
|         |           | 253.0  | 0.0060 |          |       |       |         |           | 20.6  | 0.0810 |          |       |        |
|         |           | 582.1  | 0.2933 |          |       |       |         |           | 96.5  | 0.0032 |          |       |        |
|         |           | 616.5  | 0.0126 |          |       |       |         |           |       |        |          |       |        |
|         |           | 786.2  | 0.0848 |          |       |       | 98Tc    | 4.2+6y    | 19.1  | 0.0009 | 2.419-4  | 2.651 | 1.130  |
|         |           | 820.6  | 0.0461 |          |       |       |         |           | 19.3  | 0.0016 |          |       |        |
|         |           | 835.1  | 0.2607 |          |       |       |         |           | 21.7  | 0.0005 |          |       |        |
|         |           | 1039.0 | 0.0272 |          |       |       |         |           | 652.4 | 0.9974 |          |       |        |
|         |           | 883.1  | 0.0013 |          |       |       |         |           | 745.4 | 0.9982 |          |       |        |
|         |           | 511.0  | 0.0103 |          |       |       |         |           |       |        |          |       |        |
| 96Tc    | 4.3d      | 17.4   | 0.1882 | 4.894-4  | 3.015 | 0.994 | 99Tc    | 2.1+5y    | 89.4  | 0.0000 | 1.24-10  | 0.142 | 21.144 |
|         |           | 17.5   | 0.3605 |          |       |       | 99mTc   | 6.0h      | 18.2  | 0.0210 | 3.317-5  | 0.108 | 27.731 |
|         |           | 19.6   | 0.1059 |          |       |       |         |           | 18.4  | 0.0402 |          |       |        |
|         |           | 314.3  | 0.0243 |          |       |       |         |           | 20.6  | 0.0121 |          |       |        |
|         |           | 316.5  | 0.0140 |          |       |       |         |           | 140.5 | 0.8907 |          |       |        |
|         |           | 434.7  | 0.0075 |          |       |       |         |           | 142.6 | 0.0002 |          |       |        |
|         |           | 460.0  | 0.0043 |          |       |       | 101Tc   | 14.2m     | 19.1  | 0.0043 | 6.905-5  | 1.026 | 2.920  |
|         |           | 535.8  | 0.0041 |          |       |       |         |           | 19.3  | 0.0081 |          |       |        |
|         |           | 568.9  | 0.0092 |          |       |       |         |           | 21.7  | 0.0025 |          |       |        |
|         |           | 591.2  | 0.0011 |          |       |       |         |           | 127.2 | 0.0282 |          |       |        |
|         |           | 719.6  | 0.0020 |          |       |       |         |           | 179.6 | 0.0058 |          |       |        |
|         |           | 721.5  | 0.0012 |          |       |       |         |           | 184.1 | 0.0162 |          |       |        |
|         |           | 778.2  | 0.9976 |          |       |       |         |           | 233.7 | 0.0027 |          |       |        |
|         |           | 810.2  | 0.0021 |          |       |       |         |           | 238.3 | 0.0031 |          |       |        |
|         |           | 812.5  | 0.8180 |          |       |       |         |           | 306.8 | 0.8830 |          |       |        |
|         |           | 849.9  | 0.9776 |          |       |       |         |           | 311.5 | 0.0014 |          |       |        |
|         |           | 1091.0 | 0.0110 |          |       |       |         |           | 393.3 | 0.0011 |          |       |        |
|         |           | 1127.0 | 0.1516 |          |       |       |         |           | 516.0 | 0.0011 |          |       |        |
|         |           | 1200.0 | 0.0037 |          |       |       |         |           | 531.5 | 0.0102 |          |       |        |
|         |           | 650.9  | 0.0089 |          |       |       |         |           | 545.1 | 0.0600 |          |       |        |
| 96mTc   | 51.5m     | 18.2   | 0.0975 | 4.429-5  | 1.233 | 2.430 |         |           | 627.0 | 0.0041 |          |       |        |
|         |           | 18.4   | 0.1863 |          |       |       |         |           | 694.7 | 0.0115 |          |       |        |
|         |           | 20.6   | 0.0559 |          |       |       |         |           | 715.5 | 0.0069 |          |       |        |
|         |           | 34.4   | 0.0003 |          |       |       |         |           | 720.0 | 0.0019 |          |       |        |
|         |           | 17.4   | 0.0039 |          |       |       |         |           | 842.8 | 0.0023 |          |       |        |
|         |           | 17.5   | 0.0075 |          |       |       |         |           | 928.7 | 0.0013 |          |       |        |
|         |           | 19.6   | 0.0022 |          |       |       |         |           | 617.8 | 0.0048 |          |       |        |
|         |           | 480.7  | 0.0034 |          |       |       | 97Ru    | 2.9d      | 18.2  | 0.2003 | 1.194-4  | 0.290 | 10.317 |
|         |           | 719.6  | 0.0030 |          |       |       |         |           | 18.4  | 0.3829 |          |       |        |
|         |           | 778.2  | 0.0187 |          |       |       |         |           | 20.6  | 0.1149 |          |       |        |
|         |           | 847.7  | 0.0012 |          |       |       |         |           | 108.8 | 0.0011 |          |       |        |
|         |           | 849.9  | 0.0028 |          |       |       |         |           | 215.7 | 0.8550 |          |       |        |
|         |           | 1200.0 | 0.0107 |          |       |       |         |           | 324.5 | 0.1086 |          |       |        |
|         |           | 1498.0 | 0.0012 |          |       |       |         |           | 460.6 | 0.0012 |          |       |        |
|         |           | 887.6  | 0.0056 |          |       |       |         |           | 569.3 | 0.0087 |          |       |        |
| 97Tc    | 2.6+6y    | 17.4   | 0.1886 | 7.596-5  | 0.003 | 1176. |         |           | 713.3 | 0.0031 |          |       |        |

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| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---|-------|
| 111mCd  | 48.7m     | 23.0   | 0.1170 | 8.463-5  | 0.352 | 8.502 |         |           | 945.7  | 0.0153 |          |   |       |
|         |           | 23.2   | 0.2208 |          |       |       |         |           | 949.6  | 0.0022 |          |   |       |
|         |           | 26.1   | 0.0723 |          |       |       |         |           | 952.3  | 0.0014 |          |   |       |
|         |           | 150.8  | 0.3090 |          |       |       |         |           | 963.1  | 0.0061 |          |   |       |
|         |           | 245.4  | 0.9400 |          |       |       |         |           | 969.3  | 0.0045 |          |   |       |
| 115Cd   | 2.2d      | 24.0   | 0.0100 | 4.028-5  | 1.750 | 1.712 |         |           | 1036.0 | 0.0024 |          |   |       |
|         |           | 24.2   | 0.0189 |          |       |       |         |           | 1052.0 | 0.0379 |          |   |       |
|         |           | 27.3   | 0.0063 |          |       |       |         |           | 1053.0 | 0.0073 |          |   |       |
|         |           | 35.5   | 0.0045 |          |       |       |         |           | 1117.0 | 0.0103 |          |   |       |
|         |           | 231.4  | 0.0078 |          |       |       |         |           | 1120.0 | 0.0024 |          |   |       |
|         |           | 260.9  | 0.0206 |          |       |       |         |           | 1125.0 | 0.0045 |          |   |       |
|         |           | 492.3  | 0.0851 |          |       |       |         |           | 1142.0 | 0.0167 |          |   |       |
|         |           | 527.9  | 0.2915 |          |       |       |         |           | 1144.0 | 0.0014 |          |   |       |
|         |           | 296.9  | 0.0011 |          |       |       |         |           | 1183.0 | 0.0013 |          |   |       |
| 115mCd  | 44.6d     | 484.5  | 0.0019 | 3.430-6  | 3.848 | 0.779 |         |           | 1229.0 | 0.0061 |          |   |       |
|         |           | 933.8  | 0.0133 |          |       |       |         |           | 1232.0 | 0.0028 |          |   |       |
|         |           | 1291.0 | 0.0059 |          |       |       |         |           | 1248.0 | 0.0120 |          |   |       |
|         |           | 932.7  | 0.0010 |          |       |       |         |           | 1260.0 | 0.0114 |          |   |       |
| 117Cd   | 2.5h      | 24.0   | 0.0155 | 1.735-4  | 4.083 | 0.734 |         |           | 1273.0 | 0.0073 |          |   |       |
|         |           | 24.2   | 0.0292 |          |       |       |         |           | 1291.0 | 0.0067 |          |   |       |
|         |           | 27.3   | 0.0097 |          |       |       |         |           | 1294.0 | 0.0045 |          |   |       |
|         |           | 71.1   | 0.0039 |          |       |       |         |           | 1303.0 | 0.1836 |          |   |       |
|         |           | 89.7   | 0.0326 |          |       |       |         |           | 1315.0 | 0.0059 |          |   |       |
|         |           | 160.8  | 0.0025 |          |       |       |         |           | 1338.0 | 0.0162 |          |   |       |
|         |           | 220.9  | 0.0117 |          |       |       |         |           | 1362.0 | 0.0024 |          |   |       |
|         |           | 273.4  | 0.2790 |          |       |       |         |           | 1404.0 | 0.0012 |          |   |       |
|         |           | 279.8  | 0.0011 |          |       |       |         |           | 1409.0 | 0.0128 |          |   |       |
|         |           | 292.1  | 0.0064 |          |       |       |         |           | 1422.0 | 0.0033 |          |   |       |
|         |           | 344.5  | 0.1788 |          |       |       |         |           | 1431.0 | 0.0098 |          |   |       |
|         |           | 388.0  | 0.0031 |          |       |       |         |           | 1434.0 | 0.0011 |          |   |       |
|         |           | 397.2  | 0.0020 |          |       |       |         |           | 1450.0 | 0.0061 |          |   |       |
|         |           | 419.8  | 0.0018 |          |       |       |         |           | 1476.0 | 0.0042 |          |   |       |
|         |           | 434.2  | 0.0979 |          |       |       |         |           | 1562.0 | 0.0142 |          |   |       |
|         |           | 439.4  | 0.0011 |          |       |       |         |           | 1577.0 | 0.1119 |          |   |       |
|         |           | 463.0  | 0.0075 |          |       |       |         |           | 1578.0 | 0.0014 |          |   |       |
|         |           | 497.8  | 0.0011 |          |       |       |         |           | 1652.0 | 0.0028 |          |   |       |
|         |           | 527.0  | 0.0014 |          |       |       |         |           | 1682.0 | 0.0070 |          |   |       |
|         |           | 627.0  | 0.0011 |          |       |       |         |           | 1707.0 | 0.0100 |          |   |       |
|         |           | 660.8  | 0.0011 |          |       |       |         |           | 1723.0 | 0.0201 |          |   |       |
|         |           | 699.6  | 0.0024 |          |       |       |         |           | 1739.0 | 0.0013 |          |   |       |
|         |           | 712.7  | 0.0056 |          |       |       |         |           | 1856.0 | 0.0025 |          |   |       |
|         |           | 716.4  | 0.0020 |          |       |       |         |           | 1867.0 | 0.0011 |          |   |       |
|         |           | 728.6  | 0.0024 |          |       |       |         |           | 2012.0 | 0.0011 |          |   |       |
|         |           | 748.1  | 0.0056 |          |       |       |         |           | 959.7  | 0.0212 |          |   |       |
|         |           | 831.8  | 0.0226 |          |       |       |         |           |        |        |          |   |       |
|         |           | 840.2  | 0.0081 |          |       |       |         |           |        |        |          |   |       |
|         |           | 850.7  | 0.0012 |          |       |       |         |           |        |        |          |   |       |
|         |           | 861.3  | 0.0028 |          |       |       |         |           |        |        |          |   |       |
|         |           | 862.6  | 0.0061 |          |       |       |         |           |        |        |          |   |       |
|         |           | 880.7  | 0.0396 |          |       |       |         |           |        |        |          |   |       |
| 117mCd  | 3.4h      | 24.0   | 0.0023 | 2.909-4  | 4.831 | 0.620 |         |           | 24.2   | 0.0043 |          |   |       |
|         |           | 27.3   | 0.0014 |          |       |       |         |           | 27.3   | 0.0014 |          |   |       |
|         |           | 97.7   | 0.0105 |          |       |       |         |           | 97.7   | 0.0105 |          |   |       |
|         |           | 99.4   | 0.0010 |          |       |       |         |           | 99.4   | 0.0010 |          |   |       |
|         |           | 168.6  | 0.0029 |          |       |       |         |           | 168.6  | 0.0029 |          |   |       |
|         |           | 220.9  | 0.0024 |          |       |       |         |           | 220.9  | 0.0024 |          |   |       |
|         |           | 273.4  | 0.0029 |          |       |       |         |           | 273.4  | 0.0029 |          |   |       |

| Nuclide | Half-life | E      | Prob. | $\Gamma$ | T | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|-------|----------|---|-------|---------|-----------|--------|--------|----------|-------|--------|
|         | 292.1     | 0.0010 |       |          |   |       | 111In   | 2.8d      | 23.0   | 0.2363 | 1.356-4  | 0.290 | 10.321 |
|         | 299.4     | 0.0045 |       |          |   |       |         |           | 23.2   | 0.4458 |          |       |        |
|         | 310.3     | 0.0050 |       |          |   |       |         |           | 26.1   | 0.1460 |          |       |        |
|         | 325.3     | 0.0013 |       |          |   |       |         |           | 171.3  | 0.9024 |          |       |        |
|         | 344.5     | 0.0026 |       |          |   |       |         |           | 245.4  | 0.9400 |          |       |        |
|         | 366.9     | 0.0333 |       |          |   |       |         |           | 150.8  | 0.0000 |          |       |        |
|         | 439.4     | 0.0018 |       |          |   |       | 113mIn  | 1.7h      | 24.0   | 0.0683 | 6.567-5  | 1.048 | 2.858  |
|         | 460.9     | 0.0162 |       |          |   |       |         |           | 24.2   | 0.1287 |          |       |        |
|         | 484.8     | 0.0102 |       |          |   |       |         |           | 27.3   | 0.0427 |          |       |        |
|         | 545.0     | 0.0016 |       |          |   |       |         |           | 391.7  | 0.6490 |          |       |        |
|         | 564.4     | 0.1467 |       |          |   |       |         |           |        |        |          |       |        |
|         | 597.3     | 0.0013 |       |          |   |       | 114In   | 1.2m      | 23.0   | 0.0019 | 6.168-6  | 2.207 | 1.358  |
|         | 617.5     | 0.0034 |       |          |   |       |         |           | 23.2   | 0.0037 |          |       |        |
|         | 627.3     | 0.0024 |       |          |   |       |         |           | 26.1   | 0.0012 |          |       |        |
|         | 631.8     | 0.0280 |       |          |   |       |         |           | 575.8  | 0.0506 |          |       |        |
|         | 663.5     | 0.0068 |       |          |   |       |         |           | 558.4  | 0.0001 |          |       |        |
|         | 712.7     | 0.0100 |       |          |   |       |         |           | 1300.0 | 0.0020 |          |       |        |
|         | 730.8     | 0.0010 |       |          |   |       |         |           |        |        |          |       |        |
|         | 748.1     | 0.0445 |       |          |   |       | 114mIn  | 49.5d     | 24.0   | 0.0967 | 4.066-5  | 1.289 | 2.324  |
|         | 762.7     | 0.0173 |       |          |   |       |         |           | 24.2   | 0.1821 |          |       |        |
|         | 788.2     | 0.0050 |       |          |   |       |         |           | 27.3   | 0.0605 |          |       |        |
|         | 827.6     | 0.0026 |       |          |   |       |         |           | 190.3  | 0.1595 |          |       |        |
|         | 860.4     | 0.0789 |       |          |   |       |         |           | 23.0   | 0.0091 |          |       |        |
|         | 880.7     | 0.0071 |       |          |   |       |         |           | 23.2   | 0.0173 |          |       |        |
|         | 886.0     | 0.0039 |       |          |   |       |         |           | 26.1   | 0.0056 |          |       |        |
|         | 929.3     | 0.0079 |       |          |   |       |         |           | 558.4  | 0.0448 |          |       |        |
|         | 931.4     | 0.0364 |       |          |   |       |         |           | 725.2  | 0.0449 |          |       |        |
|         | 957.2     | 0.0039 |       |          |   |       |         |           |        |        |          |       |        |
|         | 1029.0    | 0.1166 |       |          |   |       | 115mIn  | 4.4h      | 24.0   | 0.0949 | 5.329-5  | 0.710 | 4.220  |
|         | 1066.0    | 0.2306 |       |          |   |       |         |           | 24.2   | 0.1788 |          |       |        |
|         | 1171.0    | 0.0065 |       |          |   |       |         |           | 27.3   | 0.0594 |          |       |        |
|         | 1196.0    | 0.0039 |       |          |   |       |         |           | 336.3  | 0.4671 |          |       |        |
|         | 1205.0    | 0.0013 |       |          |   |       |         |           | 497.4  | 0.0005 |          |       |        |
|         | 1209.0    | 0.0013 |       |          |   |       |         |           |        |        |          |       |        |
|         | 1209.0    | 0.0018 |       |          |   |       | 116mIn  | 54.1m     | 25.0   | 0.0028 | 3.646-4  | 4.442 | 0.674  |
|         | 1235.0    | 0.1100 |       |          |   |       |         |           | 25.3   | 0.0053 |          |       |        |
|         | 1257.0    | 0.0018 |       |          |   |       |         |           | 28.5   | 0.0018 |          |       |        |
|         | 1339.0    | 0.0207 |       |          |   |       |         |           | 138.3  | 0.0330 |          |       |        |
|         | 1366.0    | 0.0165 |       |          |   |       |         |           | 263.0  | 0.0014 |          |       |        |
|         | 1433.0    | 0.1344 |       |          |   |       |         |           | 303.8  | 0.0012 |          |       |        |
|         | 1652.0    | 0.0047 |       |          |   |       |         |           | 355.4  | 0.0085 |          |       |        |
|         | 1670.0    | 0.0063 |       |          |   |       |         |           | 417.0  | 0.2780 |          |       |        |
|         | 1958.0    | 0.0016 |       |          |   |       |         |           | 463.3  | 0.0085 |          |       |        |
|         | 1997.0    | 0.2620 |       |          |   |       |         |           | 689.0  | 0.0019 |          |       |        |
|         | 2096.0    | 0.0744 |       |          |   |       |         |           | 705.7  | 0.0019 |          |       |        |
|         | 2323.0    | 0.0786 |       |          |   |       |         |           | 779.5  | 0.0027 |          |       |        |
|         | 2401.0    | 0.0076 |       |          |   |       |         |           | 781.1  | 0.0011 |          |       |        |
|         | 2417.0    | 0.0102 |       |          |   |       |         |           | 818.7  | 0.1158 |          |       |        |
|         | 2462.0    | 0.0021 |       |          |   |       |         |           | 972.5  | 0.0046 |          |       |        |
|         | 2476.0    | 0.0019 |       |          |   |       |         |           | 1097.0 | 0.5535 |          |       |        |
|         | 2541.0    | 0.0015 |       |          |   |       |         |           | 1293.0 | 0.8450 |          |       |        |
|         | 865.4     | 0.0071 |       |          |   |       |         |           | 1508.0 | 0.0989 |          |       |        |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T      | $\mu$   |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|--------|---------|
|         |           | 1752.0 | 0.0239 |          |       |        |         |           | 1089.0 | 0.0404 |          |        |         |
|         |           | 2112.0 | 0.1538 |          |       |        |         |           | 1151.0 | 0.0010 |          |        |         |
|         |           | 707.4  | 0.0134 |          |       |        |         |           | 1173.0 | 0.0019 |          |        |         |
| 117In   | 43.8m     | 25.0   | 0.0296 | 1.347-4  | 1.779 | 1.684  |         |           | 1221.0 | 0.0022 |          |        |         |
|         |           | 25.3   | 0.0556 |          |       |        |         |           | 1420.0 | 0.0046 |          |        |         |
|         |           | 28.5   | 0.0187 |          |       |        |         |           | 1806.0 | 0.0015 |          |        |         |
|         |           | 158.6  | 0.8639 |          |       |        |         |           | 2002.0 | 0.0206 |          |        |         |
|         |           | 396.6  | 0.0014 |          |       |        |         |           | 2275.0 | 0.0018 |          |        |         |
|         |           | 553.0  | 0.9930 |          |       |        |         |           | 833.2  | 0.0070 |          |        |         |
| 117mIn  | 1.9h      | 24.0   | 0.0526 | 3.060-5  | 0.534 | 5.608  | 126Sn   | 1.0+5y    | 21.6   | 0.0124 | 3.408-5  | 0.095  | 31.647  |
|         |           | 24.2   | 0.0990 |          |       |        |         |           | 23.3   | 0.0640 |          |        |         |
|         |           | 27.3   | 0.0329 |          |       |        |         |           | 26.1   | 0.0833 |          |        |         |
|         |           | 315.3  | 0.1945 |          |       |        |         |           | 26.4   | 0.1561 |          |        |         |
|         |           | 25.0   | 0.0052 |          |       |        |         |           | 29.7   | 0.0534 |          |        |         |
|         |           | 25.3   | 0.0098 |          |       |        |         |           | 42.6   | 0.0050 |          |        |         |
|         |           | 28.5   | 0.0033 |          |       |        |         |           | 64.3   | 0.0958 |          |        |         |
|         |           | 158.6  | 0.1587 |          |       |        |         |           | 86.9   | 0.0892 |          |        |         |
|         |           | 918.6  | 0.0003 |          |       |        |         |           | 87.6   | 0.3700 |          |        |         |
|         |           |        |        |          |       |        |         |           | 22.7   | 0.0010 |          |        |         |
| 113Sn   | 115.1d    | 24.0   | 0.2069 | 4.844-5  | 0.006 | 462.8  | 117Sb   | 2.8h      | 25.0   | 0.2350 | 8.214-5  | 0.210  | 14.263  |
|         |           | 24.2   | 0.3896 |          |       |        |         |           | 25.3   | 0.4409 |          |        |         |
|         |           | 27.3   | 0.1294 |          |       |        |         |           | 28.5   | 0.1487 |          |        |         |
|         |           | 255.1  | 0.0193 |          |       |        |         |           | 158.6  | 0.8610 |          |        |         |
|         |           | 638.1  | 0.0000 |          |       |        |         |           | 861.3  | 0.0031 |          |        |         |
| 117mSn  | 13.6d     | 25.0   | 0.1873 | 6.796-5  | 0.113 | 26.595 |         |           | 1005.0 | 0.0021 |          |        |         |
|         |           | 25.3   | 0.3514 |          |       |        |         |           | 1021.0 | 0.0010 |          |        |         |
|         |           | 28.5   | 0.1185 |          |       |        |         |           | 1021.0 | 0.0011 |          |        |         |
|         |           | 156.0  | 0.0211 |          |       |        |         |           | 1037.0 | 0.0026 |          |        |         |
|         |           | 158.6  | 0.8640 |          |       |        |         |           | 511.0  | 0.0340 |          |        |         |
| 119mSn  | 293.0d    | 23.9   | 0.1610 | 2.789-5  | 0.006 | 517.0  | 122Sb   | 2.7d      | 25.0   | 0.0050 | 8.140-5  | 2.141  | 1.399   |
|         |           | 25.0   | 0.0788 |          |       |        |         |           | 25.3   | 0.0094 |          |        |         |
|         |           | 25.3   | 0.1479 |          |       |        |         |           | 28.5   | 0.0032 |          |        |         |
|         |           | 28.5   | 0.0499 |          |       |        |         |           | 1140.0 | 0.0077 |          |        |         |
|         |           | 65.7   | 0.0002 |          |       |        |         |           | 27.2   | 0.0009 |          |        |         |
|         |           |        |        |          |       |        |         |           | 27.5   | 0.0016 |          |        |         |
|         |           |        |        |          |       |        |         |           | 31.0   | 0.0006 |          |        |         |
| 123Sn   | 129.2d    | 1089.0 | 0.0060 | 1.061-6  | 4.025 | 0.744  |         |           | 563.9  | 0.7065 |          |        |         |
|         |           | 992.9  | 0.0004 |          |       |        |         |           | 692.8  | 0.0374 |          |        |         |
| 125Sn   | 9.6d      | 331.9  | 0.0129 | 4.662-5  | 4.069 | 0.736  |         |           | 1257.0 | 0.0078 |          |        |         |
|         |           | 350.9  | 0.0022 |          |       |        |         |           | 1179.0 | 0.0002 |          |        |         |
|         |           | 469.7  | 0.0129 |          |       |        |         |           |        |        |          |        |         |
|         |           | 800.5  | 0.0095 |          |       |        |         |           | 124Sb  | 60.2d  | 27.2     | 0.0010 | 2.859-4 |
|         |           | 822.6  | 0.0378 |          |       |        |         |           |        |        | 27.5     | 0.0019 | 4.195   |
|         |           | 893.7  | 0.0023 |          |       |        |         |           |        |        | 31.0     | 0.0007 | 0.714   |
|         |           | 915.5  | 0.0378 |          |       |        |         |           |        |        | 400.0    | 0.0013 |         |
|         |           | 934.7  | 0.0015 |          |       |        |         |           |        |        | 444.0    | 0.0021 |         |
|         |           | 1017.0 | 0.0026 |          |       |        |         |           |        |        | 525.5    | 0.0017 |         |
|         |           | 1067.0 | 0.0860 |          |       |        |         |           |        |        | 602.7    | 0.9787 |         |
|         |           | 1087.0 | 0.0095 |          |       |        |         |           |        |        | 632.4    | 0.0015 |         |
|         |           |        |        |          |       |        |         |           |        |        | 645.9    | 0.0726 |         |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 709.3  | 0.0142 |          |       |       |         |           | 555.2  | 0.0169 |          |       |       |
|         |           | 713.8  | 0.0238 |          |       |       |         |           | 573.8  | 0.0667 |          |       |       |
|         |           | 722.8  | 0.1110 |          |       |       |         |           | 593.0  | 0.0747 |          |       |       |
|         |           | 735.7  | 0.0013 |          |       |       |         |           | 620.2  | 0.0090 |          |       |       |
|         |           | 790.7  | 0.0074 |          |       |       |         |           | 639.7  | 0.0090 |          |       |       |
|         |           | 968.2  | 0.0192 |          |       |       |         |           | 656.3  | 0.0219 |          |       |       |
|         |           | 1045.0 | 0.0186 |          |       |       |         |           | 666.3  | 0.9962 |          |       |       |
|         |           | 1325.0 | 0.0150 |          |       |       |         |           | 675.0  | 0.0369 |          |       |       |
|         |           | 1355.0 | 0.0100 |          |       |       |         |           | 695.0  | 0.9962 |          |       |       |
|         |           | 1368.0 | 0.0251 |          |       |       |         |           | 697.0  | 0.2889 |          |       |       |
|         |           | 1376.0 | 0.0044 |          |       |       |         |           | 720.5  | 0.5379 |          |       |       |
|         |           | 1437.0 | 0.0114 |          |       |       |         |           | 856.8  | 0.1763 |          |       |       |
|         |           | 1445.0 | 0.0022 |          |       |       |         |           | 954.0  | 0.0119 |          |       |       |
|         |           | 1489.0 | 0.0063 |          |       |       |         |           | 959.6  | 0.0050 |          |       |       |
|         |           | 1526.0 | 0.0040 |          |       |       |         |           | 989.3  | 0.0677 |          |       |       |
|         |           | 1580.0 | 0.0020 |          |       |       |         |           | 1035.0 | 0.0100 |          |       |       |
|         |           | 1691.0 | 0.4903 |          |       |       |         |           | 1061.0 | 0.0020 |          |       |       |
|         |           | 2091.0 | 0.0573 |          |       |       |         |           | 1064.0 | 0.0090 |          |       |       |
|         |           | 1208.0 | 0.0134 |          |       |       |         |           | 1213.0 | 0.0239 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1476.0 | 0.0028 |          |       |       |
| 125Sb   | 2.8y      | 27.2   | 0.1279 | 1.025-4  | 1.696 | 1.767 | 126mSb  | 19.0m     | 17.7   | 0.0000 | 2.814-4  | 2.366 | 1.266 |
|         |           | 27.5   | 0.2386 |          |       |       |         |           | 27.2   | 0.0039 |          |       |       |
|         |           | 31.0   | 0.0828 |          |       |       |         |           | 27.5   | 0.0072 |          |       |       |
|         |           | 35.5   | 0.0417 |          |       |       |         |           | 31.0   | 0.0025 |          |       |       |
|         |           | 116.9  | 0.0026 |          |       |       |         |           | 414.7  | 0.8567 |          |       |       |
|         |           | 172.6  | 0.0018 |          |       |       |         |           | 620.0  | 0.0154 |          |       |       |
|         |           | 176.3  | 0.0689 |          |       |       |         |           | 666.3  | 0.8567 |          |       |       |
|         |           | 204.1  | 0.0032 |          |       |       |         |           | 695.0  | 0.8567 |          |       |       |
|         |           | 208.1  | 0.0024 |          |       |       |         |           | 928.2  | 0.0128 |          |       |       |
|         |           | 227.9  | 0.0013 |          |       |       |         |           | 1035.0 | 0.0180 |          |       |       |
|         |           | 321.0  | 0.0042 |          |       |       |         |           | 1061.0 | 0.0051 |          |       |       |
|         |           | 380.4  | 0.0150 |          |       |       |         |           | 1476.0 | 0.0034 |          |       |       |
|         |           | 408.0  | 0.0018 |          |       |       |         |           |        |        |          |       |       |
|         |           | 427.9  | 0.2933 |          |       |       |         |           |        |        |          |       |       |
|         |           | 443.5  | 0.0030 |          |       |       |         |           |        |        |          |       |       |
|         |           | 463.4  | 0.1035 |          |       |       |         |           |        |        |          |       |       |
|         |           | 600.6  | 0.1777 |          |       |       |         |           |        |        |          |       |       |
|         |           | 606.6  | 0.0502 |          |       |       |         |           |        |        |          |       |       |
|         |           | 635.9  | 0.1132 |          |       |       |         |           |        |        |          |       |       |
|         |           | 671.4  | 0.0181 |          |       |       |         |           |        |        |          |       |       |
|         |           | 159.0  | 0.0007 |          |       |       |         |           |        |        |          |       |       |
|         |           |        |        |          |       |       | 127Sb   | 3.9d      | 27.2   | 0.0106 | 1.196-4  | 2.386 | 1.256 |
|         |           |        |        |          |       |       |         |           | 27.5   | 0.0197 |          |       |       |
|         |           |        |        |          |       |       |         |           | 31.0   | 0.0068 |          |       |       |
|         |           |        |        |          |       |       |         |           | 61.1   | 0.0142 |          |       |       |
|         |           |        |        |          |       |       |         |           | 154.3  | 0.0011 |          |       |       |
|         |           |        |        |          |       |       |         |           | 252.4  | 0.0839 |          |       |       |
|         |           |        |        |          |       |       |         |           | 280.4  | 0.0054 |          |       |       |
|         |           |        |        |          |       |       |         |           | 290.8  | 0.0182 |          |       |       |
|         |           |        |        |          |       |       |         |           | 293.3  | 0.0029 |          |       |       |
|         |           |        |        |          |       |       |         |           | 310.0  | 0.0020 |          |       |       |
|         |           |        |        |          |       |       |         |           | 391.8  | 0.0093 |          |       |       |
|         |           |        |        |          |       |       |         |           | 405.0  | 0.0011 |          |       |       |
|         |           |        |        |          |       |       |         |           | 411.6  | 0.0343 |          |       |       |
|         |           |        |        |          |       |       |         |           | 440.7  | 0.0025 |          |       |       |
|         |           |        |        |          |       |       |         |           | 444.9  | 0.0421 |          |       |       |
|         |           |        |        |          |       |       |         |           | 451.0  | 0.0018 |          |       |       |
|         |           |        |        |          |       |       |         |           | 456.0  | 0.0011 |          |       |       |
|         |           |        |        |          |       |       |         |           | 473.0  | 0.2503 |          |       |       |
|         |           |        |        |          |       |       |         |           | 502.8  | 0.0061 |          |       |       |
| 126Sb   | 12.4d     | 27.2   | 0.0044 | 4.841-4  | 2.556 | 1.172 |         |           |        |        |          |       |       |
|         |           | 27.5   | 0.0081 |          |       |       |         |           |        |        |          |       |       |
|         |           | 31.0   | 0.0028 |          |       |       |         |           |        |        |          |       |       |
|         |           | 149.3  | 0.0040 |          |       |       |         |           |        |        |          |       |       |
|         |           | 208.6  | 0.0050 |          |       |       |         |           |        |        |          |       |       |
|         |           | 223.8  | 0.0140 |          |       |       |         |           |        |        |          |       |       |
|         |           | 278.6  | 0.0239 |          |       |       |         |           |        |        |          |       |       |
|         |           | 296.5  | 0.0448 |          |       |       |         |           |        |        |          |       |       |
|         |           | 297.1  | 0.0050 |          |       |       |         |           |        |        |          |       |       |
|         |           | 414.7  | 0.8328 |          |       |       |         |           |        |        |          |       |       |
|         |           | 415.3  | 0.0100 |          |       |       |         |           |        |        |          |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 543.0  | 0.0264 |          |       |       |         |           | 1084.0 | 0.0055 |          |       |       |
|         |           | 584.2  | 0.0032 |          |       |       |         |           | 1104.0 | 0.0023 |          |       |       |
|         |           | 603.6  | 0.0425 |          |       |       |         |           | 1125.0 | 0.0011 |          |       |       |
|         |           | 637.8  | 0.0036 |          |       |       |         |           | 1139.0 | 0.0018 |          |       |       |
|         |           | 653.5  | 0.0025 |          |       |       |         |           | 1162.0 | 0.0011 |          |       |       |
|         |           | 666.9  | 0.0054 |          |       |       |         |           | 1168.0 | 0.0027 |          |       |       |
|         |           | 682.3  | 0.0054 |          |       |       |         |           | 1208.0 | 0.0096 |          |       |       |
|         |           | 685.2  | 0.3570 |          |       |       |         |           | 1223.0 | 0.0018 |          |       |       |
|         |           | 698.5  | 0.0339 |          |       |       |         |           | 1237.0 | 0.0027 |          |       |       |
|         |           | 722.2  | 0.0175 |          |       |       |         |           | 1257.0 | 0.0037 |          |       |       |
|         |           | 745.9  | 0.0011 |          |       |       |         |           | 1261.0 | 0.0078 |          |       |       |
|         |           | 783.8  | 0.1467 |          |       |       |         |           | 1273.0 | 0.0027 |          |       |       |
|         |           | 817.3  | 0.0027 |          |       |       |         |           | 1281.0 | 0.0059 |          |       |       |
|         |           | 820.1  | 0.0011 |          |       |       |         |           | 1300.0 | 0.0027 |          |       |       |
|         |           | 923.5  | 0.0046 |          |       |       |         |           | 1317.0 | 0.0037 |          |       |       |
|         |           | 1141.0 | 0.0036 |          |       |       |         |           | 1326.0 | 0.0055 |          |       |       |
|         |           | 1290.0 | 0.0035 |          |       |       |         |           | 1419.0 | 0.0055 |          |       |       |
|         |           | 736.7  | 0.0053 |          |       |       |         |           | 1436.0 | 0.0032 |          |       |       |
| 129Sb   | 4.4h      | 27.2   | 0.0009 | 2.307-4  | 3.566 | 0.840 |         |           | 1480.0 | 0.0050 |          |       |       |
|         |           | 27.5   | 0.0018 |          |       |       |         |           | 1526.0 | 0.0046 |          |       |       |
|         |           | 31.0   | 0.0006 |          |       |       |         |           | 1540.0 | 0.0014 |          |       |       |
|         |           | 96.1   | 0.0018 |          |       |       |         |           | 1569.0 | 0.0073 |          |       |       |
|         |           | 116.2  | 0.0018 |          |       |       |         |           | 1599.0 | 0.0055 |          |       |       |
|         |           | 146.6  | 0.0023 |          |       |       |         |           | 1621.0 | 0.0027 |          |       |       |
|         |           | 180.8  | 0.0270 |          |       |       |         |           | 1655.0 | 0.0105 |          |       |       |
|         |           | 244.7  | 0.0055 |          |       |       |         |           | 1724.0 | 0.0027 |          |       |       |
|         |           | 268.6  | 0.0027 |          |       |       |         |           | 1736.0 | 0.0635 |          |       |       |
|         |           | 295.5  | 0.0110 |          |       |       |         |           | 1842.0 | 0.0023 |          |       |       |
|         |           | 313.5  | 0.0091 |          |       |       |         |           | 1870.0 | 0.0032 |          |       |       |
|         |           | 332.5  | 0.0023 |          |       |       |         |           | 2070.0 | 0.0059 |          |       |       |
|         |           | 359.4  | 0.0302 |          |       |       |         |           | 2113.0 | 0.0037 |          |       |       |
|         |           | 363.0  | 0.0046 |          |       |       |         |           | 1311.0 | 0.0098 |          |       |       |
|         |           | 405.0  | 0.0146 |          |       |       |         |           |        |        |          |       |       |
|         |           | 453.5  | 0.0082 |          |       |       | 121Te   | 16.8d     | 26.1   | 0.2144 | 1.444-4  | 1.778 | 1.685 |
|         |           | 499.6  | 0.0023 |          |       |       |         |           | 26.4   | 0.4016 |          |       |       |
|         |           | 523.8  | 0.0169 |          |       |       |         |           | 29.7   | 0.1374 |          |       |       |
|         |           | 544.7  | 0.1906 |          |       |       |         |           | 37.1   | 0.0012 |          |       |       |
|         |           | 633.7  | 0.0293 |          |       |       |         |           | 65.6   | 0.0026 |          |       |       |
|         |           | 654.3  | 0.0320 |          |       |       |         |           | 470.5  | 0.0140 |          |       |       |
|         |           | 669.8  | 0.0087 |          |       |       |         |           | 507.6  | 0.1767 |          |       |       |
|         |           | 683.6  | 0.0073 |          |       |       |         |           | 573.1  | 0.8030 |          |       |       |
|         |           | 683.6  | 0.0544 |          |       |       |         |           |        |        |          |       |       |
|         |           | 737.1  | 0.0041 |          |       |       | 121mTe  | 154d      | 27.2   | 0.1011 | 6.703-5  | 0.518 | 5.781 |
|         |           | 761.0  | 0.0402 |          |       |       |         |           | 27.5   | 0.1886 |          |       |       |
|         |           | 773.4  | 0.0293 |          |       |       |         |           | 31.0   | 0.0655 |          |       |       |
|         |           | 786.6  | 0.0201 |          |       |       |         |           | 212.2  | 0.8147 |          |       |       |
|         |           | 812.8  | 0.4570 |          |       |       |         |           | 81.8   | 0.0005 |          |       |       |
|         |           | 876.2  | 0.0274 |          |       |       |         |           | 26.1   | 0.0459 |          |       |       |
|         |           | 914.6  | 0.2130 |          |       |       |         |           | 26.4   | 0.0859 |          |       |       |
|         |           | 939.7  | 0.0078 |          |       |       |         |           | 29.7   | 0.0294 |          |       |       |
|         |           | 966.4  | 0.0818 |          |       |       |         |           | 37.1   | 0.0094 |          |       |       |
|         |           | 995.4  | 0.0014 |          |       |       |         |           | 1102.0 | 0.0254 |          |       |       |
|         |           | 1030.0 | 0.1339 |          |       |       |         |           | 953.6  | 0.0016 |          |       |       |



| Nuclide            | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide           | Half-life | E                  | Prob.  | $\Gamma$ | T      | $\mu$               |
|--------------------|-----------|--------|--------|----------|-------|--------|-------------------|-----------|--------------------|--------|----------|--------|---------------------|
| <sup>123</sup> Te  | 1.0+13y   | 26.1   | 0.1357 | 2.687-5  | 0.007 | 429.8  | <sup>131</sup> Te | 25.0m     | 28.3               | 0.0368 | 8.065-5  | 2.491  | 1.203               |
|                    |           | 26.4   | 0.2542 |          |       |        |                   |           | 28.6               | 0.0685 |          |        |                     |
|                    |           | 29.7   | 0.0870 |          |       |        |                   |           | 32.3               | 0.0240 |          |        |                     |
| <sup>123m</sup> Te | 119.7d    | 27.2   | 0.1405 | 5.261-5  | 0.125 | 24.041 |                   |           | 149.7              | 0.6890 |          |        |                     |
|                    |           | 27.5   | 0.2620 |          |       |        |                   |           | 151.1              | 0.0017 |          |        |                     |
|                    |           | 31.0   | 0.0910 |          |       |        |                   |           | 342.9              | 0.0070 |          |        |                     |
|                    |           | 159.0  | 0.8410 |          |       |        |                   |           | 384.1              | 0.0090 |          |        |                     |
|                    |           | 89.1   | 0.0009 |          |       |        |                   |           | 452.3              | 0.1822 |          |        |                     |
| <sup>125m</sup> Te | 58d       | 27.2   | 0.3232 | 6.168-5  | 0.008 | 368.9  |                   |           | 492.7              | 0.0484 |          |        |                     |
|                    |           | 27.5   | 0.6030 |          |       |        |                   |           | 544.9              | 0.0043 |          |        |                     |
|                    |           | 31.0   | 0.2093 |          |       |        |                   |           | 567.3              | 0.0010 |          |        |                     |
|                    |           | 35.5   | 0.0649 |          |       |        |                   |           | 602.0              | 0.0420 |          |        |                     |
|                    |           | 109.3  | 0.0028 |          |       |        |                   |           | 605.5              | 0.0012 |          |        |                     |
|                    |           | 144.8  | 0.0000 |          |       |        |                   |           | 654.3              | 0.0153 |          |        |                     |
| <sup>127</sup> Te  | 9.4h      | 360.3  | 0.0013 | 9.428-7  | 1.239 | 2.419  |                   |           | 696.2              | 0.0018 |          |        |                     |
|                    |           | 417.9  | 0.0099 |          |       |        |                   |           | 727.0              | 0.0047 |          |        |                     |
|                    |           | 172.7  | 0.0013 |          |       |        |                   |           | 842.0              | 0.0020 |          |        |                     |
| <sup>127m</sup> Te | 109d      | 27.2   | 0.1041 | 1.977-5  | 0.008 | 373.5  |                   |           | 856.1              | 0.0013 |          |        |                     |
|                    |           | 27.5   | 0.1943 |          |       |        |                   |           | 898.5              | 0.0014 |          |        |                     |
|                    |           | 31.0   | 0.0674 |          |       |        |                   |           | 934.5              | 0.0087 |          |        |                     |
|                    |           | 88.3   | 0.0009 |          |       |        |                   |           | 948.5              | 0.0226 |          |        |                     |
|                    |           | 28.3   | 0.0030 |          |       |        |                   |           | 951.4              | 0.0033 |          |        |                     |
|                    |           | 28.6   | 0.0057 |          |       |        |                   |           | 997.3              | 0.0334 |          |        |                     |
|                    |           | 32.3   | 0.0020 |          |       |        |                   |           | 1008.0             | 0.0080 |          |        |                     |
|                    |           | 57.6   | 0.0038 |          |       |        |                   |           | 1098.0             | 0.0017 |          |        |                     |
|                    |           | 648.6  | 0.0001 |          |       |        |                   |           | 1147.0             | 0.0496 |          |        |                     |
| <sup>129</sup> Te  | 1.2h      | 27.8   | 0.1626 | 1.833-5  | 1.452 | 2.063  |                   |           | 1149.0             | 0.0011 |          |        |                     |
|                    |           | 209.0  | 0.0017 |          |       |        |                   |           | 1277.0             | 0.0012 |          |        |                     |
|                    |           | 250.6  | 0.0035 |          |       |        |                   |           | 1294.0             | 0.0048 |          |        |                     |
|                    |           | 278.4  | 0.0052 |          |       |        |                   |           | 1427.0             | 0.0011 |          |        |                     |
|                    |           | 281.3  | 0.0015 |          |       |        |                   |           | 1501.0             | 0.0012 |          |        |                     |
|                    |           | 459.6  | 0.0710 |          |       |        |                   |           | 696.0              | 0.0119 |          |        |                     |
|                    |           | 487.4  | 0.0131 |          |       |        |                   |           | <sup>131m</sup> Te | 1.2d   | 27.2     | 0.0359 | 2.447-4 3.344 0.896 |
|                    |           | 802.1  | 0.0018 |          |       |        |                   |           |                    |        | 27.5     | 0.0669 |                     |
|                    |           | 1084.0 | 0.0045 |          |       |        |                   |           |                    |        | 31.0     | 0.0232 |                     |
|                    |           | 1112.0 | 0.0018 |          |       |        |                   |           |                    |        | 182.3    | 0.0085 |                     |
|                    |           | 656.9  | 0.0038 |          |       |        |                   |           |                    |        | 28.3     | 0.0283 |                     |
| <sup>129m</sup> Te | 33.6d     | 27.2   | 0.0788 | 1.994-5  | 1.496 | 2.003  |                   |           |                    |        | 28.6     | 0.0527 |                     |
|                    |           | 27.5   | 0.1470 |          |       |        |                   |           |                    |        | 32.3     | 0.0185 |                     |
|                    |           | 31.0   | 0.0510 |          |       |        |                   |           |                    |        | 79.2     | 0.0013 |                     |
|                    |           | 105.5  | 0.0015 |          |       |        |                   |           |                    |        | 81.1     | 0.0407 |                     |
|                    |           | 556.6  | 0.0013 |          |       |        |                   |           |                    |        | 86.4     | 0.0015 |                     |
|                    |           | 695.9  | 0.0327 |          |       |        |                   |           |                    |        | 101.6    | 0.0017 |                     |
|                    |           | 729.6  | 0.0076 |          |       |        |                   |           |                    |        | 102.1    | 0.0794 |                     |
|                    |           | 742.5  | 0.0031 |          |       |        |                   |           |                    |        | 134.9    | 0.0071 |                     |
|                    |           |        |        |          |       |        |                   |           |                    |        | 149.7    | 0.0508 |                     |
|                    |           |        |        |          |       |        |                   |           |                    |        | 159.7    | 0.0013 |                     |
|                    |           |        |        |          |       |        |                   |           |                    |        | 182.3    | 0.0074 |                     |
|                    |           |        |        |          |       |        |                   |           |                    |        | 183.1    | 0.0015 |                     |
|                    |           |        |        |          |       |        |                   |           |                    |        | 188.1    | 0.0021 |                     |
|                    |           |        |        |          |       |        |                   |           |                    |        | 189.8    | 0.0050 |                     |
|                    |           |        |        |          |       |        |                   |           |                    |        | 190.5    | 0.0012 |                     |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|--------|----------|---|-------|---------|-----------|--------|--------|----------|-------|--------|
|         |           | 200.6  | 0.0755 |          |   |       |         |           | 1060.0 | 0.0155 |          |       |        |
|         |           | 214.0  | 0.0043 |          |   |       |         |           | 1125.0 | 0.1143 |          |       |        |
|         |           | 230.7  | 0.0019 |          |   |       |         |           | 1128.0 | 0.0097 |          |       |        |
|         |           | 240.9  | 0.0759 |          |   |       |         |           | 1149.0 | 0.0151 |          |       |        |
|         |           | 253.2  | 0.0065 |          |   |       |         |           | 1149.0 | 0.0038 |          |       |        |
|         |           | 255.4  | 0.0031 |          |   |       |         |           | 1151.0 | 0.0066 |          |       |        |
|         |           | 269.2  | 0.0011 |          |   |       |         |           | 1165.0 | 0.0014 |          |       |        |
|         |           | 278.6  | 0.0178 |          |   |       |         |           | 1207.0 | 0.0976 |          |       |        |
|         |           | 283.2  | 0.0039 |          |   |       |         |           | 1237.0 | 0.0066 |          |       |        |
|         |           | 309.5  | 0.0038 |          |   |       |         |           | 1315.0 | 0.0070 |          |       |        |
|         |           | 334.3  | 0.0957 |          |   |       |         |           | 1341.0 | 0.0010 |          |       |        |
|         |           | 335.4  | 0.0014 |          |   |       |         |           | 1395.0 | 0.0011 |          |       |        |
|         |           | 342.9  | 0.0039 |          |   |       |         |           | 1646.0 | 0.0124 |          |       |        |
|         |           | 351.3  | 0.0021 |          |   |       |         |           | 1888.0 | 0.0136 |          |       |        |
|         |           | 354.7  | 0.0023 |          |   |       |         |           | 2001.0 | 0.0202 |          |       |        |
|         |           | 365.0  | 0.0120 |          |   |       |         |           | 2168.0 | 0.0035 |          |       |        |
|         |           | 383.9  | 0.0020 |          |   |       |         |           | 2271.0 | 0.0038 |          |       |        |
|         |           | 417.4  | 0.0028 |          |   |       |         |           | 631.9  | 0.0355 |          |       |        |
|         |           | 432.4  | 0.0066 |          |   |       |         |           |        |        |          |       |        |
|         |           | 452.3  | 0.0155 |          |   |       | 132Te   | 3.3d      | 28.3   | 0.1831 | 7.549-5  | 0.296 | 10.115 |
|         |           | 462.9  | 0.0182 |          |   |       |         |           | 28.6   | 0.3409 |          |       |        |
|         |           | 468.2  | 0.0031 |          |   |       |         |           | 32.3   | 0.1195 |          |       |        |
|         |           | 492.7  | 0.0023 |          |   |       |         |           | 49.7   | 0.1311 |          |       |        |
|         |           | 524.8  | 0.0014 |          |   |       |         |           | 111.8  | 0.0185 |          |       |        |
|         |           | 530.7  | 0.0010 |          |   |       |         |           | 116.3  | 0.0194 |          |       |        |
|         |           | 541.4  | 0.0011 |          |   |       |         |           | 228.2  | 0.8800 |          |       |        |
|         |           | 586.3  | 0.0198 |          |   |       |         |           |        |        |          |       |        |
|         |           | 602.0  | 0.0031 |          |   |       | 133Te   | 12.4m     | 28.3   | 0.0063 | 1.580-4  | 3.333 | 0.899  |
|         |           | 609.4  | 0.0014 |          |   |       |         |           | 28.6   | 0.0117 |          |       |        |
|         |           | 665.1  | 0.0434 |          |   |       |         |           | 32.3   | 0.0041 |          |       |        |
|         |           | 685.9  | 0.0015 |          |   |       |         |           | 312.0  | 0.7080 |          |       |        |
|         |           | 695.6  | 0.0040 |          |   |       |         |           | 384.6  | 0.0028 |          |       |        |
|         |           | 702.5  | 0.0039 |          |   |       |         |           | 392.9  | 0.0057 |          |       |        |
|         |           | 713.1  | 0.0143 |          |   |       |         |           | 407.6  | 0.3009 |          |       |        |
|         |           | 744.2  | 0.0159 |          |   |       |         |           | 474.7  | 0.0120 |          |       |        |
|         |           | 773.7  | 0.3820 |          |   |       |         |           | 546.4  | 0.0057 |          |       |        |
|         |           | 774.1  | 0.0054 |          |   |       |         |           | 587.1  | 0.0050 |          |       |        |
|         |           | 782.5  | 0.0779 |          |   |       |         |           | 613.6  | 0.0028 |          |       |        |
|         |           | 793.7  | 0.1387 |          |   |       |         |           | 719.6  | 0.0665 |          |       |        |
|         |           | 822.8  | 0.0612 |          |   |       |         |           | 786.8  | 0.0559 |          |       |        |
|         |           | 844.9  | 0.0015 |          |   |       |         |           | 844.4  | 0.0326 |          |       |        |
|         |           | 852.2  | 0.0039 |          |   |       |         |           | 930.7  | 0.0446 |          |       |        |
|         |           | 852.2  | 0.2065 |          |   |       |         |           | 1001.0 | 0.0623 |          |       |        |
|         |           | 856.1  | 0.0062 |          |   |       |         |           | 1021.0 | 0.0269 |          |       |        |
|         |           | 865.1  | 0.0019 |          |   |       |         |           | 1062.0 | 0.0127 |          |       |        |
|         |           | 872.3  | 0.0010 |          |   |       |         |           | 1252.0 | 0.0113 |          |       |        |
|         |           | 910.0  | 0.0329 |          |   |       |         |           | 1308.0 | 0.0092 |          |       |        |
|         |           | 920.6  | 0.0120 |          |   |       |         |           | 1314.0 | 0.0078 |          |       |        |
|         |           | 923.4  | 0.0012 |          |   |       |         |           | 1333.0 | 0.0991 |          |       |        |
|         |           | 941.3  | 0.0078 |          |   |       |         |           | 1406.0 | 0.0057 |          |       |        |
|         |           | 987.8  | 0.0015 |          |   |       |         |           | 1474.0 | 0.0035 |          |       |        |
|         |           | 999.3  | 0.0017 |          |   |       |         |           | 1519.0 | 0.0050 |          |       |        |
|         |           | 1035.0 | 0.0010 |          |   |       |         |           | 1588.0 | 0.0028 |          |       |        |



| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|-------|-------|
| 122I    | 3.6m      | 27.2   | 0.0520 | 1.876-4  | 1.926 | 1.555  |         |           | 1919.0 | 0.0017 |          |       |       |
|         |           | 27.5   | 0.0970 |          |       |        |         |           | 2038.0 | 0.0033 |          |       |       |
|         |           | 31.0   | 0.0337 |          |       |        |         |           | 2079.0 | 0.0034 |          |       |       |
|         |           | 563.9  | 0.2100 |          |       |        |         |           | 2091.0 | 0.0055 |          |       |       |
|         |           | 683.5  | 0.0097 |          |       |        |         |           | 2099.0 | 0.0014 |          |       |       |
|         |           | 692.8  | 0.0149 |          |       |        |         |           | 2144.0 | 0.0011 |          |       |       |
|         |           | 793.2  | 0.0166 |          |       |        |         |           | 2232.0 | 0.0055 |          |       |       |
|         |           | 1075.0 | 0.0034 |          |       |        |         |           | 2283.0 | 0.0064 |          |       |       |
|         |           | 1257.0 | 0.0032 |          |       |        |         |           | 2747.0 | 0.0045 |          |       |       |
|         |           | 1500.0 | 0.0018 |          |       |        |         |           | 1175.0 | 0.0121 |          |       |       |
|         |           | 1747.0 | 0.0038 |          |       |        |         |           | 511.0  | 0.4654 |          |       |       |
|         |           | 1844.0 | 0.0016 |          |       |        | 125I    | 60.1d     | 27.2   | 0.3923 | 7.432-5  | 0.008 | 373.1 |
|         |           | 2193.0 | 0.0036 |          |       |        |         |           | 27.5   | 0.7320 |          |       |       |
|         |           | 1862.0 | 0.0081 |          |       |        |         |           | 31.0   | 0.2541 |          |       |       |
|         |           | 511.0  | 1.5210 |          |       |        |         |           | 35.5   | 0.0649 |          |       |       |
| 123I    | 13.1h     | 27.2   | 0.2463 | 7.475-5  | 0.154 | 19.422 | 126I    | 12.9d     | 27.2   | 0.1269 | 1.052-4  | 2.061 | 1.454 |
|         |           | 27.5   | 0.4595 |          |       |        |         |           | 27.5   | 0.2368 |          |       |       |
|         |           | 31.0   | 0.1595 |          |       |        |         |           | 31.0   | 0.0822 |          |       |       |
|         |           | 159.0  | 0.8340 |          |       |        |         |           | 666.3  | 0.4020 |          |       |       |
|         |           | 346.4  | 0.0013 |          |       |        |         |           | 753.8  | 0.0506 |          |       |       |
|         |           | 440.0  | 0.0043 |          |       |        |         |           | 1420.0 | 0.0036 |          |       |       |
|         |           | 505.3  | 0.0032 |          |       |        |         |           | 1781.0 | 0.0001 |          |       |       |
|         |           | 529.0  | 0.0139 |          |       |        |         |           | 511.0  | 0.0215 |          |       |       |
|         |           | 538.5  | 0.0038 |          |       |        |         |           | 29.5   | 0.0011 |          |       |       |
|         |           | 494.4  | 0.0048 |          |       |        |         |           | 29.8   | 0.0021 |          |       |       |
|         |           |        |        |          |       |        |         |           | 33.6   | 0.0008 |          |       |       |
| 124I    | 4.2d      | 27.2   | 0.1650 | 2.033-4  | 3.093 | 0.969  |         |           | 388.6  | 0.2908 |          |       |       |
|         |           | 27.5   | 0.3078 |          |       |        |         |           | 491.2  | 0.0243 |          |       |       |
|         |           | 31.0   | 0.1068 |          |       |        |         |           | 879.9  | 0.0064 |          |       |       |
|         |           | 541.2  | 0.0018 |          |       |        |         |           |        |        |          |       |       |
|         |           | 554.0  | 0.0010 |          |       |        | 128I    | 25.0m     | 27.2   | 0.0106 | 1.614-5  | 1.506 | 1.989 |
|         |           | 602.7  | 0.5900 |          |       |        |         |           | 27.5   | 0.0198 |          |       |       |
|         |           | 645.9  | 0.0092 |          |       |        |         |           | 31.0   | 0.0069 |          |       |       |
|         |           | 695.0  | 0.0018 |          |       |        |         |           | 743.2  | 0.0013 |          |       |       |
|         |           | 713.8  | 0.0011 |          |       |        |         |           | 29.5   | 0.0004 |          |       |       |
|         |           | 722.8  | 0.0974 |          |       |        |         |           | 29.8   | 0.0007 |          |       |       |
|         |           | 968.2  | 0.0040 |          |       |        |         |           | 33.6   | 0.0003 |          |       |       |
|         |           | 976.3  | 0.0010 |          |       |        |         |           | 442.9  | 0.1425 |          |       |       |
|         |           | 1045.0 | 0.0041 |          |       |        |         |           | 526.6  | 0.0137 |          |       |       |
|         |           | 1054.0 | 0.0012 |          |       |        |         |           | 969.4  | 0.0034 |          |       |       |
|         |           | 1325.0 | 0.0140 |          |       |        |         |           | 1051.0 | 0.0001 |          |       |       |
|         |           | 1368.0 | 0.0028 |          |       |        |         |           |        |        |          |       |       |
|         |           | 1376.0 | 0.0162 |          |       |        | 129I    | 1.6+7y    | 29.5   | 0.1996 | 3.401-5  | 0.010 | 287.0 |
|         |           | 1489.0 | 0.0018 |          |       |        |         |           | 29.8   | 0.3702 |          |       |       |
|         |           | 1509.0 | 0.0292 |          |       |        |         |           | 33.6   | 0.1316 |          |       |       |
|         |           | 1560.0 | 0.0016 |          |       |        |         |           | 39.6   | 0.0752 |          |       |       |
|         |           | 1638.0 | 0.0019 |          |       |        |         |           |        |        |          |       |       |
|         |           | 1676.0 | 0.0011 |          |       |        | 130I    | 12.4h     | 29.5   | 0.0042 | 3.768-4  | 2.525 | 1.186 |
|         |           | 1691.0 | 0.1015 |          |       |        |         |           | 29.8   | 0.0078 |          |       |       |
|         |           | 1720.0 | 0.0017 |          |       |        |         |           | 33.6   | 0.0028 |          |       |       |
|         |           | 1851.0 | 0.0020 |          |       |        |         |           | 418.0  | 0.3415 |          |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 457.7  | 0.0024 |          |       |       |         |           | 505.9  | 0.0503 |          |       |       |
|         |           | 510.3  | 0.0085 |          |       |       |         |           | 522.7  | 0.1609 |          |       |       |
|         |           | 536.1  | 0.9900 |          |       |       |         |           | 535.5  | 0.0052 |          |       |       |
|         |           | 539.1  | 0.0140 |          |       |       |         |           | 540.0  | 0.0011 |          |       |       |
|         |           | 553.9  | 0.0066 |          |       |       |         |           | 547.1  | 0.0125 |          |       |       |
|         |           | 586.0  | 0.0169 |          |       |       |         |           | 500.0  | 0.0014 |          |       |       |
|         |           | 603.5  | 0.0061 |          |       |       |         |           | 620.8  | 0.0039 |          |       |       |
|         |           | 668.5  | 0.9613 |          |       |       |         |           | 621.2  | 0.0158 |          |       |       |
|         |           | 686.0  | 0.0107 |          |       |       |         |           | 630.2  | 0.1372 |          |       |       |
|         |           | 739.5  | 0.8227 |          |       |       |         |           | 650.6  | 0.0267 |          |       |       |
|         |           | 800.2  | 0.0010 |          |       |       |         |           | 659.0  | 0.0039 |          |       |       |
|         |           | 808.3  | 0.0024 |          |       |       |         |           | 667.7  | 0.9870 |          |       |       |
|         |           | 877.3  | 0.0019 |          |       |       |         |           | 669.8  | 0.0494 |          |       |       |
|         |           | 967.0  | 0.0088 |          |       |       |         |           | 671.6  | 0.0523 |          |       |       |
|         |           | 1097.0 | 0.0055 |          |       |       |         |           | 727.0  | 0.0316 |          |       |       |
|         |           | 1122.0 | 0.0025 |          |       |       |         |           | 727.2  | 0.0217 |          |       |       |
|         |           | 1158.0 | 0.1131 |          |       |       |         |           | 728.5  | 0.0109 |          |       |       |
|         |           | 1223.0 | 0.0018 |          |       |       |         |           | 764.5  | 0.0039 |          |       |       |
|         |           | 1272.0 | 0.0075 |          |       |       |         |           | 772.6  | 0.7620 |          |       |       |
|         |           | 1404.0 | 0.0034 |          |       |       |         |           | 780.2  | 0.0123 |          |       |       |
|         |           | 809.9  | 0.0085 |          |       |       |         |           | 784.5  | 0.0042 |          |       |       |
|         |           |        |        |          |       |       |         |           | 809.8  | 0.0286 |          |       |       |
| 131I    | 8.0d      | 29.5   | 0.0135 | 7.640-5  | 1.242 | 2.412 |         |           | 812.2  | 0.0563 |          |       |       |
|         |           | 29.8   | 0.0250 |          |       |       |         |           | 863.3  | 0.0058 |          |       |       |
|         |           | 33.6   | 0.0089 |          |       |       |         |           | 876.8  | 0.0108 |          |       |       |
|         |           | 80.2   | 0.0262 |          |       |       |         |           | 910.3  | 0.0092 |          |       |       |
|         |           | 177.2  | 0.0026 |          |       |       |         |           | 927.6  | 0.0041 |          |       |       |
|         |           | 284.3  | 0.0605 |          |       |       |         |           | 954.6  | 0.1806 |          |       |       |
|         |           | 325.8  | 0.0025 |          |       |       |         |           | 983.7  | 0.0056 |          |       |       |
|         |           | 364.5  | 0.8116 |          |       |       |         |           | 1035.0 | 0.0047 |          |       |       |
|         |           | 503.0  | 0.0036 |          |       |       |         |           | 1136.0 | 0.0296 |          |       |       |
|         |           | 637.0  | 0.0726 |          |       |       |         |           | 1143.0 | 0.0135 |          |       |       |
|         |           | 642.7  | 0.0022 |          |       |       |         |           | 1147.0 | 0.0028 |          |       |       |
|         |           | 722.9  | 0.0180 |          |       |       |         |           | 1173.0 | 0.0109 |          |       |       |
|         |           | 329.4  | 0.0023 |          |       |       |         |           | 1273.0 | 0.0018 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1291.0 | 0.0114 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1295.0 | 0.0197 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1298.0 | 0.0089 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1318.0 | 0.0012 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1372.0 | 0.0247 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1399.0 | 0.0711 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1443.0 | 0.0142 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1477.0 | 0.0014 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1757.0 | 0.0030 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1921.0 | 0.0118 |          |       |       |
|         |           |        |        |          |       |       |         |           | 2002.0 | 0.0109 |          |       |       |
|         |           |        |        |          |       |       |         |           | 2087.0 | 0.0024 |          |       |       |
|         |           |        |        |          |       |       |         |           | 2173.0 | 0.0020 |          |       |       |
|         |           |        |        |          |       |       |         |           | 2223.0 | 0.0012 |          |       |       |
|         |           |        |        |          |       |       |         |           | 2391.0 | 0.0017 |          |       |       |
|         |           |        |        |          |       |       |         |           | 1015.0 | 0.0320 |          |       |       |
| 132I    | 2.3h      | 29.5   | 0.0017 | 3.841-4  | 3.081 | 0.972 |         |           |        |        |          |       |       |
|         |           | 29.8   | 0.0032 |          |       |       |         |           |        |        |          |       |       |
|         |           | 33.6   | 0.0011 |          |       |       |         |           |        |        |          |       |       |
|         |           | 147.2  | 0.0024 |          |       |       |         |           |        |        |          |       |       |
|         |           | 183.3  | 0.0014 |          |       |       |         |           |        |        |          |       |       |
|         |           | 254.8  | 0.0019 |          |       |       |         |           |        |        |          |       |       |
|         |           | 262.7  | 0.0144 |          |       |       |         |           |        |        |          |       |       |
|         |           | 284.8  | 0.0072 |          |       |       |         |           |        |        |          |       |       |
|         |           | 316.5  | 0.0014 |          |       |       |         |           |        |        |          |       |       |
|         |           | 363.5  | 0.0049 |          |       |       |         |           |        |        |          |       |       |
|         |           | 387.8  | 0.0030 |          |       |       |         |           |        |        |          |       |       |
|         |           | 416.8  | 0.0047 |          |       |       |         |           |        |        |          |       |       |
|         |           | 431.9  | 0.0048 |          |       |       |         |           |        |        |          |       |       |
|         |           | 446.0  | 0.0060 |          |       |       |         |           |        |        |          |       |       |
|         |           | 473.9  | 0.0018 |          |       |       |         |           |        |        |          |       |       |
|         |           | 478.5  | 0.0015 |          |       |       |         |           |        |        |          |       |       |
|         |           | 488.2  | 0.0041 |          |       |       |         |           |        |        |          |       |       |
|         |           |        |        |          |       |       | 133I    | 20.8h     | 29.5   | 0.0015 | 1.092-4  | 2.273 | 1.318 |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T       | $\mu$ |       |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---------|-------|-------|
|         |           | 29.8   | 0.0028 |          |       |       |         |           | 730.7  | 0.0191 |          |         |       |       |
|         |           | 33.6   | 0.0010 |          |       |       |         |           | 739.2  | 0.0076 |          |         |       |       |
|         |           | 262.7  | 0.0036 |          |       |       |         |           | 766.7  | 0.0410 |          |         |       |       |
|         |           | 267.2  | 0.0012 |          |       |       |         |           | 816.4  | 0.0652 |          |         |       |       |
|         |           | 345.4  | 0.0010 |          |       |       |         |           | 847.0  | 0.9541 |          |         |       |       |
|         |           | 361.1  | 0.0011 |          |       |       |         |           | 857.3  | 0.0696 |          |         |       |       |
|         |           | 418.0  | 0.0015 |          |       |       |         |           | 864.0  | 0.0019 |          |         |       |       |
|         |           | 422.9  | 0.0031 |          |       |       |         |           | 884.1  | 0.6526 |          |         |       |       |
|         |           | 510.5  | 0.0181 |          |       |       |         |           | 922.6  | 0.0014 |          |         |       |       |
|         |           | 529.9  | 0.8632 |          |       |       |         |           | 947.9  | 0.0404 |          |         |       |       |
|         |           | 618.0  | 0.0054 |          |       |       |         |           | 966.9  | 0.0035 |          |         |       |       |
|         |           | 680.2  | 0.0064 |          |       |       |         |           | 974.7  | 0.0468 |          |         |       |       |
|         |           | 706.6  | 0.0149 |          |       |       |         |           | 1040.0 | 0.0191 |          |         |       |       |
|         |           | 768.4  | 0.0046 |          |       |       |         |           | 1073.0 | 0.1527 |          |         |       |       |
|         |           | 820.5  | 0.0015 |          |       |       |         |           | 1100.0 | 0.0069 |          |         |       |       |
|         |           | 856.3  | 0.0123 |          |       |       |         |           | 1103.0 | 0.0073 |          |         |       |       |
|         |           | 875.3  | 0.0447 |          |       |       |         |           | 1136.0 | 0.0973 |          |         |       |       |
|         |           | 909.7  | 0.0021 |          |       |       |         |           | 1159.0 | 0.0035 |          |         |       |       |
|         |           | 1052.0 | 0.0055 |          |       |       |         |           | 1164.0 | 0.0013 |          |         |       |       |
|         |           | 1060.0 | 0.0014 |          |       |       |         |           | 1190.0 | 0.0035 |          |         |       |       |
|         |           | 1236.0 | 0.0149 |          |       |       |         |           | 1239.0 | 0.0021 |          |         |       |       |
|         |           | 1298.0 | 0.0233 |          |       |       |         |           | 1269.0 | 0.0056 |          |         |       |       |
|         |           | 1350.0 | 0.0015 |          |       |       |         |           | 1322.0 | 0.0010 |          |         |       |       |
|         |           | 535.2  | 0.0061 |          |       |       |         |           | 1336.0 | 0.0014 |          |         |       |       |
| 134I    | 52.6m     | 29.5   | 0.0043 | 4.240-4  | 3.545 | 0.845 |         |           | 1353.0 | 0.0045 |          |         |       |       |
|         |           | 29.8   | 0.0080 |          |       |       |         |           | 1414.0 | 0.0022 |          |         |       |       |
|         |           | 33.6   | 0.0029 |          |       |       |         |           | 1428.0 | 0.0017 |          |         |       |       |
|         |           | 135.4  | 0.0376 |          |       |       |         |           | 1431.0 | 0.0017 |          |         |       |       |
|         |           | 139.0  | 0.0069 |          |       |       |         |           | 1455.0 | 0.0229 |          |         |       |       |
|         |           | 152.0  | 0.0011 |          |       |       |         |           | 1470.0 | 0.0077 |          |         |       |       |
|         |           | 162.5  | 0.0026 |          |       |       |         |           | 1505.0 | 0.0011 |          |         |       |       |
|         |           | 188.5  | 0.0070 |          |       |       |         |           | 1542.0 | 0.0051 |          |         |       |       |
|         |           | 217.0  | 0.0025 |          |       |       |         |           | 1614.0 | 0.0436 |          |         |       |       |
|         |           | 235.5  | 0.0198 |          |       |       |         |           | 1629.0 | 0.0026 |          |         |       |       |
|         |           | 278.8  | 0.0013 |          |       |       |         |           | 1644.0 | 0.0040 |          |         |       |       |
|         |           | 319.8  | 0.0052 |          |       |       |         |           | 1655.0 | 0.0023 |          |         |       |       |
|         |           | 351.1  | 0.0050 |          |       |       |         |           | 1741.0 | 0.0267 |          |         |       |       |
|         |           | 405.4  | 0.0735 |          |       |       |         |           | 1807.0 | 0.0573 |          |         |       |       |
|         |           | 411.0  | 0.0061 |          |       |       |         |           | 1926.0 | 0.0018 |          |         |       |       |
|         |           | 433.4  | 0.0419 |          |       |       |         |           | 2021.0 | 0.0017 |          |         |       |       |
|         |           | 458.9  | 0.0130 |          |       |       |         |           | 2160.0 | 0.0021 |          |         |       |       |
|         |           | 465.5  | 0.0036 |          |       |       |         |           | 2312.0 | 0.0024 |          |         |       |       |
|         |           | 488.9  | 0.0141 |          |       |       |         |           | 2467.0 | 0.0015 |          |         |       |       |
|         |           | 514.4  | 0.0234 |          |       |       |         |           | 1787.0 | 0.0134 |          |         |       |       |
|         |           | 540.8  | 0.0782 |          |       |       |         |           |        |        |          |         |       |       |
|         |           | 565.5  | 0.0088 |          |       |       |         | 135I      | 6.6h   | 29.5   | 0.0007   | 2.317-4 | 4.511 | 0.664 |
|         |           | 570.7  | 0.0021 |          |       |       |         |           |        | 29.8   | 0.0013   |         |       |       |
|         |           | 595.4  | 0.1135 |          |       |       |         |           |        | 33.6   | 0.0005   |         |       |       |
|         |           | 621.8  | 0.1059 |          |       |       |         |           |        | 220.5  | 0.0175   |         |       |       |
|         |           | 628.0  | 0.0237 |          |       |       |         |           |        | 229.7  | 0.0023   |         |       |       |
|         |           | 677.3  | 0.0849 |          |       |       |         |           |        | 264.3  | 0.0018   |         |       |       |
|         |           | 706.7  | 0.0083 |          |       |       |         |           |        | 288.5  | 0.0309   |         |       |       |
|         |           |        |        |          |       |       |         |           |        | 290.3  | 0.0030   |         |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---|-------|
|         |           | 361.9  | 0.0019 |          |       |       |         |           | 682.7  | 0.0019 |          |   |       |
|         |           | 403.0  | 0.0023 |          |       |       |         |           | 812.6  | 0.0090 |          |   |       |
|         |           | 414.8  | 0.0030 |          |       |       |         |           | 865.5  | 0.0067 |          |   |       |
|         |           | 417.6  | 0.0352 |          |       |       |         |           | 976.5  | 0.0278 |          |   |       |
|         |           | 429.9  | 0.0030 |          |       |       |         |           | 994.2  | 0.0168 |          |   |       |
|         |           | 433.7  | 0.0055 |          |       |       |         |           | 1057.0 | 0.0030 |          |   |       |
|         |           | 451.6  | 0.0031 |          |       |       |         |           | 1101.0 | 0.0050 |          |   |       |
|         |           | 546.6  | 0.0012 |          |       |       |         |           | 1179.0 | 0.0023 |          |   |       |
|         |           | 576.0  | 0.0013 |          |       |       |         |           | 1223.0 | 0.0016 |          |   |       |
|         |           | 649.9  | 0.0045 |          |       |       |         |           | 1247.0 | 0.0236 |          |   |       |
|         |           | 690.1  | 0.0013 |          |       |       |         |           | 1313.0 | 0.0694 |          |   |       |
|         |           | 707.9  | 0.0066 |          |       |       |         |           | 1321.0 | 0.2582 |          |   |       |
|         |           | 785.5  | 0.0015 |          |       |       |         |           | 1400.0 | 0.0011 |          |   |       |
|         |           | 797.7  | 0.0017 |          |       |       |         |           | 1536.0 | 0.0135 |          |   |       |
|         |           | 836.8  | 0.0667 |          |       |       |         |           | 1556.0 | 0.0049 |          |   |       |
|         |           | 961.5  | 0.0015 |          |       |       |         |           | 1583.0 | 0.0026 |          |   |       |
|         |           | 972.0  | 0.0089 |          |       |       |         |           | 1625.0 | 0.0024 |          |   |       |
|         |           | 972.6  | 0.0120 |          |       |       |         |           | 1635.0 | 0.0039 |          |   |       |
|         |           | 995.1  | 0.0015 |          |       |       |         |           | 1640.0 | 0.0019 |          |   |       |
|         |           | 1039.0 | 0.0792 |          |       |       |         |           | 1666.0 | 0.0018 |          |   |       |
|         |           | 1102.0 | 0.0160 |          |       |       |         |           | 1686.0 | 0.0032 |          |   |       |
|         |           | 1124.0 | 0.0360 |          |       |       |         |           | 1689.0 | 0.0027 |          |   |       |
|         |           | 1132.0 | 0.2252 |          |       |       |         |           | 1709.0 | 0.0072 |          |   |       |
|         |           | 1160.0 | 0.0010 |          |       |       |         |           | 1738.0 | 0.0017 |          |   |       |
|         |           | 1169.0 | 0.0087 |          |       |       |         |           | 1820.0 | 0.0022 |          |   |       |
|         |           | 1241.0 | 0.0090 |          |       |       |         |           | 1962.0 | 0.0237 |          |   |       |
|         |           | 1260.0 | 0.2861 |          |       |       |         |           | 1968.0 | 0.0017 |          |   |       |
|         |           | 1368.0 | 0.0061 |          |       |       |         |           | 1980.0 | 0.0014 |          |   |       |
|         |           | 1448.0 | 0.0031 |          |       |       |         |           | 2039.0 | 0.0017 |          |   |       |
|         |           | 1458.0 | 0.0864 |          |       |       |         |           | 2228.0 | 0.0011 |          |   |       |
|         |           | 1503.0 | 0.0107 |          |       |       |         |           | 2290.0 | 0.1083 |          |   |       |
|         |           | 1566.0 | 0.0129 |          |       |       |         |           | 2383.0 | 0.0022 |          |   |       |
|         |           | 1678.0 | 0.0953 |          |       |       |         |           | 2415.0 | 0.0708 |          |   |       |
|         |           | 1707.0 | 0.0409 |          |       |       |         |           | 2428.0 | 0.0019 |          |   |       |
|         |           | 1791.0 | 0.0770 |          |       |       |         |           | 2480.0 | 0.0014 |          |   |       |
|         |           | 1831.0 | 0.0058 |          |       |       |         |           | 2548.0 | 0.0013 |          |   |       |
|         |           | 1927.0 | 0.0029 |          |       |       |         |           | 2602.0 | 0.0012 |          |   |       |
|         |           | 2046.0 | 0.0087 |          |       |       |         |           | 2634.0 | 0.0701 |          |   |       |
|         |           | 2255.0 | 0.0061 |          |       |       |         |           | 2828.0 | 0.0010 |          |   |       |
|         |           | 2409.0 | 0.0095 |          |       |       |         |           | 2369.0 | 0.0410 |          |   |       |
|         |           | 1073.0 | 0.0149 |          |       |       |         |           | 2956.0 | 0.0075 |          |   |       |
|         |           |        |        |          |       |       |         |           | 2979.0 | 0.0032 |          |   |       |
| 136I    | 1.4m      | 219.3  | 0.0085 | 3.370-4  | 5.164 | 0.580 |         |           | 3141.0 | 0.0072 |          |   |       |
|         |           | 240.5  | 0.0024 |          |       |       |         |           | 3195.0 | 0.0017 |          |   |       |
|         |           | 270.2  | 0.0022 |          |       |       |         |           | 3212.0 | 0.0053 |          |   |       |
|         |           | 309.1  | 0.0035 |          |       |       |         |           | 3349.0 | 0.0020 |          |   |       |
|         |           | 344.7  | 0.0250 |          |       |       |         |           | 3626.0 | 0.0017 |          |   |       |
|         |           | 362.5  | 0.0013 |          |       |       |         |           | 3635.0 | 0.0012 |          |   |       |
|         |           | 381.4  | 0.0086 |          |       |       |         |           | 3674.0 | 0.0017 |          |   |       |
|         |           | 396.0  | 0.0044 |          |       |       |         |           | 4064.0 | 0.0017 |          |   |       |
|         |           | 431.4  | 0.0021 |          |       |       |         |           | 4269.0 | 0.0037 |          |   |       |
|         |           | 434.2  | 0.0083 |          |       |       |         |           | 4474.0 | 0.0014 |          |   |       |
|         |           | 597.8  | 0.0037 |          |       |       |         |           | 4739.0 | 0.0011 |          |   |       |





| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 618.4  | 0.0001 |          |       |        |         |           | 2850.0 | 0.0018 |          |       |       |
|         |           |        |        |          |       |        |         |           | 1491.0 | 0.0131 |          |       |       |
| 129mXe  | 8.9d      | 29.5   | 0.3612 | 6.165-5  | 0.013 | 235.8  | 138Xe   | 14.1m     | 10.9   | 0.0070 | 1.664-4  | 4.894 | 0.612 |
|         |           | 29.8   | 0.6701 |          |       |        |         |           | 30.6   | 0.0103 |          |       |       |
|         |           | 33.6   | 0.2382 |          |       |        |         |           | 31.0   | 0.0190 |          |       |       |
|         |           | 39.6   | 0.0752 |          |       |        |         |           | 35.0   | 0.0068 |          |       |       |
|         |           | 196.6  | 0.0474 |          |       |        |         |           | 153.8  | 0.0595 |          |       |       |
| 131mXe  | 11.8d     | 29.5   | 0.1548 | 2.533-5  | 0.011 | 263.7  |         |           | 242.6  | 0.0350 |          |       |       |
|         |           | 29.8   | 0.2872 |          |       |        |         |           | 258.3  | 0.3150 |          |       |       |
|         |           | 33.6   | 0.1021 |          |       |        |         |           | 282.5  | 0.0043 |          |       |       |
|         |           | 163.9  | 0.0196 |          |       |        |         |           | 335.3  | 0.0011 |          |       |       |
|         |           |        |        |          |       |        |         |           | 371.4  | 0.0050 |          |       |       |
| 133Xe   | 5.2d      | 30.6   | 0.1364 | 2.783-5  | 0.075 | 39.877 |         |           | 396.4  | 0.0630 |          |       |       |
|         |           | 31.0   | 0.2526 |          |       |        |         |           | 401.4  | 0.0217 |          |       |       |
|         |           | 35.0   | 0.0906 |          |       |        |         |           | 434.5  | 0.2032 |          |       |       |
|         |           | 79.6   | 0.0022 |          |       |        |         |           | 500.2  | 0.0036 |          |       |       |
|         |           | 81.0   | 0.3648 |          |       |        |         |           | 530.1  | 0.0025 |          |       |       |
|         |           | 177.7  | 0.0007 |          |       |        |         |           | 537.8  | 0.0012 |          |       |       |
| 133mXe  | 2.2d      | 29.5   | 0.1601 | 3.034-5  | 0.152 | 19.750 |         |           | 555.9  | 0.0012 |          |       |       |
|         |           | 29.8   | 0.2970 |          |       |        |         |           | 568.5  | 0.0031 |          |       |       |
|         |           | 33.6   | 0.1056 |          |       |        |         |           | 588.8  | 0.0012 |          |       |       |
|         |           | 233.2  | 0.1030 |          |       |        |         |           | 654.1  | 0.0014 |          |       |       |
|         |           |        |        |          |       |        |         |           | 865.8  | 0.0030 |          |       |       |
| 135Xe   | 9.1h      | 30.6   | 0.0145 | 5.118-5  | 0.596 | 5.026  |         |           | 869.4  | 0.0062 |          |       |       |
|         |           | 31.0   | 0.0268 |          |       |        |         |           | 896.9  | 0.0013 |          |       |       |
|         |           | 35.0   | 0.0096 |          |       |        |         |           | 912.5  | 0.0033 |          |       |       |
|         |           | 158.2  | 0.0029 |          |       |        |         |           | 917.1  | 0.0092 |          |       |       |
|         |           | 249.8  | 0.8990 |          |       |        |         |           | 936.4  | 0.0014 |          |       |       |
|         |           | 358.4  | 0.0022 |          |       |        |         |           | 941.3  | 0.0023 |          |       |       |
|         |           | 408.0  | 0.0036 |          |       |        |         |           | 1094.0 | 0.0041 |          |       |       |
|         |           | 608.2  | 0.0290 |          |       |        |         |           | 1099.0 | 0.0021 |          |       |       |
|         |           | 684.3  | 0.0021 |          |       |        |         |           | 1102.0 | 0.0011 |          |       |       |
|         |           |        |        |          |       |        |         |           | 1114.0 | 0.0147 |          |       |       |
| 135mXe  | 15.4m     | 29.5   | 0.0384 | 8.540-5  | 1.796 | 1.668  |         |           | 1142.0 | 0.0051 |          |       |       |
|         |           | 29.8   | 0.0713 |          |       |        |         |           | 1145.0 | 0.0013 |          |       |       |
|         |           | 33.6   | 0.0254 |          |       |        |         |           | 1572.0 | 0.0026 |          |       |       |
|         |           | 526.6  | 0.8100 |          |       |        |         |           | 1615.0 | 0.0024 |          |       |       |
|         |           |        |        |          |       |        |         |           | 1768.0 | 0.1673 |          |       |       |
| 137Xe   | 3.8m      | 30.6   | 0.0009 | 3.340-5  | 2.393 | 1.252  |         |           | 1812.0 | 0.0018 |          |       |       |
|         |           | 31.0   | 0.0017 |          |       |        |         |           | 1851.0 | 0.0142 |          |       |       |
|         |           | 35.0   | 0.0006 |          |       |        |         |           | 1925.0 | 0.0056 |          |       |       |
|         |           | 298.0  | 0.0012 |          |       |        |         |           | 2005.0 | 0.0536 |          |       |       |
|         |           | 393.4  | 0.0014 |          |       |        |         |           | 2016.0 | 0.1225 |          |       |       |
|         |           | 455.5  | 0.3070 |          |       |        |         |           | 2079.0 | 0.0144 |          |       |       |
|         |           | 849.0  | 0.0061 |          |       |        |         |           | 2252.0 | 0.0229 |          |       |       |
|         |           | 982.2  | 0.0021 |          |       |        |         |           | 2322.0 | 0.0062 |          |       |       |
|         |           | 1119.0 | 0.0011 |          |       |        |         |           | 2475.0 | 0.0031 |          |       |       |
|         |           | 1273.0 | 0.0022 |          |       |        |         |           | 2498.0 | 0.0017 |          |       |       |
|         |           | 1577.0 | 0.0010 |          |       |        |         |           | 1119.0 | 0.0266 |          |       |       |
|         |           | 1613.0 | 0.0012 |          |       |        |         |           |        |        |          |       |       |
|         |           | 1783.0 | 0.0041 |          |       |        |         |           |        |        |          |       |       |
|         |           |        |        |          |       |        | 126Cs   | 1.6m      | 29.5   | 0.0410 | 2.141-4  | 1.895 | 1.581 |
|         |           |        |        |          |       |        |         |           | 29.8   | 0.0760 |          |       |       |



| Nuclide           | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|-------------------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---|-------|
| <sup>138</sup> Cs | 32.2m     | 31.8   | 0.0029 | 3.395-4  | 4.716 | 0.635 |         |           | 532.0  | 0.0021 |          |   |       |
|                   |           | 32.2   | 0.0053 |          |       |       |         |           | 567.7  | 0.0013 |          |   |       |
|                   |           | 36.4   | 0.0019 |          |       |       |         |           | 627.2  | 0.0154 |          |   |       |
|                   |           | 112.6  | 0.0013 |          |       |       |         |           | 827.5  | 0.0011 |          |   |       |
|                   |           | 138.1  | 0.0149 |          |       |       |         |           | 929.2  | 0.0023 |          |   |       |
|                   |           | 192.0  | 0.0050 |          |       |       |         |           | 946.5  | 0.0010 |          |   |       |
|                   |           | 193.9  | 0.0033 |          |       |       |         |           | 1190.0 | 0.0018 |          |   |       |
|                   |           | 212.3  | 0.0017 |          |       |       |         |           | 1283.0 | 0.0720 |          |   |       |
|                   |           | 227.8  | 0.0151 |          |       |       |         |           | 1306.0 | 0.0011 |          |   |       |
|                   |           | 324.9  | 0.0029 |          |       |       |         |           | 1308.0 | 0.0037 |          |   |       |
|                   |           | 363.9  | 0.0024 |          |       |       |         |           | 1322.0 | 0.0023 |          |   |       |
|                   |           | 365.3  | 0.0019 |          |       |       |         |           | 1411.0 | 0.0015 |          |   |       |
|                   |           | 409.0  | 0.0466 |          |       |       |         |           | 1421.0 | 0.0080 |          |   |       |
|                   |           | 421.6  | 0.0043 |          |       |       |         |           | 1621.0 | 0.0042 |          |   |       |
|                   |           | 462.8  | 0.3075 |          |       |       |         |           | 1681.0 | 0.0060 |          |   |       |
|                   |           | 516.7  | 0.0043 |          |       |       |         |           | 1699.0 | 0.0018 |          |   |       |
|                   |           | 546.9  | 0.1076 |          |       |       |         |           | 1877.0 | 0.0034 |          |   |       |
|                   |           | 683.6  | 0.0011 |          |       |       |         |           | 1888.0 | 0.0022 |          |   |       |
|                   |           | 766.1  | 0.0015 |          |       |       |         |           | 1904.0 | 0.0012 |          |   |       |
|                   |           | 773.3  | 0.0023 |          |       |       |         |           | 1934.0 | 0.0024 |          |   |       |
|                   |           | 782.1  | 0.0033 |          |       |       |         |           | 2021.0 | 0.0013 |          |   |       |
|                   |           | 871.8  | 0.0511 |          |       |       |         |           | 2090.0 | 0.0014 |          |   |       |
|                   |           | 880.8  | 0.0011 |          |       |       |         |           | 2111.0 | 0.0066 |          |   |       |
|                   |           | 935.0  | 0.0018 |          |       |       |         |           | 2174.0 | 0.0020 |          |   |       |
|                   |           | 1010.0 | 0.2983 |          |       |       |         |           | 2350.0 | 0.0056 |          |   |       |
|                   |           | 1054.0 | 0.0016 |          |       |       |         |           | 2381.0 | 0.0019 |          |   |       |
|                   |           | 1147.0 | 0.0124 |          |       |       |         |           | 2532.0 | 0.0042 |          |   |       |
|                   |           | 1199.0 | 0.0017 |          |       |       |         |           | 2606.0 | 0.0024 |          |   |       |
|                   |           | 1204.0 | 0.0040 |          |       |       |         |           | 2649.0 | 0.0017 |          |   |       |
|                   |           | 1265.0 | 0.0014 |          |       |       |         |           | 2848.0 | 0.0010 |          |   |       |
|                   |           | 1344.0 | 0.0114 |          |       |       |         |           | 3464.0 | 0.0011 |          |   |       |
|                   |           | 1416.0 | 0.0037 |          |       |       |         |           | 3666.0 | 0.0014 |          |   |       |
|                   |           | 1436.0 | 0.7630 |          |       |       |         |           | 1569.0 | 0.0387 |          |   |       |
|                   |           | 1445.0 | 0.0097 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 1496.0 | 0.0018 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 1555.0 | 0.0037 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 1614.0 | 0.0014 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 1717.0 | 0.0011 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 1728.0 | 0.0011 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 1778.0 | 0.0014 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 2024.0 | 0.0012 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 2062.0 | 0.0011 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 2211.0 | 0.0021 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 2218.0 | 0.1518 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 2499.0 | 0.0017 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 2583.0 | 0.0024 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 2640.0 | 0.0763 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 2731.0 | 0.0012 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 3339.0 | 0.0015 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 3367.0 | 0.0023 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 1613.0 | 0.0162 |          |       |       |         |           |        |        |          |   |       |
| <sup>139</sup> Cs | 9.4m      | 454.7  | 0.0013 | 4.218-5  | 5.022 | 0.596 |         |           |        |        |          |   |       |
|                   |           |        |        |          |       |       |         |           |        |        |          |   |       |
|                   |           |        |        |          |       |       |         |           |        |        |          |   |       |
|                   |           |        |        |          |       |       |         |           |        |        |          |   |       |
|                   |           |        |        |          |       |       |         |           |        |        |          |   |       |
|                   |           |        |        |          |       |       |         |           |        |        |          |   |       |
|                   |           |        |        |          |       |       |         |           |        |        |          |   |       |
|                   |           |        |        |          |       |       |         |           |        |        |          |   |       |
| <sup>131</sup> Ba | 11.8d     | 30.6   | 0.2766 | 1.244-4  | 1.338 | 2.239 |         |           |        |        |          |   |       |
|                   |           | 31.0   | 0.5121 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 35.0   | 0.1838 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 78.8   | 0.0073 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 92.3   | 0.0064 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 123.8  | 0.2897 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 133.6  | 0.0216 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 157.1  | 0.0019 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 216.1  | 0.1975 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 239.6  | 0.0240 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 246.9  | 0.0064 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 249.4  | 0.0282 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 294.5  | 0.0017 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 351.2  | 0.0010 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 373.2  | 0.1399 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 404.0  | 0.0131 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 461.3  | 0.0010 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 480.4  | 0.0032 |          |       |       |         |           |        |        |          |   |       |
|                   |           | 486.5  | 0.0207 |          |       |       |         |           |        |        |          |   |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|-------|--------|----------|-------|-------|
|         |           | 496.3  | 0.4680 |          |       |       |         |           | 162.6 | 0.0673 |          |       |       |
|         |           | 572.7  | 0.0016 |          |       |       |         |           | 304.8 | 0.0451 |          |       |       |
|         |           | 585.0  | 0.0122 |          |       |       |         |           | 423.7 | 0.0325 |          |       |       |
|         |           | 620.1  | 0.0136 |          |       |       |         |           | 437.6 | 0.0199 |          |       |       |
|         |           | 674.4  | 0.0013 |          |       |       |         |           | 467.6 | 0.0015 |          |       |       |
|         |           | 696.5  | 0.0015 |          |       |       |         |           | 537.3 | 0.2550 |          |       |       |
|         |           | 831.6  | 0.0023 |          |       |       |         |           | 118.8 | 0.0007 |          |       |       |
|         |           | 923.8  | 0.0073 |          |       |       |         |           |       |        |          |       |       |
|         |           | 1048.0 | 0.0117 |          |       |       | 141Ba   | 18.3m     | 33.0  | 0.0195 | 1.559-4  | 3.049 | 0.982 |
|         |           | 501.9  | 0.0058 |          |       |       |         |           | 33.4  | 0.0360 |          |       |       |
|         |           |        |        |          |       |       |         |           | 37.8  | 0.0132 |          |       |       |
| 133Ba   | 10.5y     | 30.6   | 0.3425 | 1.231-4  | 0.715 | 4.188 |         |           | 112.9 | 0.0099 |          |       |       |
|         |           | 31.0   | 0.6342 |          |       |       |         |           | 163.0 | 0.0047 |          |       |       |
|         |           | 35.0   | 0.2276 |          |       |       |         |           | 180.5 | 0.0052 |          |       |       |
|         |           | 53.1   | 0.0214 |          |       |       |         |           | 190.2 | 0.4860 |          |       |       |
|         |           | 79.6   | 0.0255 |          |       |       |         |           | 277.0 | 0.2459 |          |       |       |
|         |           | 81.0   | 0.3297 |          |       |       |         |           | 304.2 | 0.2658 |          |       |       |
|         |           | 160.6  | 0.0060 |          |       |       |         |           | 343.7 | 0.1502 |          |       |       |
|         |           | 223.1  | 0.0044 |          |       |       |         |           | 349.3 | 0.0030 |          |       |       |
|         |           | 276.4  | 0.0690 |          |       |       |         |           | 364.4 | 0.0061 |          |       |       |
|         |           | 302.8  | 0.1779 |          |       |       |         |           | 381.3 | 0.0012 |          |       |       |
|         |           | 356.0  | 0.6050 |          |       |       |         |           | 389.8 | 0.0140 |          |       |       |
|         |           | 383.8  | 0.0867 |          |       |       |         |           | 457.6 | 0.0505 |          |       |       |
|         |           |        |        |          |       |       |         |           | 462.1 | 0.0505 |          |       |       |
| 133mBa  | 1.6d      | 12.3   | 0.0135 | 3.372-5  | 0.341 | 8.792 |         |           | 467.3 | 0.0578 |          |       |       |
|         |           | 31.8   | 0.1509 |          |       |       |         |           | 522.2 | 0.0046 |          |       |       |
|         |           | 32.2   | 0.2784 |          |       |       |         |           | 524.2 | 0.0043 |          |       |       |
|         |           | 36.4   | 0.1013 |          |       |       |         |           | 527.4 | 0.0040 |          |       |       |
|         |           | 276.1  | 0.1800 |          |       |       |         |           | 561.9 | 0.0010 |          |       |       |
|         |           |        |        |          |       |       |         |           | 572.1 | 0.0027 |          |       |       |
| 135mBa  | 1.2d      | 31.8   | 0.1535 | 2.974-5  | 0.316 | 9.494 |         |           | 572.1 | 0.0027 |          |       |       |
|         |           | 32.2   | 0.2832 |          |       |       |         |           | 599.3 | 0.0025 |          |       |       |
|         |           | 36.4   | 0.1030 |          |       |       |         |           | 608.9 | 0.0026 |          |       |       |
|         |           | 268.2  | 0.1600 |          |       |       |         |           | 625.2 | 0.0345 |          |       |       |
|         |           |        |        |          |       |       |         |           | 636.0 | 0.0030 |          |       |       |
| 137mBa  | 2.6m      | 31.8   | 0.0207 | 1.075-4  | 2.447 | 1.224 |         |           | 641.4 | 0.0038 |          |       |       |
|         |           | 32.2   | 0.0382 |          |       |       |         |           | 647.9 | 0.0593 |          |       |       |
|         |           | 36.4   | 0.0139 |          |       |       |         |           | 670.0 | 0.0019 |          |       |       |
|         |           | 661.7  | 0.8998 |          |       |       |         |           | 674.2 | 0.0011 |          |       |       |
|         |           |        |        |          |       |       |         |           | 675.7 | 0.0023 |          |       |       |
| 139Ba   | 1.4h      | 33.0   | 0.0093 | 7.714-6  | 1.062 | 2.820 |         |           | 685.7 | 0.0014 |          |       |       |
|         |           | 33.4   | 0.0172 |          |       |       |         |           | 687.8 | 0.0011 |          |       |       |
|         |           | 37.8   | 0.0063 |          |       |       |         |           | 698.5 | 0.0030 |          |       |       |
|         |           | 165.8  | 0.1736 |          |       |       |         |           | 700.0 | 0.0022 |          |       |       |
|         |           | 1420.0 | 0.0028 |          |       |       |         |           | 704.8 | 0.0032 |          |       |       |
|         |           | 1271.0 | 0.0011 |          |       |       |         |           | 739.1 | 0.0452 |          |       |       |
|         |           |        |        |          |       |       |         |           | 753.9 | 0.0010 |          |       |       |
| 140Ba   | 12.8d     | 13.8   | 0.0117 | 4.413-5  | 1.583 | 1.892 |         |           | 762.2 | 0.0015 |          |       |       |
|         |           | 30.0   | 0.1403 |          |       |       |         |           | 778.2 | 0.0011 |          |       |       |
|         |           | 33.0   | 0.0053 |          |       |       |         |           | 805.4 | 0.0010 |          |       |       |
|         |           | 33.4   | 0.0098 |          |       |       |         |           | 826.3 | 0.0035 |          |       |       |
|         |           | 37.8   | 0.0036 |          |       |       |         |           | 831.7 | 0.0160 |          |       |       |
|         |           | 132.8  | 0.0021 |          |       |       |         |           | 832.6 | 0.0017 |          |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---|-------|
|         |           | 867.9  | 0.0016 |          |       |       |         |           | 77.6   | 0.0961 |          |   |       |
|         |           | 876.3  | 0.0360 |          |       |       |         |           | 122.9  | 0.0093 |          |   |       |
|         |           | 880.6  | 0.0021 |          |       |       |         |           | 154.2  | 0.0052 |          |   |       |
|         |           | 908.8  | 0.0013 |          |       |       |         |           | 162.0  | 0.0011 |          |   |       |
|         |           | 929.5  | 0.0073 |          |       |       |         |           | 176.8  | 0.0148 |          |   |       |
|         |           | 943.3  | 0.0077 |          |       |       |         |           | 216.3  | 0.0020 |          |   |       |
|         |           | 981.6  | 0.0082 |          |       |       |         |           | 222.6  | 0.0027 |          |   |       |
|         |           | 996.6  | 0.0013 |          |       |       |         |           | 231.5  | 0.1015 |          |   |       |
|         |           | 1012.0 | 0.0011 |          |       |       |         |           | 242.7  | 0.0016 |          |   |       |
|         |           | 1035.0 | 0.0031 |          |       |       |         |           | 255.1  | 0.1780 |          |   |       |
|         |           | 1040.0 | 0.0010 |          |       |       |         |           | 269.3  | 0.0068 |          |   |       |
|         |           | 1046.0 | 0.0036 |          |       |       |         |           | 283.9  | 0.0018 |          |   |       |
|         |           | 1094.0 | 0.0023 |          |       |       |         |           | 286.2  | 0.0093 |          |   |       |
|         |           | 1161.0 | 0.0025 |          |       |       |         |           | 309.0  | 0.0226 |          |   |       |
|         |           | 1161.0 | 0.0097 |          |       |       |         |           | 334.8  | 0.0125 |          |   |       |
|         |           | 1198.0 | 0.0486 |          |       |       |         |           | 337.1  | 0.0025 |          |   |       |
|         |           | 1225.0 | 0.0043 |          |       |       |         |           | 346.7  | 0.0014 |          |   |       |
|         |           | 1236.0 | 0.0015 |          |       |       |         |           | 363.8  | 0.0392 |          |   |       |
|         |           | 1264.0 | 0.0087 |          |       |       |         |           | 379.1  | 0.0046 |          |   |       |
|         |           | 1274.0 | 0.0054 |          |       |       |         |           | 417.8  | 0.0034 |          |   |       |
|         |           | 1278.0 | 0.0069 |          |       |       |         |           | 425.0  | 0.0498 |          |   |       |
|         |           | 1309.0 | 0.0025 |          |       |       |         |           | 432.3  | 0.0098 |          |   |       |
|         |           | 1311.0 | 0.0063 |          |       |       |         |           | 434.4  | 0.0030 |          |   |       |
|         |           | 1324.0 | 0.0100 |          |       |       |         |           | 448.1  | 0.0021 |          |   |       |
|         |           | 1345.0 | 0.0023 |          |       |       |         |           | 457.3  | 0.0039 |          |   |       |
|         |           | 1358.0 | 0.0017 |          |       |       |         |           | 473.4  | 0.0030 |          |   |       |
|         |           | 1377.0 | 0.0074 |          |       |       |         |           | 488.3  | 0.0011 |          |   |       |
|         |           | 1406.0 | 0.0029 |          |       |       |         |           | 513.3  | 0.0023 |          |   |       |
|         |           | 1437.0 | 0.0087 |          |       |       |         |           | 537.5  | 0.0011 |          |   |       |
|         |           | 1459.0 | 0.0071 |          |       |       |         |           | 558.3  | 0.0030 |          |   |       |
|         |           | 1502.0 | 0.0033 |          |       |       |         |           | 590.7  | 0.0025 |          |   |       |
|         |           | 1551.0 | 0.0033 |          |       |       |         |           | 599.8  | 0.0160 |          |   |       |
|         |           | 1569.0 | 0.0027 |          |       |       |         |           | 604.2  | 0.0032 |          |   |       |
|         |           | 1654.0 | 0.0079 |          |       |       |         |           | 769.4  | 0.0061 |          |   |       |
|         |           | 1682.0 | 0.0141 |          |       |       |         |           | 786.4  | 0.0025 |          |   |       |
|         |           | 1713.0 | 0.0018 |          |       |       |         |           | 792.2  | 0.0021 |          |   |       |
|         |           | 1736.0 | 0.0019 |          |       |       |         |           | 823.4  | 0.0041 |          |   |       |
|         |           | 1741.0 | 0.0033 |          |       |       |         |           | 840.2  | 0.0303 |          |   |       |
|         |           | 1796.0 | 0.0051 |          |       |       |         |           | 894.9  | 0.1104 |          |   |       |
|         |           | 1913.0 | 0.0014 |          |       |       |         |           | 948.8  | 0.0890 |          |   |       |
|         |           | 1990.0 | 0.0019 |          |       |       |         |           | 1001.0 | 0.0783 |          |   |       |
|         |           | 2027.0 | 0.0040 |          |       |       |         |           | 1033.0 | 0.0048 |          |   |       |
|         |           | 2137.0 | 0.0012 |          |       |       |         |           | 1078.0 | 0.0926 |          |   |       |
|         |           | 2165.0 | 0.0017 |          |       |       |         |           | 1094.0 | 0.0221 |          |   |       |
|         |           | 2279.0 | 0.0010 |          |       |       |         |           | 1123.0 | 0.0030 |          |   |       |
|         |           | 2469.0 | 0.0019 |          |       |       |         |           | 1127.0 | 0.0153 |          |   |       |
|         |           | 1252.0 | 0.0173 |          |       |       |         |           | 1148.0 | 0.0039 |          |   |       |
|         |           |        |        |          |       |       |         |           | 1202.0 | 0.0534 |          |   |       |
| 142Ba   | 10.7m     | 33.0   | 0.0555 | 1.536-4  | 3.564 | 0.841 |         |           | 1204.0 | 0.1371 |          |   |       |
|         |           | 33.4   | 0.1021 |          |       |       |         |           | 1283.0 | 0.0016 |          |   |       |
|         |           | 37.8   | 0.0375 |          |       |       |         |           | 1380.0 | 0.0340 |          |   |       |
|         |           | 69.4   | 0.0036 |          |       |       |         |           | 685.1  | 0.0016 |          |   |       |
|         |           | 76.8   | 0.0089 |          |       |       |         |           |        |        |          |   |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---|-------|
| 140La   | 1.7d      | 34.3   | 0.0047 | 3.424-4  | 4.532 | 0.661 |         |           | 1074.0 | 0.0011 |          |   |       |
|         |           | 34.7   | 0.0087 |          |       |       |         |           | 1089.0 | 0.0026 |          |   |       |
|         |           | 39.3   | 0.0032 |          |       |       |         |           | 1113.0 | 0.0011 |          |   |       |
|         |           | 109.4  | 0.0019 |          |       |       |         |           | 1117.0 | 0.0011 |          |   |       |
|         |           | 131.1  | 0.0055 |          |       |       |         |           | 1131.0 | 0.0052 |          |   |       |
|         |           | 173.6  | 0.0012 |          |       |       |         |           | 1144.0 | 0.0016 |          |   |       |
|         |           | 242.0  | 0.0043 |          |       |       |         |           | 1160.0 | 0.0194 |          |   |       |
|         |           | 266.6  | 0.0049 |          |       |       |         |           | 1174.0 | 0.0016 |          |   |       |
|         |           | 328.8  | 0.2053 |          |       |       |         |           | 1191.0 | 0.0042 |          |   |       |
|         |           | 432.5  | 0.0294 |          |       |       |         |           | 1231.0 | 0.0032 |          |   |       |
|         |           | 487.0  | 0.4555 |          |       |       |         |           | 1233.0 | 0.0205 |          |   |       |
|         |           | 751.8  | 0.0440 |          |       |       |         |           | 1242.0 | 0.0021 |          |   |       |
|         |           | 815.9  | 0.2349 |          |       |       |         |           | 1265.0 | 0.0011 |          |   |       |
|         |           | 867.8  | 0.0563 |          |       |       |         |           | 1270.0 | 0.0011 |          |   |       |
|         |           | 919.6  | 0.0288 |          |       |       |         |           | 1288.0 | 0.0011 |          |   |       |
|         |           | 925.2  | 0.0708 |          |       |       |         |           | 1323.0 | 0.0037 |          |   |       |
|         |           | 950.9  | 0.0053 |          |       |       |         |           | 1332.0 | 0.0011 |          |   |       |
|         |           | 1597.0 | 0.9549 |          |       |       |         |           | 1355.0 | 0.0011 |          |   |       |
|         |           | 2349.0 | 0.0085 |          |       |       |         |           | 1363.0 | 0.0236 |          |   |       |
|         |           | 2522.0 | 0.0346 |          |       |       |         |           | 1374.0 | 0.0021 |          |   |       |
|         |           | 2547.0 | 0.0010 |          |       |       |         |           | 1389.0 | 0.0047 |          |   |       |
|         |           | 1208.0 | 0.0042 |          |       |       |         |           | 1395.0 | 0.0021 |          |   |       |
| 141La   | 3.9h      | 1354.0 | 0.0262 | 6.088-6  | 4.731 | 0.633 |         |           | 1402.0 | 0.0016 |          |   |       |
|         |           | 1693.0 | 0.0012 |          |       |       |         |           | 1446.0 | 0.0016 |          |   |       |
|         |           | 1674.0 | 0.0031 |          |       |       |         |           | 1455.0 | 0.0011 |          |   |       |
|         |           |        |        |          |       |       |         |           | 1494.0 | 0.0016 |          |   |       |
| 142La   | 1.6h      | 34.3   | 0.0006 | 3.586-4  | 5.354 | 0.560 |         |           | 1516.0 | 0.0047 |          |   |       |
|         |           | 34.7   | 0.0012 |          |       |       |         |           | 1535.0 | 0.0026 |          |   |       |
|         |           | 39.3   | 0.0004 |          |       |       |         |           | 1540.0 | 0.0052 |          |   |       |
|         |           | 106.1  | 0.0016 |          |       |       |         |           | 1546.0 | 0.0331 |          |   |       |
|         |           | 174.1  | 0.0011 |          |       |       |         |           | 1618.0 | 0.0032 |          |   |       |
|         |           | 367.3  | 0.0011 |          |       |       |         |           | 1651.0 | 0.0021 |          |   |       |
|         |           | 393.7  | 0.0011 |          |       |       |         |           | 1688.0 | 0.0026 |          |   |       |
|         |           | 420.8  | 0.0026 |          |       |       |         |           | 1723.0 | 0.0168 |          |   |       |
|         |           | 433.3  | 0.0042 |          |       |       |         |           | 1752.0 | 0.0011 |          |   |       |
|         |           | 514.7  | 0.0016 |          |       |       |         |           | 1756.0 | 0.0331 |          |   |       |
|         |           | 532.0  | 0.0016 |          |       |       |         |           | 1768.0 | 0.0021 |          |   |       |
|         |           | 578.1  | 0.0136 |          |       |       |         |           | 1771.0 | 0.0021 |          |   |       |
|         |           | 619.5  | 0.0016 |          |       |       |         |           | 1794.0 | 0.0011 |          |   |       |
|         |           | 641.2  | 0.5250 |          |       |       |         |           | 1806.0 | 0.0016 |          |   |       |
|         |           | 861.6  | 0.0199 |          |       |       |         |           | 1817.0 | 0.0011 |          |   |       |
|         |           | 878.2  | 0.0021 |          |       |       |         |           | 1885.0 | 0.0058 |          |   |       |
|         |           | 894.9  | 0.0940 |          |       |       |         |           | 1901.0 | 0.0871 |          |   |       |
|         |           | 946.5  | 0.0011 |          |       |       |         |           | 1923.0 | 0.0026 |          |   |       |
|         |           | 962.2  | 0.0042 |          |       |       |         |           | 1934.0 | 0.0016 |          |   |       |
|         |           | 991.2  | 0.0011 |          |       |       |         |           | 1948.0 | 0.0052 |          |   |       |
|         |           | 1007.0 | 0.0026 |          |       |       |         |           | 1961.0 | 0.0016 |          |   |       |
|         |           | 1011.0 | 0.0436 |          |       |       |         |           | 2004.0 | 0.0105 |          |   |       |
|         |           | 1039.0 | 0.0011 |          |       |       |         |           | 2026.0 | 0.0136 |          |   |       |
|         |           | 1044.0 | 0.0305 |          |       |       |         |           | 2039.0 | 0.0110 |          |   |       |
|         |           | 1062.0 | 0.0016 |          |       |       |         |           | 2050.0 | 0.0052 |          |   |       |
|         |           | 1070.0 | 0.0016 |          |       |       |         |           | 2055.0 | 0.0294 |          |   |       |
|         |           |        |        |          |       |       |         |           | 2077.0 | 0.0073 |          |   |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.   | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|---------|----------|-------|--------|
|         |           | 2086.0 | 0.0042 |          |       |        |         |           | 40.7   | 0.0336  |          |       |        |
|         |           | 2100.0 | 0.0105 |          |       |        |         |           | 145.4  | 0.4840  |          |       |        |
|         |           | 2126.0 | 0.0037 |          |       |        |         |           |        |         |          |       |        |
|         |           | 2139.0 | 0.0058 |          |       |        | 143Ce   | 1.4d      | 35.5   | 0.1768  | 6.886-5  | 1.418 | 2.113  |
|         |           | 2180.0 | 0.0058 |          |       |        |         |           | 36.0   | 0.3239  |          |       |        |
|         |           | 2187.0 | 0.0583 |          |       |        |         |           | 40.7   | 0.1217  |          |       |        |
|         |           | 2290.0 | 0.0037 |          |       |        |         |           | 57.4   | 0.1176  |          |       |        |
|         |           | 2358.0 | 0.0084 |          |       |        |         |           | 169.0  | 0.0029  |          |       |        |
|         |           | 2364.0 | 0.0047 |          |       |        |         |           | 216.0  | 0.0021  |          |       |        |
|         |           | 2398.0 | 0.1627 |          |       |        |         |           | 231.6  | 0.0202  |          |       |        |
|         |           | 2420.0 | 0.0021 |          |       |        |         |           | 293.3  | 0.4200  |          |       |        |
|         |           | 2459.0 | 0.0042 |          |       |        |         |           | 338.0  | 0.0029  |          |       |        |
|         |           | 2513.0 | 0.0016 |          |       |        |         |           | 350.6  | 0.0336  |          |       |        |
|         |           | 2532.0 | 0.0011 |          |       |        |         |           | 433.0  | 0.0013  |          |       |        |
|         |           | 2539.0 | 0.0079 |          |       |        |         |           | 439.0  | 0.0012  |          |       |        |
|         |           | 2543.0 | 0.1123 |          |       |        |         |           | 490.4  | 0.0197  |          |       |        |
|         |           | 2663.0 | 0.0079 |          |       |        |         |           | 587.3  | 0.0024  |          |       |        |
|         |           | 2667.0 | 0.0189 |          |       |        |         |           | 664.6  | 0.0525  |          |       |        |
|         |           | 2673.0 | 0.0021 |          |       |        |         |           | 722.0  | 0.0512  |          |       |        |
|         |           | 2782.0 | 0.0032 |          |       |        |         |           | 880.4  | 0.0092  |          |       |        |
|         |           | 2801.0 | 0.0063 |          |       |        |         |           | 1103.0 | 0.0037  |          |       |        |
|         |           | 2818.0 | 0.0084 |          |       |        |         |           | 610.6  | 0.0089  |          |       |        |
|         |           | 2829.0 | 0.0026 |          |       |        |         |           |        |         |          |       |        |
|         |           | 2970.0 | 0.0079 |          |       |        | 144Ce   | 284.3d    | 33.6   | 0.0028  | 6.302-6  | 0.084 | 35.806 |
|         |           | 2972.0 | 0.0331 |          |       |        |         |           | 35.5   | 0.0243  |          |       |        |
|         |           | 2992.0 | 0.0011 |          |       |        |         |           | 36.0   | 0.0445  |          |       |        |
|         |           | 3000.0 | 0.0052 |          |       |        |         |           | 40.7   | 0.0167  |          |       |        |
|         |           | 3007.0 | 0.0021 |          |       |        |         |           | 40.9   | 0.0039  |          |       |        |
|         |           | 3013.0 | 0.0073 |          |       |        |         |           | 80.1   | 0.0160  |          |       |        |
|         |           | 3022.0 | 0.0011 |          |       |        |         |           | 133.5  | 0.1080  |          |       |        |
|         |           | 3034.0 | 0.0058 |          |       |        |         |           | 67.3   | 0.0013  |          |       |        |
|         |           | 3047.0 | 0.0042 |          |       |        |         |           |        |         |          |       |        |
|         |           | 3076.0 | 0.0016 |          |       |        | 142Pr   | 19.1h     | 1576.0 | 0.0370  | 8.050-6  | 5.069 | 0.591  |
|         |           | 3155.0 | 0.0021 |          |       |        |         |           | 508.8  | 0.0002  |          |       |        |
|         |           | 3181.0 | 0.0032 |          |       |        |         |           |        |         |          |       |        |
|         |           | 3237.0 | 0.0032 |          |       |        | 143Pr   | 13.6d     | 742.0  | 0.00001 | 1.52+-12 | 2.824 | 1.061  |
|         |           | 3242.0 | 0.0021 |          |       |        |         |           |        |         |          |       |        |
|         |           | 3273.0 | 0.0016 |          |       |        | 144Pr   | 17.3m     | 696.5  | 0.0148  | 4.545-6  | 4.825 | 0.621  |
|         |           | 3315.0 | 0.0136 |          |       |        |         |           | 1489.0 | 0.0030  |          |       |        |
|         |           | 3402.0 | 0.0032 |          |       |        |         |           | 2186.0 | 0.0077  |          |       |        |
|         |           | 3459.0 | 0.0037 |          |       |        |         |           | 1059.0 | 0.0002  |          |       |        |
|         |           | 3612.0 | 0.0089 |          |       |        |         |           |        |         |          |       |        |
|         |           | 3633.0 | 0.0115 |          |       |        | 144mPr  | 7.2m      | 35.5   | 0.0868  | 9.933-6  | 0.016 | 184.2  |
|         |           | 3719.0 | 0.0032 |          |       |        |         |           | 36.0   | 0.1589  |          |       |        |
|         |           | 3850.0 | 0.0026 |          |       |        |         |           | 40.7   | 0.0597  |          |       |        |
|         |           |        |        |          |       |        |         |           | 59.0   | 0.0008  |          |       |        |
| 139Ce   | 137.7d    | 33.0   | 0.2248 | 5.554-5  | 0.134 | 22.320 |         |           |        |         |          |       |        |
|         |           | 33.4   | 0.4140 |          |       |        | 147Nd   | 11.0d     | 38.2   | 0.1296  | 3.751-5  | 1.337 | 2.241  |
|         |           | 37.8   | 0.1521 |          |       |        |         |           | 38.7   | 0.2356  |          |       |        |
|         |           | 165.8  | 0.8035 |          |       |        |         |           | 43.8   | 0.0906  |          |       |        |
|         |           |        |        |          |       |        |         |           | 91.1   | 0.2800  |          |       |        |
| 141Ce   | 32.5d     | 35.5   | 0.0488 | 1.979-5  | 0.115 | 26.056 |         |           | 120.5  | 0.0040  |          |       |        |
|         |           | 36.0   | 0.0894 |          |       |        |         |           | 196.6  | 0.0020  |          |       |        |

| Nuclide           | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide           | Half-life           | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|-------------------|-----------|-------|--------|----------|-------|-------|-------------------|---------------------|--------|--------|----------|-------|-------|
|                   |           | 275.4 | 0.0080 |          |       |       |                   |                     | 443.5  | 0.0150 |          |       |       |
|                   |           | 319.4 | 0.0196 |          |       |       |                   |                     | 538.2  | 0.0011 |          |       |       |
|                   |           | 398.2 | 0.0087 |          |       |       |                   |                     | 540.5  | 0.0767 |          |       |       |
|                   |           | 410.5 | 0.0014 |          |       |       |                   |                     | 556.4  | 0.0117 |          |       |       |
|                   |           | 439.9 | 0.0120 |          |       |       |                   |                     | 630.2  | 0.0022 |          |       |       |
|                   |           | 489.2 | 0.0015 |          |       |       |                   |                     | 635.5  | 0.0011 |          |       |       |
|                   |           | 531.0 | 0.1313 |          |       |       |                   |                     | 654.8  | 0.0732 |          |       |       |
|                   |           | 594.8 | 0.0027 |          |       |       |                   |                     | 686.9  | 0.0010 |          |       |       |
|                   |           | 685.9 | 0.0081 |          |       |       |                   |                     | 696.3  | 0.0017 |          |       |       |
|                   |           | 616.6 | 0.0007 |          |       |       |                   |                     | 808.8  | 0.0017 |          |       |       |
|                   |           |       |        |          |       |       |                   |                     | 923.9  | 0.0011 |          |       |       |
| <sup>149</sup> Nd | 1.7h      | 38.2  | 0.0909 | 8.093-5  | 1.359 | 2.205 |                   |                     | 979.0  | 0.0011 |          |       |       |
|                   |           | 38.7  | 0.1652 |          |       |       |                   |                     | 1023.0 | 0.0012 |          |       |       |
|                   |           | 43.8  | 0.0635 |          |       |       |                   |                     | 1234.0 | 0.0029 |          |       |       |
|                   |           | 58.9  | 0.0151 |          |       |       |                   |                     | 634.0  | 0.0158 |          |       |       |
|                   |           | 74.3  | 0.0125 |          |       |       |                   |                     |        |        |          |       |       |
|                   |           | 74.7  | 0.0098 |          |       |       | <sup>143</sup> Pm | 265.0d              | 36.8   | 0.2169 | 7.201-5  | 2.452 | 1.222 |
|                   |           | 75.7  | 0.0033 |          |       |       |                   |                     | 37.4   | 0.3958 |          |       |       |
|                   |           | 97.0  | 0.0152 |          |       |       |                   |                     | 42.3   | 0.1507 |          |       |       |
|                   |           | 114.3 | 0.1877 |          |       |       |                   |                     | 742.0  | 0.3830 |          |       |       |
|                   |           | 116.9 | 0.0012 |          |       |       |                   |                     |        |        |          |       |       |
|                   |           | 122.4 | 0.0023 |          |       |       | <sup>144</sup> Pm | 363.0d              | 36.8   | 0.2219 | 2.944-4  | 2.305 | 1.300 |
|                   |           | 126.6 | 0.0011 |          |       |       |                   |                     | 37.4   | 0.4049 |          |       |       |
|                   |           | 139.2 | 0.0048 |          |       |       |                   |                     | 42.3   | 0.1542 |          |       |       |
|                   |           | 155.9 | 0.0604 |          |       |       |                   |                     | 301.7  | 0.0018 |          |       |       |
|                   |           | 177.8 | 0.0016 |          |       |       |                   |                     | 476.8  | 0.4199 |          |       |       |
|                   |           | 188.6 | 0.0199 |          |       |       |                   |                     | 582.4  | 0.0019 |          |       |       |
|                   |           | 192.0 | 0.0059 |          |       |       |                   |                     | 618.0  | 0.9860 |          |       |       |
|                   |           | 198.9 | 0.0144 |          |       |       |                   |                     | 694.0  | 0.0055 |          |       |       |
|                   |           | 208.2 | 0.0291 |          |       |       |                   |                     | 696.5  | 0.9949 |          |       |       |
|                   |           | 211.3 | 0.2720 |          |       |       |                   |                     | 778.6  | 0.0151 |          |       |       |
|                   |           | 213.9 | 0.0041 |          |       |       |                   |                     | 814.1  | 0.0055 |          |       |       |
|                   |           | 226.8 | 0.0016 |          |       |       |                   |                     | 897.6  | 0.0004 |          |       |       |
|                   |           | 240.2 | 0.0394 |          |       |       |                   |                     |        |        |          |       |       |
|                   |           | 245.7 | 0.0103 |          |       |       | <sup>145</sup> Pm | 17.7y               | 36.8   | 0.2174 | 2.418-5  | 0.020 | 152.1 |
|                   |           | 258.1 | 0.0038 |          |       |       |                   |                     | 37.4   | 0.3967 |          |       |       |
|                   |           | 267.7 | 0.0604 |          |       |       |                   |                     | 42.3   | 0.1511 |          |       |       |
|                   |           | 270.2 | 0.1061 |          |       |       |                   |                     | 67.2   | 0.0069 |          |       |       |
|                   |           | 273.2 | 0.0023 |          |       |       |                   |                     | 72.4   | 0.0231 |          |       |       |
|                   |           | 275.4 | 0.0060 |          |       |       |                   |                     |        |        |          |       |       |
|                   |           | 277.0 | 0.0032 |          |       |       | <sup>146</sup> Pm | 5.5y <sub>1/2</sub> | 36.8   | 0.1394 | 1.460-4  | 2.275 | 1.317 |
|                   |           | 282.5 | 0.0061 |          |       |       |                   |                     | 37.4   | 0.2543 |          |       |       |
|                   |           | 288.2 | 0.0067 |          |       |       |                   |                     | 42.3   | 0.0969 |          |       |       |
|                   |           | 294.8 | 0.0058 |          |       |       |                   |                     | 146.2  | 0.0021 |          |       |       |



| Nuclide | Half-life | E      | Prob.   | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E     | Prob.   | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|---------|----------|-------|--------|---------|-----------|-------|---------|----------|---|-------|
| 147Pm   | 2.6y      | 121.3  | 0.00007 | 2.3+-10  | 0.082 | 36.526 |         |           |       |         |          |   |       |
| 148Pm   | 5.4d      | 39.5   | 0.0005  | 8.885-5  | 4.046 | 0.740  |         |           | 64.9  | 0.0197  |          |   |       |
|         |           | 40.1   | 0.0009  |          |       |        |         |           | 65.8  | 0.0117  |          |   |       |
|         |           | 45.4   | 0.0003  |          |       |        |         |           | 69.7  | 0.0048  |          |   |       |
|         |           | 550.3  | 0.2200  |          |       |        |         |           | 76.2  | 0.0021  |          |   |       |
|         |           | 592.8  | 0.0035  |          |       |        |         |           | 98.0  | 0.0037  |          |   |       |
|         |           | 611.3  | 0.0102  |          |       |        |         |           | 100.0 | 0.0256  |          |   |       |
|         |           | 874.2  | 0.0024  |          |       |        |         |           | 101.9 | 0.0131  |          |   |       |
|         |           | 896.4  | 0.0098  |          |       |        |         |           | 104.8 | 0.0355  |          |   |       |
|         |           | 914.8  | 0.1145  |          |       |        |         |           | 139.3 | 0.0051  |          |   |       |
|         |           | 1465.0 | 0.2220  |          |       |        |         |           | 143.2 | 0.0022  |          |   |       |
|         |           | 1294.0 | 0.0030  |          |       |        |         |           | 147.5 | 0.0015  |          |   |       |
|         |           |        |         |          |       |        |         |           | 156.2 | 0.0015  |          |   |       |
|         |           |        |         |          |       |        |         |           | 162.9 | 0.0089  |          |   |       |
| 148mPm  | 41.3d     | 38.2   | 0.0073  | 3.544-4  | 2.477 | 1.209  |         |           | 163.6 | 0.0163  |          |   |       |
|         |           | 38.7   | 0.0132  |          |       |        |         |           | 167.8 | 0.00879 |          |   |       |
|         |           | 43.8   | 0.0051  |          |       |        |         |           | 168.4 | 0.0092  |          |   |       |
|         |           | 75.7   | 0.0093  |          |       |        |         |           | 176.5 | 0.0087  |          |   |       |
|         |           | 61.5   | 0.0000  |          |       |        |         |           | 177.2 | 0.0387  |          |   |       |
|         |           | 39.5   | 0.0167  |          |       |        |         |           | 186.6 | 0.0017  |          |   |       |
|         |           | 40.1   | 0.0303  |          |       |        |         |           | 202.0 | 0.0094  |          |   |       |
|         |           | 45.4   | 0.0117  |          |       |        |         |           | 204.2 | 0.0013  |          |   |       |
|         |           | 98.5   | 0.0247  |          |       |        |         |           | 209.0 | 0.0179  |          |   |       |
|         |           | 189.6  | 0.0110  |          |       |        |         |           | 227.2 | 0.0034  |          |   |       |
|         |           | 288.1  | 0.1256  |          |       |        |         |           | 232.4 | 0.0105  |          |   |       |
|         |           | 311.6  | 0.0392  |          |       |        |         |           | 236.6 | 0.0016  |          |   |       |
|         |           | 362.1  | 0.0018  |          |       |        |         |           | 236.7 | 0.0020  |          |   |       |
|         |           | 414.1  | 0.1866  |          |       |        |         |           | 237.0 | 0.0053  |          |   |       |
|         |           | 432.8  | 0.0535  |          |       |        |         |           | 240.1 | 0.0389  |          |   |       |
|         |           | 460.6  | 0.0042  |          |       |        |         |           | 254.3 | 0.0016  |          |   |       |
|         |           | 501.3  | 0.0675  |          |       |        |         |           | 258.1 | 0.0060  |          |   |       |
|         |           | 550.3  | 0.9487  |          |       |        |         |           | 275.2 | 0.0717  |          |   |       |
|         |           | 553.2  | 0.0040  |          |       |        |         |           | 280.1 | 0.0023  |          |   |       |
|         |           | 572.0  | 0.0021  |          |       |        |         |           | 290.8 | 0.0088  |          |   |       |
|         |           | 599.7  | 0.1254  |          |       |        |         |           | 306.7 | 0.0024  |          |   |       |
|         |           | 611.3  | 0.0548  |          |       |        |         |           | 323.9 | 0.0121  |          |   |       |
|         |           | 630.0  | 0.8900  |          |       |        |         |           | 325.8 | 0.0011  |          |   |       |
|         |           | 725.7  | 0.3284  |          |       |        |         |           | 329.8 | 0.0021  |          |   |       |
|         |           | 915.3  | 0.1717  |          |       |        |         |           | 340.1 | 0.2290  |          |   |       |
|         |           | 1014.0 | 0.2028  |          |       |        |         |           | 344.9 | 0.0218  |          |   |       |
|         |           | 714.8  | 0.0019  |          |       |        |         |           | 349.8 | 0.0014  |          |   |       |
|         |           |        |         |          |       |        |         |           | 353.3 | 0.0011  |          |   |       |
| 149Pm   | 2.2d      | 39.5   | 0.0006  | 2.314-6  | 1.205 | 2.485  |         |           | 379.9 | 0.0097  |          |   |       |
|         |           | 40.1   | 0.0012  |          |       |        |         |           | 407.0 | 0.0019  |          |   |       |
|         |           | 45.4   | 0.0005  |          |       |        |         |           | 440.9 | 0.0153  |          |   |       |
|         |           | 285.9  | 0.0310  |          |       |        |         |           | 445.7 | 0.0408  |          |   |       |
|         |           | 859.4  | 0.0010  |          |       |        |         |           | 451.4 | 0.0030  |          |   |       |
|         |           | 598.3  | 0.0030  |          |       |        |         |           | 490.3 | 0.0013  |          |   |       |
|         |           |        |         |          |       |        |         |           | 516.3 | 0.0020  |          |   |       |
| 151Pm   | 1.2d      | 25.7   | 0.0094  | 7.080-5  | 1.522 | 1.969  |         |           | 565.0 | 0.0036  |          |   |       |
|         |           | 39.5   | 0.0807  |          |       |        |         |           | 575.0 | 0.0012  |          |   |       |
|         |           | 40.1   | 0.1462  |          |       |        |         |           | 636.2 | 0.0147  |          |   |       |
|         |           | 45.4   | 0.0567  |          |       |        |         |           | 654.3 | 0.0025  |          |   |       |
|         |           | 62.9   | 0.0022  |          |       |        |         |           | 668.7 | 0.0036  |          |   |       |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|-------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 669.2 | 0.0029 |          |       |        |         |           | 963.4  | 0.0011 |          |       |       |
|         |           | 671.3 | 0.0093 |          |       |        |         |           | 964.0  | 0.1444 |          |       |       |
|         |           | 704.2 | 0.0035 |          |       |        |         |           | 1005.0 | 0.0066 |          |       |       |
|         |           | 709.3 | 0.0015 |          |       |        |         |           | 1085.0 | 0.0025 |          |       |       |
|         |           | 712.0 | 0.0011 |          |       |        |         |           | 1086.0 | 0.0996 |          |       |       |
|         |           | 717.8 | 0.0412 |          |       |        |         |           | 1112.0 | 0.1330 |          |       |       |
|         |           | 736.1 | 0.0049 |          |       |        |         |           | 1213.0 | 0.0138 |          |       |       |
|         |           | 752.8 | 0.0133 |          |       |        |         |           | 1250.0 | 0.0018 |          |       |       |
|         |           | 769.1 | 0.0011 |          |       |        |         |           | 1293.0 | 0.0010 |          |       |       |
|         |           | 772.8 | 0.0096 |          |       |        |         |           | 1408.0 | 0.2075 |          |       |       |
|         |           | 785.1 | 0.0023 |          |       |        |         |           | 1458.0 | 0.0049 |          |       |       |
|         |           | 807.9 | 0.0053 |          |       |        |         |           | 1528.0 | 0.0026 |          |       |       |
|         |           | 817.7 | 0.0017 |          |       |        |         |           | 685.8  | 0.0087 |          |       |       |
|         |           | 848.7 | 0.0030 |          |       |        |         |           | 42.3   | 0.0022 |          |       |       |
|         |           | 877.7 | 0.0010 |          |       |        |         |           | 43.0   | 0.0039 |          |       |       |
|         |           | 948.7 | 0.0036 |          |       |        |         |           | 48.7   | 0.0016 |          |       |       |
|         |           | 428.8 | 0.0436 |          |       |        |         |           | 344.3  | 0.2649 |          |       |       |
|         |           |       |        |          |       |        |         |           | 367.7  | 0.0086 |          |       |       |
| 151Sm   | 90y       | 21.5  | 0.0003 | 2.442-8  | 0.004 | 776.6  |         |           | 411.1  | 0.0221 |          |       |       |
|         |           |       |        |          |       |        |         |           | 503.4  | 0.0015 |          |       |       |
| 153Sm   | 1.9d      | 40.9  | 0.1726 | 2.440-5  | 0.043 | 69.799 |         |           | 586.3  | 0.0045 |          |       |       |
|         |           | 41.5  | 0.3122 |          |       |        |         |           | 678.6  | 0.0047 |          |       |       |
|         |           | 47.0  | 0.1222 |          |       |        |         |           | 764.8  | 0.0017 |          |       |       |
|         |           | 69.7  | 0.0517 |          |       |        |         |           | 778.9  | 0.1274 |          |       |       |
|         |           | 75.4  | 0.0019 |          |       |        |         |           | 1090.0 | 0.0168 |          |       |       |
|         |           | 83.4  | 0.0020 |          |       |        |         |           | 1109.0 | 0.0017 |          |       |       |
|         |           | 89.5  | 0.0016 |          |       |        |         |           | 1299.0 | 0.0160 |          |       |       |
|         |           | 97.4  | 0.0072 |          |       |        |         |           | 631.4  | 0.0071 |          |       |       |
|         |           | 103.2 | 0.2830 |          |       |        |         |           |        |        |          |       |       |
|         |           | 422.7 | 0.0028 |          |       |        |         |           |        |        |          |       |       |
|         |           |       |        |          |       |        | 152mEu  | 9.3h      | 39.5   | 0.0788 | 5.742-5  | 3.290 | 0.911 |
|         |           |       |        |          |       |        |         |           | 40.1   | 0.1428 |          |       |       |
|         |           |       |        |          |       |        |         |           | 45.4   | 0.0554 |          |       |       |
|         |           |       |        |          |       |        |         |           | 121.8  | 0.0745 |          |       |       |
|         |           |       |        |          |       |        |         |           | 562.9  | 0.0023 |          |       |       |
|         |           |       |        |          |       |        |         |           | 841.5  | 0.1508 |          |       |       |
|         |           |       |        |          |       |        |         |           | 961.1  | 0.0021 |          |       |       |
|         |           |       |        |          |       |        |         |           | 963.4  | 0.1241 |          |       |       |
|         |           |       |        |          |       |        |         |           | 1389.0 | 0.0088 |          |       |       |
|         |           |       |        |          |       |        |         |           | 794.2  | 0.0041 |          |       |       |
|         |           |       |        |          |       |        |         |           | 344.3  | 0.0249 |          |       |       |
|         |           |       |        |          |       |        |         |           | 970.4  | 0.0062 |          |       |       |
|         |           |       |        |          |       |        |         |           | 1315.0 | 0.0098 |          |       |       |
|         |           |       |        |          |       |        |         |           | 676.7  | 0.0032 |          |       |       |
| 152Eu   | 13.6y     | 39.5  | 0.2083 | 2.009-4  | 3.604 | 0.831  |         |           |        |        |          |       |       |
|         |           | 40.1  | 0.3773 |          |       |        |         |           |        |        |          |       |       |
|         |           | 45.4  | 0.1464 |          |       |        |         |           |        |        |          |       |       |
|         |           | 121.8 | 0.2843 |          |       |        |         |           |        |        |          |       |       |
|         |           | 244.7 | 0.0749 |          |       |        |         |           |        |        |          |       |       |
|         |           | 295.9 | 0.0043 |          |       |        |         |           |        |        |          |       |       |
|         |           | 329.3 | 0.0012 |          |       |        |         |           |        |        |          |       |       |
|         |           | 415.9 | 0.0010 |          |       |        |         |           |        |        |          |       |       |
|         |           | 444.0 | 0.0281 |          |       |        |         |           |        |        |          |       |       |
|         |           | 444.0 | 0.0030 |          |       |        |         |           |        |        |          |       |       |
|         |           | 488.7 | 0.0041 |          |       |        |         |           |        |        |          |       |       |
|         |           | 564.0 | 0.0048 |          |       |        |         |           |        |        |          |       |       |
|         |           | 566.4 | 0.0013 |          |       |        |         |           |        |        |          |       |       |
|         |           | 656.4 | 0.0014 |          |       |        |         |           |        |        |          |       |       |
|         |           | 674.6 | 0.0015 |          |       |        |         |           |        |        |          |       |       |
|         |           | 688.6 | 0.0084 |          |       |        |         |           |        |        |          |       |       |
|         |           | 719.3 | 0.0027 |          |       |        |         |           |        |        |          |       |       |
|         |           | 810.4 | 0.0031 |          |       |        |         |           |        |        |          |       |       |
|         |           | 841.5 | 0.0016 |          |       |        |         |           |        |        |          |       |       |
|         |           | 867.3 | 0.0416 |          |       |        |         |           |        |        |          |       |       |
|         |           | 919.3 | 0.0040 |          |       |        |         |           |        |        |          |       |       |
|         |           | 926.2 | 0.0026 |          |       |        |         |           |        |        |          |       |       |
|         |           |       |        |          |       |        |         |           |        |        |          |       |       |
|         |           |       |        |          |       |        | 154Eu   | 8.8y      | 42.3   | 0.0726 | 2.039-4  | 3.734 | 0.802 |
|         |           |       |        |          |       |        |         |           | 43.0   | 0.1309 |          |       |       |
|         |           |       |        |          |       |        |         |           | 48.7   | 0.0517 |          |       |       |
|         |           |       |        |          |       |        |         |           | 123.1  | 0.4046 |          |       |       |
|         |           |       |        |          |       |        |         |           | 188.3  | 0.0023 |          |       |       |
|         |           |       |        |          |       |        |         |           | 247.9  | 0.0660 |          |       |       |
|         |           |       |        |          |       |        |         |           | 401.3  | 0.0021 |          |       |       |
|         |           |       |        |          |       |        |         |           | 444.5  | 0.0050 |          |       |       |
|         |           |       |        |          |       |        |         |           | 478.3  | 0.0022 |          |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T      | $\mu$                |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|--------|----------------------|
|         |           | 557.6  | 0.0026 |          |       |        |         |           | 811.8  | 0.1040 |          |        |                      |
|         |           | 582.0  | 0.0084 |          |       |        |         |           | 820.4  | 0.0016 |          |        |                      |
|         |           | 591.8  | 0.0483 |          |       |        |         |           | 841.1  | 0.0023 |          |        |                      |
|         |           | 625.2  | 0.0031 |          |       |        |         |           | 858.4  | 0.0013 |          |        |                      |
|         |           | 676.6  | 0.0014 |          |       |        |         |           | 866.0  | 0.0016 |          |        |                      |
|         |           | 692.4  | 0.0169 |          |       |        |         |           | 867.0  | 0.0140 |          |        |                      |
|         |           | 715.8  | 0.0017 |          |       |        |         |           | 944.4  | 0.0139 |          |        |                      |
|         |           | 723.3  | 0.1970 |          |       |        |         |           | 947.5  | 0.0031 |          |        |                      |
|         |           | 756.9  | 0.0433 |          |       |        |         |           | 960.5  | 0.0162 |          |        |                      |
|         |           | 815.6  | 0.0047 |          |       |        |         |           | 961.0  | 0.0016 |          |        |                      |
|         |           | 845.4  | 0.0055 |          |       |        |         |           | 969.8  | 0.0039 |          |        |                      |
|         |           | 850.6  | 0.0023 |          |       |        |         |           | 1012.0 | 0.0034 |          |        |                      |
|         |           | 873.2  | 0.1150 |          |       |        |         |           | 1027.0 | 0.0012 |          |        |                      |
|         |           | 892.7  | 0.0046 |          |       |        |         |           | 1040.0 | 0.0053 |          |        |                      |
|         |           | 904.0  | 0.0082 |          |       |        |         |           | 1065.0 | 0.0524 |          |        |                      |
|         |           | 996.3  | 0.1029 |          |       |        |         |           | 1076.0 | 0.0037 |          |        |                      |
|         |           | 1005.0 | 0.1789 |          |       |        |         |           | 1079.0 | 0.0489 |          |        |                      |
|         |           | 1047.0 | 0.0014 |          |       |        |         |           | 1130.0 | 0.0014 |          |        |                      |
|         |           | 1118.0 | 0.0010 |          |       |        |         |           | 1141.0 | 0.0030 |          |        |                      |
|         |           | 1128.0 | 0.0027 |          |       |        |         |           | 1153.0 | 0.0718 |          |        |                      |
|         |           | 1141.0 | 0.0022 |          |       |        |         |           | 1154.0 | 0.0530 |          |        |                      |
|         |           | 1242.0 | 0.0013 |          |       |        |         |           | 1156.0 | 0.0014 |          |        |                      |
|         |           | 1246.0 | 0.0090 |          |       |        |         |           | 1169.0 | 0.0029 |          |        |                      |
|         |           | 1274.0 | 0.3549 |          |       |        |         |           | 1231.0 | 0.0894 |          |        |                      |
|         |           | 1494.0 | 0.0065 |          |       |        |         |           | 1242.0 | 0.0676 |          |        |                      |
|         |           | 1593.0 | 0.0103 |          |       |        |         |           | 1277.0 | 0.0321 |          |        |                      |
|         |           | 1597.0 | 0.0185 |          |       |        |         |           | 1366.0 | 0.0176 |          |        |                      |
|         |           | 710.5  | 0.0159 |          |       |        |         |           | 1682.0 | 0.0030 |          |        |                      |
|         |           |        |        |          |       |        |         |           | 1857.0 | 0.0025 |          |        |                      |
| 155Eu   | 5.0y      | 26.5   | 0.0032 | 1.804-5  | 0.105 | 28.563 |         |           | 1877.0 | 0.0173 |          |        |                      |
|         |           | 42.3   | 0.0647 |          |       |        |         |           | 1938.0 | 0.0214 |          |        |                      |
|         |           | 43.0   | 0.1166 |          |       |        |         |           | 1946.0 | 0.0019 |          |        |                      |
|         |           | 45.3   | 0.0129 |          |       |        |         |           | 1966.0 | 0.0420 |          |        |                      |
|         |           | 48.7   | 0.0460 |          |       |        |         |           | 2027.0 | 0.0354 |          |        |                      |
|         |           | 60.0   | 0.0111 |          |       |        |         |           | 2033.0 | 0.0013 |          |        |                      |
|         |           | 86.1   | 0.0015 |          |       |        |         |           | 2098.0 | 0.0427 |          |        |                      |
|         |           | 86.5   | 0.3090 |          |       |        |         |           | 2116.0 | 0.0013 |          |        |                      |
|         |           | 105.3  | 0.2067 |          |       |        |         |           | 2181.0 | 0.0243 |          |        |                      |
|         |           | 69.3   | 0.0018 |          |       |        |         |           | 2187.0 | 0.0395 |          |        |                      |
|         |           |        |        |          |       |        |         |           | 2205.0 | 0.0100 |          |        |                      |
| 156Eu   | 15.2d     | 42.3   | 0.0383 | 1.978-4  | 4.622 | 0.648  |         |           | 2270.0 | 0.0112 |          |        |                      |
|         |           | 43.0   | 0.0691 |          |       |        |         |           | 1103.0 | 0.0159 |          |        |                      |
|         |           | 48.7   | 0.0273 |          |       |        |         |           |        |        |          |        |                      |
|         |           | 89.0   | 0.0905 |          |       |        |         |           | 153Gd  | 241.6d | 40.9     | 0.3579 | 4.659-5 0.038 79.273 |
|         |           | 199.2  | 0.0079 |          |       |        |         |           |        |        | 41.5     | 0.6472 |                      |
|         |           | 434.4  | 0.0022 |          |       |        |         |           |        |        | 47.0     | 0.2533 |                      |
|         |           | 472.7  | 0.0015 |          |       |        |         |           |        |        | 69.7     | 0.0257 |                      |
|         |           | 490.3  | 0.0018 |          |       |        |         |           |        |        | 83.4     | 0.0022 |                      |
|         |           | 599.5  | 0.0231 |          |       |        |         |           |        |        | 97.4     | 0.3130 |                      |
|         |           | 646.3  | 0.0709 |          |       |        |         |           |        |        | 103.2    | 0.2222 |                      |
|         |           | 709.9  | 0.0092 |          |       |        |         |           |        |        | 95.8     | 0.0022 |                      |
|         |           | 723.5  | 0.0602 |          |       |        |         |           |        |        |          |        |                      |
|         |           | 797.7  | 0.0011 |          |       |        |         |           | 159Gd  | 18.6h  | 43.7     | 0.0430 | 1.059-5 0.827 3.622  |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|--------|
|         |           | 44.5   | 0.0773 |          |       |       |         |           | 622.5  | 0.0088 |          |       |        |
|         |           | 50.4   | 0.0307 |          |       |       |         |           | 697.4  | 0.0254 |          |       |        |
|         |           | 58.0   | 0.0176 |          |       |       |         |           | 807.5  | 0.4210 |          |       |        |
|         |           | 226.0  | 0.0016 |          |       |       |         |           | 882.3  | 0.1322 |          |       |        |
|         |           | 348.2  | 0.0017 |          |       |       |         |           | 888.2  | 0.3806 |          |       |        |
|         |           | 363.6  | 0.0840 |          |       |       |         |           | 1068.0 | 0.0055 |          |       |        |
|         |           | 364.0  | 0.0021 |          |       |       |         |           | 1288.0 | 0.0015 |          |       |        |
|         |           |        |        |          |       |       |         |           | 1611.0 | 0.0014 |          |       |        |
| 162Gd   | 9.7m      | 38.8   | 0.0647 | 8.341-5  | 1.313 | 2.282 |         |           | 1453.0 | 0.0114 |          |       |        |
|         |           | 43.7   | 0.0016 |          |       |       | 157Dy   | 8.1h      | 43.7   | 0.2354 | 8.357-5  | 0.739 | 4.053  |
|         |           | 44.5   | 0.0029 |          |       |       |         |           | 44.5   | 0.4227 |          |       |        |
|         |           | 50.4   | 0.0012 |          |       |       |         |           | 50.4   | 0.1678 |          |       |        |
|         |           | 402.8  | 0.4620 |          |       |       |         |           | 60.8   | 0.0045 |          |       |        |
|         |           | 441.6  | 0.5313 |          |       |       |         |           | 83.0   | 0.0058 |          |       |        |
| 157Tb   | 150y      | 42.3   | 0.0265 | 2.426-6  | 0.026 | 1.144 |         |           | 182.2  | 0.0165 |          |       |        |
|         |           | 43.0   | 0.0478 |          |       |       |         |           | 265.3  | 0.0018 |          |       |        |
|         |           | 48.7   | 0.0189 |          |       |       |         |           | 326.2  | 0.9380 |          |       |        |
|         |           | 54.5   | 0.0001 |          |       |       |         |           | 594.2  | 0.0047 |          |       |        |
| 160Tb   | 72.3d     | 45.2   | 0.0603 | 1.787-4  | 3.594 | 0.834 | 165Dy   | 2.3h      | 46.7   | 0.0259 | 6.177-6  | 1.646 | 1.820  |
|         |           | 46.0   | 0.1079 |          |       |       |         |           | 47.5   | 0.0461 |          |       |        |
|         |           | 52.1   | 0.0432 |          |       |       |         |           | 53.9   | 0.0186 |          |       |        |
|         |           | 86.8   | 0.1330 |          |       |       |         |           | 94.7   | 0.0358 |          |       |        |
|         |           | 197.0  | 0.0490 |          |       |       |         |           | 279.8  | 0.0050 |          |       |        |
|         |           | 215.7  | 0.0371 |          |       |       |         |           | 361.7  | 0.0084 |          |       |        |
|         |           | 298.6  | 0.2707 |          |       |       |         |           | 545.8  | 0.0016 |          |       |        |
|         |           | 309.6  | 0.0082 |          |       |       |         |           | 565.7  | 0.0013 |          |       |        |
|         |           | 337.3  | 0.0033 |          |       |       |         |           | 633.4  | 0.0057 |          |       |        |
|         |           | 392.5  | 0.0128 |          |       |       |         |           | 715.3  | 0.0053 |          |       |        |
|         |           | 682.3  | 0.0055 |          |       |       |         |           | 641.4  | 0.0067 |          |       |        |
|         |           | 765.3  | 0.0193 |          |       |       | 166Dy   | 3.4d      | 28.2   | 0.0101 | 1.550-5  | 0.129 | 23.196 |
|         |           | 872.0  | 0.0018 |          |       |       |         |           | 46.7   | 0.1358 |          |       |        |
|         |           | 879.4  | 0.2850 |          |       |       |         |           | 47.5   | 0.2420 |          |       |        |
|         |           | 962.3  | 0.0902 |          |       |       |         |           | 53.9   | 0.0978 |          |       |        |
|         |           | 966.1  | 0.2422 |          |       |       |         |           | 54.2   | 0.0070 |          |       |        |
|         |           | 1003.0 | 0.0097 |          |       |       |         |           | 82.5   | 0.1290 |          |       |        |
|         |           | 1103.0 | 0.0052 |          |       |       |         |           | 371.7  | 0.0049 |          |       |        |
|         |           | 1115.0 | 0.0150 |          |       |       |         |           | 426.0  | 0.0054 |          |       |        |
|         |           | 1178.0 | 0.1444 |          |       |       |         |           | 333.9  | 0.0007 |          |       |        |
|         |           | 1200.0 | 0.0236 |          |       |       | 166Ho   | 1.1d      | 48.2   | 0.0286 | 6.258-6  | 3.403 | 0.880  |
|         |           | 1272.0 | 0.0703 |          |       |       |         |           | 49.1   | 0.0508 |          |       |        |
|         |           | 1312.0 | 0.0285 |          |       |       |         |           | 55.7   | 0.0207 |          |       |        |
|         |           | 651.4  | 0.0051 |          |       |       |         |           | 80.6   | 0.0620 |          |       |        |
| 162Tb   | 7.8m      | 45.2   | 0.0507 | 1.923-4  | 2.944 | 1.018 |         |           | 1379.0 | 0.0093 |          |       |        |
|         |           | 46.0   | 0.0908 |          |       |       |         |           | 1582.0 | 0.0018 |          |       |        |
|         |           | 52.1   | 0.0364 |          |       |       |         |           | 1662.0 | 0.0012 |          |       |        |
|         |           | 80.7   | 0.0850 |          |       |       |         |           | 1182.0 | 0.0008 |          |       |        |
|         |           | 185.0  | 0.0265 |          |       |       | 166mHo  | 1200.0y   | 48.2   | 0.1089 | 2.863-4  | 2.627 | 1.140  |
|         |           | 185.3  | 0.1419 |          |       |       |         |           | 49.1   | 0.1934 |          |       |        |
|         |           | 260.1  | 0.7873 |          |       |       |         |           |        |        |          |       |        |
|         |           | 543.2  | 0.0011 |          |       |       |         |           |        |        |          |       |        |

| Nuclide | Half-life | E      | Prob.   | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|---------|----------|-------|--------|---------|-----------|-------|--------|----------|-------|--------|
|         |           | 49.1   | 0.1934  |          |       |        |         |           | 237.1 | 0.0030 |          |       |        |
|         |           | 55.7   | 0.0789  |          |       |        |         |           | 277.4 | 0.0058 |          |       |        |
|         |           | 80.6   | 0.1270  |          |       |        |         |           | 295.9 | 0.2890 |          |       |        |
|         |           | 94.6   | 0.0016  |          |       |        |         |           | 308.3 | 0.6440 |          |       |        |
|         |           | 119.0  | 0.0016  |          |       |        |         |           | 372.0 | 0.0026 |          |       |        |
|         |           | 121.2  | 0.0024  |          |       |        |         |           | 670.7 | 0.0025 |          |       |        |
|         |           | 184.4  | 0.7260  |          |       |        |         |           | 676.1 | 0.0029 |          |       |        |
|         |           | 190.7  | 0.0022  |          |       |        |         |           | 784.1 | 0.0024 |          |       |        |
|         |           | 214.8  | 0.0043  |          |       |        |         |           | 796.6 | 0.0064 |          |       |        |
|         |           | 215.9  | 0.0257  |          |       |        |         |           | 907.7 | 0.0063 |          |       |        |
|         |           | 231.3  | 0.0021  |          |       |        |         |           | 535.2 | 0.0109 |          |       |        |
|         |           | 259.7  | 0.0105  |          |       |        |         |           |       |        |          |       |        |
|         |           | 280.4  | 0.2962  |          |       |        | 170Tm   | 128.6d    | 51.3  | 0.0127 | 1.673-6  | 0.103 | 29.074 |
|         |           | 300.7  | 0.0372  |          |       |        |         |           | 52.4  | 0.0225 |          |       |        |
|         |           | 339.8  | 0.0017  |          |       |        |         |           | 59.4  | 0.0093 |          |       |        |
|         |           | 365.8  | 0.0242  |          |       |        |         |           | 84.2  | 0.0326 |          |       |        |
|         |           | 410.9  | 0.1111  |          |       |        |         |           |       |        |          |       |        |
|         |           | 451.5  | 0.0292  |          |       |        | 171Tm   | 1.9y      | 51.3  | 0.0029 | 2.597-7  | 0.050 | 60.086 |
|         |           | 464.8  | 0.0120  |          |       |        |         |           | 52.4  | 0.0052 |          |       |        |
|         |           | 496.7  | 0.0022  |          |       |        |         |           | 59.4  | 0.0021 |          |       |        |
|         |           | 529.8  | 0.0951  |          |       |        |         |           | 66.7  | 0.0016 |          |       |        |
|         |           | 571.0  | 0.0547  |          |       |        |         |           |       |        |          |       |        |
|         |           | 594.4  | 0.0056  |          |       |        | 169Yb   | 32.0d     | 20.7  | 0.0021 | 8.837-5  | 0.225 | 13.301 |
|         |           | 611.5  | 0.0142  |          |       |        |         |           | 49.8  | 0.5278 |          |       |        |
|         |           | 644.5  | 0.0015  |          |       |        |         |           | 50.7  | 0.9341 |          |       |        |
|         |           | 670.5  | 0.0535  |          |       |        |         |           | 57.5  | 0.3830 |          |       |        |
|         |           | 691.2  | 0.0136  |          |       |        |         |           | 63.1  | 0.4375 |          |       |        |
|         |           | 711.7  | 0.5409  |          |       |        |         |           | 93.6  | 0.0266 |          |       |        |
|         |           | 712.4  | 0.0022  |          |       |        |         |           | 109.8 | 0.1736 |          |       |        |
|         |           | 736.7  | 0.0037  |          |       |        |         |           | 118.2 | 0.0188 |          |       |        |
|         |           | 752.3  | 0.1205  |          |       |        |         |           | 130.5 | 0.1106 |          |       |        |
|         |           | 778.8  | 0.0303  |          |       |        |         |           | 177.2 | 0.2144 |          |       |        |
|         |           | 810.3  | 0.5714  |          |       |        |         |           | 198.0 | 0.3492 |          |       |        |
|         |           | 830.6  | 0.0966  |          |       |        |         |           | 240.3 | 0.0012 |          |       |        |
|         |           | 875.6  | 0.0072  |          |       |        |         |           | 261.1 | 0.0177 |          |       |        |
|         |           | 950.9  | 0.0269  |          |       |        |         |           | 307.7 | 0.1081 |          |       |        |
|         |           | 1120.0 | 0.0024  |          |       |        |         |           | 344.1 | 0.0018 |          |       |        |
|         |           | 1147.0 | 0.9020  |          |       |        |         |           |       |        |          |       |        |
|         |           | 1241.0 | 0.0083  |          |       |        | 175Yb   | 4.2d      | 53.0  | 0.0109 | 8.233-6  | 0.999 | 2.998  |
|         |           | 1282.0 | 0.0018  |          |       |        |         |           | 54.1  | 0.0191 |          |       |        |
|         |           | 1401.0 | 0.0050  |          |       |        |         |           | 61.3  | 0.0079 |          |       |        |
|         |           | 1427.0 | 0.0048  |          |       |        |         |           | 113.8 | 0.0189 |          |       |        |
|         |           | 416.6  | 0.0063  |          |       |        |         |           | 137.7 | 0.0010 |          |       |        |
|         |           |        |         |          |       |        |         |           | 144.9 | 0.0034 |          |       |        |
| 169Er   | 9.4d      | 110.5  | 0.00003 | 3.406-10 | 0.065 | 46.160 |         |           | 282.5 | 0.0302 |          |       |        |
|         |           |        |         |          |       |        |         |           | 396.3 | 0.0650 |          |       |        |
|         |           |        |         |          |       |        |         |           | 251.5 | 0.0009 |          |       |        |
| 171Er   | 7.5h      | 49.8   | 0.1309  | 8.008-5  | 0.757 | 3.960  |         |           |       |        |          |       |        |
|         |           | 50.7   | 0.2316  |          |       |        |         |           |       |        |          |       |        |
|         |           | 57.5   | 0.0950  |          |       |        | 177Lu   | 6.7d      | 54.6  | 0.0163 | 7.636-6  | 0.274 | 10.935 |
|         |           | 111.6  | 0.2050  |          |       |        |         |           | 55.8  | 0.0285 |          |       |        |
|         |           | 116.7  | 0.0230  |          |       |        |         |           | 63.2  | 0.0120 |          |       |        |
|         |           | 124.0  | 0.0910  |          |       |        |         |           | 71.7  | 0.0016 |          |       |        |
|         |           | 210.6  | 0.0064  |          |       |        |         |           | 113.0 | 0.0638 |          |       |        |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|-------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 208.4 | 0.1100 |          |       |       |         |           | 426.3  | 0.0041 |          |       |       |
|         |           | 249.7 | 0.0021 |          |       |       |         |           | 466.0  | 0.0233 |          |       |       |
|         |           | 321.3 | 0.0022 |          |       |       |         |           | 157.9  | 0.0020 |          |       |       |
|         |           | 136.7 | 0.0005 |          |       |       |         |           |        |        |          |       |       |
| 177mLu  | 160.1d    | 53.0  | 0.0514 | 2.112-4  | 0.752 | 3.983 | 181Hf   | 42.4d     | 56.3   | 0.0861 | 1.061-4  | 1.441 | 2.078 |
|         |           | 54.1  | 0.0903 |          |       |       |         |           | 57.5   | 0.1503 |          |       |       |
|         |           | 61.3  | 0.0376 |          |       |       |         |           | 65.2   | 0.0634 |          |       |       |
|         |           | 115.8 | 0.0068 |          |       |       |         |           | 133.0  | 0.4167 |          |       |       |
|         |           | 121.6 | 0.0586 |          |       |       |         |           | 136.3  | 0.0523 |          |       |       |
|         |           | 147.2 | 0.0359 |          |       |       |         |           | 136.9  | 0.0076 |          |       |       |
|         |           | 171.9 | 0.0490 |          |       |       |         |           | 345.8  | 0.1715 |          |       |       |
|         |           | 195.6 | 0.0086 |          |       |       |         |           | 476.0  | 0.0042 |          |       |       |
|         |           | 218.1 | 0.0299 |          |       |       |         |           | 482.0  | 0.8285 |          |       |       |
|         |           | 268.8 | 0.0338 |          |       |       |         |           | 615.5  | 0.0014 |          |       |       |
|         |           | 319.0 | 0.1028 |          |       |       |         |           | 522.6  | 0.0003 |          |       |       |
|         |           | 367.4 | 0.0297 |          |       |       | 182Ta   | 114.7d    | 31.7   | 0.0080 | 2.084-4  | 3.987 | 0.751 |
|         |           | 413.6 | 0.1638 |          |       |       |         |           | 42.7   | 0.0025 |          |       |       |
|         |           | 54.6  | 0.3337 |          |       |       |         |           | 58.0   | 0.1041 |          |       |       |
|         |           | 55.1  | 0.0120 |          |       |       |         |           | 59.3   | 0.1810 |          |       |       |
|         |           | 55.8  | 0.5845 |          |       |       |         |           | 65.7   | 0.0280 |          |       |       |
|         |           | 63.2  | 0.2452 |          |       |       |         |           | 67.2   | 0.0770 |          |       |       |
|         |           | 71.7  | 0.0089 |          |       |       |         |           | 67.7   | 0.4235 |          |       |       |
|         |           | 105.3 | 0.1201 |          |       |       |         |           | 84.7   | 0.0274 |          |       |       |
|         |           | 113.0 | 0.2150 |          |       |       |         |           | 100.1  | 0.1407 |          |       |       |
|         |           | 117.0 | 0.0024 |          |       |       |         |           | 113.7  | 0.0190 |          |       |       |
|         |           | 128.5 | 0.1525 |          |       |       |         |           | 116.4  | 0.0044 |          |       |       |
|         |           | 136.7 | 0.0137 |          |       |       |         |           | 152.4  | 0.0717 |          |       |       |
|         |           | 145.6 | 0.0090 |          |       |       |         |           | 156.4  | 0.0272 |          |       |       |
|         |           | 153.3 | 0.1802 |          |       |       |         |           | 179.4  | 0.0319 |          |       |       |
|         |           | 159.9 | 0.0060 |          |       |       |         |           | 198.4  | 0.0151 |          |       |       |
|         |           | 174.4 | 0.1261 |          |       |       |         |           | 222.1  | 0.0756 |          |       |       |
|         |           | 177.0 | 0.0347 |          |       |       |         |           | 229.3  | 0.0364 |          |       |       |
|         |           | 204.1 | 0.1429 |          |       |       |         |           | 264.1  | 0.0364 |          |       |       |
|         |           | 208.4 | 0.6125 |          |       |       |         |           | 928.0  | 0.0062 |          |       |       |
|         |           | 214.4 | 0.0661 |          |       |       |         |           | 959.7  | 0.0035 |          |       |       |
|         |           | 228.4 | 0.3723 |          |       |       |         |           | 1002.0 | 0.0209 |          |       |       |
|         |           | 233.9 | 0.0565 |          |       |       |         |           | 1044.0 | 0.0024 |          |       |       |
|         |           | 249.7 | 0.0613 |          |       |       |         |           | 1113.0 | 0.0044 |          |       |       |
|         |           | 281.8 | 0.1405 |          |       |       |         |           | 1121.0 | 0.3500 |          |       |       |
|         |           | 283.4 | 0.0052 |          |       |       |         |           | 1157.0 | 0.0063 |          |       |       |
|         |           | 291.4 | 0.0101 |          |       |       |         |           | 1158.0 | 0.0035 |          |       |       |
|         |           | 292.5 | 0.0080 |          |       |       |         |           | 1189.0 | 0.1631 |          |       |       |
|         |           | 296.5 | 0.0541 |          |       |       |         |           | 1221.0 | 0.2705 |          |       |       |
|         |           | 299.0 | 0.0172 |          |       |       |         |           | 1224.0 | 0.0021 |          |       |       |
|         |           | 305.5 | 0.0174 |          |       |       |         |           | 1231.0 | 0.1151 |          |       |       |
|         |           | 313.7 | 0.0138 |          |       |       |         |           | 1257.0 | 0.0149 |          |       |       |
|         |           | 321.3 | 0.0139 |          |       |       |         |           | 1274.0 | 0.0065 |          |       |       |
|         |           | 327.7 | 0.1753 |          |       |       |         |           | 1289.0 | 0.0135 |          |       |       |
|         |           | 341.6 | 0.0179 |          |       |       |         |           | 1343.0 | 0.0025 |          |       |       |
|         |           | 378.5 | 0.2786 |          |       |       |         |           | 1374.0 | 0.0022 |          |       |       |
|         |           | 385.0 | 0.0294 |          |       |       |         |           | 943.0  | 0.0040 |          |       |       |
|         |           | 418.5 | 0.2006 |          |       |       |         |           |        |        |          |       |       |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|-------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|---|-------|
| 181W    | 120.9d    | 56.3  | 0.1869 | 1.389-5  | 0.058 | 52.038 |         |           |        |        |          |   |       |
|         |           | 57.5  | 0.3263 |          |       |        |         |           | 148.8  | 0.0171 |          |   |       |
|         |           | 65.2  | 0.1375 |          |       |        |         |           | 149.4  | 0.0088 |          |   |       |
|         |           | 147.7 | 0.0014 |          |       |        |         |           | 150.3  | 0.0049 |          |   |       |
| 185W    | 75.1d     | 125.4 | 0.0002 | 5.465-9  | 0.089 | 33.602 |         |           | 151.1  | 0.0043 |          |   |       |
| 187W    | 23.8h     | 59.7  | 0.0760 | 8.853-5  | 2.213 | 1.353  |         |           | 152.4  | 0.0833 |          |   |       |
|         |           | 61.1  | 0.1317 |          |       |        |         |           | 153.9  | 0.0024 |          |   |       |
|         |           | 69.3  | 0.0563 |          |       |        |         |           | 156.4  | 0.0741 |          |   |       |
|         |           | 72.1  | 0.1191 |          |       |        |         |           | 160.1  | 0.0023 |          |   |       |
|         |           | 134.2 | 0.0946 |          |       |        |         |           | 169.2  | 0.1167 |          |   |       |
|         |           | 206.3 | 0.0015 |          |       |        |         |           | 172.9  | 0.0348 |          |   |       |
|         |           | 246.2 | 0.0013 |          |       |        |         |           | 178.4  | 0.0220 |          |   |       |
|         |           | 479.5 | 0.2335 |          |       |        |         |           | 179.4  | 0.0293 |          |   |       |
|         |           | 511.8 | 0.0069 |          |       |        |         |           | 187.3  | 0.0031 |          |   |       |
|         |           | 551.5 | 0.0544 |          |       |        |         |           | 188.5  | 0.0013 |          |   |       |
|         |           | 589.1 | 0.0013 |          |       |        |         |           | 189.6  | 0.0038 |          |   |       |
|         |           | 618.4 | 0.0671 |          |       |        |         |           | 191.4  | 0.0769 |          |   |       |
|         |           | 625.5 | 0.0116 |          |       |        |         |           | 198.4  | 0.0408 |          |   |       |
|         |           | 685.8 | 0.2917 |          |       |        |         |           | 203.3  | 0.0045 |          |   |       |
|         |           | 745.2 | 0.0032 |          |       |        |         |           | 206.0  | 0.0049 |          |   |       |
|         |           | 772.9 | 0.0440 |          |       |        |         |           | 208.2  | 0.0060 |          |   |       |
|         |           | 864.5 | 0.0036 |          |       |        |         |           | 209.4  | 0.0047 |          |   |       |
|         |           | 879.4 | 0.0015 |          |       |        |         |           | 214.3  | 0.0107 |          |   |       |
|         |           | 276.2 | 0.0042 |          |       |        |         |           | 215.7  | 0.0075 |          |   |       |
| 188W    | 69.4d     | 63.6  | 0.0011 | 3.615-7  | 0.562 | 5.327  |         |           | 217.5  | 0.0318 |          |   |       |
|         |           | 227.1 | 0.0022 |          |       |        |         |           | 221.6  | 0.0620 |          |   |       |
|         |           | 290.7 | 0.0040 |          |       |        |         |           | 222.1  | 0.0833 |          |   |       |
|         |           | 165.2 | 0.0002 |          |       |        |         |           | 226.2  | 0.0325 |          |   |       |
| 182Re   | 2.7d      | 31.7  | 0.0043 | 3.076-4  | 3.512 | 0.853  |         |           | 229.3  | 0.2500 |          |   |       |
|         |           | 39.1  | 0.0030 |          |       |        |         |           | 247.4  | 0.0489 |          |   |       |
|         |           | 42.7  | 0.0028 |          |       |        |         |           | 256.4  | 0.1004 |          |   |       |
|         |           | 58.0  | 0.4990 |          |       |        |         |           | 264.1  | 0.0376 |          |   |       |
|         |           | 59.3  | 0.8678 |          |       |        |         |           | 276.3  | 0.0902 |          |   |       |
|         |           | 60.6  | 0.0011 |          |       |        |         |           | 281.4  | 0.0579 |          |   |       |
|         |           | 65.7  | 0.0265 |          |       |        |         |           | 286.6  | 0.0752 |          |   |       |
|         |           | 67.2  | 0.3690 |          |       |        |         |           | 295.7  | 0.0019 |          |   |       |
|         |           | 67.7  | 0.2201 |          |       |        |         |           | 300.0  | 0.0122 |          |   |       |
|         |           | 84.7  | 0.0267 |          |       |        |         |           | 300.5  | 0.0165 |          |   |       |
|         |           | 100.1 | 0.1453 |          |       |        |         |           | 313.9  | 0.0060 |          |   |       |
|         |           | 107.2 | 0.0137 |          |       |        |         |           | 323.4  | 0.0190 |          |   |       |
|         |           | 108.6 | 0.0077 |          |       |        |         |           | 339.1  | 0.0560 |          |   |       |
|         |           | 110.4 | 0.0010 |          |       |        |         |           | 342.0  | 0.0103 |          |   |       |
|         |           | 111.1 | 0.0020 |          |       |        |         |           | 345.4  | 0.0047 |          |   |       |
|         |           | 113.7 | 0.0438 |          |       |        |         |           | 351.1  | 0.1032 |          |   |       |
|         |           | 116.4 | 0.0049 |          |       |        |         |           | 357.1  | 0.0051 |          |   |       |
|         |           | 130.8 | 0.0727 |          |       |        |         |           | 928.0  | 0.0036 |          |   |       |
|         |           | 131.3 | 0.0016 |          |       |        |         |           | 943.0  | 0.0022 |          |   |       |
|         |           | 133.8 | 0.0246 |          |       |        |         |           | 959.7  | 0.0019 |          |   |       |
|         |           | 145.4 | 0.0064 |          |       |        |         |           | 1002.0 | 0.0242 |          |   |       |
|         |           | 147.6 | 0.0088 |          |       |        |         |           | 1044.0 | 0.0028 |          |   |       |
|         |           |       |        |          |       |        |         |           | 1076.0 | 0.1019 |          |   |       |
|         |           |       |        |          |       |        |         |           | 1088.0 | 0.0019 |          |   |       |
|         |           |       |        |          |       |        |         |           | 1113.0 | 0.0457 |          |   |       |
|         |           |       |        |          |       |        |         |           | 1121.0 | 0.2137 |          |   |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|--------|
|         |           | 1157.0 | 0.0038 |          |       |       |         |           | 1002.0 | 0.0022 |          |       |        |
|         |           | 1158.0 | 0.0085 |          |       |       |         |           | 1044.0 | 0.0018 |          |       |        |
|         |           | 1181.0 | 0.0054 |          |       |       |         |           | 1121.0 | 0.3190 |          |       |        |
|         |           | 1189.0 | 0.0876 |          |       |       |         |           | 1157.0 | 0.0070 |          |       |        |
|         |           | 1221.0 | 0.1652 |          |       |       |         |           | 1181.0 | 0.0022 |          |       |        |
|         |           | 1224.0 | 0.0019 |          |       |       |         |           | 1189.0 | 0.1515 |          |       |        |
|         |           | 1231.0 | 0.1440 |          |       |       |         |           | 1221.0 | 0.2466 |          |       |        |
|         |           | 1257.0 | 0.0103 |          |       |       |         |           | 1231.0 | 0.0131 |          |       |        |
|         |           | 1274.0 | 0.0092 |          |       |       |         |           | 1257.0 | 0.0140 |          |       |        |
|         |           | 1289.0 | 0.0074 |          |       |       |         |           | 1274.0 | 0.0054 |          |       |        |
|         |           | 1292.0 | 0.0023 |          |       |       |         |           | 1289.0 | 0.0121 |          |       |        |
|         |           | 1294.0 | 0.0158 |          |       |       |         |           | 1294.0 | 0.0018 |          |       |        |
|         |           | 1331.0 | 0.0035 |          |       |       |         |           | 1374.0 | 0.0019 |          |       |        |
|         |           | 1343.0 | 0.0269 |          |       |       |         |           | 1771.0 | 0.0029 |          |       |        |
|         |           | 1374.0 | 0.0029 |          |       |       |         |           | 1819.0 | 0.0011 |          |       |        |
|         |           | 1387.0 | 0.0026 |          |       |       |         |           | 1871.0 | 0.0029 |          |       |        |
|         |           | 1410.0 | 0.0028 |          |       |       |         |           | 1957.0 | 0.0046 |          |       |        |
|         |           | 1427.0 | 0.0945 |          |       |       |         |           | 2016.0 | 0.0078 |          |       |        |
|         |           | 1439.0 | 0.0016 |          |       |       |         |           | 2047.0 | 0.0012 |          |       |        |
|         |           | 1074.0 | 0.0040 |          |       |       |         |           | 2057.0 | 0.0083 |          |       |        |
|         |           |        |        |          |       |       |         |           | 2208.0 | 0.0010 |          |       |        |
|         |           |        |        |          |       |       |         |           | 1790.0 | 0.0089 |          |       |        |
|         |           |        |        |          |       |       |         |           | 511.0  | 0.0379 |          |       |        |
| 182mRe  | 12.7h     | 31.7   | 0.0073 | 1.990-4  | 3.936 | 0.761 |         |           |        |        |          |       |        |
|         |           | 42.7   | 0.0023 |          |       |       |         |           |        |        |          |       |        |
|         |           | 58.0   | 0.3004 |          |       |       |         |           |        |        |          |       |        |
|         |           | 59.3   | 0.5225 |          |       |       | 183Re   | 70d       | 46.5   | 0.0798 | 4.257-5  | 0.176 | 17.021 |
|         |           | 65.7   | 0.0025 |          |       |       |         |           | 52.6   | 0.0222 |          |       |        |
|         |           | 67.2   | 0.2222 |          |       |       |         |           | 58.0   | 0.3424 |          |       |        |
|         |           | 67.7   | 0.3924 |          |       |       |         |           | 59.3   | 0.5954 |          |       |        |
|         |           | 84.7   | 0.0265 |          |       |       |         |           | 67.2   | 0.2532 |          |       |        |
|         |           | 100.1  | 0.1435 |          |       |       |         |           | 82.9   | 0.0025 |          |       |        |
|         |           | 113.7  | 0.0041 |          |       |       |         |           | 84.7   | 0.0088 |          |       |        |
|         |           | 116.4  | 0.0035 |          |       |       |         |           | 99.1   | 0.0269 |          |       |        |
|         |           | 152.4  | 0.0670 |          |       |       |         |           | 107.9  | 0.0218 |          |       |        |
|         |           | 156.4  | 0.0041 |          |       |       |         |           | 109.7  | 0.0290 |          |       |        |
|         |           | 179.4  | 0.0024 |          |       |       |         |           | 144.1  | 0.0012 |          |       |        |
|         |           | 198.4  | 0.0018 |          |       |       |         |           | 160.5  | 0.0059 |          |       |        |
|         |           | 222.1  | 0.0067 |          |       |       |         |           | 161.3  | 0.0036 |          |       |        |
|         |           | 229.3  | 0.0214 |          |       |       |         |           | 162.3  | 0.2336 |          |       |        |
|         |           | 264.1  | 0.0026 |          |       |       |         |           | 192.6  | 0.0026 |          |       |        |
|         |           | 470.3  | 0.0198 |          |       |       |         |           | 205.1  | 0.0011 |          |       |        |
|         |           | 536.0  | 0.0021 |          |       |       |         |           | 208.8  | 0.0298 |          |       |        |
|         |           | 555.0  | 0.0011 |          |       |       |         |           | 209.9  | 0.0026 |          |       |        |
|         |           | 598.6  | 0.0040 |          |       |       |         |           | 244.3  | 0.0041 |          |       |        |
|         |           | 649.7  | 0.0035 |          |       |       |         |           | 245.2  | 0.0026 |          |       |        |
|         |           | 734.5  | 0.0038 |          |       |       |         |           | 246.1  | 0.0132 |          |       |        |
|         |           | 787.1  | 0.0026 |          |       |       |         |           | 291.7  | 0.0317 |          |       |        |
|         |           | 800.0  | 0.0015 |          |       |       |         |           | 313.0  | 0.0042 |          |       |        |
|         |           | 810.2  | 0.0038 |          |       |       |         |           | 354.0  | 0.0054 |          |       |        |
|         |           | 836.0  | 0.0048 |          |       |       |         |           | 254.0  | 0.0019 |          |       |        |
|         |           | 894.9  | 0.0210 |          |       |       |         |           |        |        |          |       |        |
|         |           | 900.8  | 0.0035 |          |       |       | 184Re   | 38.0d     | 58.0   | 0.2548 | 1.572-4  | 3.112 | 0.963  |
|         |           | 928.0  | 0.0051 |          |       |       |         |           | 59.3   | 0.4431 |          |       |        |
|         |           | 959.7  | 0.0038 |          |       |       |         |           | 67.2   | 0.1884 |          |       |        |



| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|-------|--------|
|         |           | 111.2  | 0.1714 |          |       |        |         |           | 71.4   | 0.0086 |          |       |        |
|         |           | 252.8  | 0.0302 |          |       |        |         |           | 137.2  | 0.0952 |          |       |        |
|         |           | 539.2  | 0.0033 |          |       |        |         |           | 702.2  | 0.0006 |          |       |        |
|         |           | 641.9  | 0.0194 |          |       |        |         |           |        |        |          |       |        |
|         |           | 769.8  | 0.0067 |          |       |        | 188Re   | 17.0h     | 61.5   | 0.0136 | 1.093-5  | 2.261 | 1.325  |
|         |           | 792.1  | 0.3746 |          |       |        |         |           | 63.0   | 0.0235 |          |       |        |
|         |           | 894.8  | 0.1559 |          |       |        |         |           | 71.4   | 0.0101 |          |       |        |
|         |           | 903.3  | 0.3788 |          |       |        |         |           | 155.0  | 0.1497 |          |       |        |
|         |           | 1023.0 | 0.0052 |          |       |        |         |           | 478.0  | 0.0105 |          |       |        |
|         |           | 1275.0 | 0.0012 |          |       |        |         |           | 633.1  | 0.0126 |          |       |        |
|         |           | 1386.0 | 0.0010 |          |       |        |         |           | 635.0  | 0.0015 |          |       |        |
|         |           | 787.3  | 0.0029 |          |       |        |         |           | 672.5  | 0.0011 |          |       |        |
| 184mRe  | 169.0d    | 59.7   | 0.1407 | 7.671-5  | 2.473 | 1.211  |         |           | 829.5  | 0.0041 |          |       |        |
|         |           | 61.1   | 0.2438 |          |       |        |         |           | 931.3  | 0.0056 |          |       |        |
|         |           | 69.3   | 0.1042 |          |       |        |         |           | 1134.0 | 0.0072 |          |       |        |
|         |           | 104.7  | 0.1328 |          |       |        |         |           |        |        |          |       |        |
|         |           | 87.3   | 0.0001 |          |       |        | 185Os   | 93.6d     | 59.7   | 0.2099 | 1.310-4  | 2.457 | 1.219  |
|         |           | 55.3   | 0.0236 |          |       |        |         |           | 61.1   | 0.3638 |          |       |        |
|         |           | 58.0   | 0.0851 |          |       |        |         |           | 69.3   | 0.1555 |          |       |        |
|         |           | 59.3   | 0.1480 |          |       |        |         |           | 71.3   | 0.0025 |          |       |        |
|         |           | 63.7   | 0.0038 |          |       |        |         |           | 125.4  | 0.0035 |          |       |        |
|         |           | 67.2   | 0.0629 |          |       |        |         |           | 162.9  | 0.0056 |          |       |        |
|         |           | 87.5   | 0.0024 |          |       |        |         |           | 234.2  | 0.0041 |          |       |        |
|         |           | 91.3   | 0.0026 |          |       |        |         |           | 592.1  | 0.0131 |          |       |        |
|         |           | 111.2  | 0.0592 |          |       |        |         |           | 646.1  | 0.8020 |          |       |        |
|         |           | 124.1  | 0.0015 |          |       |        |         |           | 717.4  | 0.0408 |          |       |        |
|         |           | 161.3  | 0.0664 |          |       |        |         |           | 874.8  | 0.0654 |          |       |        |
|         |           | 215.3  | 0.0284 |          |       |        |         |           | 880.3  | 0.0495 |          |       |        |
|         |           | 216.5  | 0.0963 |          |       |        |         |           | 910.0  | 0.0006 |          |       |        |
|         |           | 226.8  | 0.0151 |          |       |        |         |           |        |        |          |       |        |
|         |           | 252.8  | 0.1091 |          |       |        | 190mOs  | 9.9m      | 61.5   | 0.0570 | 2.995-4  | 1.777 | 1.686  |
|         |           | 318.0  | 0.0588 |          |       |        |         |           | 63.0   | 0.0984 |          |       |        |
|         |           | 384.2  | 0.0320 |          |       |        |         |           | 71.4   | 0.0423 |          |       |        |
|         |           | 536.7  | 0.0337 |          |       |        |         |           | 186.7  | 0.7020 |          |       |        |
|         |           | 641.9  | 0.0035 |          |       |        |         |           | 361.1  | 0.9488 |          |       |        |
|         |           | 769.8  | 0.0024 |          |       |        |         |           | 502.6  | 0.9778 |          |       |        |
|         |           | 792.1  | 0.0377 |          |       |        |         |           | 616.1  | 0.9862 |          |       |        |
|         |           | 857.2  | 0.0017 |          |       |        |         |           | 38.9   | 0.0010 |          |       |        |
|         |           | 894.8  | 0.0281 |          |       |        |         |           |        |        |          |       |        |
|         |           | 903.3  | 0.0382 |          |       |        | 191Os   | 15.4d     | 63.3   | 0.1601 | 1.837-5  | 0.086 | 34.767 |
|         |           | 920.9  | 0.0831 |          |       |        |         |           | 64.9   | 0.2756 |          |       |        |
|         |           | 1023.0 | 0.0018 |          |       |        |         |           | 73.6   | 0.1190 |          |       |        |
|         |           | 1110.0 | 0.0060 |          |       |        |         |           | 129.4  | 0.2590 |          |       |        |
|         |           | 1174.0 | 0.0124 |          |       |        |         |           | 73.2   | 0.0004 |          |       |        |
|         |           | 647.4  | 0.0027 |          |       |        |         |           |        |        |          |       |        |
| 186Re   | 3.8d      | 58.0   | 0.0160 | 4.909-6  | 0.105 | 28.583 | 191mOs  | 13.0h     | 61.5   | 0.0199 | 1.449-6  | 0.073 | 40.828 |
|         |           | 59.3   | 0.0278 |          |       |        |         |           | 63.0   | 0.0345 |          |       |        |
|         |           | 67.2   | 0.0118 |          |       |        |         |           | 71.4   | 0.0148 |          |       |        |
|         |           | 122.3  | 0.0070 |          |       |        |         |           | 74.4   | 0.0006 |          |       |        |
|         |           | 61.5   | 0.0116 |          |       |        | 193Os   | 1.2d      | 63.3   | 0.0357 | 1.412-5  | 1.221 | 2.454  |
|         |           | 63.0   | 0.0200 |          |       |        |         |           | 64.9   | 0.0614 |          |       |        |
|         |           |        |        |          |       |        |         |           | 73.0   | 0.0348 |          |       |        |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|-------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|--------|
|         |           | 73.6  | 0.0265 |          |       |       |         |           | 605.1  | 0.3780 |          |       |        |
|         |           | 96.8  | 0.0010 |          |       |       |         |           | 615.4  | 0.0044 |          |       |        |
|         |           | 107.0 | 0.0064 |          |       |       |         |           | 628.4  | 0.0071 |          |       |        |
|         |           | 138.9 | 0.0432 |          |       |       |         |           | 630.9  | 0.0279 |          |       |        |
|         |           | 180.0 | 0.0018 |          |       |       |         |           | 631.0  | 0.0080 |          |       |        |
|         |           | 181.8 | 0.0020 |          |       |       |         |           | 656.0  | 0.0110 |          |       |        |
|         |           | 219.1 | 0.0028 |          |       |       |         |           | 690.0  | 0.0027 |          |       |        |
|         |           | 251.6 | 0.0022 |          |       |       |         |           | 726.2  | 0.0359 |          |       |        |
|         |           | 280.4 | 0.0126 |          |       |       |         |           | 740.2  | 0.0018 |          |       |        |
|         |           | 288.8 | 0.0014 |          |       |       |         |           | 768.6  | 0.0210 |          |       |        |
|         |           | 298.8 | 0.0019 |          |       |       |         |           | 821.8  | 0.0031 |          |       |        |
|         |           | 321.6 | 0.0129 |          |       |       |         |           | 828.0  | 0.0054 |          |       |        |
|         |           | 361.8 | 0.0030 |          |       |       |         |           | 829.0  | 0.0328 |          |       |        |
|         |           | 387.5 | 0.0128 |          |       |       |         |           | 839.1  | 0.0108 |          |       |        |
|         |           | 420.3 | 0.0017 |          |       |       |         |           | 916.8  | 0.0012 |          |       |        |
|         |           | 460.5 | 0.0400 |          |       |       |         |           | 1036.0 | 0.0229 |          |       |        |
|         |           | 484.3 | 0.0017 |          |       |       |         |           | 1134.0 | 0.0041 |          |       |        |
|         |           | 557.4 | 0.0132 |          |       |       |         |           | 1147.0 | 0.0013 |          |       |        |
|         |           | 559.3 | 0.0049 |          |       |       |         |           | 1200.0 | 0.0042 |          |       |        |
|         |           | 400.7 | 0.0067 |          |       |       |         |           | 1324.0 | 0.0046 |          |       |        |
|         |           |       |        |          |       |       |         |           | 1387.0 | 0.0015 |          |       |        |
| 190Ir   | 11.8d     | 61.5  | 0.2471 | 2.666-4  | 1.895 | 1.581 |         |           | 1397.0 | 0.0014 |          |       |        |
|         |           | 63.0  | 0.4267 |          |       |       |         |           | 823.7  | 0.0079 |          |       |        |
|         |           | 71.4  | 0.1833 |          |       |       |         |           |        |        |          |       |        |
|         |           | 186.7 | 0.4968 |          |       |       | 190mIr  | 1.2h      | 26.3   | 0.0000 | 6.12-11  | 0.007 | 456.9  |
|         |           | 190.5 | 0.0013 |          |       |       |         |           |        |        |          |       |        |
|         |           | 196.9 | 0.0324 |          |       |       | 190mIr  | 3.2h      | 63.3   | 0.0031 | 1.499-5  | 0.073 | 40.962 |
|         |           | 198.1 | 0.0184 |          |       |       |         |           | 64.9   | 0.0053 |          |       |        |
|         |           | 199.3 | 0.0022 |          |       |       |         |           | 73.6   | 0.0023 |          |       |        |
|         |           | 207.9 | 0.0032 |          |       |       |         |           | 148.7  | 0.0001 |          |       |        |
|         |           | 207.9 | 0.0112 |          |       |       |         |           | 61.5   | 0.2051 |          |       |        |
|         |           | 223.8 | 0.0354 |          |       |       |         |           | 63.0   | 0.3542 |          |       |        |
|         |           | 235.5 | 0.0040 |          |       |       |         |           | 71.4   | 0.1521 |          |       |        |
|         |           | 248.2 | 0.0011 |          |       |       |         |           |        |        |          |       |        |
|         |           | 282.9 | 0.0045 |          |       |       | 192Ir   | 74.0d     | 61.5   | 0.0113 | 1.597-4  | 1.284 | 2.334  |
|         |           | 288.2 | 0.0156 |          |       |       |         |           | 63.0   | 0.0195 |          |       |        |
|         |           | 294.7 | 0.0616 |          |       |       |         |           | 71.4   | 0.0084 |          |       |        |
|         |           | 361.1 | 0.1233 |          |       |       |         |           | 201.3  | 0.0047 |          |       |        |
|         |           | 371.2 | 0.2160 |          |       |       |         |           | 205.8  | 0.0329 |          |       |        |
|         |           | 380.0 | 0.0192 |          |       |       |         |           | 283.3  | 0.0026 |          |       |        |
|         |           | 397.4 | 0.0620 |          |       |       |         |           | 374.5  | 0.0073 |          |       |        |
|         |           | 407.2 | 0.0432 |          |       |       |         |           | 484.6  | 0.0316 |          |       |        |
|         |           | 407.2 | 0.2268 |          |       |       |         |           | 489.1  | 0.0040 |          |       |        |
|         |           | 420.6 | 0.0156 |          |       |       |         |           | 423.1  | 0.0008 |          |       |        |
|         |           | 431.6 | 0.0259 |          |       |       |         |           | 65.1   | 0.0264 |          |       | </     |

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| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T       | $\mu$        |
|---------|-----------|--------|--------|----------|---|-------|---------|-----------|--------|--------|----------|---------|--------------|
|         |           | 203.0  | 0.0034 |          |   |       |         |           | 1563.0 | 0.0033 |          |         |              |
|         |           | 290.8  | 0.0011 |          |   |       |         |           | 1592.0 | 0.0109 |          |         |              |
|         |           | 293.5  | 0.1116 |          |   |       |         |           | 1594.0 | 0.0064 |          |         |              |
|         |           | 300.7  | 0.0092 |          |   |       |         |           | 1596.0 | 0.0185 |          |         |              |
|         |           | 318.1  | 0.0033 |          |   |       |         |           | 1602.0 | 0.0026 |          |         |              |
|         |           | 328.5  | 0.6380 |          |   |       |         |           | 1618.0 | 0.0022 |          |         |              |
|         |           | 364.9  | 0.0154 |          |   |       |         |           | 1622.0 | 0.0020 |          |         |              |
|         |           | 449.4  | 0.0017 |          |   |       |         |           | 1633.0 | 0.0025 |          |         |              |
|         |           | 482.9  | 0.0119 |          |   |       |         |           | 1671.0 | 0.0018 |          |         |              |
|         |           | 528.8  | 0.0172 |          |   |       |         |           | 1676.0 | 0.0014 |          |         |              |
|         |           | 530.2  | 0.0056 |          |   |       |         |           | 1690.0 | 0.0018 |          |         |              |
|         |           | 589.2  | 0.0026 |          |   |       |         |           | 1715.0 | 0.0071 |          |         |              |
|         |           | 593.3  | 0.0035 |          |   |       |         |           | 1735.0 | 0.0030 |          |         |              |
|         |           | 594.3  | 0.0017 |          |   |       |         |           | 1785.0 | 0.0041 |          |         |              |
|         |           | 607.5  | 0.0032 |          |   |       |         |           | 1797.0 | 0.0061 |          |         |              |
|         |           | 621.3  | 0.0080 |          |   |       |         |           | 1803.0 | 0.0019 |          |         |              |
|         |           | 622.0  | 0.0147 |          |   |       |         |           | 1806.0 | 0.0019 |          |         |              |
|         |           | 645.2  | 0.0231 |          |   |       |         |           | 1829.0 | 0.0025 |          |         |              |
|         |           | 668.3  | 0.0012 |          |   |       |         |           | 1835.0 | 0.0042 |          |         |              |
|         |           | 703.5  | 0.0045 |          |   |       |         |           | 1886.0 | 0.0185 |          |         |              |
|         |           | 736.2  | 0.0013 |          |   |       |         |           | 1887.0 | 0.0160 |          |         |              |
|         |           | 810.6  | 0.0020 |          |   |       |         |           | 1911.0 | 0.0013 |          |         |              |
|         |           | 843.9  | 0.0013 |          |   |       |         |           | 1924.0 | 0.0210 |          |         |              |
|         |           | 855.8  | 0.0011 |          |   |       |         |           | 1959.0 | 0.0017 |          |         |              |
|         |           | 890.0  | 0.0017 |          |   |       |         |           | 1970.0 | 0.0046 |          |         |              |
|         |           | 925.3  | 0.0031 |          |   |       |         |           | 2044.0 | 0.0383 |          |         |              |
|         |           | 938.7  | 0.0119 |          |   |       |         |           | 2114.0 | 0.0028 |          |         |              |
|         |           | 948.3  | 0.0237 |          |   |       |         |           | 2215.0 | 0.0018 |          |         |              |
|         |           | 1000.0 | 0.0022 |          |   |       |         |           | 2312.0 | 0.0018 |          |         |              |
|         |           | 1039.0 | 0.0033 |          |   |       |         |           | 1020.0 | 0.0233 |          |         |              |
|         |           | 1049.0 | 0.0090 |          |   |       |         |           | 511.0  | 0.0332 |          |         |              |
|         |           | 1104.0 | 0.0216 |          |   |       |         |           |        |        |          |         |              |
|         |           | 1120.0 | 0.0013 |          |   |       |         | 195Au     | 183d   | 30.9   | 0.0075   | 2.362-5 | 0.083 36 236 |
|         |           | 1151.0 | 0.0146 |          |   |       |         |           |        | 65.1   | 0.2895   |         |              |
|         |           | 1157.0 | 0.0045 |          |   |       |         |           |        | 66.8   | 0.4966   |         |              |
|         |           | 1175.0 | 0.0210 |          |   |       |         |           |        | 75.7   | 0.2162   |         |              |
|         |           | 1184.0 | 0.0066 |          |   |       |         |           |        | 98.9   | 0.1086   |         |              |
|         |           | 1219.0 | 0.0119 |          |   |       |         |           |        | 129.8  | 0.0081   |         |              |
|         |           | 1292.0 | 0.0011 |          |   |       |         |           |        | 206.1  | 0.0002   |         |              |
|         |           | 1294.0 | 0.0018 |          |   |       |         |           |        |        |          |         |              |
|         |           | 1302.0 | 0.0028 |          |   |       |         | 195mAu    | 30.6s  | 61.5   | 0.0016   | 4.132-5 | 0.485 6 174  |
|         |           | 1308.0 | 0.0016 |          |   |       |         |           |        | 67.0   | 0.0668   |         |              |
|         |           | 1340.0 | 0.0031 |          |   |       |         |           |        | 68.8   | 0.1142   |         |              |
|         |           | 1342.0 | 0.0125 |          |   |       |         |           |        | 78.0   | 0.0500   |         |              |
|         |           | 1422.0 | 0.0034 |          |   |       |         |           |        | 200.4  | 0.0156   |         |              |
|         |           | 1431.0 | 0.0015 |          |   |       |         |           |        | 261.7  | 0.6820   |         |              |
|         |           | 1442.0 | 0.0019 |          |   |       |         |           |        | 196.7  | 0.0007   |         |              |
|         |           | 1450.0 | 0.0034 |          |   |       |         |           |        |        |          |         |              |
|         |           | 1463.0 | 0.0077 |          |   |       |         | 196Au     | 6.2d   | 65.1   | 0.2199   | 9.922-5 | 0.919 3.261  |
|         |           | 1469.0 | 0.0676 |          |   |       |         |           |        | 66.8   | 0.3773   |         |              |
|         |           | 1487.0 | 0.0014 |          |   |       |         |           |        | 75.7   | 0.1642   |         |              |
|         |           | 1492.0 | 0.0019 |          |   |       |         |           |        | 333.0  | 0.2307   |         |              |
|         |           | 1500.0 | 0.0040 |          |   |       |         |           |        | 355.7  | 0.8772   |         |              |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T      | $\mu$                |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|--------|----------------------|
|         |           | 1091.0 | 0.0015 |          |       |        |         |           | 252.0  | 0.0038 |          |        |                      |
|         |           | 543.0  | 0.0017 |          |       |        |         |           | 289.4  | 0.0052 |          |        |                      |
|         |           | 68.9   | 0.0005 |          |       |        |         |           | 309.2  | 0.0026 |          |        |                      |
|         |           | 70.8   | 0.0009 |          |       |        |         |           | 367.9  | 0.8730 |          |        |                      |
|         |           | 80.3   | 0.0004 |          |       |        |         |           | 387.4  | 0.0016 |          |        |                      |
|         |           | 426.1  | 0.0666 |          |       |        |         |           | 476.8  | 0.0032 |          |        |                      |
| 198Au   | 2.7d      | 68.9   | 0.0081 | 7.881-5  | 1.293 | 2.317  |         |           | 521.4  | 0.0025 |          |        |                      |
|         |           | 70.8   | 0.0137 |          |       |        |         |           | 579.3  | 0.1379 |          |        |                      |
|         |           | 80.3   | 0.0060 |          |       |        |         |           | 591.7  | 0.0029 |          |        |                      |
|         |           | 411.8  | 0.9551 |          |       |        |         |           | 612.1  | 0.0024 |          |        |                      |
|         |           | 675.9  | 0.0106 |          |       |        |         |           | 628.8  | 0.0100 |          |        |                      |
|         |           | 1088.0 | 0.0023 |          |       |        |         |           | 661.4  | 0.0228 |          |        |                      |
| 199Au   | 3.1d      | 49.8   | 0.0033 | 1.866-5  | 0.186 | 16.121 |         |           | 688.9  | 0.0011 |          |        |                      |
|         |           | 68.9   | 0.0484 |          |       |        |         |           | 701.6  | 0.0129 |          |        |                      |
|         |           | 70.8   | 0.0824 |          |       |        |         |           | 711.7  | 0.0027 |          |        |                      |
|         |           | 80.3   | 0.0362 |          |       |        |         |           | 783.7  | 0.0057 |          |        |                      |
|         |           | 158.4  | 0.3680 |          |       |        |         |           | 787.1  | 0.0103 |          |        |                      |
|         |           | 208.2  | 0.0835 |          |       |        |         |           | 828.3  | 0.1083 |          |        |                      |
| 197Hg   | 2.7d      | 67.0   | 0.2069 | 1.874-5  | 0.100 | 29.898 |         |           | 886.2  | 0.0202 |          |        |                      |
|         |           | 68.8   | 0.3536 |          |       |        |         |           | 898.6  | 0.0062 |          |        |                      |
|         |           | 77.4   | 0.1850 |          |       |        |         |           | 1147.0 | 0.0012 |          |        |                      |
|         |           | 78.0   | 0.1547 |          |       |        |         |           | 1167.0 | 0.0010 |          |        |                      |
|         |           | 191.4  | 0.0050 |          |       |        |         |           | 1181.0 | 0.0011 |          |        |                      |
|         |           | 268.7  | 0.0004 |          |       |        |         |           | 1202.0 | 0.0011 |          |        |                      |
| 197mHg  | 23.8h     | 68.9   | 0.0977 | 2.059-5  | 0.206 | 14.537 |         |           | 1206.0 | 0.2994 |          |        |                      |
|         |           | 70.8   | 0.1661 |          |       |        |         |           | 1225.0 | 0.0336 |          |        |                      |
|         |           | 80.3   | 0.0731 |          |       |        |         |           | 1254.0 | 0.0093 |          |        |                      |
|         |           | 133.9  | 0.3404 |          |       |        |         |           | 1263.0 | 0.0079 |          |        |                      |
|         |           | 165.0  | 0.0027 |          |       |        |         |           | 1274.0 | 0.0332 |          |        |                      |
|         |           | 67.0   | 0.0191 |          |       |        |         |           | 1291.0 | 0.0060 |          |        |                      |
|         |           | 68.8   | 0.0326 |          |       |        |         |           | 1350.0 | 0.0015 |          |        |                      |
|         |           | 78.0   | 0.0143 |          |       |        |         |           | 1363.0 | 0.0341 |          |        |                      |
|         |           | 130.4  | 0.0023 |          |       |        |         |           | 1367.0 | 0.0087 |          |        |                      |
|         |           | 279.0  | 0.0498 |          |       |        |         |           | 1408.0 | 0.0145 |          |        |                      |
|         |           | 194.6  | 0.0010 |          |       |        |         |           | 1478.0 | 0.0015 |          |        |                      |
| 203Hg   | 46.6d     | 10.3   | 0.0722 | 6.841-5  | 0.499 | 6.007  |         |           | 1515.0 | 0.0402 |          |        |                      |
|         |           | 70.8   | 0.0475 |          |       |        |         |           | 1571.0 | 0.0027 |          |        |                      |
|         |           | 72.9   | 0.0804 |          |       |        |         |           | 1604.0 | 0.0117 |          |        |                      |
|         |           | 82.6   | 0.0356 |          |       |        |         |           | 1718.0 | 0.0033 |          |        |                      |
|         |           | 279.2  | 0.7730 |          |       |        |         |           | 1759.0 | 0.0018 |          |        |                      |
| 200Tl   | 1.1d      | 68.9   | 0.2362 | 2.249-4  | 3.380 | 0.886  |         |           | 1906.0 | 0.0011 |          |        |                      |
|         |           | 70.8   | 0.4018 |          |       |        |         |           | 771.5  | 0.0312 |          |        |                      |
|         |           | 80.3   | 0.1767 |          |       |        |         |           | 511.0  | 0.0075 |          |        |                      |
|         |           | 116.5  | 0.0011 |          |       |        |         |           | 201Tl  | 3.0d   | 30.6     | 0.0022 | 2.372-5 0.114 26.211 |
|         |           | 140.9  | 0.0017 |          |       |        |         |           |        |        | 32.2     | 0.0022 |                      |
|         |           | 151.9  | 0.0015 |          |       |        |         |           |        |        | 68.9     | 0.2736 |                      |
|         |           | 164.5  | 0.0021 |          |       |        |         |           |        |        | 70.8     | 0.4652 |                      |
|         |           |        |        |          |       |        |         |           |        |        | 80.3     | 0.2046 |                      |
|         |           |        |        |          |       |        |         |           |        |        | 135.3    | 0.0265 |                      |
|         |           |        |        |          |       |        |         |           |        |        | 165.9    | 0.0016 |                      |
|         |           |        |        |          |       |        |         |           |        |        | 167.4    | 0.1000 |                      |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T      | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T      | $\mu$   |         |       |        |
|---------|-----------|-------|--------|----------|--------|--------|---------|-----------|--------|--------|----------|--------|---------|---------|-------|--------|
| 202Tl   | 12.2d     | 68.9  | 0.2284 | 9.437-5  | 1.319  | 2.272  |         |           | 910.0  | 0.0297 |          |        |         |         |       |        |
|         |           | 70.8  | 0.3884 |          |        |        | 1060.0  | 0.1187    |        |        |          |        |         |         |       |        |
|         |           | 80.3  | 0.1708 |          |        |        | 1110.0  | 0.0693    |        |        |          |        |         |         |       |        |
|         |           | 439.6 | 0.9150 |          |        |        | 1210.0  | 0.1682    |        |        |          |        |         |         |       |        |
|         |           | 520.1 | 0.0091 |          |        |        | 1310.0  | 0.2078    |        |        |          |        |         |         |       |        |
|         |           | 959.7 | 0.0012 |          |        |        | 1410.0  | 0.0495    |        |        |          |        |         |         |       |        |
| 204Tl   | 3.8y      | 68.9  | 0.0042 | 3.014-7  | 0.100  | 30.056 |         |           | 1490.0 | 0.0198 |          |        |         |         |       |        |
|         |           | 70.8  | 0.0072 |          |        |        | 1540.0  | 0.0198    |        |        |          |        |         |         |       |        |
|         |           | 80.3  | 0.0032 |          |        |        | 1590.0  | 0.0198    |        |        |          |        |         |         |       |        |
|         |           |       |        |          |        |        | 1650.0  | 0.0198    |        |        |          |        |         |         |       |        |
| 207Tl   | 4.8m      | 897.8 | 0.0024 | 3.523-7  | 3.442  | 0.870  |         |           | 2010.0 | 0.0693 |          |        |         |         |       |        |
|         |           |       |        |          |        |        | 2090.0  | 0.0495    |        |        |          |        |         |         |       |        |
|         |           | 459.1 | 0.0000 |          |        |        | 2280.0  | 0.0297    |        |        |          |        |         |         |       |        |
|         |           |       |        |          |        |        | 2360.0  | 0.0792    |        |        |          |        |         |         |       |        |
| 208Tl   | 3.1m      | 10.6  | 0.0289 | 4.497-4  | 5.397  | 0.555  |         |           | 2430.0 | 0.0891 |          |        |         |         |       |        |
|         |           | 72.8  | 0.0203 |          |        |        |         |           |        |        |          |        |         |         |       |        |
|         |           | 75.0  | 0.0343 |          |        |        | 203Pb   | 2.2d      | 10.3   | 0.3920 | 1.828-4  | 0.343  | 8.724   |         |       |        |
|         |           | 84.9  | 0.0152 |          |        |        |         |           | 70.8   | 0.2663 |          |        |         |         |       |        |
|         |           | 211.4 | 0.0017 |          |        |        |         |           | 72.9   | 0.4513 |          |        |         |         |       |        |
|         |           | 233.4 | 0.0031 |          |        |        |         |           | 82.6   | 0.1995 |          |        |         |         |       |        |
|         |           | 252.6 | 0.0080 |          |        |        |         |           | 279.2  | 0.7680 |          |        |         |         |       |        |
|         |           | 277.4 | 0.0679 |          |        |        |         |           | 401.3  | 0.0330 |          |        |         |         |       |        |
|         |           | 510.8 | 0.2156 |          |        |        |         |           | 680.5  | 0.0067 |          |        |         |         |       |        |
|         |           | 583.1 | 0.8423 |          |        |        |         |           |        |        |          |        |         |         |       |        |
|         |           | 722.0 | 0.0020 |          |        |        | 204mPb  | 1.1h      | 10.6   | 0.0492 |          |        |         | 3.649-4 | 3.167 | 0.946  |
|         |           | 763.1 | 0.0164 |          |        |        |         |           | 72.8   | 0.0258 |          |        |         |         |       |        |
|         |           | 860.4 | 0.1245 |          |        |        |         |           | 75.0   | 0.0436 |          |        |         |         |       |        |
|         |           | 927.6 | 0.0012 |          |        |        |         |           | 84.9   | 0.0194 |          |        |         |         |       |        |
| 982.7   | 0.0020    |       |        | 289.3    | 0.0017 |        |         |           |        |        |          |        |         |         |       |        |
| 1094.0  | 0.0037    |       |        | 374.7    | 0.9411 |        |         |           |        |        |          |        |         |         |       |        |
| 2615.0  | 0.9980    |       |        | 622.2    | 0.0022 |        |         |           |        |        |          |        |         |         |       |        |
| 840.4   | 0.0036    |       |        | 899.2    | 0.9916 |        |         |           |        |        |          |        |         |         |       |        |
| 209Tl   | 2.2m      | 10.6  | 0.0872 | 3.482-4  | 4.248  | 0.705  |         |           | 911.7  | 0.9113 |          |        |         |         |       |        |
|         |           | 72.8  | 0.0588 |          |        |        |         |           | 779.7  | 0.0006 |          |        |         |         |       |        |
|         |           | 75.0  | 0.0993 |          |        |        | 205Pb   | 1.5+7y    | 10.3   | 0.2268 | 6.789-5  | 0.002  | 1345.   |         |       |        |
|         |           | 84.9  | 0.0441 |          |        |        |         |           |        |        |          |        |         |         |       |        |
|         |           | 117.2 | 0.7700 |          |        |        | 210Pb   | 22.3y     | 10.8   | 0.2432 |          |        |         | 6.801-5 | 0.003 | 1098.  |
|         |           | 465.1 | 0.9658 |          |        |        |         |           | 46.5   | 0.0405 |          |        |         |         |       |        |
| 1567.0  | 0.9969    |       |        |          |        |        |         |           |        |        |          |        |         |         |       |        |
| 210Tl   | 1.3m      | 10.6  | 0.1260 | 4.575-4  | 4.057  | 0.738  |         |           | 211Pb  | 36.1m  | 10.8     | 0.0028 | 9.822-6 | 2.474   | 1.211 |        |
|         |           | 72.8  | 0.0255 |          |        |        |         |           | 74.8   | 0.0012 |          |        |         |         |       |        |
|         |           | 75.0  | 0.0431 |          |        |        |         |           | 77.1   | 0.0021 |          |        |         |         |       |        |
|         |           | 84.9  | 0.0191 |          |        |        |         |           | 87.3   | 0.0009 |          |        |         |         |       |        |
|         |           | 97.0  | 0.0396 |          |        |        |         |           | 404.8  | 0.0294 |          |        |         |         |       |        |
|         |           | 298.0 | 0.7917 |          |        |        |         |           | 427.1  | 0.0132 |          |        |         |         |       |        |
|         |           | 356.0 | 0.0396 |          |        |        |         |           | 704.6  | 0.0037 |          |        |         |         |       |        |
|         |           | 382.0 | 0.0297 |          |        |        |         |           | 766.5  | 0.0054 |          |        |         |         |       |        |
|         |           | 480.0 | 0.0198 |          |        |        |         |           | 832.0  | 0.0286 |          |        |         |         |       |        |
|         |           | 670.0 | 0.0198 |          |        |        |         |           | 517.6  | 0.0047 |          |        |         |         |       |        |
|         |           | 799.7 | 0.9896 |          |        |        |         |           |        |        |          |        |         |         |       |        |
|         |           | 860.0 | 0.0693 |          |        |        | 212Pb   | 10.6h     | 10.8   | 0.1549 | 7.389-5  | 0.270  |         |         |       | 11.112 |
|         |           |       |        |          |        |        |         |           | 74.8   | 0.1069 |          |        |         |         |       |        |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|-------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 77.1  | 0.1800 |          |       |       |         |           | 1019.0 | 0.0759 |          |       |       |
|         |           | 87.3  | 0.0803 |          |       |       |         |           | 1098.0 | 0.1350 |          |       |       |
|         |           | 115.2 | 0.0060 |          |       |       |         |           | 1142.0 | 0.0011 |          |       |       |
|         |           | 238.6 | 0.4465 |          |       |       |         |           | 1195.0 | 0.0028 |          |       |       |
|         |           | 300.1 | 0.0341 |          |       |       |         |           | 1203.0 | 0.0010 |          |       |       |
|         |           | 176.7 | 0.0005 |          |       |       |         |           | 1332.0 | 0.0028 |          |       |       |
|         |           |       |        |          |       |       |         |           | 1405.0 | 0.0143 |          |       |       |
| 214Pb   | 26.8m     | 10.8  | 0.1350 | 8.741-5  | 0.744 | 4.029 |         |           | 1496.0 | 0.0018 |          |       |       |
|         |           | 53.2  | 0.0110 |          |       |       |         |           | 1560.0 | 0.0038 |          |       |       |
|         |           | 74.8  | 0.0621 |          |       |       |         |           | 1565.0 | 0.0030 |          |       |       |
|         |           | 77.1  | 0.1046 |          |       |       |         |           | 1595.0 | 0.0501 |          |       |       |
|         |           | 87.3  | 0.0467 |          |       |       |         |           | 1719.0 | 0.3184 |          |       |       |
|         |           | 242.0 | 0.0749 |          |       |       |         |           | 1844.0 | 0.0057 |          |       |       |
|         |           | 258.8 | 0.0055 |          |       |       |         |           | 1879.0 | 0.0201 |          |       |       |
|         |           | 274.5 | 0.0033 |          |       |       |         |           | 1904.0 | 0.0035 |          |       |       |
|         |           | 295.2 | 0.1925 |          |       |       |         |           | 2600.0 | 0.0013 |          |       |       |
|         |           | 351.9 | 0.3721 |          |       |       |         |           | 928.7  | 0.0124 |          |       |       |
|         |           | 462.1 | 0.0017 |          |       |       |         |           |        |        |          |       |       |
|         |           | 480.4 | 0.0034 |          |       |       | 207Bi   | 33.4y     | 10.6   | 0.3590 | 3.590-4  | 2.952 | 1.015 |
|         |           | 487.1 | 0.0044 |          |       |       |         |           | 72.8   | 0.2178 |          |       |       |
|         |           | 533.7 | 0.0019 |          |       |       |         |           | 75.0   | 0.3679 |          |       |       |
|         |           | 580.2 | 0.0037 |          |       |       |         |           | 84.9   | 0.1634 |          |       |       |
|         |           | 785.9 | 0.0110 |          |       |       |         |           | 569.7  | 0.9772 |          |       |       |
|         |           | 839.0 | 0.0059 |          |       |       |         |           | 897.8  | 0.0015 |          |       |       |
|         |           | 280.7 | 0.0033 |          |       |       |         |           | 1064.0 | 0.7485 |          |       |       |
|         |           |       |        |          |       |       |         |           | 1442.0 | 0.0015 |          |       |       |
| 206Bi   | 6.2d      | 10.6  | 0.4894 | 6.797-4  | 3.167 | 0.946 |         |           | 1770.0 | 0.0684 |          |       |       |
|         |           | 72.8  | 0.3205 |          |       |       |         |           | 328.2  | 0.0000 |          |       |       |
|         |           | 75.0  | 0.5415 |          |       |       |         |           |        |        |          |       |       |
|         |           | 84.9  | 0.2405 |          |       |       | 208Bi   | 3.7+5y    | 10.6   | 0.3101 | 4.014-4  | 5.613 | 0.534 |
|         |           | 184.0 | 0.1582 |          |       |       |         |           | 72.8   | 0.1232 |          |       |       |
|         |           | 234.3 | 0.0024 |          |       |       |         |           | 75.0   | 0.2080 |          |       |       |
|         |           | 262.7 | 0.0302 |          |       |       |         |           | 84.9   | 0.0924 |          |       |       |
|         |           | 313.7 | 0.0036 |          |       |       |         |           | 2615.0 | 0.9980 |          |       |       |
|         |           | 343.5 | 0.2344 |          |       |       |         |           |        |        |          |       |       |
|         |           | 386.2 | 0.0052 |          |       |       | 211Bi   | 2.1m      | 10.3   | 0.0105 | 1.274-5  | 0.828 | 3.617 |
|         |           | 398.0 | 0.1074 |          |       |       |         |           | 70.8   | 0.0075 |          |       |       |
|         |           | 452.8 | 0.0016 |          |       |       |         |           | 72.9   | 0.0127 |          |       |       |
|         |           | 497.1 | 0.1531 |          |       |       |         |           | 82.6   | 0.0056 |          |       |       |
|         |           | 516.2 | 0.4074 |          |       |       |         |           | 351.1  | 0.1295 |          |       |       |
|         |           | 537.5 | 0.3046 |          |       |       |         |           |        |        |          |       |       |
|         |           | 576.4 | 0.0011 |          |       |       | 212Bi   | 1.0h      | 10.3   | 0.0769 | 5.252-5  | 2.872 | 1.043 |
|         |           | 582.0 | 0.0048 |          |       |       |         |           | 39.9   | 0.0102 |          |       |       |
|         |           | 620.5 | 0.0576 |          |       |       |         |           | 70.8   | 0.0003 |          |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|---|-------|
|         |           | 89.8   | 0.0003 |          |       |       |         |           | 1238.0 | 0.0594 |          |   |       |
|         |           | 727.2  | 0.1183 |          |       |       |         |           | 1281.0 | 0.0148 |          |   |       |
|         |           | 785.5  | 0.0197 |          |       |       |         |           | 1304.0 | 0.0012 |          |   |       |
|         |           | 893.4  | 0.0065 |          |       |       |         |           | 1378.0 | 0.0411 |          |   |       |
|         |           | 952.1  | 0.0031 |          |       |       |         |           | 1385.0 | 0.0078 |          |   |       |
|         |           | 1079.0 | 0.0095 |          |       |       |         |           | 1401.0 | 0.0139 |          |   |       |
|         |           | 1513.0 | 0.0056 |          |       |       |         |           | 1408.0 | 0.0249 |          |   |       |
|         |           | 1621.0 | 0.0275 |          |       |       |         |           | 1509.0 | 0.0222 |          |   |       |
|         |           | 1679.0 | 0.0012 |          |       |       |         |           | 1538.0 | 0.0041 |          |   |       |
|         |           | 1806.0 | 0.0020 |          |       |       |         |           | 1543.0 | 0.0036 |          |   |       |
|         |           | 1074.0 | 0.0003 |          |       |       |         |           | 1583.0 | 0.0072 |          |   |       |
|         |           |        |        |          |       |       |         |           | 1595.0 | 0.0027 |          |   |       |
| 213Bi   | 45.6m     | 323.8  | 0.0013 | 3.140-5  | 1.430 | 2.094 |         |           | 1599.0 | 0.0034 |          |   |       |
|         |           | 11.1   | 0.0183 |          |       |       |         |           | 1661.0 | 0.0115 |          |   |       |
|         |           | 76.9   | 0.0125 |          |       |       |         |           | 1684.0 | 0.0024 |          |   |       |
|         |           | 79.3   | 0.0210 |          |       |       |         |           | 1730.0 | 0.0297 |          |   |       |
|         |           | 89.8   | 0.0094 |          |       |       |         |           | 1764.0 | 0.1584 |          |   |       |
|         |           | 292.9  | 0.0074 |          |       |       |         |           | 1838.0 | 0.0038 |          |   |       |
|         |           | 440.4  | 0.2797 |          |       |       |         |           | 1847.0 | 0.0209 |          |   |       |
|         |           | 659.8  | 0.0015 |          |       |       |         |           | 1873.0 | 0.0023 |          |   |       |
|         |           | 807.4  | 0.0044 |          |       |       |         |           | 1896.0 | 0.0018 |          |   |       |
|         |           | 1100.0 | 0.0048 |          |       |       |         |           | 2119.0 | 0.0117 |          |   |       |
|         |           |        |        |          |       |       |         |           | 2204.0 | 0.0498 |          |   |       |
|         |           |        |        |          |       |       |         |           | 2293.0 | 0.0033 |          |   |       |
|         |           |        |        |          |       |       |         |           | 2448.0 | 0.0156 |          |   |       |
|         |           |        |        |          |       |       |         |           | 1158.0 | 0.0351 |          |   |       |
| 214Bi   | 19.9m     | 11.1   | 0.0052 | 2.251-4  | 4.451 | 0.673 |         |           |        |        |          |   |       |
|         |           | 76.9   | 0.0036 |          |       |       |         |           |        |        |          |   |       |
|         |           | 79.3   | 0.0060 |          |       |       |         |           |        |        |          |   |       |
|         |           | 89.8   | 0.0027 |          |       |       |         |           |        |        |          |   |       |
|         |           | 273.7  | 0.0018 |          |       |       |         |           |        |        |          |   |       |
|         |           | 387.0  | 0.0037 |          |       |       |         |           |        |        |          |   |       |
|         |           | 389.1  | 0.0041 |          |       |       |         |           |        |        |          |   |       |
|         |           | 405.7  | 0.0017 |          |       |       |         |           |        |        |          |   |       |
|         |           | 426.5  | 0.0011 |          |       |       |         |           |        |        |          |   |       |
|         |           | 454.8  | 0.0032 |          |       |       |         |           |        |        |          |   |       |
|         |           | 469.7  | 0.0013 |          |       |       |         |           |        |        |          |   |       |
|         |           | 474.4  | 0.0012 |          |       |       |         |           |        |        |          |   |       |
|         |           | 609.3  | 0.4628 |          |       |       |         |           |        |        |          |   |       |
|         |           | 665.4  | 0.0157 |          |       |       |         |           |        |        |          |   |       |
|         |           | 703.1  | 0.0047 |          |       |       |         |           |        |        |          |   |       |
|         |           | 719.9  | 0.0040 |          |       |       |         |           |        |        |          |   |       |
|         |           | 752.8  | 0.0013 |          |       |       |         |           |        |        |          |   |       |
|         |           | 768.4  | 0.0504 |          |       |       |         |           |        |        |          |   |       |
|         |           | 786.1  | 0.0032 |          |       |       |         |           |        |        |          |   |       |
|         |           | 806.2  | 0.0123 |          |       |       |         |           |        |        |          |   |       |
|         |           | 821.2  | 0.0015 |          |       |       |         |           |        |        |          |   |       |
|         |           | 904.3  | 0.0011 |          |       |       |         |           |        |        |          |   |       |
|         |           | 934.1  | 0.0321 |          |       |       |         |           |        |        |          |   |       |
|         |           | 964.1  | 0.0038 |          |       |       |         |           |        |        |          |   |       |
|         |           | 1052.0 | 0.0032 |          |       |       |         |           |        |        |          |   |       |
|         |           | 1070.0 | 0.0029 |          |       |       |         |           |        |        |          |   |       |
|         |           | 1120.0 | 0.1515 |          |       |       |         |           |        |        |          |   |       |
|         |           | 1134.0 | 0.0026 |          |       |       |         |           |        |        |          |   |       |
|         |           | 1155.0 | 0.0170 |          |       |       |         |           |        |        |          |   |       |
|         |           | 1208.0 | 0.0046 |          |       |       |         |           |        |        |          |   |       |
| 209Po   | 102.0y    | 10.6   | 0.0004 | 9.809-7  | 2.326 | 1.288 |         |           |        |        |          |   |       |
|         |           | 72.8   | 0.0004 |          |       |       |         |           |        |        |          |   |       |
|         |           | 75.0   | 0.0006 |          |       |       |         |           |        |        |          |   |       |
|         |           | 84.9   | 0.0003 |          |       |       |         |           |        |        |          |   |       |
|         |           | 260.5  | 0.0026 |          |       |       |         |           |        |        |          |   |       |
|         |           | 266.1  | 0.0009 |          |       |       |         |           |        |        |          |   |       |
|         |           | 10.8   | 0.0009 |          |       |       |         |           |        |        |          |   |       |
|         |           | 74.8   | 0.0006 |          |       |       |         |           |        |        |          |   |       |
|         |           | 77.1   | 0.0010 |          |       |       |         |           |        |        |          |   |       |
|         |           | 87.3   | 0.0004 |          |       |       |         |           |        |        |          |   |       |
|         |           | 896.4  | 0.0025 |          |       |       |         |           |        |        |          |   |       |
| 210Po   | 138.4d    | 803.1  | 0.0000 | 1.422-9  | 3.097 | 0.967 |         |           |        |        |          |   |       |
| 211Po   | 0.5s      | 569.7  | 0.0054 | 1.321-6  | 2.953 | 1.015 |         |           |        |        |          |   |       |
|         |           | 897.8  | 0.0052 |          |       |       |         |           |        |        |          |   |       |
|         |           | 328.2  | 0.0000 |          |       |       |         |           |        |        |          |   |       |
| 213Po   | 4.2-6s    | 779.0  | 0.0000 | 5.136-9  | 2.990 | 1.002 |         |           |        |        |          |   |       |
| 214Po   | 1.6-4s    | 797.3  | 0.0001 | 1.396-8  | 3.073 | 0.975 |         |           |        |        |          |   |       |
| 215Po   | 1.8-3s    | 438.7  | 0.0003 | 2.861-8  | 1.392 | 1.152 |         |           |        |        |          |   |       |
| 216Po   | 0.1s      | 804.9  | 0.0000 | 2.421-9  | 3.104 | 0.965 |         |           |        |        |          |   |       |



| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T       | $\mu$  |         |       |
|---------|-----------|-------|--------|----------|-------|--------|---------|-----------|-------|--------|----------|---------|--------|---------|-------|
| 211At   | 7.2h      | 685.2 | 0.0000 | 6.120-5  | 0.049 | 61.112 | 222Ra   | 38.0s     | 11.7  | 0.0011 | 2.115-6  | 0.766   | 3.912  |         |       |
|         |           | 11.1  | 0.1973 |          |       |        |         |           |       |        |          |         |        |         |       |
|         |           | 76.9  | 0.1270 |          |       |        |         |           |       |        |          |         |        |         |       |
|         |           | 79.3  | 0.2128 |          |       |        |         |           |       |        |          |         |        |         |       |
|         |           | 89.8  | 0.0955 |          |       |        |         |           |       |        |          |         |        |         |       |
|         |           | 687.0 | 0.0025 |          |       |        |         |           |       |        |          |         |        |         |       |
| 217At   | 0.032s    | 594.5 | 0.0004 | 4.292-8  | 2.174 | 1.378  | 223Ra   | 11.4d     | 11.7  | 0.2474 | 8.789-5  | 0.278   | 10.794 |         |       |
|         |           |       |        |          |       |        |         |           |       |        |          |         |        |         |       |
| 218Rn   | 0.035s    | 609.3 | 0.0012 | 1.356-7  | 2.241 | 1.337  |         |           | 80.2  | 0.0020 |          |         |        |         |       |
| 219Rn   | 4.0s      | 11.1  | 0.0104 | 1.419-5  | 0.843 | 3.555  |         |           | 81.1  | 0.1487 |          |         |        |         |       |
|         |           | 76.9  | 0.0053 |          |       |        |         |           | 83.8  | 0.2470 |          |         |        |         |       |
|         |           | 79.3  | 0.0088 |          |       |        |         |           | 94.9  | 0.1120 |          |         |        |         |       |
|         |           | 89.8  | 0.0039 |          |       |        |         |           | 98.2  | 0.0045 |          |         |        |         |       |
|         |           |       |        |          |       |        |         |           | 122.3 | 0.0119 |          |         |        |         |       |
|         |           | 130.6 | 0.0012 |          |       |        |         |           | 144.2 | 0.0324 |          |         |        |         |       |
|         |           | 271.2 | 0.1060 |          |       |        |         |           | 154.2 | 0.0558 |          |         |        |         |       |
|         |           | 401.8 | 0.0650 |          |       |        |         |           | 158.6 | 0.0068 |          |         |        |         |       |
| 220Rn   | 55.6s     | 549.7 | 0.0010 | 9.607-8  | 1.953 | 1.534  |         |           | 179.5 | 0.0014 |          |         |        |         |       |
|         |           |       |        |          |       |        |         |           | 269.5 | 0.1360 |          |         |        |         |       |
|         |           |       |        |          |       |        |         |           | 288.2 | 0.0015 |          |         |        |         |       |
|         |           |       |        |          |       |        |         |           | 323.9 | 0.0388 |          |         |        |         |       |
|         |           |       |        |          |       |        |         |           | 328.4 | 0.0020 |          |         |        |         |       |
|         |           |       |        |          |       |        |         |           | 338.3 | 0.0273 |          |         |        |         |       |
|         |           |       |        |          |       |        |         |           | 342.9 | 0.0022 |          |         |        |         |       |
|         |           |       |        |          |       |        |         |           | 349.8 | 0.0034 |          |         |        |         |       |
| 221Fr   | 4.8m      | 11.4  | 0.0227 | 1.193-5  | 0.260 | 11.522 |         |           | 371.7 | 0.0047 |          |         |        |         |       |
|         |           | 78.9  | 0.0077 |          |       |        |         |           | 445.0 | 0.0118 |          |         |        |         |       |
|         |           | 81.5  | 0.0129 |          |       |        |         |           | 292.1 | 0.0135 |          |         |        |         |       |
|         |           | 92.3  | 0.0058 |          |       |        |         |           |       |        |          |         |        |         |       |
|         |           | 99.5  | 0.0016 |          |       |        |         |           |       |        |          |         |        |         |       |
|         |           | 217.6 | 0.1250 |          |       |        |         |           |       |        |          |         |        |         |       |
|         |           | 412.0 | 0.0010 |          |       |        |         |           |       |        |          |         |        |         |       |
|         |           | 211.0 | 0.0033 |          |       |        |         |           |       |        |          |         |        |         |       |
| 223Fr   | 21.8m     | 12.3  | 0.3368 | 8.930-5  | 0.044 | 67.594 | 224Ra   | 3.6d      | 11.7  | 0.0040 | 2.967-6  | 0.369   | 8.124  |         |       |
|         |           | 20.3  | 0.0076 |          |       |        |         |           |       | 81.1   | 0.0013   |         |        |         |       |
|         |           | 49.9  | 0.0076 |          |       |        |         |           |       | 83.8   | 0.0021   |         |        |         |       |
|         |           | 50.1  | 0.3170 |          |       |        |         |           |       | 94.9   | 0.0009   |         |        |         |       |
|         |           | 68.7  | 0.0038 |          |       |        |         |           | 241.0 | 0.0395 |          |         |        |         |       |
|         |           | 79.8  | 0.0761 |          |       |        |         |           | 465.0 | 0.0002 |          |         |        |         |       |
|         |           | 85.4  | 0.0147 |          |       |        |         | 225Ra     | 14.8d | 12.7   | 0.1576   | 4.164-5 | 0.010  | 304.4   |       |
|         |           | 88.5  | 0.0242 |          |       |        |         |           |       |        | 40.0     | 0.3100  |        |         |       |
|         |           | 100.0 | 0.0111 |          |       |        |         |           |       | 226Ra  | 1600y    | 11.7    | 0.0080 | 3.274-6 | 0.160 |
|         |           | 100.4 | 0.0095 |          |       |        |         |           | 81.1  |        |          | 0.0018  |        |         |       |
| 134.6   | 0.0051    |       |        |          |       |        | 83.8    | 0.0030    |       |        |          |         |        |         |       |
| 173.4   | 0.0013    |       |        |          |       |        | 94.9    | 0.0014    |       |        |          |         |        |         |       |
|         |           | 184.8 | 0.0029 |          |       |        |         |           | 186.2 | 0.0328 |          |         |        |         |       |
|         |           | 205.0 | 0.0108 |          |       |        |         |           | 309.7 | 0.0001 |          |         |        |         |       |
|         |           | 234.9 | 0.0282 |          |       |        |         | 225Ac     | 10.0d | 12.0   | 0.2129   | 5.172-5 | 0.006  | 472.6   |       |
|         |           | 289.5 | 0.0023 |          |       |        |         |           |       |        | 62.9     | 0.0055  |        |         |       |
|         |           | 319.4 | 0.0051 |          |       |        |         |           |       |        | 73.8     | 0.0032  |        |         |       |
|         |           | 369.4 | 0.0010 |          |       |        |         |           |       |        | 82.9     | 0.0015  |        |         |       |
|         |           | 775.3 | 0.0039 |          |       |        |         |           | 83.2  | 0.0102 |          |         |        |         |       |
|         |           | 482.0 | 0.0123 |          |       |        |         |           | 86.1  | 0.0168 |          |         |        |         |       |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|-------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 87.4  | 0.0029 |          |       |       |         |           | 570.7  | 0.0018 |          |       |       |
|         |           | 94.9  | 0.0016 |          |       |       |         |           | 572.1  | 0.0016 |          |       |       |
|         |           | 97.5  | 0.0077 |          |       |       |         |           | 583.2  | 0.0014 |          |       |       |
|         |           | 99.6  | 0.0065 |          |       |       |         |           | 623.8  | 0.0011 |          |       |       |
|         |           | 99.8  | 0.0170 |          |       |       |         |           | 701.5  | 0.0019 |          |       |       |
|         |           | 108.4 | 0.0028 |          |       |       |         |           | 707.1  | 0.0015 |          |       |       |
|         |           | 111.5 | 0.0032 |          |       |       |         |           | 727.0  | 0.0078 |          |       |       |
|         |           | 123.8 | 0.0019 |          |       |       |         |           | 755.2  | 0.0105 |          |       |       |
|         |           | 138.2 | 0.0020 |          |       |       |         |           | 772.2  | 0.0155 |          |       |       |
|         |           | 145.0 | 0.0013 |          |       |       |         |           | 782.0  | 0.0053 |          |       |       |
|         |           | 150.1 | 0.0071 |          |       |       |         |           | 794.7  | 0.0463 |          |       |       |
|         |           | 154.0 | 0.0019 |          |       |       |         |           | 830.5  | 0.0059 |          |       |       |
|         |           | 157.2 | 0.0031 |          |       |       |         |           | 835.5  | 0.0175 |          |       |       |
|         |           | 188.0 | 0.0046 |          |       |       |         |           | 840.0  | 0.0094 |          |       |       |
|         |           | 195.7 | 0.0014 |          |       |       |         |           | 904.5  | 0.0083 |          |       |       |
|         |           | 253.5 | 0.0010 |          |       |       |         |           | 911.1  | 0.2770 |          |       |       |
|         |           | 452.4 | 0.0011 |          |       |       |         |           | 944.1  | 0.0010 |          |       |       |
|         |           | 165.2 | 0.0093 |          |       |       |         |           | 948.0  | 0.0012 |          |       |       |
| 227Ac   | 21.8y     | 115.4 | 0.0010 | 2.364-6  | 0.004 | 690.1 |         |           | 958.5  | 0.0030 |          |       |       |
|         |           | 13.0  | 0.0115 |          |       |       |         |           | 964.6  | 0.0521 |          |       |       |
|         |           | 17.4  | 0.0004 |          |       |       |         |           | 969.1  | 0.1662 |          |       |       |
|         |           |       |        |          |       |       |         |           | 987.8  | 0.0018 |          |       |       |
| 228Ac   | 6.1h      | 13.0  | 0.3911 | 2.278-4  | 3.039 | 0.986 |         |           | 1033.0 | 0.0022 |          |       |       |
|         |           | 57.8  | 0.0050 |          |       |       |         |           | 1065.0 | 0.0014 |          |       |       |
|         |           | 90.3  | 0.0213 |          |       |       |         |           | 1096.0 | 0.0013 |          |       |       |
|         |           | 93.3  | 0.0349 |          |       |       |         |           | 1110.0 | 0.0033 |          |       |       |
|         |           | 99.4  | 0.0130 |          |       |       |         |           | 1154.0 | 0.0015 |          |       |       |
|         |           | 105.0 | 0.0161 |          |       |       |         |           | 1246.0 | 0.0054 |          |       |       |
|         |           | 129.1 | 0.0277 |          |       |       |         |           | 1288.0 | 0.0011 |          |       |       |
|         |           | 146.1 | 0.0021 |          |       |       |         |           | 1459.0 | 0.0100 |          |       |       |
|         |           | 154.2 | 0.0094 |          |       |       |         |           | 1496.0 | 0.0100 |          |       |       |
|         |           | 191.2 | 0.0012 |          |       |       |         |           | 1502.0 | 0.0055 |          |       |       |
|         |           | 199.7 | 0.0033 |          |       |       |         |           | 1557.0 | 0.0019 |          |       |       |
|         |           | 204.4 | 0.0016 |          |       |       |         |           | 1580.0 | 0.0069 |          |       |       |
|         |           | 209.3 | 0.0443 |          |       |       |         |           | 1588.0 | 0.0355 |          |       |       |
|         |           | 210.0 | 0.0022 |          |       |       |         |           | 1625.0 | 0.0030 |          |       |       |
|         |           | 270.2 | 0.0360 |          |       |       |         |           | 1630.0 | 0.0186 |          |       |       |
|         |           | 279.0 | 0.0022 |          |       |       |         |           | 1638.0 | 0.0053 |          |       |       |
|         |           | 321.7 | 0.0024 |          |       |       |         |           | 1666.0 | 0.0020 |          |       |       |
|         |           | 327.6 | 0.0321 |          |       |       |         |           | 1887.0 | 0.0011 |          |       |       |
|         |           | 332.4 | 0.0044 |          |       |       |         |           | 947.8  | 0.0412 |          |       |       |
|         |           | 338.3 | 0.1136 |          |       |       | 226Th   | 30.9m     | 12.3   | 0.0755 | 1.818-5  | 0.022 | 13    |
|         |           | 341.1 | 0.0042 |          |       |       |         |           | 85.4   | 0.0028 |          |       |       |
|         |           | 409.5 | 0.0213 |          |       |       |         |           | 88.5   | 0.0046 |          |       |       |
|         |           | 440.3 | 0.0014 |          |       |       |         |           | 100.0  | 0.0021 |          |       |       |
|         |           | 463.0 | 0.0443 |          |       |       |         |           | 111.1  | 0.0329 |          |       |       |
|         |           | 478.2 | 0.0023 |          |       |       |         |           | 131.0  | 0.0028 |          |       |       |
|         |           | 503.6 | 0.0021 |          |       |       |         |           | 190.3  | 0.0011 |          |       |       |
|         |           | 509.6 | 0.0047 |          |       |       |         |           | 206.2  | 0.0019 |          |       |       |
|         |           | 523.0 | 0.0012 |          |       |       |         |           | 242.1  | 0.0087 |          |       |       |
|         |           | 546.3 | 0.0021 |          |       |       |         |           | 556.1  | 0.0000 |          |       |       |
|         |           | 562.3 | 0.0094 |          |       |       |         |           |        |        |          |       |       |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|-------|--------|----------|-------|--------|---------|-----------|-------|--------|----------|-------|-------|
| 227Th   | 18.7d     | 12.3  | 0.4180 | 1.145-4  | 0.203 | 14.730 |         |           | 68.2  | 0.0010 |          |       |       |
|         |           | 20.3  | 0.0020 |          |       |        |         |           | 68.9  | 0.0011 |          |       |       |
|         |           | 29.9  | 0.0010 |          |       |        |         |           | 75.2  | 0.0052 |          |       |       |
|         |           | 43.7  | 0.0023 |          |       |        |         |           | 85.4  | 0.1647 |          |       |       |
|         |           | 49.9  | 0.0020 |          |       |        |         |           | 86.3  | 0.0038 |          |       |       |
|         |           | 50.1  | 0.0840 |          |       |        |         |           | 86.4  | 0.0306 |          |       |       |
|         |           | 62.2  | 0.0024 |          |       |        |         |           | 88.5  | 0.2714 |          |       |       |
|         |           | 79.8  | 0.0200 |          |       |        |         |           | 100.0 | 0.1243 |          |       |       |
|         |           | 85.4  | 0.0141 |          |       |        |         |           | 107.2 | 0.0084 |          |       |       |
|         |           | 88.5  | 0.0232 |          |       |        |         |           | 124.5 | 0.0122 |          |       |       |
|         |           | 94.0  | 0.0140 |          |       |        |         |           | 124.7 | 0.0061 |          |       |       |
|         |           | 100.0 | 0.0106 |          |       |        |         |           | 132.0 | 0.0033 |          |       |       |
|         |           | 113.1 | 0.0017 |          |       |        |         |           | 137.0 | 0.0163 |          |       |       |
|         |           | 113.1 | 0.0054 |          |       |        |         |           | 142.9 | 0.0043 |          |       |       |
|         |           | 117.2 | 0.0018 |          |       |        |         |           | 148.3 | 0.0139 |          |       |       |
|         |           | 141.2 | 0.0014 |          |       |        |         |           | 154.4 | 0.0066 |          |       |       |
|         |           | 204.3 | 0.0023 |          |       |        |         |           | 156.5 | 0.0112 |          |       |       |
|         |           | 205.0 | 0.0017 |          |       |        |         |           | 172.9 | 0.0022 |          |       |       |
|         |           | 206.0 | 0.0026 |          |       |        |         |           | 179.8 | 0.0051 |          |       |       |
|         |           | 210.6 | 0.0126 |          |       |        |         |           | 184.0 | 0.0023 |          |       |       |
|         |           | 234.9 | 0.0046 |          |       |        |         |           | 193.6 | 0.0459 |          |       |       |
|         |           | 236.0 | 0.1150 |          |       |        |         |           | 211.0 | 0.0326 |          |       |       |
|         |           | 250.1 | 0.0049 |          |       |        |         |           | 218.1 | 0.0014 |          |       |       |
|         |           | 252.5 | 0.0011 |          |       |        |         |           | 130.8 | 0.0007 |          |       |       |
|         |           | 254.7 | 0.0091 |          |       |        |         |           |       |        |          |       |       |
|         |           | 256.2 | 0.0630 |          |       |        | 230Th   | 7.7+4y    | 12.3  | 0.0843 | 1.861-5  | 0.004 | 815.2 |
|         |           | 262.9 | 0.0010 |          |       |        |         |           | 67.7  | 0.0037 |          |       |       |
|         |           | 273.0 | 0.0049 |          |       |        |         |           | 168.1 | 0.0007 |          |       |       |
|         |           | 281.3 | 0.0017 |          |       |        |         |           |       |        |          |       |       |
|         |           | 286.1 | 0.0160 |          |       |        | 231Th   | 1.1d      | 13.3  | 0.7077 | 1.473-4  | 0.004 | 679.1 |
|         |           | 296.6 | 0.0042 |          |       |        |         |           | 17.2  | 0.0019 |          |       |       |
|         |           | 299.8 | 0.0184 |          |       |        |         |           | 25.6  | 0.1465 |          |       |       |
|         |           | 300.3 | 0.0028 |          |       |        |         |           | 58.6  | 0.0048 |          |       |       |
|         |           | 304.4 | 0.0135 |          |       |        |         |           | 72.8  | 0.0025 |          |       |       |
|         |           | 312.6 | 0.0043 |          |       |        |         |           | 81.2  | 0.0088 |          |       |       |
|         |           | 314.8 | 0.0042 |          |       |        |         |           | 82.1  | 0.0040 |          |       |       |
|         |           | 329.7 | 0.0290 |          |       |        |         |           | 84.2  | 0.0644 |          |       |       |
|         |           | 334.2 | 0.0115 |          |       |        |         |           | 90.0  | 0.0093 |          |       |       |
|         |           | 342.4 | 0.0034 |          |       |        |         |           | 92.3  | 0.0035 |          |       |       |
|         |           | 350.5 | 0.0012 |          |       |        |         |           | 95.9  | 0.0057 |          |       |       |
|         |           | 185.0 | 0.0173 |          |       |        |         |           | 99.3  | 0.0012 |          |       |       |
|         |           |       |        |          |       |        |         |           | 102.3 | 0.0041 |          |       |       |
| 228Th   | 1.9y      | 12.3  | 0.0956 | 2.142-5  | 0.004 | 731.9  |         |           | 108.0 | 0.0026 |          |       |       |
|         |           | 84.4  | 0.0121 |          |       |        |         |           | 163.1 | 0.0015 |          |       |       |
|         |           | 131.6 | 0.0012 |          |       |        |         |           | 114.3 | 0.0057 |          |       |       |
|         |           | 216.0 | 0.0024 |          |       |        |         |           |       |        |          |       |       |
|         |           | 172.5 | 0.0011 |          |       |        | 232Th   | 1.4+10y   | 12.3  | 0.0839 | 1.848-5  | 0.004 | 828.7 |
|         |           |       |        |          |       |        |         |           | 59.0  | 0.0019 |          |       |       |
|         |           |       |        |          |       |        |         |           | 125.0 | 0.0004 |          |       |       |
| 229Th   | 7340.0y   | 12.3  | 0.8105 | 1.989-4  | 0.029 | 103.7  |         |           |       |        |          |       |       |
|         |           | 17.4  | 0.0017 |          |       |        |         |           |       |        |          |       |       |
|         |           | 31.3  | 0.0408 |          |       |        | 233Th   | 22.3m     | 13.3  | 0.0956 | 2.586-5  | 0.812 | 3.689 |
|         |           | 42.8  | 0.0016 |          |       |        |         |           | 29.4  | 0.0250 |          |       |       |
|         |           | 56.6  | 0.0033 |          |       |        |         |           | 86.5  | 0.0270 |          |       |       |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|-------|--------|----------|-------|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 88.0  | 0.0030 |          |       |       |         |           |        |        |          |       |       |
|         |           | 92.3  | 0.0023 |          |       |       |         |           | 918.5  | 0.0814 |          |       |       |
|         |           | 94.7  | 0.0080 |          |       |       |         |           | 952.0  | 0.2878 |          |       |       |
|         |           | 95.9  | 0.0038 |          |       |       |         |           | 953.0  | 0.0016 |          |       |       |
|         |           | 108.0 | 0.0018 |          |       |       |         |           | 956.3  | 0.0157 |          |       |       |
|         |           | 162.5 | 0.0017 |          |       |       |         |           | 959.3  | 0.0049 |          |       |       |
|         |           | 162.5 | 0.0015 |          |       |       |         |           | 1010.0 | 0.0107 |          |       |       |
|         |           | 169.2 | 0.0034 |          |       |       |         |           | 1026.0 | 0.0144 |          |       |       |
|         |           | 170.7 | 0.0013 |          |       |       |         |           | 1075.0 | 0.0074 |          |       |       |
|         |           | 190.5 | 0.0013 |          |       |       |         |           | 537.2  | 0.0092 |          |       |       |
|         |           | 195.1 | 0.0016 |          |       |       |         |           | 13.6   | 0.0382 |          |       |       |
|         |           | 359.9 | 0.0012 |          |       |       |         |           | 314.8  | 0.0011 |          |       |       |
|         |           | 441.0 | 0.0023 |          |       |       |         |           | 297.6  | 0.0017 |          |       |       |
|         |           | 447.7 | 0.0015 |          |       |       |         |           |        |        |          |       |       |
|         |           | 459.2 | 0.0140 |          |       |       | 231Pa   | 3.3+4y    | 12.7   | 0.4321 | 1.011-4  | 0.023 | 131.7 |
|         |           | 490.8 | 0.0017 |          |       |       |         |           | 18.9   | 0.0035 |          |       |       |
|         |           | 499.0 | 0.0021 |          |       |       |         |           | 27.4   | 0.0930 |          |       |       |
|         |           | 595.2 | 0.0016 |          |       |       |         |           | 38.2   | 0.0015 |          |       |       |
|         |           | 669.8 | 0.0068 |          |       |       |         |           | 46.4   | 0.0021 |          |       |       |
|         |           | 764.4 | 0.0012 |          |       |       |         |           | 87.7   | 0.0062 |          |       |       |
|         |           | 890.1 | 0.0014 |          |       |       |         |           | 90.9   | 0.0102 |          |       |       |
|         |           | 484.7 | 0.0188 |          |       |       |         |           | 102.0  | 0.0047 |          |       |       |
|         |           |       |        |          |       |       |         |           | 255.8  | 0.0010 |          |       |       |
| 234Th   | 24.1d     | 13.3  | 0.0957 | 2.038-5  | 0.020 | 148.5 |         |           | 260.2  | 0.0017 |          |       |       |
|         |           | 63.3  | 0.0381 |          |       |       |         |           | 283.7  | 0.0160 |          |       |       |
|         |           | 92.4  | 0.0273 |          |       |       |         |           | 300.1  | 0.0230 |          |       |       |
|         |           | 92.8  | 0.0269 |          |       |       |         |           | 302.7  | 0.0230 |          |       |       |
|         |           | 112.8 | 0.0024 |          |       |       |         |           | 330.1  | 0.0130 |          |       |       |
|         |           | 76.8  | 0.0013 |          |       |       |         |           | 340.8  | 0.0017 |          |       |       |
|         |           |       |        |          |       |       |         |           | 357.2  | 0.0017 |          |       |       |
| 230Pa   | 17.4d     | 13.0  | 0.5974 | 2.386-4  | 2.266 | 1.322 |         |           | 165.5  | 0.0145 |          |       |       |
|         |           | 53.2  | 0.0024 |          |       |       |         |           |        |        |          |       |       |
|         |           | 90.0  | 0.1877 |          |       |       | 233Pa   | 27.0d     | 13.6   | 0.4892 | 1.335-4  | 0.426 | 7.024 |
|         |           | 93.3  | 0.3068 |          |       |       |         |           | 75.3   | 0.0126 |          |       |       |
|         |           | 105.0 | 0.1419 |          |       |       |         |           | 86.6   | 0.0189 |          |       |       |
|         |           | 120.9 | 0.0034 |          |       |       |         |           | 94.7   | 0.1084 |          |       |       |
|         |           | 316.8 | 0.0016 |          |       |       |         |           | 98.4   | 0.1757 |          |       |       |
|         |           | 380.2 | 0.0030 |          |       |       |         |           | 103.9  | 0.0074 |          |       |       |
|         |           | 397.8 | 0.0185 |          |       |       |         |           | 111.0  | 0.0818 |          |       |       |
|         |           | 400.0 | 0.0062 |          |       |       |         |           | 271.5  | 0.0030 |          |       |       |
|         |           | 440.8 | 0.0011 |          |       |       |         |           | 300.1  | 0.0664 |          |       |       |
|         |           | 443.8 | 0.0543 |          |       |       |         |           | 312.0  | 0.3860 |          |       |       |
|         |           | 454.9 | 0.0619 |          |       |       |         |           | 340.5  | 0.0452 |          |       |       |
|         |           | 463.6 | 0.0081 |          |       |       |         |           | 375.5  | 0.0062 |          |       |       |
|         |           | 508.0 | 0.0022 |          |       |       |         |           | 398.6  | 0.0127 |          |       |       |
|         |           | 508.2 | 0.0353 |          |       |       |         |           | 415.8  | 0.0162 |          |       |       |
|         |           | 518.5 | 0.0195 |          |       |       |         |           | 120.5  | 0.0021 |          |       |       |
|         |           | 556.0 | 0.0020 |          |       |       |         |           |        |        |          |       |       |
|         |           | 571.1 | 0.0107 |          |       |       | 234Pa   | 6.7h      | 13.6   | 1.1360 | 5.348-4  | 2.705 | 1.107 |
|         |           | 581.8 | 0.0013 |          |       |       |         |           | 43.5   | 0.0012 |          |       |       |
|         |           | 619.7 | 0.0016 |          |       |       |         |           | 63.0   | 0.0326 |          |       |       |
|         |           | 728.2 | 0.0187 |          |       |       |         |           | 69.9   | 0.0023 |          |       |       |
|         |           | 781.4 | 0.0147 |          |       |       |         |           | 79.7   | 0.0012 |          |       |       |
|         |           | 898.6 | 0.0576 |          |       |       |         |           | 94.7   | 0.1570 |          |       |       |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T | $\mu$ | Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|-------|--------|----------|---|-------|---------|-----------|-------|--------|----------|---|-------|
|         |           | 98.4  | 0.2544 |          |   |       |         |           | 506.8 | 0.0163 |          |   |       |
|         |           | 99.9  | 0.0490 |          |   |       |         |           | 513.7 | 0.0133 |          |   |       |
|         |           | 103.4 | 0.0012 |          |   |       |         |           | 520.2 | 0.0061 |          |   |       |
|         |           | 111.0 | 0.1185 |          |   |       |         |           | 521.0 | 0.0092 |          |   |       |
|         |           | 125.4 | 0.0102 |          |   |       |         |           | 528.0 | 0.0061 |          |   |       |
|         |           | 131.2 | 0.2040 |          |   |       |         |           | 533.2 | 0.0020 |          |   |       |
|         |           | 134.4 | 0.0021 |          |   |       |         |           | 537.1 | 0.0016 |          |   |       |
|         |           | 137.7 | 0.0015 |          |   |       |         |           | 557.0 | 0.0027 |          |   |       |
|         |           | 140.3 | 0.0092 |          |   |       |         |           | 565.9 | 0.0143 |          |   |       |
|         |           | 144.0 | 0.0036 |          |   |       |         |           | 568.7 | 0.0306 |          |   |       |
|         |           | 150.2 | 0.0020 |          |   |       |         |           | 569.5 | 0.1091 |          |   |       |
|         |           | 152.7 | 0.0683 |          |   |       |         |           | 574.0 | 0.0204 |          |   |       |
|         |           | 159.1 | 0.0071 |          |   |       |         |           | 585.8 | 0.0015 |          |   |       |
|         |           | 170.7 | 0.0051 |          |   |       |         |           | 596.6 | 0.0051 |          |   |       |
|         |           | 174.6 | 0.0020 |          |   |       |         |           | 602.8 | 0.0092 |          |   |       |
|         |           | 186.0 | 0.0204 |          |   |       |         |           | 611.5 | 0.0082 |          |   |       |
|         |           | 193.6 | 0.0061 |          |   |       |         |           | 616.2 | 0.0020 |          |   |       |
|         |           | 199.7 | 0.0049 |          |   |       |         |           | 623.6 | 0.0082 |          |   |       |
|         |           | 201.0 | 0.0112 |          |   |       |         |           | 627.5 | 0.0082 |          |   |       |
|         |           | 203.0 | 0.0122 |          |   |       |         |           | 630.6 | 0.0041 |          |   |       |
|         |           | 219.8 | 0.0020 |          |   |       |         |           | 634.5 | 0.0031 |          |   |       |
|         |           | 226.4 | 0.0602 |          |   |       |         |           | 639.7 | 0.0020 |          |   |       |
|         |           | 227.2 | 0.0561 |          |   |       |         |           | 643.2 | 0.0020 |          |   |       |
|         |           | 245.2 | 0.0092 |          |   |       |         |           | 646.0 | 0.0031 |          |   |       |
|         |           | 248.9 | 0.0286 |          |   |       |         |           | 653.7 | 0.0092 |          |   |       |
|         |           | 267.1 | 0.0017 |          |   |       |         |           | 655.0 | 0.0061 |          |   |       |
|         |           | 272.1 | 0.0102 |          |   |       |         |           | 658.0 | 0.0092 |          |   |       |
|         |           | 275.5 | 0.0027 |          |   |       |         |           | 660.6 | 0.0031 |          |   |       |
|         |           | 286.1 | 0.0014 |          |   |       |         |           | 664.8 | 0.0133 |          |   |       |
|         |           | 289.6 | 0.0011 |          |   |       |         |           | 666.7 | 0.0163 |          |   |       |
|         |           | 293.7 | 0.0398 |          |   |       |         |           | 669.9 | 0.0143 |          |   |       |
|         |           | 309.6 | 0.0010 |          |   |       |         |           | 683.3 | 0.0024 |          |   |       |
|         |           | 312.5 | 0.0031 |          |   |       |         |           | 687.0 | 0.0029 |          |   |       |
|         |           | 316.3 | 0.0012 |          |   |       |         |           | 692.7 | 0.0153 |          |   |       |
|         |           | 320.7 | 0.0012 |          |   |       |         |           | 699.0 | 0.0469 |          |   |       |
|         |           | 328.0 | 0.0031 |          |   |       |         |           | 706.1 | 0.0316 |          |   |       |
|         |           | 330.6 | 0.0061 |          |   |       |         |           | 711.2 | 0.0020 |          |   |       |
|         |           | 351.9 | 0.0061 |          |   |       |         |           | 713.8 | 0.0016 |          |   |       |
|         |           | 369.8 | 0.0296 |          |   |       |         |           | 733.0 | 0.0877 |          |   |       |
|         |           | 372.4 | 0.0133 |          |   |       |         |           | 738.0 | 0.0102 |          |   |       |
|         |           | 409.8 | 0.0041 |          |   |       |         |           | 742.8 | 0.0245 |          |   |       |
|         |           | 416.3 | 0.0010 |          |   |       |         |           | 746.5 | 0.0013 |          |   |       |
|         |           | 426.8 | 0.0061 |          |   |       |         |           | 755.6 | 0.0143 |          |   |       |
|         |           | 446.5 | 0.0012 |          |   |       |         |           | 760.0 | 0.0016 |          |   |       |
|         |           | 458.8 | 0.0153 |          |   |       |         |           | 766.4 | 0.0031 |          |   |       |
|         |           | 461.8 | 0.0016 |          |   |       |         |           | 768.7 | 0.0057 |          |   |       |
|         |           | 467.5 | 0.0041 |          |   |       |         |           | 777.9 | 0.0020 |          |   |       |
|         |           | 472.1 | 0.0024 |          |   |       |         |           | 780.7 | 0.0112 |          |   |       |
|         |           | 473.5 | 0.0018 |          |   |       |         |           | 783.1 | 0.0051 |          |   |       |
|         |           | 478.7 | 0.0031 |          |   |       |         |           | 786.3 | 0.0143 |          |   |       |
|         |           | 480.4 | 0.0041 |          |   |       |         |           | 793.6 | 0.0153 |          |   |       |
|         |           | 482.5 | 0.0031 |          |   |       |         |           | 796.3 | 0.0388 |          |   |       |
|         |           | 498.9 | 0.0010 |          |   |       |         |           | 804.3 | 0.0041 |          |   |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|--------|--------|----------|---|-------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 805.6  | 0.0337 |          |   |       |         |           | 1516.0 | 0.0041 |          |       |       |
|         |           | 812.5  | 0.0051 |          |   |       |         |           | 1549.0 | 0.0010 |          |       |       |
|         |           | 819.6  | 0.0265 |          |   |       |         |           | 1580.0 | 0.0017 |          |       |       |
|         |           | 826.3  | 0.0408 |          |   |       |         |           | 1585.0 | 0.0026 |          |       |       |
|         |           | 831.6  | 0.0561 |          |   |       |         |           | 1594.0 | 0.0061 |          |       |       |
|         |           | 841.9  | 0.0014 |          |   |       |         |           | 1628.0 | 0.0015 |          |       |       |
|         |           | 844.0  | 0.0051 |          |   |       |         |           | 1638.0 | 0.0041 |          |       |       |
|         |           | 851.7  | 0.0012 |          |   |       |         |           | 1656.0 | 0.0015 |          |       |       |
|         |           | 872.9  | 0.0012 |          |   |       |         |           | 1668.0 | 0.0122 |          |       |       |
|         |           | 876.4  | 0.0408 |          |   |       |         |           | 1686.0 | 0.0051 |          |       |       |
|         |           | 880.5  | 0.0102 |          |   |       |         |           | 1695.0 | 0.0122 |          |       |       |
|         |           | 880.5  | 0.1224 |          |   |       |         |           | 1700.0 | 0.0015 |          |       |       |
|         |           | 883.2  | 0.1224 |          |   |       |         |           | 1738.0 | 0.0010 |          |       |       |
|         |           | 899.0  | 0.0418 |          |   |       |         |           | 1742.0 | 0.0010 |          |       |       |
|         |           | 904.4  | 0.0051 |          |   |       |         |           | 1756.0 | 0.0026 |          |       |       |
|         |           | 920.0  | 0.0041 |          |   |       |         |           | 1772.0 | 0.0010 |          |       |       |
|         |           | 925.0  | 0.0296 |          |   |       |         |           | 1797.0 | 0.0031 |          |       |       |
|         |           | 926.0  | 0.1122 |          |   |       |         |           | 1890.0 | 0.0019 |          |       |       |
|         |           | 926.7  | 0.0918 |          |   |       |         |           | 1897.0 | 0.0015 |          |       |       |
|         |           | 946.0  | 0.1224 |          |   |       |         |           | 1905.0 | 0.0029 |          |       |       |
|         |           | 949.0  | 0.0816 |          |   |       |         |           | 1926.0 | 0.0051 |          |       |       |
|         |           | 960.0  | 0.0010 |          |   |       |         |           | 955.6  | 0.0101 |          |       |       |
|         |           | 966.0  | 0.0061 |          |   |       |         |           |        |        |          |       |       |
|         |           | 978.8  | 0.0143 |          |   |       | 234mPa  | 1.2m      | 13.3   | 0.0006 | 2.776-6  | 3.062 | 0.978 |
|         |           | 980.5  | 0.0306 |          |   |       |         |           | 73.9   | 0.0001 |          |       |       |
|         |           | 980.5  | 0.0204 |          |   |       |         |           | 13.6   | 0.0044 |          |       |       |
|         |           | 984.0  | 0.0194 |          |   |       |         |           | 94.7   | 0.0012 |          |       |       |
|         |           | 1023.0 | 0.0061 |          |   |       |         |           | 98.4   | 0.0019 |          |       |       |
|         |           | 1028.0 | 0.0082 |          |   |       |         |           | 111.0  | 0.0009 |          |       |       |
|         |           | 1045.0 | 0.0051 |          |   |       |         |           | 766.4  | 0.0021 |          |       |       |
|         |           | 1075.0 | 0.0026 |          |   |       |         |           | 1001.0 | 0.0059 |          |       |       |
|         |           | 1083.0 | 0.0076 |          |   |       |         |           | 926.2  | 0.0037 |          |       |       |
|         |           | 1108.0 | 0.0031 |          |   |       |         |           |        |        |          |       |       |
|         |           | 1122.0 | 0.0051 |          |   |       | 230U    | 20.8d     | 13.0   | 0.1223 | 2.463-5  | 0.004 | 683.5 |
|         |           | 1126.0 | 0.0082 |          |   |       |         |           | 72.2   | 0.0060 |          |       |       |
|         |           | 1153.0 | 0.0031 |          |   |       |         |           | 154.2  | 0.0013 |          |       |       |
|         |           | 1171.0 | 0.0024 |          |   |       |         |           | 230.4  | 0.0012 |          |       |       |
|         |           | 1208.0 | 0.0031 |          |   |       |         |           | 169.9  | 0.0008 |          |       |       |
|         |           | 1217.0 | 0.0038 |          |   |       |         |           |        |        |          |       |       |
|         |           | 1229.0 | 0.0031 |          |   |       | 231U    | 4.2d      | 13.3   | 0.9894 | 2.120-4  | 0.010 | 293.4 |
|         |           | 1241.0 | 0.0051 |          |   |       |         |           | 25.6   | 0.1200 |          |       |       |
|         |           | 1251.0 | 0.0031 |          |   |       |         |           | 58.6   | 0.0044 |          |       |       |
|         |           | 1277.0 | 0.0020 |          |   |       |         |           | 84.2   | 0.0700 |          |       |       |
|         |           | 1293.0 | 0.0061 |          |   |       |         |           | 92.3   | 0.1730 |          |       |       |
|         |           | 1353.0 | 0.0173 |          |   |       |         |           | 95.9   | 0.2818 |          |       |       |
|         |           | 1358.0 | 0.0012 |          |   |       |         |           | 108.0  | 0.1305 |          |       |       |
|         |           | 1394.0 | 0.0306 |          |   |       |         |           | 217.9  | 0.0080 |          |       |       |
|         |           | 1400.0 | 0.0023 |          |   |       |         |           | 236.0  | 0.0018 |          |       |       |
|         |           | 1427.0 | 0.0020 |          |   |       |         |           | 178.4  | 0.0022 |          |       |       |
|         |           | 1446.0 | 0.0041 |          |   |       |         |           |        |        |          |       |       |
|         |           | 1453.0 | 0.0102 |          |   |       | 232U    | 72y       | 13.0   | 0.1202 | 2.403-5  | 0.004 | 716.7 |
|         |           | 1460.0 | 0.0031 |          |   |       |         |           | 57.8   | 0.0020 |          |       |       |
|         |           | 1494.0 | 0.0020 |          |   |       |         |           | 142.0  | 0.0007 |          |       |       |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|-------|--------|----------|-------|--------|---------|-----------|-------|--------|----------|-------|--------|
| 233U    | 1.59+5y   | 13.0  | 0.0392 | 7.866-6  | 0.004 | 705.0  |         |           | 662.2 | 0.0018 |          |       |        |
|         |           | 114.5 | 0.0018 |          |       |        |         |           | 748.1 | 0.0010 |          |       |        |
|         |           |       |        |          |       |        |         |           | 819.2 | 0.0014 |          |       |        |
|         |           |       |        |          |       |        |         |           | 844.1 | 0.0016 |          |       |        |
|         |           |       |        |          |       |        |         |           | 611.2 | 0.0092 |          |       |        |
| 234U    | 2.44+5y   | 13.0  | 0.1050 | 2.097-5  | 0.004 | 720.7  | 240U    | 14.1h     | 13.9  | 0.4309 | 7.686-5  | 0.003 | 921.3  |
|         |           | 53.2  | 0.0012 |          |       |        |         |           | 44.1  | 0.0165 |          |       |        |
|         |           | 121.4 | 0.0004 |          |       |        |         |           |       |        |          |       |        |
| 235U    | 7.04+8y   | 13.0  | 0.3091 | 9.159-5  | 0.132 | 22.675 | 235Np   | 1.1y      | 13.6  | 0.3766 | 6.979-5  | 0.004 | 840.5  |
|         |           | 72.7  | 0.0011 |          |       |        |         |           | 94.7  | 0.0051 |          |       |        |
|         |           | 90.0  | 0.0273 |          |       |        |         |           | 98.4  | 0.0083 |          |       |        |
|         |           | 93.3  | 0.0446 |          |       |        |         |           | 111.0 | 0.0039 |          |       |        |
|         |           | 105.0 | 0.0206 |          |       |        |         |           |       |        |          |       |        |
|         |           | 109.1 | 0.0150 |          |       |        |         |           |       |        |          |       |        |
|         |           | 120.0 | 0.0015 |          |       |        | 236Np   | 1.15+5y   | 13.6  | 1.3100 | 2.833-4  | 0.019 | 161.6  |
|         |           | 140.8 | 0.0022 |          |       |        |         |           | 45.2  | 0.0015 |          |       |        |
|         |           | 143.8 | 0.1050 |          |       |        |         |           | 94.7  | 0.2070 |          |       |        |
|         |           | 163.3 | 0.0470 |          |       |        |         |           | 98.4  | 0.3355 |          |       |        |
|         |           | 182.7 | 0.0040 |          |       |        |         |           | 104.2 | 0.0747 |          |       |        |
|         |           | 183.7 | 0.5400 |          |       |        |         |           | 111.0 | 0.1562 |          |       |        |
|         |           | 194.9 | 0.0059 |          |       |        |         |           | 160.3 | 0.2760 |          |       |        |
|         |           | 202.1 | 0.0100 |          |       |        |         |           | 14.3  | 0.0884 |          |       |        |
|         |           | 205.3 | 0.0470 |          |       |        |         |           | 99.6  | 0.0008 |          |       |        |
|         |           | 221.4 | 0.0010 |          |       |        |         |           | 100.0 | 0.0052 |          |       |        |
|         |           | 190.3 | 0.0092 |          |       |        |         |           | 103.8 | 0.0013 |          |       |        |
|         |           |       |        |          |       |        |         |           | 117.0 | 0.0006 |          |       |        |
| 236U    | 2.34+7y   | 13.0  | 0.0998 | 1.992-5  | 0.004 | 722.4  |         |           | 160.0 | 0.0142 |          |       |        |
|         |           | 68.2  | 0.0011 |          |       |        |         |           | 44.6  | 0.0001 |          |       |        |
| 237U    | 6.7d      | 13.8  | 0.0010 | 1.589-4  | 0.055 | 54.239 | 236mNp  | 22.5h     | 13.6  | 0.2559 | 6.389-5  | 0.043 | 69.991 |
|         |           | 13.9  | 0.7114 |          |       |        |         |           | 94.7  | 0.1126 |          |       |        |
|         |           | 26.3  | 0.0228 |          |       |        |         |           | 98.4  | 0.1824 |          |       |        |
|         |           | 33.2  | 0.0011 |          |       |        |         |           | 111.0 | 0.0849 |          |       |        |
|         |           | 51.0  | 0.0021 |          |       |        |         |           | 642.3 | 0.0138 |          |       |        |
|         |           | 59.5  | 0.3402 |          |       |        |         |           | 687.5 | 0.0037 |          |       |        |
|         |           | 64.8  | 0.0118 |          |       |        |         |           | 304.5 | 0.0003 |          |       |        |
|         |           | 97.1  | 0.1629 |          |       |        |         |           | 14.3  | 0.0363 |          |       |        |
|         |           | 101.1 | 0.2632 |          |       |        |         |           | 44.6  | 0.0001 |          |       |        |
|         |           | 114.0 | 0.1231 |          |       |        |         |           |       |        |          |       |        |
|         |           | 164.6 | 0.0186 |          |       |        | 237Np   | 2.1+6y    | 13.3  | 0.5919 | 1.251-4  | 0.006 | 512.0  |
|         |           | 208.0 | 0.2202 |          |       |        |         |           | 29.4  | 0.1399 |          |       |        |
|         |           | 267.5 | 0.0072 |          |       |        |         |           | 46.5  | 0.0014 |          |       |        |
|         |           | 332.4 | 0.0122 |          |       |        |         |           | 57.1  | 0.0042 |          |       |        |
|         |           | 370.9 | 0.0011 |          |       |        |         |           | 86.5  | 0.1260 |          |       |        |
|         |           | 262.9 | 0.0025 |          |       |        |         |           | 88.0  | 0.0016 |          |       |        |
|         |           |       |        |          |       |        |         |           | 92.3  | 0.0158 |          |       |        |
| 238U    | 4.47+9y   | 13.0  | 0.0883 | 1.763-5  | 0.004 | 722.5  |         |           | 94.7  | 0.0083 |          |       |        |
|         |           | 66.4  | 0.0010 |          |       |        |         |           | 95.9  | 0.0258 |          |       |        |
| 239U    | 23.4m     | 13.9  | 0.1326 | 3.629-5  | 0.124 | 24.144 |         |           | 108.0 | 0.0119 |          |       |        |
|         |           | 43.5  | 0.0427 |          |       |        |         |           | 117.7 | 0.0017 |          |       |        |
|         |           | 74.7  | 0.4800 |          |       |        |         |           | 143.2 | 0.0042 |          |       |        |
|         |           | 117.7 | 0.0014 |          |       |        |         |           | 151.4 | 0.0025 |          |       |        |
|         |           |       |        |          |       |        |         |           | 195.1 | 0.0021 |          |       |        |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T      | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |       |        |         |       |       |
|---------|-----------|--------|--------|----------|--------|--------|---------|-----------|--------|--------|----------|-------|--------|-------|--------|---------|-------|-------|
| 238Np   | 2.1d      | 212.4  | 0.0016 | 1.497-4  | 3.104  | 0.965  |         |           | 280.2  | 0.0037 | 1.497-4  | 3.104 | 0.965  |       |        |         |       |       |
|         |           | 163.6  | 0.0105 |          |        |        |         |           | 295.0  | 0.0065 |          |       |        |       |        |         |       |       |
|         |           | 14.3   | 0.3739 |          |        |        |         |           | 307.0  | 0.0139 |          |       |        |       |        |         |       |       |
|         |           | 44.1   | 0.0010 |          |        |        |         |           | 448.2  | 0.1669 |          |       |        |       |        |         |       |       |
|         |           | 99.6   | 0.0011 |          |        |        |         |           | 462.2  | 0.0139 |          |       |        |       |        |         |       |       |
|         |           | 101.9  | 0.0021 |          |        |        |         |           | 467.4  | 0.0204 |          |       |        |       |        |         |       |       |
|         |           | 103.8  | 0.0018 |          |        |        |         |           | 507.2  | 0.0185 |          |       |        |       |        |         |       |       |
|         |           | 117.0  | 0.0008 |          |        |        |         |           | 566.4  | 0.2688 |          |       |        |       |        |         |       |       |
|         |           | 561.2  | 0.0010 |          |        |        |         |           | 601.1  | 0.2039 |          |       |        |       |        |         |       |       |
|         |           | 882.6  | 0.0076 |          |        |        |         |           | 606.1  | 0.0158 |          |       |        |       |        |         |       |       |
|         |           | 918.7  | 0.0051 |          |        |        |         |           | 847.0  | 0.0464 |          |       |        |       |        |         |       |       |
|         |           | 924.0  | 0.0248 |          |        |        |         |           | 867.4  | 0.0834 |          |       |        |       |        |         |       |       |
|         |           | 936.6  | 0.0033 |          |        |        |         |           | 884.9  | 0.0371 |          |       |        |       |        |         |       |       |
|         |           | 941.4  | 0.0045 |          |        |        |         |           | 888.8  | 0.0111 |          |       |        |       |        |         |       |       |
|         |           | 962.8  | 0.0061 |          |        |        |         |           | 896.5  | 0.1298 |          |       |        |       |        |         |       |       |
|         |           | 984.4  | 0.2380 |          |        |        |         |           | 916.0  | 0.0139 |          |       |        |       |        |         |       |       |
|         |           | 1026.0 | 0.0821 |          |        |        |         |           | 959.1  | 0.0232 |          |       |        |       |        |         |       |       |
|         |           | 1028.0 | 0.1737 |          |        |        |         |           | 973.9  | 0.2132 |          |       |        |       |        |         |       |       |
|         |           | 503.9  | 0.0054 |          |        |        |         |           | 987.8  | 0.0464 |          |       |        |       |        |         |       |       |
| 239Np   | 2.4d      | 14.3   | 0.6179 | 1.386-4  | 0.154  | 19.449 |         |           | 1074.0 | 0.0093 | 1.386-4  | 0.154 | 19.449 |       |        |         |       |       |
|         |           | 49.4   | 0.0010 |          |        |        |         |           | 1089.0 | 0.0046 |          |       |        |       |        |         |       |       |
|         |           | 57.3   | 0.0015 |          |        |        |         |           | 1131.0 | 0.0065 |          |       |        |       |        |         |       |       |
|         |           | 61.5   | 0.0096 |          |        |        |         |           | 1163.0 | 0.0065 |          |       |        |       |        |         |       |       |
|         |           | 99.6   | 0.1471 |          |        |        |         |           | 1168.0 | 0.0464 |          |       |        |       |        |         |       |       |
|         |           | 103.8  | 0.2365 |          |        |        |         |           | 1180.0 | 0.0065 |          |       |        |       |        |         |       |       |
|         |           | 106.1  | 0.2270 |          |        |        |         |           | 1223.0 | 0.0046 |          |       |        |       |        |         |       |       |
|         |           | 117.0  | 0.1113 |          |        |        |         |           | 422.7  | 0.0020 |          |       |        |       |        |         |       |       |
|         |           | 181.7  | 0.0011 |          |        |        |         |           | 240mNp | 7.4m   |          |       |        | 20.0  | 0.0000 | 1.140-4 | 2.010 | 1.490 |
|         |           | 209.7  | 0.0324 |          |        |        |         |           |        |        |          |       |        | 14.3  | 0.3374 |         |       |       |
|         |           | 226.4  | 0.0034 |          |        |        |         |           |        |        |          |       |        | 66.5  | 0.0027 |         |       |       |
|         |           | 228.2  | 0.1072 |          |        |        |         |           |        |        |          |       |        | 98.9  | 0.0017 |         |       |       |
|         |           | 254.4  | 0.0010 |          |        |        |         |           |        |        |          |       |        | 99.6  | 0.0013 |         |       |       |
|         |           | 277.6  | 0.1410 |          |        |        |         |           |        |        |          |       |        | 103.8 | 0.0021 |         |       |       |
|         |           | 285.4  | 0.0078 |          |        |        |         |           |        |        |          |       |        | 117.0 | 0.0010 |         |       |       |
|         |           | 315.9  | 0.0159 |          |        |        |         |           |        |        |          |       |        | 189.5 | 0.0025 |         |       |       |
|         |           | 334.3  | 0.0203 |          |        |        |         |           |        |        |          |       |        | 251.5 | 0.0096 |         |       |       |
|         |           | 151.5  | 0.0039 |          |        |        |         |           |        |        |          |       |        | 263.4 | 0.0117 |         |       |       |
|         |           | 240Np  | 65m    |          |        |        |         |           |        |        |          |       |        | 14.3  | 1.0940 |         |       |       |
| 42.8    | 0.0011    |        |        | 507.2    | 0.0079 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 98.9    | 0.0510    |        |        | 554.6    | 0.2237 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 99.6    | 0.0852    |        |        | 597.4    | 0.1249 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 103.8   | 0.1370    |        |        | 606.1    | 0.0074 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 117.0   | 0.0644    |        |        | 758.6    | 0.0119 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 134.6   | 0.0037    |        |        | 789.6    | 0.0021 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 147.2   | 0.0139    |        |        | 813.4    | 0.0021 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 152.2   | 0.0834    |        |        | 817.9    | 0.0124 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 175.0   | 0.0603    |        |        | 841.1    | 0.0017 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 182.6   | 0.0093    |        |        | 857.5    | 0.0047 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 192.7   | 0.0677    |        |        | 900.5    | 0.0013 |        |         |           |        |        |          |       |        |       |        |         |       |       |
| 270.8   | 0.0834    |        |        | 910.1    | 0.0017 |        |         |           |        |        |          |       |        |       |        |         |       |       |
|         |           |        |        | 916.0    | 0.0104 |        |         |           |        |        |          |       |        |       |        |         |       |       |
|         |           |        |        | 928.6    | 0.0017 |        |         |           |        |        |          |       |        |       |        |         |       |       |



| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T | $\mu$ |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|---|-------|
|         |           | 938.0  | 0.0129 |          |       |        |         |           | 280.3  | 0.0137 |          |   |       |
|         |           | 942.4  | 0.0011 |          |       |        |         |           | 308.1  | 0.0522 |          |   |       |
|         |           | 961.6  | 0.0014 |          |       |        |         |           | 327.3  | 0.2700 |          |   |       |
|         |           | 1445.0 | 0.0036 |          |       |        |         |           | 341.0  | 0.0011 |          |   |       |
|         |           | 1488.0 | 0.0021 |          |       |        |         |           | 348.7  | 0.0103 |          |   |       |
|         |           | 1497.0 | 0.0131 |          |       |        |         |           | 376.6  | 0.0342 |          |   |       |
|         |           | 1540.0 | 0.0079 |          |       |        |         |           | 387.9  | 0.0031 |          |   |       |
|         |           | 1633.0 | 0.0014 |          |       |        |         |           | 395.9  | 0.0011 |          |   |       |
|         |           | 1020.0 | 0.0126 |          |       |        |         |           | 411.7  | 0.0052 |          |   |       |
| 236Pu   | 2.9y      | 13.6   | 0.1304 | 2.405-5  | 0.003 | 863.6  |         |           | 428.5  | 0.0056 |          |   |       |
|         |           | 60.9   | 0.0008 |          |       |        |         |           | 445.3  | 0.0032 |          |   |       |
| 237Pu   | 45.3d     | 13.9   | 0.5254 | 1.039-4  | 0.014 | 212.7  |         |           | 491.5  | 0.0288 |          |   |       |
|         |           | 26.3   | 0.0024 |          |       |        |         |           | 514.6  | 0.0018 |          |   |       |
|         |           | 59.5   | 0.0328 |          |       |        |         |           | 525.1  | 0.0029 |          |   |       |
|         |           | 97.1   | 0.1278 |          |       |        |         |           | 560.0  | 0.0576 |          |   |       |
|         |           | 101.1  | 0.2064 |          |       |        |         |           | 591.6  | 0.0018 |          |   |       |
|         |           | 114.0  | 0.0966 |          |       |        |         |           | 598.8  | 0.0013 |          |   |       |
|         |           | 33.2   | 0.0008 |          |       |        |         |           | 624.4  | 0.0023 |          |   |       |
| 238Pu   | 87.8y     | 13.6   | 0.1158 | 2.135-5  | 0.003 | 864.8  |         |           | 630.0  | 0.0288 |          |   |       |
|         |           | 55.3   | 0.0005 |          |       |        |         |           | 657.2  | 0.0014 |          |   |       |
| 239Pu   | 2.41+4y   | 13.6   | 0.0441 | 8.145-6  | 0.003 | 860.3  |         |           | 660.2  | 0.0090 |          |   |       |
|         |           | 112.9  | 0.0005 |          |       |        |         |           | 669.3  | 0.0036 |          |   |       |
| 240Pu   | 6569y     | 13.6   | 0.1101 | 2.030-5  | 0.003 | 864.4  |         |           | 708.0  | 0.0029 |          |   |       |
|         |           | 54.3   | 0.0005 |          |       |        |         |           | 730.4  | 0.0020 |          |   |       |
| 242Pu   | 3.76+5y   | 13.6   | 0.0913 | 1.684-5  | 0.003 | 864.4  |         |           | 738.0  | 0.0023 |          |   |       |
|         |           | 56.4   | 0.0004 |          |       |        |         |           | 740.2  | 0.0014 |          |   |       |
| 243Pu   | 5.0h      | 14.6   | 0.1188 | 2.509-5  | 0.084 | 35.548 |         |           | 743.7  | 0.0016 |          |   |       |
|         |           | 41.8   | 0.0076 |          |       |        |         |           | 762.7  | 0.0076 |          |   |       |
|         |           | 67.0   | 0.0023 |          |       |        |         |           | 766.6  | 0.0038 |          |   |       |
|         |           | 84.0   | 0.2300 |          |       |        |         |           | 776.7  | 0.0022 |          |   |       |
|         |           | 102.1  | 0.0009 |          |       |        |         |           | 786.5  | 0.0040 |          |   |       |
|         |           | 106.5  | 0.0015 |          |       |        |         |           | 796.4  | 0.0027 |          |   |       |
|         |           | 109.3  | 0.0016 |          |       |        |         |           | 799.9  | 0.0167 |          |   |       |
|         |           | 120.0  | 0.0007 |          |       |        |         |           | 817.0  | 0.0090 |          |   |       |
|         |           | 356.4  | 0.0013 |          |       |        |         |           | 833.1  | 0.0056 |          |   |       |
|         |           | 381.7  | 0.0055 |          |       |        |         |           | 840.6  | 0.0137 |          |   |       |
|         |           | 137.2  | 0.0018 |          |       |        |         |           | 859.5  | 0.0054 |          |   |       |
| 244Pu   | 8.3+7y    | 13.6   | 0.0793 | 1.462-5  | 0.003 | 865.1  |         |           | 868.8  | 0.0013 |          |   |       |
|         |           | 44.0   | 0.0003 |          |       |        |         |           | 874.2  | 0.0014 |          |   |       |
| 245Pu   | 10.6h     | 14.6   | 0.1750 | 1.045-4  | 2.092 | 1.432  |         |           | 887.1  | 0.0076 |          |   |       |
|         |           | 28.0   | 0.0072 |          |       |        |         |           | 910.5  | 0.0148 |          |   |       |
|         |           | 102.1  | 0.0753 |          |       |        |         |           | 938.4  | 0.0108 |          |   |       |
|         |           | 106.5  | 0.1208 |          |       |        |         |           | 941.0  | 0.0027 |          |   |       |
|         |           | 120.0  | 0.0571 |          |       |        |         |           | 957.6  | 0.0104 |          |   |       |
|         |           |        |        |          |       |        |         |           | 975.0  | 0.0027 |          |   |       |
|         |           |        |        |          |       |        |         |           | 977.2  | 0.0041 |          |   |       |
|         |           |        |        |          |       |        |         |           | 987.6  | 0.0140 |          |   |       |
|         |           |        |        |          |       |        |         |           | 996.0  | 0.0022 |          |   |       |
|         |           |        |        |          |       |        |         |           | 1005.0 | 0.0029 |          |   |       |
|         |           |        |        |          |       |        |         |           | 1007.0 | 0.0043 |          |   |       |
|         |           |        |        |          |       |        |         |           | 1013.0 | 0.0011 |          |   |       |
|         |           |        |        |          |       |        |         |           | 1018.0 | 0.0110 |          |   |       |
|         |           |        |        |          |       |        |         |           | 1023.0 | 0.0058 |          |   |       |

| Nuclide | Half-life | E     | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$ |
|---------|-----------|-------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|-------|-------|
|         |           | 762.0 | 0.0235 |          |       |        |         |           | 104.6  | 0.0363 |          |       |       |
| 246Pu   | 10.9d     | 27.6  | 0.0423 | 2.727-5  | 0.278 | 10.771 |         |           | 109.3  | 0.0578 |          |       |       |
|         |           | 43.8  | 0.3000 |          |       |        |         |           | 123.0  | 0.0275 |          |       |       |
|         |           | 66.6  | 0.0031 |          |       |        |         |           | 241.2  | 0.0034 |          |       |       |
|         |           | 75.6  | 0.0022 |          |       |        |         |           | 252.8  | 0.0610 |          |       |       |
|         |           | 179.9 | 0.1164 |          |       |        |         |           | 295.8  | 0.0022 |          |       |       |
|         |           | 216.5 | 0.0014 |          |       |        |         |           | 95.9   | 0.0010 |          |       |       |
|         |           | 223.7 | 0.2820 |          |       |        | 246Am   | 25.0m     | 15.0   | 0.3906 | 2.148-4  | 3.331 | 0.899 |
|         |           | 255.5 | 0.0028 |          |       |        |         |           | 99.2   | 0.0017 |          |       |       |
|         |           | 202.9 | 0.0030 |          |       |        |         |           | 104.6  | 0.0088 |          |       |       |
| 241Am   | 432.2y    | 13.9  | 0.4270 | 8.479-5  | 0.011 | 260.7  |         |           | 109.3  | 0.0141 |          |       |       |
|         |           | 26.3  | 0.0240 |          |       |        |         |           | 123.0  | 0.0067 |          |       |       |
|         |           | 33.2  | 0.0011 |          |       |        |         |           | 237.2  | 0.0014 |          |       |       |
|         |           | 59.5  | 0.3590 |          |       |        |         |           | 238.6  | 0.0015 |          |       |       |
|         |           | 69.2  | 0.0018 |          |       |        |         |           | 244.0  | 0.0068 |          |       |       |
| 242Am   | 16.0h     | 14.3  | 0.1266 | 5.476-5  | 0.004 | 720.7  |         |           | 261.7  | 0.0016 |          |       |       |
|         |           | 99.6  | 0.0366 |          |       |        |         |           | 270.1  | 0.0103 |          |       |       |
|         |           | 103.8 | 0.0589 |          |       |        |         |           | 287.8  | 0.0013 |          |       |       |
|         |           | 117.0 | 0.0277 |          |       |        |         |           | 401.7  | 0.0027 |          |       |       |
|         |           | 44.5  | 0.0001 |          |       |        |         |           | 493.5  | 0.0011 |          |       |       |
|         |           | 15.0  | 0.1965 |          |       |        |         |           | 602.5  | 0.0023 |          |       |       |
|         |           | 42.2  | 0.0004 |          |       |        |         |           | 649.5  | 0.0037 |          |       |       |
| 242mAm  | 152y      | 14.6  | 0.2993 | 4.950-5  | 0.003 | 1132.  |         |           | 684.3  | 0.0059 |          |       |       |
|         |           | 48.6  | 0.0000 |          |       |        |         |           | 698.3  | 0.0012 |          |       |       |
|         |           | 13.9  | 0.0038 |          |       |        |         |           | 717.2  | 0.0025 |          |       |       |
|         |           | 49.4  | 0.0020 |          |       |        |         |           | 724.8  | 0.0021 |          |       |       |
|         |           | 108.3 | 0.0016 |          |       |        |         |           | 734.4  | 0.0117 |          |       |       |
| 243Am   | 7380.0y   | 13.9  | 0.3905 | 8.456-5  | 0.042 | 71.213 |         |           | 745.1  | 0.0024 |          |       |       |
|         |           | 43.5  | 0.0554 |          |       |        |         |           | 752.1  | 0.0082 |          |       |       |
|         |           | 74.7  | 0.6600 |          |       |        |         |           | 759.6  | 0.0064 |          |       |       |
|         |           | 86.7  | 0.0034 |          |       |        |         |           | 781.3  | 0.0017 |          |       |       |
|         |           | 117.7 | 0.0055 |          |       |        |         |           | 798.8  | 0.2490 |          |       |       |
|         |           | 142.2 | 0.0013 |          |       |        |         |           | 833.6  | 0.0179 |          |       |       |
|         |           | 48.4  | 0.0016 |          |       |        |         |           | 986.0  | 0.0096 |          |       |       |
| 244Am   | 10.1h     | 15.0  | 1.1690 | 3.166-4  | 2.108 | 1.421  |         |           | 1036.0 | 0.1275 |          |       |       |
|         |           | 99.4  | 0.0483 |          |       |        |         |           | 1062.0 | 0.1721 |          |       |       |
|         |           | 104.6 | 0.0226 |          |       |        |         |           | 1079.0 | 0.2789 |          |       |       |
|         |           | 109.3 | 0.0361 |          |       |        |         |           | 1081.0 | 0.0025 |          |       |       |
|         |           | 123.0 | 0.0171 |          |       |        |         |           | 1085.0 | 0.0153 |          |       |       |
|         |           | 154.0 | 0.1800 |          |       |        |         |           | 1124.0 | 0.0026 |          |       |       |
|         |           | 206.0 | 0.0026 |          |       |        |         |           | 1207.0 | 0.0015 |          |       |       |
|         |           | 540.0 | 0.0038 |          |       |        |         |           | 1250.0 | 0.0015 |          |       |       |
|         |           | 746.0 | 0.6700 |          |       |        |         |           | 1275.0 | 0.0027 |          |       |       |
|         |           | 900.0 | 0.2800 |          |       |        |         |           | 1349.0 | 0.0012 |          |       |       |
|         |           | 42.9  | 0.0009 |          |       |        |         |           | 1479.0 | 0.0023 |          |       |       |
| 245Am   | 2.0h      | 15.0  | 0.1100 | 2.341-5  | 0.169 | 17.710 |         |           | 1529.0 | 0.0022 |          |       |       |
|         |           |       |        |          |       |        |         |           | 1551.0 | 0.0027 |          |       |       |
|         |           |       |        |          |       |        |         |           | 1591.0 | 0.0052 |          |       |       |
|         |           |       |        |          |       |        |         |           | 1604.0 | 0.0010 |          |       |       |
|         |           |       |        |          |       |        |         |           | 1619.0 | 0.0012 |          |       |       |
|         |           |       |        |          |       |        |         |           | 1638.0 | 0.0016 |          |       |       |
|         |           |       |        |          |       |        |         |           | 1662.0 | 0.0023 |          |       |       |

| Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  | Nuclide | Half-life | E      | Prob.  | $\Gamma$ | T     | $\mu$  |
|---------|-----------|--------|--------|----------|-------|--------|---------|-----------|--------|--------|----------|-------|--------|
|         |           | 1738.0 | 0.0011 |          |       |        |         |           | 126.0  | 0.0006 |          |       |        |
|         |           | 914.7  | 0.0296 |          |       |        |         |           | 368.8  | 0.0035 |          |       |        |
| 242Cm   | 163.2d    | 14.3   | 0.1154 | 1.949-5  | 0.003 | 1054.  |         |           | 560.4  | 0.0084 |          |       |        |
|         |           | 59.3   | 0.0004 |          |       |        |         |           | 621.9  | 0.0018 |          |       |        |
| 243Cm   | 28.5y     | 14.3   | 0.6081 | 1.286-4  | 0.129 | 23.268 |         |           | 634.3  | 0.0150 |          |       |        |
|         |           | 44.7   | 0.0012 |          |       |        |         |           | 652.8  | 0.0014 |          |       |        |
|         |           | 57.3   | 0.0014 |          |       |        | 250Bk   | 3.2h      | 15.7   | 0.3118 | 1.834-4  | 3.474 | 0.862  |
|         |           | 67.8   | 0.0014 |          |       |        |         |           | 98.2   | 0.0012 |          |       |        |
|         |           | 99.6   | 0.1429 |          |       |        |         |           | 109.9  | 0.0026 |          |       |        |
|         |           | 103.8  | 0.2297 |          |       |        |         |           | 115.1  | 0.0041 |          |       |        |
|         |           | 106.1  | 0.0026 |          |       |        |         |           | 129.0  | 0.0020 |          |       |        |
|         |           | 117.0  | 0.1080 |          |       |        |         |           | 890.0  | 0.0164 |          |       |        |
|         |           | 209.7  | 0.0329 |          |       |        |         |           | 929.3  | 0.0137 |          |       |        |
|         |           | 228.2  | 0.1058 |          |       |        |         |           | 989.0  | 0.4510 |          |       |        |
|         |           | 254.4  | 0.0011 |          |       |        |         |           | 1029.0 | 0.0439 |          |       |        |
|         |           | 277.6  | 0.1397 |          |       |        |         |           | 1032.0 | 0.3510 |          |       |        |
|         |           | 285.4  | 0.0073 |          |       |        |         |           | 42.2   | 0.0004 |          |       |        |
|         |           | 167.0  | 0.0033 |          |       |        |         |           |        |        |          |       |        |
| 244Cm   | 18.1y     | 14.3   | 0.1031 | 1.741-5  | 0.003 | 1054   | 248Cf   | 333.5d    | 15.0   | 0.0791 | 1.229-5  | 0.002 | 1271   |
|         |           | 56.9   | 0.0003 |          |       |        |         |           | 42.9   | 0.0002 |          |       |        |
| 245Cm   | 8500.0y   | 14.3   | 0.6361 | 1.220-4  | 0.021 | 140.0  | 249Cf   | 350.6y    | 15.0   | 0.3026 | 1.119-4  | 0.866 | 3.459  |
|         |           | 42.0   | 0.0012 |          |       |        |         |           | 54.7   | 0.0021 |          |       |        |
|         |           | 99.6   | 0.1361 |          |       |        |         |           | 92.3   | 0.0030 |          |       |        |
|         |           | 103.8  | 0.2188 |          |       |        |         |           | 104.6  | 0.0219 |          |       |        |
|         |           | 117.0  | 0.1029 |          |       |        |         |           | 109.3  | 0.0350 |          |       |        |
|         |           | 133.0  | 0.0627 |          |       |        |         |           | 123.0  | 0.0166 |          |       |        |
|         |           | 174.0  | 0.0640 |          |       |        |         |           | 241.2  | 0.0022 |          |       |        |
| 246Cm   | 4750.0y   | 14.3   | 0.0918 | 1.551-5  | 0.003 | 1054   |         |           | 252.8  | 0.0273 |          |       |        |
|         |           | 44.5   | 0.0003 |          |       |        |         |           | 266.7  | 0.0075 |          |       |        |
| 247Cm   | 1.6+7y    | 14.3   | 0.0593 | 7.217-5  | 1.120 | 2.675  |         |           | 295.8  | 0.0014 |          |       |        |
|         |           | 99.6   | 0.0120 |          |       |        |         |           | 333.4  | 0.1551 |          |       |        |
|         |           | 103.8  | 0.0193 |          |       |        |         |           | 388.0  | 0.6600 |          |       |        |
|         |           | 117.0  | 0.0091 |          |       |        |         |           | 283.7  | 0.0043 |          |       |        |
|         |           | 275.1  | 0.0052 |          |       |        | 250Cf   | 13.1y     | 15.0   | 0.0780 | 1.212-5  | 0.002 | 1269.  |
|         |           | 278.0  | 0.0340 |          |       |        |         |           | 76.6   | 0.0003 |          |       |        |
|         |           | 287.4  | 0.0200 |          |       |        |         |           |        |        |          |       |        |
|         |           | 333.0  | 0.0034 |          |       |        | 251Cf   | 900.0y    | 15.0   | 0.5962 | 1.162-4  | 0.068 | 43.960 |
|         |           | 346.0  | 0.0130 |          |       |        |         |           | 61.5   | 0.0056 |          |       |        |
|         |           | 402.6  | 0.7200 |          |       |        |         |           | 68.0   | 0.0020 |          |       |        |
|         |           | 116.1  | 0.0011 |          |       |        |         |           | 73.0   | 0.0030 |          |       |        |
| 248Cm   | 3.4+5y    | 14.3   | 0.0727 | 1.227-5  | 0.003 | 1054.  |         |           | 83.0   | 0.0010 |          |       |        |
|         |           | 56.2   | 0.0003 |          |       |        |         |           | 104.6  | 0.1548 |          |       |        |
| 249Cm   | 1.1h      | 15.3   | 0.0038 | 3.982-6  | 2.025 | 1.479  |         |           | 109.3  | 0.2469 |          |       |        |
|         |           | 107.2  | 0.0008 |          |       |        |         |           | 123.0  | 0.1173 |          |       |        |
|         |           | 112.1  | 0.0012 |          |       |        |         |           | 135.0  | 0.0010 |          |       |        |
|         |           |        |        |          |       |        |         |           | 144.0  | 0.0010 |          |       |        |
|         |           |        |        |          |       |        |         |           | 154.0  | 0.0020 |          |       |        |
|         |           |        |        |          |       |        |         |           | 176.6  | 0.1770 |          |       |        |
|         |           |        |        |          |       |        |         |           | 214.0  | 0.0020 |          |       |        |

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